

Critical local smoothing for the three-dimensional wave equation

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September 24, 2026

Abstract

We prove the critical L^3 local smoothing estimate for the wave equation in three spatial dimensions, with every positive Sobolev loss. This resolves Sogge's Euclidean local smoothing conjecture in dimension three.

1 Introduction

Let

$$\widehat{f}(\xi) = \int_{\mathbb{R}^3} e^{-ix \cdot \xi} f(x) dx, \quad Uf(x, t) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{i(x \cdot \xi + t|\xi|)} \widehat{f}(\xi) d\xi.$$

For $\alpha \in \mathbb{R}$, write $J^\alpha = (1 - \Delta)^{\alpha/2}$, with Fourier multiplier $(1 + |\xi|^2)^{\alpha/2}$. For $p > 2$, the sharp fixed-time L^p estimate has Sobolev loss $2(1/2 - 1/p)$ in three dimensions [23, 17]. Time integration can recover part of that loss. Sogge's Euclidean local smoothing conjecture [28] predicts, for $2 < p < \infty$, the strict range $\alpha > \max\{0, 1 - 3/p\}$; its critical exponent is $p = 3$. Our main result treats that exponent.

Theorem 1.1 (Critical local smoothing). *For every $\varepsilon > 0$ there exists $C_\varepsilon < \infty$ such that*

$$\|Uf\|_{L^3(\mathbb{R}^3 \times [1, 2])} \leq C_\varepsilon \|J^\varepsilon f\|_{L^3(\mathbb{R}^3)} \quad (f \in \mathcal{S}(\mathbb{R}^3)). \quad (1.1)$$

The critical estimate supplies the interpolation point for the full Euclidean range: the L^2 energy estimate gives the exponents below $p = 3$, and an annular L^∞ kernel bound gives those above it. Frequency summation then yields Corollary 10.6:

$$\|Uf\|_{L^p(\mathbb{R}^3 \times [1, 2])} \leq C_{p, \alpha} \|J^\alpha f\|_{L^p(\mathbb{R}^3)}, \quad 2 < p < \infty, \quad \alpha > \max\{0, 1 - 3/p\}. \quad (1.2)$$

Together, these estimates resolve Sogge's Euclidean local smoothing conjecture in three spatial dimensions. At $p = 3$, the estimate holds with every positive Sobolev loss; the lossless estimate $\varepsilon = 0$ is not addressed.

There is also a pointwise consequence for the wave evolution. Define the finite-time maximal half-wave by $U_*f(x) = \sup_{0 < t < 1} |Uf(x, t)|$. For $3 \leq p < \infty$ and $s > 1 - 2/p$, Corollary 10.7 gives $\|U_*f\|_p \leq C_{p, s} \|J^s f\|_p$ and almost-everywhere convergence $Uf(x, t) \rightarrow f(x)$ as $t \downarrow 0$ for data with $J^s f \in L^p$. The established implication from local smoothing uses Sobolev embedding in time. Its extra cost explains the stronger regularity hypothesis: at $p = 3$, spacetime smoothing allows every positive loss, whereas this maximal conclusion requires $s > 1/3$.

The local-smoothing estimate also has a consequence for Fourier summation. For $R > 0$, $\delta > 0$, and $f \in \mathcal{S}(\mathbb{R}^3)$, define the Bochner–Riesz means and their maximal operator by

$$\begin{aligned} B_R^\delta f(x) &= \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{ix \cdot \xi} \left(1 - \frac{|\xi|^2}{R^2}\right)_+^\delta \widehat{f}(\xi) d\xi, \\ B_*^\delta f(x) &= \sup_{R>0} |B_R^\delta f(x)|. \end{aligned} \tag{1.3}$$

Propositions 3.2 and 3.3 of Beltran–Hickman–Sogge [2] turn local smoothing into Bochner–Riesz estimates. Section 11 applies them to (1.2): Theorem 11.1 bounds B_*^δ strongly on L^p and gives $B_R^\delta f \rightarrow f$ almost everywhere for $3 \leq p < \infty$ and $\delta > 1 - 3/p$. Separately, Corollary 11.2 gives uniform bounds for the individual means throughout the strict range $\delta > \max\{3|1/p - 1/2| - 1/2, 0\}$, $1 \leq p \leq \infty$. The individual bounds control Fourier summation uniformly in its radius. The stronger maximal bound controls all radii simultaneously and makes convergence stable under L^p approximation; this is what permits almost-everywhere recovery of arbitrary L^p data. Here the strict condition on the summation order absorbs the positive loss in local smoothing, so no Sobolev regularity of the input is required.

1.1 History and significance

Sogge introduced local smoothing in his study of circular maximal operators and their variable-coefficient analogues [28]. He proved a positive gain from time averaging in two spatial dimensions and proposed the critical higher-dimensional formulation in [28, Section 4, especially (4.5)]. The conjecture measures the improvement over the sharp fixed-time estimates of Peral and Miyachi [23, 17]. In three dimensions it asks to recover almost $1/p$ derivatives when $p \geq 3$; at $p = 3$, time averaging should remove all but an arbitrarily small part of the fixed-time loss of $1/3$. The connection with Fourier summation was already part of this formulation: Sogge showed how critical smoothing implies Bochner–Riesz bounds.

One approach separates the angular decomposition of the wave from the geometry of its rays. Mockenhaupt–Seeger–Sogge developed square-function estimates and their relation to local smoothing, including a framework for Fourier integral operators satisfying cinematic curvature [18, 19]. A square function sums the squared magnitudes of the angular pieces at each spacetime point before taking a norm. It therefore retains information about where those pieces overlap. Geometric estimates for thin tubes enter when this overlap is related to the initial data. This approach gave partial smoothing gains and remained important near the critical exponent.

Wolff obtained the conjectured gain for large p in two spatial dimensions by wave-packet localization and induction on scales [32]. His decoupling approach estimates the sum of angular pieces using their separate L^p norms. Łaba–Wolff extended this method to higher dimensions [15], and Garrigós–Seeger combined the induction scheme with bilinear restriction estimates to improve cone plate-decomposition inequalities [10]. These results distinguished sharp derivative gain from the full conjectured range of exponents. Bourgain–Demeter then proved sharp cone decoupling in every dimension [6, Theorem 1.2]. Its local-smoothing consequence gives the conjectured gain for $p \geq 4$ in three spatial dimensions, leaving the critical $p = 3$ estimate beyond that range.

Beltran–Hickman–Sogge extended the decoupling method to variable coefficients and proved sharp smoothing gains for wave equations on compact Riemannian manifolds and a broader class of Fourier integral operators [3]. The geometric obstruction depends on the setting. Minicozzi–Sogge

constructed metrics with concentrating families of geodesics [16]; in spatial dimension three these examples require $p \geq 10/3$ for the full smoothing gain on general manifolds [3, Section 1.1]. For the broader cinematic-curvature class, the $p \geq 4$ range proved by Beltran–Hickman–Sogge is sharp in dimension three [3, Theorem 1.2 and Proposition 1.3]. These distinctions explain the Euclidean qualification in the conjecture considered here. In the broader setting of uniformly elliptic, time-independent wave equations with real coefficients of bounded $C^{1,1}$ norm, Rozendaal–Schipa obtain a diagonal $p = 3$ estimate for $\alpha > 2/9$ [26, Corollary 1.2].

The two-dimensional completion illustrates the importance of retaining spatial overlap information. There Bourgain–Demeter’s theorem gives the full smoothing gain for $p \geq 6$; Guth–Wang–Zhang reached the critical exponent $p = 4$ through a sharp cone square-function estimate [12]. Their stronger wave-envelope estimate tracks how packets occupy regions at intermediate angular scales and exploits sparse packet configurations. This has a direct role in the present proof: we use the wave-envelope theorem as a lower-dimensional input, while the three-dimensional argument must control the additional direction and the changes of geometry between scales.

A further route uses transverse interactions between wave packets. The multilinear estimates of Bennett–Carbery–Tao and the multilinear-to-linear reduction of Bourgain–Guth [4, 7] lead to a broad–narrow strategy: separate contributions from many distinct directions from those concentrated near a lower-dimensional direction space. Guth’s polynomial-partitioning method provides a weaker k -broad estimate for the former, while the latter requires rescaling and induction [11]. Ou–Wang adapt this architecture to the cone, combining broad estimates with narrow decoupling and Lorentz rescaling [22].

Gao–Liu–Miao–Xi use this broad–narrow architecture for local smoothing. Rescaling changes the class of phases to be controlled; their argument treats general positively curved cones and combines broad estimates with narrow decoupling in scale-dependent phase classes [9]. In three spatial dimensions they obtain, at $p = 3$, every Sobolev exponent $\alpha > 1/9$, or equivalently a smoothing gain below $2/9$ [9, Corollary 1.3].

Gan–He–Li–Wu combine refined decoupling with estimates for the local L^2 energy and incidence geometry of wave packets [8]. Their packet density measures energy in windows compatible with Lorentz rescaling. A simultaneous broad and two-ends reduction retains transverse contributions and controls concentration in short portions of packets; in dimension three, slab coverings sharpen the energy and incidence estimates together. Theorem 1.3 of their January 2026 version gives the conjectured gain for $p \geq 10/3$, but does not include $p = 3$. Interpolation of the $p = 10/3$ estimate with the L^2 energy bound gives $\alpha > 1/12$ at $p = 3$, improving the preceding $1/9$ threshold. The remaining fixed positive loss is the gap addressed by Theorem 1.1.

Our proof also studies packet concentration across scales, but its central quantity is the entropy of a finite phase-space record. The records retain the changing shear frames as well as the time and direction information. The proof uses two external quantitative inputs: the planar Furstenberg theorem of Ren–Wang [25, Theorem 4.1] and the cone wave-envelope theorem of Guth–Wang–Zhang [12, Theorem 1.3]. The former follows a line of discretized incidence and projection work including Katz–Tao, Orponen–Shmerkin, and Shmerkin–Wang [14, 21, 27]. Section 4 states these inputs and proves the specialized probabilistic and Hilbert-valued consequences used below. The extremal reductions and entropy inequalities then connect this lower-dimensional information to the three-dimensional wave estimate.

1.2 Structure of the argument

We first seek the estimate on one frequency annulus $|\xi| \asymp H^{-1}$: the physical half-wave should map L^3 into spacetime L^3 with a loss of at most $H^{-\delta}$ for every $\delta > 0$. Summing these estimates gives Theorem 1.1. On a fixed angular chart, null coordinates turn the light rays into rays with velocities $(A, |A|^2)$, $A \in \mathbb{R}^2$. We analyze the wave into packets and measure their energy by squared L^2 norms.

At the terminal time length H , the packet positions have width H and their angular aperture has size one. Write $m_{j,Q}$ for the packet energy assigned to a spatial H -cube Q at the start of the j th time interval. The synthesis estimate (10.21) controls the physical cubed L^3 norm by $H^{-1/2} \sum_{j,Q} m_{j,Q}^{3/2}$, with an arbitrarily small power loss and a negligible error. Thus the main task is to bound a cubic sum of packet masses. Energy is copied by evolution into descendant time intervals; those descendants are not orthogonal pieces of the original input.

The intermediate scales require more flexible position and direction windows. For the round model $(A, |A|^2)$, a window of angular radius r has position widths ℓr and ℓr^2 at time length ℓ ; a parameter $f \leq 1$ measures its relative thickness near a hyperquadric. Thin round windows lead to the cylinder (X, X^2, N) . There r controls the curved base direction, while f is the normal velocity width, with normal position width ℓf after a polynomial shear. The models therefore use different test weights: $r^4 f^{1-\eta}$ in the round model and $r^{3+\eta} f^{1-\eta}$ in the cylinder, where $0 < \eta < 1$. These windows and their weights are defined in Definitions 3.2 and 8.2.

Suppose the initial packet measure has total energy e and obeys $\mu(Q) \leq \kappa w(Q)$ for every test. At the final cut, let m_λ be the energy assigned to a test of weight w_λ . The index λ includes both its time interval and its test; assignments may be fractional, but their sum at each interval is bounded by that interval's row energy. The normalized cubic sum is

$$\mathcal{B}_s = \frac{s^{-1/2}}{e\sqrt{\kappa}} \sum_{\lambda} \frac{m_{\lambda}^{3/2}}{\sqrt{w_{\lambda}}}. \quad (1.4)$$

Here s is the ratio between final and initial time lengths, and the sum includes all final intervals. In the physical application $s = H$ and the terminal tests have weights comparable to one, recovering the mass sum above. Theorem 9.1 bounds $\mathcal{B}_s$ with a scale loss that tends to zero with η and the subsequent losses. Section 10 verifies the required initial regularity on the physical time slice and completes the frequency summation.

The model bounds are proved by contradiction. Suppose that the cubic sum has a growth exponent larger than the permitted η -dependent loss. In the cylinder model, an extremal reduction selects homogeneous mass and weight data. Any ordinary regularity split that preserves the extremal exponent must attain the corresponding energy and regularity bounds at the level of exponents. Comparing the retained tests then controls both their polynomial shears and their widths at successive cuts. After a further reduction, the width exponents are affine functions of depth. The lower-dimensional wave and incidence estimates exclude the configurations in which both normalized shear costs vanish.

For the remaining configurations, we study a finite phase name at time depth a and horizontal resolution v . It records bins for time, direction, base position, and the normal phase after subtracting a recorded shear. From its normalized entropy we subtract the baseline prescribed by regularity, the test weight, and the extremal mass data. The resulting entropy excess is nonnegative. At specified boundary resolutions, recovery of the name from a retained test label gives the matching upper bound, so the excess vanishes there. Conditional sampling supplies

auxiliary probes, but the quantity ultimately estimated remains the entropy of this one phase name. Planar incidence and local projection estimates give directions in which the excess must decrease when followed backwards from those boundary points. Constructing a path that stays in the domain of the applicable inequalities gives the contradiction to nonnegativity. The admissible directions depend on the shear profile; unit-cost boundaries and intervals with no angular entropy growth require separate estimates.

To pass from the cylinder to the round velocity surface $(A, |A|^2)$, $A \in \mathbb{R}^2$, we transfer thin round tests to cylindrical ones. If one coordinate remains quantum, this transfer retains its one-dimensional Schrödinger evolution. The remaining round configurations are excluded by a separate entropy argument, first for classical rays and then for waves. Finally, input regularity is verified on the original physical time slice before summing the dyadic frequency estimates.

The paper is organized as follows. Definitions 3.1, 3.2, and 3.3 specify the cylinder states, tests, and fixed setups; Proposition 3.16 gives the extremal configurations used in the subsequent argument. Section 4 supplies the lower-dimensional estimates. Definition 5.2 introduces the canonical names, and Proposition 5.4 defines their entropy excess. Section 6 derives the local inequalities, which Section 7 uses in the proof of Theorem 3.6. Definitions 8.1 and 8.2 specify the round experiment and its tests. Proposition 8.8 is the thin-shape reduction used in Theorem 9.1; Section 10 then makes the passage to the physical wave. Section 11 derives the maximal and nonmaximal Fourier-summation consequences and records their standard restriction and Keakeya implications.

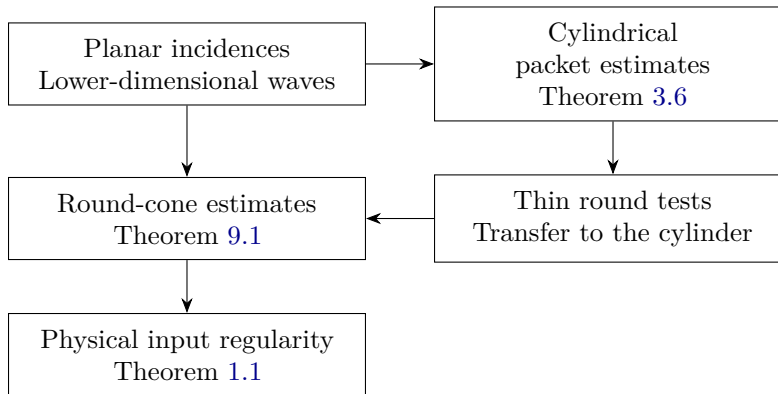


Figure 1: Dependencies of the main estimates. Arrows indicate which estimates are used in subsequent arguments.

The principal technical steps are the extremal reduction with compatible shears, the phase-name calculus, and the uniform transfer of wave packets on thin gaps. These intermediate results are formulated for the packet models considered here, independently of the final frequency summation. Throughout, finite operator stacks and polynomial cutoff bounds are fixed before taking a scale limit; Section 2 specifies the order of all subsequent limits.

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2 Scales, energies, and limiting arguments

The arguments below compare experiments whose parameters and probability spaces vary with scale. This section records the meaning of those comparisons. It does not assert that a geometrically defined subexperiment is admissible: its support, regularity, and operator bounds must be checked in the section where it is constructed.

2.1 Fixed setups and exponent bounds

We use dyadic scales $s = 2^{-n}$ tending to zero. A depth a represents the scale s^a . Replacing it by the nearest dyadic scale changes a positive quantity by a bounded factor and hence changes its normalized logarithm by $O(1/\log(1/s))$. Width exponents and growth exponents have opposite sign conventions:

$$\text{width exponent}(r) = \frac{\log r}{\log s}, \quad \text{growth exponent}(B) = \frac{\log B}{\log(1/s)}.$$

The growth exponent of a zero quantity is $-\infty$.

Definition 2.1 (Fixed polynomial setup). A fixed polynomial setup specifies all structural constants, the finite upper bounds on powers of s^{-1} occurring in its hypotheses, and a finite bound on the number of operations along each branch. These data are fixed before $s \rightarrow 0$. A sequence in this setup may have parameters, operators, measures, and Hilbert spaces depending on s , provided the specified bounds hold uniformly along the sequence.

An error r_s is negligible relative to a positive reference quantity a_s if, for every $N > 0$,

$$|r_s| \leq C_N s^N a_s$$

for all sufficiently small scales, with C_N independent of scale. For a family of choices at each scale, uniform negligibility means that the same constants and scale thresholds work for every choice. The reference quantity is always specified; it may be an input energy or an operator norm. Polynomially many errors can be added without losing negligibility when their reference quantities and polynomial bounds are controlled uniformly.

For a nonnegative functional $\mathcal{B}_s$, define its upper exponent to be

$$\Gamma = \sup_{\mathcal{S}} \sup_{(s_k, \mathcal{E}_k)} \limsup_{k \rightarrow \infty} \frac{\log \mathcal{B}_{s_k}(\mathcal{E}_k)}{\log(1/s_k)}. \quad (2.1)$$

Here $\mathcal{S}$ ranges over fixed polynomial setups, and the inner supremum ranges over admissible sequences in that setup with $s_k \rightarrow 0$. The particular model specifies which setups are allowed. The finiteness of Γ must be proved for that model.

Lemma 2.2 (Uniform bound within a fixed setup). *Suppose $\Gamma \in \mathbb{R}$ in (2.1). Fix a family $\mathcal{A}_s$ such that every sequence of choices $\mathcal{E}_k \in \mathcal{A}_{s_k}$, $s_k \rightarrow 0$, is admissible in one fixed polynomial setup. In particular, all negligible tail hypotheses are uniform over this family. For every $\varepsilon > 0$ there is $s_0 > 0$ such that*

$$\mathcal{B}_s(\mathcal{E}) \leq s^{-\Gamma-\varepsilon} \quad (0 < s < s_0, \mathcal{E} \in \mathcal{A}_s). \quad (2.2)$$

If the functional is also uniformly bounded on this family at each of the finitely many larger dyadic scales under consideration, then (2.2) holds on the entire scale range after inserting a constant $C_{\varepsilon, \mathcal{S}}$.

Proof. Failure of the first assertion provides $s_k \rightarrow 0$ and $\mathcal{E}_k \in \mathcal{A}_{s_k}$ with $\mathcal{B}_{s_k}(\mathcal{E}_k) > s_k^{-\Gamma-\varepsilon}$. The resulting sequence is admissible in the fixed setup and has upper exponent at least $\Gamma + \varepsilon$, contradicting (2.1). For the second assertion, take the maximum of the finitely many required constants at the remaining scales and 1. $\square$

The constant in Lemma 2.2 may depend on every bound fixed in Definition 2.1. It is not uniform when those bounds grow along an outer approximation. Conversely, a failure of an estimate with a fixed positive exponent slack can be disproved within one fixed setup, even when that setup was selected late in an outer argument.

Remark 2.3 (Small normalized pieces). Changing the reference energy can change a tail hypothesis. For example, an error negligible relative to e is negligible relative to e' whenever $e'/e \geq s^C$ for some fixed C , but this implication does not hold for arbitrary $e' > 0$. Applications to a collection of subexperiments must therefore either verify the hypotheses relative to each new energy, or use a direct estimate on pieces with superpolynomially small energy. No uniformity assertion in Lemma 2.2 supplies this missing verification.

2.2 Energy and finite extraction

Energy means a squared Hilbert norm. Input energy at a time interval means the energy before the evolutions and masks being estimated. When a time interval is subdivided, its input is copied into its descendants. Consequently the sum of input energies over distinct starting intervals need not be bounded by the original energy. This sum will always be distinguished from the energy available to one interval.

Lemma 2.4 (Disjoint row restrictions and copies). *Let $V : \mathcal{H} \rightarrow \mathcal{K}$ be an isometry and let $P_1, \dots, P_N$ be mutually orthogonal projections on $\mathcal{K}$. For $g_j = V^* P_j V g$,*

$$\sum_{j=1}^N \|g_j\|_{\mathcal{H}}^2 \leq \sum_{j=1}^N \|P_j V g\|_{\mathcal{K}}^2 \leq \|g\|_{\mathcal{H}}^2. \quad (2.3)$$

If $\sum_j P_j V g = V g$, then $\sum_j g_j = g$. By contrast, for N contractions T_j applied to separate copies of g , the general bound is only $\sum_j \|T_j g\|^2 \leq N \|g\|^2$.

Proof. The adjoint V^* has norm one. Apply this fact to each $P_j V g$ and then use orthogonality. The reconstruction assertion follows from $V^* V = \text{Id}$. The assertion about copies follows by applying $\|T_j g\| \leq \|g\|$ separately to each j . $\square$

The projections in Lemma 2.4 may be multiplication by disjoint measurable row sets in a direct integral. Their synthesized images need not be orthogonal. For the weighted sums used here, this is accommodated by a norm inequality, not by an orthogonality claim.

Lemma 2.5 (Extraction from finitely many classes). *Let $q \geq 1$, and suppose that a nonnegative functional F satisfies*

$$F(g_1 + \dots + g_N)^{1/q} \leq \sum_{j=1}^N F(g_j)^{1/q}.$$

Then some j satisfies $F(g_j) \geq N^{-q} F(\sum_i g_i)$. Consequently, if $N = N_s$ obeys $\log N_s / \log(1/s) \rightarrow 0$, passing to a maximizing class costs zero growth exponent. The same conclusion holds for a fixed

finite number of successive such extractions. To use the conclusion for a model exponent, each selected class must itself be admissible in a fixed setup.

In particular, for linear maps A_λ into Hilbert spaces and positive weights w_λ ,

$$F(g) = \sum_{\lambda} w_{\lambda}^{-1/2} \|A_{\lambda}g\|^3 \quad (2.4)$$

has the required inequality with $q = 3$.

Proof. The right side of the assumed inequality is at most $N \max_j F(g_j)^{1/q}$. Raising to the q th power proves the extraction estimate. Its logarithmic loss is $q \log N_s / \log(1/s)$; finitely many such losses still tend to zero. For (2.4), $F(g)^{1/3}$ is the norm of $(w_{\lambda}^{-1/6} A_{\lambda}g)_{\lambda}$ in the corresponding ℓ^3 sum of Hilbert spaces, and Minkowski's inequality applies. For a countable label set, apply the inequality first to finite subsets and then use monotone convergence. $\square$

A separate elementary extraction will be used when energies are already summable: if $0 < E < \infty$, $A > 0$, $a_j \geq 0$, $e_j \geq 0$, $\sum_j e_j \leq E$, and $\sum_j a_j \geq A$, then $\sup_{j:e_j>0} a_j/e_j \geq A/E$, provided $a_j = 0$ whenever $e_j = 0$. Indeed the contrary strict upper bound, multiplied by e_j and summed, contradicts the two hypotheses. Any additional count or regularity constraints on the chosen class require their own discard estimates.

2.3 Closure of exponent data

For a finite list of positive quantities, exponent data are the limits of their normalized logarithms along admissible sequences, when these limits exist and are finite. We may also take the closure of such data in a fixed finite-dimensional Euclidean space. This outer closure permits a different fixed polynomial setup for each approximating sequence. It does not turn these sequences into one admissible sequence with unbounded setup parameters.

Lemma 2.6 (Extrema of bounded exponent data). *Let $K \subset \mathbb{R}^{m+1}$ be a nonempty compact set of feasible limiting data $(g, x_1, \dots, x_m)$. Set $G = \max_K g$, minimize x_1 on $K \cap \{g = G\}$, and then successively minimize x_j on the slice where all preceding extrema are attained. Every slice is nonempty and compact, and every stated minimum is attained.*

The attainment assertions, and nonemptiness and closedness of the successive slices, remain valid for a possibly noncompact closed set K under the following hypotheses: g has a finite supremum and a bounded maximizing sequence in K , and each successive coordinate has a finite infimum and a bounded minimizing sequence within the preceding extremal slice. This alternative does not assert compactness of the slices. A new coordinate may be added to the tuple only after the required boundedness and feasibility have been proved for the enlarged tuple.

Proof. A continuous coordinate function attains its extremum on a nonempty compact set. The set where that extremum is attained is closed in the same compact set and is nonempty. Induction proves the first assertion. For the alternative, pass first to a convergent subsequence of the bounded maximizing sequence. Closedness and continuity attain the supremum of g . At each subsequent step, pass to a convergent subsequence of the stipulated bounded minimizing sequence. Closedness preserves feasibility, and continuity of the coordinates preserves the preceding equalities and attains the next infimum. Each resulting slice is a nonempty intersection of closed sets. $\square$

Remark 2.7. Boundedness of a new coordinate cannot be inferred from compactness of previously chosen coordinates. For example, the closure of the projection of $\{(p, c) : c \geq 1, p = 1/c\}$ onto the p axis contains $p = 0$, although the closure of the full set has no finite point with $p = 0$. Thus a geometric bound on the curvature cost at the already chosen growth and mass exponents is an essential part of a curvature minimization argument.

Lemma 2.8 (Strict improvement in a closure). *Let $K \subset \mathbb{R}^{k+1}$ be a closed set of feasible data. Suppose that c_0 is the minimum of the last coordinate c on the nonempty slice $K \cap \{z = z_0\}$. There cannot be data $(z_n, c_n) \in K$ with $z_n \rightarrow z_0$ and $\limsup_n c_n < c_0$, provided (c_n) is bounded below and has a bounded subsequence. In particular, a construction contradicts minimality when its preceding exponents converge to the specified extrema and its cost improves by a fixed positive amount in the units of the constructed experiment.*

Proof. Pass to a convergent subsequence of (c_n) . Its limit c is strictly less than c_0 . Closedness places (z_0, c) in the minimizing slice, which is a contradiction. $\square$

These lemmas assert neither an exactly extremizing input nor an infinite chain of operators. A construction at limiting exponent data must be performed on approximating finite experiments. The estimates must then imply the asserted bounds on a convergent subsequence of the finite tuples actually used.

2.4 Ordered limits and changes of scale

At a depth gap $h > 0$, the new basic scale is $\delta = s^h$. A factor $s^{-\alpha}$ has growth exponent α/h in these new units. Consequently a loss tending to zero in the original units is harmless on a fixed gap, but it is harmless on shrinking gaps only after it has been made $o(h)$.

All uses of mesh refinement or tangent gaps have the following order. First fix the positive gap, the finite list of cuts and observations, and all cutoff depths needed for that comparison. In each fixed polynomial setup take the scale limit. Next make the preceding approximation errors and mesh errors as small as required relative to that fixed gap. Only then shrink the gap or enlarge the finite list. When further tangent layers are used, their positive lengths and required accuracies are fixed before the preceding tangent limit is taken. Each application of a model exponent is to an admissible sequence at a fixed stage of these outer choices.

Lemma 2.9 (Pullback of a strict exponent contradiction). *Let $\Gamma \in \mathbb{R}$ be an upper exponent as in (2.1). Suppose that, for each integer m , a construction produces an admissible sequence in a fixed setup $\mathcal{S}_m$ with upper exponent at least $\Gamma + \tau - r_m$, where $\tau > 0$ is fixed and $r_m \rightarrow 0$. Then such a construction is impossible. The setup need not be uniform in m .*

Proof. Choose one fixed m with $r_m < \tau/2$. The sequence in $\mathcal{S}_m$ then contradicts the definition of Γ . $\square$

Having one favorable scale for each m would not satisfy the hypothesis of Lemma 2.9. For example, the quantities $B_m(2^{-n})$, equal to 2^n when $n = m$ and equal to 1 otherwise, have upper exponent zero for every fixed m but exponent one along the diagonal $m = n$. We will therefore use a diagonal of scales for probability comparisons only after the necessary model estimates with strict slack have been established in their fixed setups.

2.5 Elementary probability comparisons

All random variables below are evaluated under the probability law specified at that point of the argument. A conditional probability may condition on a continuous variable or a sigma-algebra. We use natural logarithms. Under a law P , for a finite-valued name X , write

$$H_P(X | Y) = -\mathbb{E}_P \log P(X | Y).$$

Relative entropy is $\text{KL}(P||Q) = \int \log(dP/dQ) dP$ when $P \ll Q$, and is $+\infty$ otherwise. Conditional mutual information is

$$I_P(X; V | Y) = \mathbb{E}_{P_Y} \text{KL}(P_{X,V|Y} || P_{X|Y} \otimes P_{V|Y}).$$

This definition also permits continuous variables; when X is finite, it equals $H_P(X | Y) - H_P(X | V, Y)$. For $0 < \rho < 1$, the scale-normalized quantities are $H_\rho = H_P / \log(1/\rho)$ and $I_\rho = I_P / \log(1/\rho)$, with the law indicated when it changes. For a finite-valued name X and conditioning data Y , write

$$i_s(X | Y) = -\frac{\log \mathbb{P}(X | Y)}{\log(1/s)}, \quad (2.5)$$

where the conditional probability is evaluated at the sampled value of X . Values on null events are immaterial. Thus $H_s(X | Y) = \mathbb{E}_P i_s(X | Y)$: the score is random, whereas entropy and mutual information are averaged quantities.

Lemma 2.10 (Capacity tails). *Suppose that, given Y , the variable X has at most N possible values almost surely. For every $u \geq 0$,*

$$\mathbb{P}\{\mathbb{P}(X | Y) < e^{-u}/N\} \leq e^{-u}. \quad (2.6)$$

If $N \leq s^{-C}$ for a fixed C , then $\mathbb{P}\{i_s(X | Y) > C + t\} \leq s^t$ for $t \geq 0$. These normalized scores are uniformly integrable as $s \rightarrow 0$. In particular, convergence in probability to a finite constant implies convergence of their expectations to that constant.

Proof. For each conditioning value, sum the conditional probabilities of the at most N values below e^{-u}/N , and then integrate. This proves (2.6) and its normalized version. For $K > C$, the tail integral identity gives

$$\mathbb{E}[i_s(X | Y) \mathbf{1}_{\{i_s(X | Y) > K\}}] \leq s^{K-C} \left(K + \frac{1}{\log(1/s)} \right).$$

For $s \leq 1/2$ the right side tends to zero uniformly as $K \rightarrow \infty$. This is uniform integrability; truncation at K proves the final assertion. $\square$

The same estimate applies to a relative name with at most N refinements after its parent has been included in the conditioning. It does not require a bounded number of values for the conditioning data. When absolute names have unbounded capacity exponents along an outer limit, the lemma supplies uniform integrability only for the relative names whose capacities have been bounded.

Lemma 2.11 (Stability under restriction). *Let $Q = P(\cdot | E)$, where $P(E) = q > 0$, and let X be a finite name with arbitrary conditioning data Y . If $q \geq s^\alpha$, then, outside a set of Q -probability at most $2s^\tau$,*

$$|i_s^Q(X | Y) - i_s^P(X | Y)| \leq \alpha + \tau \quad (\tau > 0). \quad (2.7)$$

Consequently, restrictions with $q = s^{o(1)}$ preserve deterministic limiting scores, provided their previous exceptional probabilities are negligible relative to q . On a gap with scale s^h , the corresponding condition is $\log(1/q) = o(h \log(1/s))$ and all errors are divided by h .

Proof. For the marginal of any variable W , the likelihood ratio $L_W = dQ_W/dP_W$ is $P(E | W)/q$, hence is at most $1/q$. Moreover $Q\{L_W < s^\tau\} \leq s^\tau$ by integrating L_W against P_W . Apply these two facts to $W = (X, Y)$ and $W = Y$. Except on the union of the two exceptional sets, both ratios belong to $[s^\tau, s^{-\alpha}]$. The conditional likelihood ratio is $L_{(X,Y)}/L_Y$, so it belongs to $[s^{\alpha+\tau}, s^{-\alpha-\tau}]$. Taking logarithms proves (2.7). An old exceptional event of P -probability r has Q -probability at most r/q , proving the stated qualification and the consequences. $\square$

Lemma 2.12 (Negligible information and positive restrictions). *Let U, V be finite names, with arbitrary conditioning data Y . Put $L = \log(1/s)$ and*

$$j = \log \frac{P(U | V, Y)}{P(U | Y)}.$$

Then $P\{j < -r\} \leq e^{-r}$ for $r \geq 0$, and $\mathbb{E}j = I_P(U; V | Y)$. If $I_P(U; V | Y) = o(L)$, then $j/L \rightarrow 0$ in probability. If $Q = P(\cdot | E)$ and $P(E) \geq q_0 > 0$, then

$$I_Q(U; V | Y) \leq \frac{I_P(U; V | Y) + \log 2}{q_0}. \quad (2.8)$$

In particular such restrictions preserve $o(L)$ information budgets. No such conclusion is asserted for arbitrary events of probability $s^{o(1)}$.

Proof. Condition on Y and compare the joint law of (U, V) with the product of its conditional marginals. At actual samples their likelihood ratio is e^j , so integrating over the set where it is less than e^{-r} proves the tail bound. Therefore $\mathbb{E}(j_-) \leq 1$, while the definition of conditional mutual information gives $\mathbb{E}j = I_P(U; V | Y)$. It follows that $\mathbb{E}(j_+) \leq I_P(U; V | Y) + 1$. Markov's inequality and the negative tail bound prove convergence in probability after division by L .

Let $B = \mathbf{1}_E$. The chain rule gives

$$I_P(U; V | Y, B) \leq I_P(U; V, B | Y) \leq I_P(U; V | Y) + H_P(B | V, Y) \leq I_P(U; V | Y) + \log 2.$$

The left side is the average, over the two values of B , of the corresponding conditional mutual informations. Its contribution from $B = 1$ is $P(E)I_Q(U; V | Y)$, proving (2.8). $\square$

The elementary comparisons in this subsection refer to actual laws and finite lists of names. Later score limits may be obtained from joint weak subsequences of those finite lists; no convergence of conditional sampling kernels is required or asserted. Extending the limits to a dense set of depth queries requires the specific nesting and list bounds proved for those names. In particular, these probability lemmas neither couple input and output packet marginals through a quantum mask nor transfer a profile between different measuring laws without a separate argument.

3 Cylinder experiments and extremal configurations

This section defines the two cylinder models and constructs the configurations used in the cylinder estimates. A state is measured by its squared Hilbert norm. Time subdivision copies that state into its descendant intervals; it does not divide its energy among them. This distinction explains the growth factor in the normalization below.

3.1 States, tests, and assignments

Fix $0 < \eta < 1$ and $s = 2^{-n}$. A cut of length $\ell = 2^{-k}$, with $0 \leq k \leq n$, consists of the dyadic subintervals of $[0, 1)$ of that length. The final cut has length s and contains exactly s^{-1} intervals. We measure an interval at its left endpoint T . There are finitely many operator cuts, in nonincreasing order of length; repeated cuts are permitted. Their number is bounded on each polynomial setup. Every child receives the freely evolved state of its parent, with the operators specified on that branch.

The exact base index is $F = (X, B, Y)$, with

$$B_T = B + TX, \quad Y_T = Y + TX^2. \quad (3.1)$$

Additional exact indices range over standard Borel spaces; arbitrary separable Hilbert multiplicities are allowed. All integrals and squared norms include these indices. One may equivalently use a measurable direct integral of separable Hilbert spaces over such an index space.

Definition 3.1 (Cylinder states). In the *classical cylinder*, (N, D) is another exact index, $D_T = D + TN$, and free evolution leaves the state unchanged in these moving coordinates. In the *hybrid cylinder*, $H > 0$ is fixed, there is an exact index $\sigma \in [1, 2]$, and a fiber state belongs to $L^2(\mathbb{R}_D; \mathcal{H})$. Its free propagator $U_t^{H, \sigma}$ is

$$\widehat{U_t^{H, \sigma}} g(p) = e^{-itHp^2/(2\sigma)} \widehat{g}(p).$$

At length ℓ , set

$$d_\ell = (H/\ell)^{1/2}, \quad q_\ell = (H\ell)^{1/2}, \quad \phi_{D, N}^{\ell, \sigma}(z) = (\pi q_\ell^2)^{-1/4} e^{-(z-D)^2/(2q_\ell^2) + i\sigma Nz/H}.$$

The analysis operator is

$$(V_\ell g)(D, N) = \langle g, \phi_{D, N}^{\ell, \sigma} \rangle, \quad d\nu_\sigma(D, N) = \frac{\sigma}{2\pi H} dD dN.$$

For the classical model, $V_\ell = \text{Id}$ and $d_\ell = 0$. An admissible operator at a cut is $V_\ell^* M V_\ell$, where M is a measurable pointwise contraction on the multiplicity space and preserves all exact indices. The row measure μ_I is the squared norm of the analyzed state, possibly restricted by row assignments.

With the Fourier normalization of Section 1, Plancherel in N gives

$$\int_{\mathbb{R}} |\langle g, \phi_{D, N}^{\ell, \sigma} \rangle|^2 \frac{\sigma dN}{2\pi H} = \int_{\mathbb{R}} \|g(z)\|_{\mathcal{H}}^2 (\pi q_\ell^2)^{-1/2} e^{-(z-D)^2/q_\ell^2} dz.$$

Integration in D proves $V_\ell^* V_\ell = \text{Id}$. In particular, every admissible operator is a contraction. The packet parameter D in the hybrid model is an observation coordinate, not an exact classical index.

Definition 3.2 (Cylinder tests). A test Q at (T, ℓ) has widths $r > 0$, $f \geq d_\ell$, with $f > 0$. For a hybrid row observed at the cut T , D_T in the following inequalities denotes its instantaneous analyzed center D , not an exact classical trajectory. The test is specified by

$$|X - x| \leq r, \quad |B_T - b| \leq \ell r, \quad |Y_T - y - 2x(B_T - b)| \leq \ell r^2, \quad (3.2)$$

$$|N - c - \alpha X - \beta(T)X^2 - K(Y_T - 2XB_T)| \leq f, \quad (3.3)$$

$$|D_T - d - \alpha B_T - \beta(T)Y_T + KB_T^2| \leq \ell f. \quad (3.4)$$

All centers and coefficients are real. Extend the coefficients by

$$\beta(t) = \beta(T) + (t - T)K, \quad d(t) = d + (t - T)c.$$

The test weight is $w(Q) = r^{3+\eta}f^{1-\eta}$. A row measure is κ -regular if $\mu(Q) \leq \kappa w(Q)$ for every admissible test.

If the expressions subtracted from N, D_T in Equations (3.3)–(3.4) are $A(F)$ and $S_t(F)$, respectively, direct differentiation gives $\partial_t A = 0$ and $\partial_t S_t = A$. Also

$$P(F) = \alpha + 2\beta(t)X - 2KB_t \tag{3.5}$$

is independent of t .

An assignment to the final cut is a family of nonnegative row measures $\mu_\lambda \leq \mu_I$, each supported on a test Q_λ at its own interval, such that $\sum_{\lambda \text{ at } I} \mu_\lambda \leq \mu_I$. Fractional assignments are allowed. Put $m_\lambda = \mu_\lambda(Q_\lambda)$ and $w_\lambda = w(Q_\lambda)$. If $e > 0$ is the root energy and the root row measure is κ -regular, define

$$\mathcal{B}_s = \frac{s^{-1/2}}{e\sqrt{\kappa}} \sum_{\lambda} \frac{m_\lambda^{3/2}}{\sqrt{w_\lambda}}. \tag{3.6}$$

The sum includes all final intervals. Bounded enlargements of the defining inequalities are interchangeable with a bounded change of the weight: enlarge r, f by fixed constants or cover by finitely many translates.

3.2 Cutoffs and localization

Definition 3.3 (Polynomial cylinder setup). The quantities $e/\kappa, \kappa/e$ and, in the hybrid case, H, H^{-1} are bounded by fixed powers of s^{-1} . The initial base support lies in a box of fixed-power size. The initial normal phase has negligible mass relative to e outside such a box. The marginal base density and the joint density in (X, B, Y, N, D) are bounded by e times a fixed power of s^{-1} , with respect to Lebesgue measure. Negligibility is uniform within each family to which an exponent bound is applied. All these powers, and the bound on the number of operators on a branch, are fixed before $s \rightarrow 0$.

The exponent closure may use successively larger fixed powers, in the ordered sense specified in the framework. It never replaces the inner scale limit by isolated scales with increasing cutoff powers.

Lemma 3.4 (Symmetries and Gaussian localization). *The models are invariant under translations, base boosts, dilation in base units (r, r, r^2) , normal dilation, affine time normalization, and subtraction of one common normal shear of Equations (3.3)–(3.4). All operators and assignments must be transformed with the state.*

For $\ell' \leq \ell$, $|t| \leq C\ell$, and fixed $\varepsilon > 0$, the hybrid frame transfer is negligible in operator norm outside

$$|N' - N| \leq s^{-\varepsilon} d_{\ell'}, \quad |D' - D - tN| \leq s^{-\varepsilon} q_{\ell}. \tag{3.7}$$

This remains true along any fixed finite stack of masks and after summing over a polynomial number of intervals. At a repeated cut, synthesis and reanalysis of a restricted κ -regular row measure produce a $C_\eta \kappa$ -regular measure. The same assertion holds for truncated families of tests whenever the normal enlargements in the proof belong to that family or are covered by translates belonging to it.

Proof. A base boost replaces X by $X - x_0$ and replaces the last base position by its appropriate linear tilt in B_t ; Equation (3.1) then has the same form. A common normal shear subtracts $a(F)$ from velocity and $b(F) + ta(F)$ from position. On each hybrid fiber it is a Weyl translation, and the free propagator intertwines the two translations up to a scalar phase. The scalar phase can be included in the conjugated mask. If a time interval of length L is normalized to unit length, positions are divided by L , velocities are unchanged, and H becomes H/L . Dividing normal coordinates and velocities by F_0 changes H to H/F_0^2 . Dividing base coordinates by (r_0, r_0, r_0^2) and normal coordinates by F_0 divides all weights by $A = r_0^{3+\eta} F_0^{1-\eta}$ and multiplies the regularity constant by A . Energy is preserved by the corresponding unitary coordinate changes. Thus Equation (3.6) is unchanged.

The scalar transfer kernel is $\langle U_t^{H,\sigma} \phi_{D,N}^{\ell,\sigma}, \phi_{D',N'}^{\ell',\sigma} \rangle$. Free propagation changes the first packet's position width to at most Cq_ℓ , since $|t|H/q_\ell \leq Cq_\ell$, and leaves its Fourier width comparable to q_ℓ^{-1} . Integrating two Gaussians, in physical space for the position difference and in Fourier space for the momentum difference, gives an upper bound of the form

$$C_s \exp \left[-c \left(\frac{|D' - D - tN|^2}{q_\ell^2} + \frac{|N' - N|^2}{d_\ell^2} \right) \right],$$

where C_s and the Schur integration volumes have only fixed-power costs on the setup. Each of the two separate Gaussian bounds implies the displayed joint bound after reducing c . Schur's test therefore makes the discarded kernel smaller than every power of s . A finite product and a polynomial sum preserve this conclusion. If no chain of the correspondences connects two row sets, the operator between those sets is negligible; this statement does not require a positive-kernel description of its main part.

At a repeated cut the absolute kernel is exactly

$$\exp \left(-\frac{(D - D')^2}{4q_\ell^2} - \frac{\sigma^2(N - N')^2}{4d_\ell^2} \right).$$

Its Schur norms are bounded independently of H, ℓ . Cauchy-Schwarz against this kernel bounds output energy on a test by a Gaussian-weighted sum of input energies on its normal enlargements. Since $f \geq d_\ell$ and $\ell f \geq q_\ell$, the enlargement in the j th Gaussian shell is obtained by multiplying the normal width by $C(1+j)$. The resulting regularity bound is $C\kappa w(Q) \sum_{j \geq 0} e^{-cj^2} (1+j)^{1-\eta}$. Exact base indices are never mixed. This proves the last assertion. $\square$

Lemma 3.5 (Polynomial test reduction). *At every stage of a polynomial setup, the base and normal cutoff and density conditions hold with enlarged fixed powers. To any prescribed polynomial regularity floor or error tolerance, it suffices to test widths, reciprocal widths, and absolute test parameters bounded by fixed powers. Discarding output tests whose mass-to-weight ratio is below ρ costs at most $\sqrt{\rho}$ times total assigned mass in the cubic readout, independently of the number of tests.*

Proof. Base indices move exactly and masks are contractions on each fiber. In the hybrid model, the pointwise packet bound gives a joint phase density at most CH^{-1} times the marginal base energy density. Lemma 3.4 propagates normal tails. The density and base-support bounds give, uniformly in shear parameters,

$$\mu(Q) \leq es^{-C} \min(r^4, 1) \min(f^2, 1). \quad (3.8)$$

For $f \leq 1$ use joint density, and for $f > 1$ use marginal base density. Dividing by $r^{3+\eta}f^{1-\eta}$ shows that either sufficiently small width gives an arbitrarily small prescribed power: the factors are $r^{1-\eta}$ and $f^{1+\eta}$, respectively. If neither width is small and one is sufficiently large, the total-energy bound suffices.

We record an elementary sublevel estimate to handle coefficients. For a nonzero real polynomial Q of degree at most two in $d \leq 3$ variables, on a fixed cube,

$$\text{vol}\{|Q| \leq \rho \|Q\|_{\text{coeff}}\} \leq C_d \rho^{2^{-d}}, \quad 0 < \rho < 1. \quad (3.9)$$

Here the coefficient norm is the maximum absolute coefficient. In one variable, three points in a set of measure m , separated by at least cm , and Lagrange interpolation show $\|Q\|_{\text{coeff}} \leq C\rho \|Q\|_{\text{coeff}}/m^2$. For the induction, write Q as a polynomial in the last variable and choose a coefficient polynomial with coefficient norm equal to $\|Q\|_{\text{coeff}}$. Outside the set where this coefficient is smaller than $\rho^{1/2}\|Q\|_{\text{coeff}}$, the one-dimensional bound is $C\rho^{1/4}$; the exceptional base set has measure at most $C\rho^{2^{-d}}$ by induction. This proves Equation (3.9), with harmless changes of constants.

For widths in a fixed polynomial range, discard negligible normal tails and clip base centers to a fixed-power box. The two normal shear polynomials have coefficient norm comparable, by a fixed-power change of coordinates, to the maximum of $|c|, |d|, |\alpha|, |\beta(T)|, |K|$. If that maximum is sufficiently large, at least one of their simultaneous sublevel conditions has arbitrarily small prescribed polynomial volume by Equation (3.9). The density bound handles its mass. Finally, $m_\lambda^{3/2}/\sqrt{w_\lambda} = m_\lambda \sqrt{m_\lambda/w_\lambda}$, so small ratios are summed against assigned mass, not against label cardinality. $\square$

We shall also use the following convention for very small pieces. An experiment with input energy e' and prescribed regularity κ' satisfies the direct estimate

$$\mathcal{B}_s \leq s^{-3/2} \max_\lambda \sqrt{e' / (\kappa' w_\lambda)}. \quad (3.10)$$

Indeed $m_\lambda \leq e'$ and total assigned mass is at most $s^{-1}e'$. For uniform application to a family of pieces one uses a fixed threshold, not a scale-dependent notion of smallness. Let the old reference energy be e_0 , suppose the summed piece energies have a polynomial bound relative to e_0 , and let $w_{\min}$ be a common polynomial lower bound for relevant output weights. Pieces with $e_G \leq s^A e_0$ have total cubic readout at most

$$s^{-1} w_{\min}^{-1/2} \sum_G e_G^{3/2} \leq s^{-1} w_{\min}^{-1/2} (s^A e_0)^{1/2} \sum_G e_G.$$

Taking the fixed A sufficiently large makes this smaller than any specified polynomial comparison scale. Every remaining piece has uniform cutoff and negligible-tail bounds relative to its own energy, after enlarging fixed powers. Prescribed regularity constants in the applications are also polynomial relative to e_0 , so both energy to regularity ratios are polynomial. The uniform model exponent from Lemma 2.2 therefore applies to this remaining family.

3.3 The extremal exponent and ordinary splitting

For each cylinder model, take the supremum of the upper $\log_{1/s}$ -exponents of Equation (3.6) over its polynomial setups. Denote these exponents by Γ_{cl} and Γ_{hyb} . The uniform interpretation on each fixed setup, and all closure extrema below, use the ordered conventions of Section 2.

Theorem 3.6 (Cylinder exponent bound). *For the classical and hybrid cylinder experiments with the regularity weight parameter $0 < \eta < 1$,*

$$\Gamma_{\text{cl}} \leq \eta/2, \quad \Gamma_{\text{hyb}} \leq \eta/2.$$

The proof occupies the rest of this section and Sections 4–7. We first bound the growth exponent, then study experiments whose cubic readout approaches it. Any ordinary regularity split that preserves this extremal growth forces its energy and regularity bounds to be sharp at the level of exponents. We then choose the retained tests with compatible shears and describe how their widths vary with depth. These properties supply the geometric input for the entropy argument.

Fix either model and write Γ for its exponent. We begin by proving that it is finite.

Lemma 3.7 (One-time bound). *For a polynomial test at a cut of length ℓ , an experiment with κ -regular input satisfies*

$$\mu_I(Q) \leq s^{-o(1)} \kappa w(Q) \ell^{1+\eta/2}. \quad (3.11)$$

Consequently Γ is finite.

Proof. Subtract the test's shear. Backward packet correspondences place its normal predecessors in root conditions of width $s^{-o(1)}f$, since every intervening $d_{\ell'}$ is at most $d_{\ell} \leq f$. Split its X -range into $O(\ell^{-1/2})$ intervals of width $r\sqrt{\ell}$. For each such interval, backflow of the two base position conditions is covered by a bounded number of root tests of base width $r\sqrt{\ell}$: the horizontal error is $O(r\sqrt{\ell})$ and the curvature error in the tilted coordinate is $O(\ell r^2)$. Summing their weights gives

$$O(\ell^{-1/2})(r\sqrt{\ell})^{3+\eta} f^{1-\eta} = O(1)w(Q)\ell^{1+\eta/2}.$$

Contraction on this predecessor region and Lemma 3.4 prove the claim; polynomial test bounds absorb the negligible errors. Total output mass is at most $s^{-1}e$, so Equation (3.11) also gives $\mathcal{B}_s \leq s^{-1+\eta/4-o(1)}$. $\square$

Suppose henceforth that $\Gamma > 0$. Dyadic binning of widths and mass-to-weight ratios produces extremizing families with common widths and

$$\frac{m_{\lambda}}{\kappa w_{\lambda}} = s^{p+o(1)}, \quad \sum_{\lambda} m_{\lambda} = es^{-d+o(1)}, \quad \Gamma = d + \frac{1-p}{2}, \quad p \geq 1, \quad 0 < d \leq 1. \quad (3.12)$$

Thus p measures the decay of each assigned mass relative to its regularity bound, whereas d measures the total mass read across the copied time intervals. The identity for Γ follows by substituting the two mass relations into Equation (3.6). Negligible ratio classes are first removed by Lemma 3.5; there are subpower many remaining bins. Choose the smallest possible $p = p_0$ in the closure of these extremal data. This selects the largest normalized assigned mass-to-weight ratios compatible with extremal growth. In particular $p < 3$. The output label family then has

$$\sum_{\lambda \text{ allowed}} w_{\lambda} \leq (e/\kappa)s^{-(p+d)+o(1)}. \quad (3.13)$$

Lemma 3.8 (Regularity peeling). *At a fixed cut, a state can be decomposed coherently into subpower many row projections followed by synthesis. Apart from a remainder at any prescribed polynomial floor, each level has a number $b > 0$ with the following properties. Its raw row measure is $O(b)$ -regular; it has disjoint assignments to tests of masses between bw and Cbw ; if z is its*

total raw energy, their total weight is at most z/b . The synthesized energies sum to at most z , and each synthesized state is $s^{-o(1)}O(b)$ -regular. Levels may be subdivided by subpower many statistics without altering these upper bounds.

Proof. Start above the fixed-power upper bound furnished by Lemma 3.5, and descend through dyadic thresholds. At threshold b , remove the entire remaining part of a test whenever its mass exceeds bw . Its mass is at most $2bw$, by regularity of the preceding remainder. A polynomial lower bound on the allowed weights makes the removal procedure finite by total mass. The union removed at this threshold is dominated by that preceding remainder and is therefore $2b$ -regular. The unremoved measure is b -regular. Continue to the prescribed floor. The removed row sets are disjoint and their projections sum to the identity. Synthesis is a contraction, and Lemma 3.4 gives post-synthesis regularity.

For a fixed terminal assignment, its cubic readout to the power $1/3$ is a weighted ℓ^3 norm of Hilbert norms. Hence coherent decomposition costs at most a fixed power of the number of levels by the triangle inequality. That number is subpower, and at least one level preserves any extremal output exponent. $\square$

Proposition 3.9 (Saturation of a preserving split). *Retain the terminal family in Equation (3.13). After any finite sequence of inserted operators or splittings that preserves its extremal exponent, every dominant homogeneous terminal class again has the exponents p, d in Equation (3.12). Suppose a non-remainder level supplied by Lemma 3.8 at depth $a \in (0, 1)$ preserves this exponent. Let b be its peeling threshold and z its total raw energy, summed over the intervals of that cut. Then*

$$b = \kappa s^{pa+o(1)}, \quad z = e s^{-da+o(1)}. \quad (3.14)$$

The summed post-synthesis energy has the same exponent as z . The conclusion holds for a further sublevel that preserves the output.

Proof. A new terminal class has $p' \geq p$ by minimality and $d' = \Gamma + (p' - 1)/2$. Its assigned mass and the fixed weight count imply $p' + d' \leq p + d$. Both inequalities force $p' = p$ and $d' = d$.

For a non-remainder peeling level at $\ell = s^{a+o(1)}$, apply the prefix exponent to its raw heavy-test assignments. Lemma 3.8 gives

$$\ell^{-1/2} z \sqrt{b/\kappa} \leq \ell^{-\Gamma-o(1)} e. \quad (3.15)$$

The tail exponent, applied separately to each interval and summed with its own post-energy, gives cubic readout at most

$$(s/\ell)^{1/2-\Gamma-o(1)} z \sqrt{b}.$$

Combining this with Equation (3.15) recovers the full extremal upper exponent. Since the chosen level preserves that exponent, both estimates, and the synthesis energy bound, must be saturated in exponent. The floor remainder is negligible: its summed input energy is at most $\ell^{-1}e$, and a sufficiently small regularity floor makes the tail readout smaller by a fixed power.

Write $b/\kappa = s^{P+o(1)}$. The saturated prefix has mass-to-weight exponent P/a , so minimality implies $P/a \geq p$. For the tail first select a homogeneous dominant terminal mass class, then an interval whose normalized readout is near maximal. That tail realizes Γ in the closure and has mass-to-weight exponent $(p - P)/(1 - a)$; hence $(p - P)/(1 - a) \geq p$. The one-time bound bounds P on the fixed gap, so these are legitimate finite closure data. Thus $P = pa$, and saturation in Equation (3.15) gives $z = e s^{-da+o(1)}$.

For a sublevel, use the unpartitioned level in the prefix argument. It bounds the sublevel's raw energy from above. The preserving tail bounds its post-energy from below by the same exponent. At the root, the given input regularity and energy replace the prefix estimate. At the last cut a repeated-cut regularity estimate replaces the tail. $\square$

For later use put

$$\ell_i = s^i, \quad E_i = es^{-di}, \quad \kappa_i = \kappa s^{pi}. \quad (3.16)$$

Depths and widths are dyadically rounded. All exponent upper and lower bounds below permit $s^{\mp o(1)}$ factors. On a fixed gap $h > 0$, these are also vanishing losses in $\log_{s^h}$ units.

3.4 Matched gates

Ordinary splitting controls energy and regularity at successive cuts. We now also align the shear frames of the retained tests. This compatibility will allow a later phase name to be compared with earlier names along a retained path.

A *documented gate* is a row projection followed by synthesis, whose retained rows are partitioned among tests. A retained row belongs to its assigned test; the associated test is its documented label. The last gate is simply the final assignment and need not be synthesized. Statements about reachability refer to chains of Equation (3.7) through the fixed retained mask/test row sets. They do not impose the original state's positive-norm support at intermediate cuts, which need not persist after an input restriction. They assert geometric incidence, not a joint probability law for the input and output of a quantum operator.

Proposition 3.10 (Matched-chain construction). *On any fixed finite depth mesh $0 \leq i_0 < \dots < i_b = 1$, an extremizing family can be modified by finitely many masks, preserving its terminal exponent on the original label family, so that the following hold.*

- (i) *At depth i , the documented tests have common widths r_i, f_i and total weight at most $s^{-o(1)}E_i/\kappa_i$.*
- (ii) *Ordinary regularization may be imposed immediately before each gate, in forward order, giving summed energy at most $s^{-o(1)}E_i$ and regularity at most $s^{-o(1)}\kappa_i$ in each interval.*
- (iii) *The total mass read on the documented rows is at least $s^{o(1)}E_i$. This lower bound persists after any further restrictions that preserve the terminal exponent and retain these gates in series.*
- (iv) *If j is the next mesh depth and $\delta = s^{j-i}$, then, for $L_* = L_*(\eta)$ independent of the mesh,*

$$\delta^{L_*} \lesssim r_i/r_j \leq 1, \quad \delta^{L_*} \lesssim f_i/f_j \lesssim \delta^{-L_*}.$$

On every connecting packet chain between the two documented labels,

$$\begin{aligned} |\beta_i(T_i) - \beta_j(T_i)| + \ell_i |K_i - K_j| &\lesssim (f_j/r_j^2)\delta^{-L_*}, \\ |P_i(F) - P_j(F)| &\lesssim (f_j/r_j)\delta^{-L_*}. \end{aligned} \quad (M)$$

These comparisons survive insertion of any fixed finite intervening operator stack, with a sufficiently small correspondence truncation exponent relative to the mesh gaps.

We first prove the geometric estimate used in its construction.

Lemma 3.11 (Two-shear intersection). *Normalize one starting interval and the later test widths to $\ell_i = r_j = f_j = 1$, and put $\delta = \ell_j/\ell_i$. Let $Q^-, \tilde{Q}^-$ be two projected tests, with bounded base footprints and normal widths $O(\delta^{-1})$. At the first base center let M be the maximum of 1 and the differences of $P, \beta(T_i), K$. For a κ_i -regular starting measure,*

$$\mu(Q^- \cap \tilde{Q}^-) \leq s^{-o(1)} \delta^{-C} \kappa_i w_j / M. \quad (\text{I})$$

If every retained descendant label has mass at least $\delta^D \kappa_i w_j$, then a fixed projected label has at most $s^{-o(1)} \delta^{-C_D} M_0^{1-\eta}$ intersecting partners in the shell $M \asymp M_0$.

Proof. Subtract the first shear and center its base chart. The intersection imposes, on bounded (X, B, Y) ,

$$|\alpha X + \beta X^2 + k(Y - 2XB) + c'| \lesssim \delta^{-1}, \quad |\alpha B + \beta Y - kB^2 + d'| \lesssim \delta^{-1},$$

where $\max(1, |\alpha|, |\beta|, |k|) \asymp M$. If $\max(|\beta|, |k|) \gtrsim M$, multiplying the first expression by β and subtracting k times the second gives

$$|t^2 + \alpha t + \text{const}| \lesssim M \delta^{-1}, \quad t = \beta X - kB.$$

On a dyadic shell $R \asymp \max(\sqrt{M}, |2t + \alpha|)$, its admissible t -set lies in at most two intervals of length $O(M \delta^{-1}/R)$. In the lowest shell the same conclusion follows from proximity to the quadratic vertex. Since $\max(|\beta|, |k|) \gtrsim M$, the horizontal set is covered by $O(\delta^{-1}R)$ squares of side $1/R$.

In such a square use the coordinate $Y - 2x_{\text{cell}}B$. After the vertical terms are removed, the horizontal derivatives of the two expressions are $O(R + M/R)$, so their oscillation across the square is $O(1)$ because $R \geq \sqrt{M}$. At least one vertical coefficient has size comparable to M . Therefore $O(\delta^{-1}R^2/M)$ vertical intervals of length R^{-2} suffice. The intersection is covered by $O(\delta^{-2}R^3/M)$ Heisenberg base cells of width $O(1/R)$, together with the first normal conditions enlarged by $O(\delta^{-1})$. Regularity bounds their total mass by

$$C \delta^{-C} \kappa_i w_j \frac{R^3}{M} R^{-3-\eta} \leq C \delta^{-C} \kappa_i w_j / M.$$

There are only logarithmically many shells on a polynomial setup.

Otherwise $|\alpha| \asymp M$ and $|\beta|, |k|$ are sufficiently small relative to M on the bounded base box. For fixed Y , the second expression has a B derivative comparable to M , and the first then has an X derivative comparable to M . The volume is $O(\delta^{-2}M^{-2})$. Its Heisenberg neighborhood of width $1/M$ obeys the same estimate, since the expressions change only by a constant there. It can consequently be covered by $O(\delta^{-2}M^2)$ cells of weight $O(M^{-3-\eta})$, proving the same bound. Bounded M needs no subdivision.

For the partner count, every intersecting partner in the shell has its whole projected footprint in a bounded enlargement of the first base box, and in normal width $O(\delta^{-1}M_0)$ about the first shear. This follows by extending the difference polynomials from an intersection point across bounded base coordinates. At each descendant time, contraction and starting regularity bound the total mass of these assigned rows by $s^{-o(1)} \kappa_i w_j O(\delta^{-1}M_0)^{1-\eta}$. Divide by the assumed mass per label and sum over at most $O(\delta^{-1})$ descendant times. $\square$

Proof of Proposition 3.10. Read mass at a gate. First observe that the asserted mass lower bound follows whenever the gate and weight count have been constructed. Insert an ordinary

split immediately after the gate in a thought experiment preserving the terminal exponent. Proposition 3.9 requires summed post-energy $s^{o(1)}E_i$, and contraction bounds it by the raw gate mass. At the final cut use the fixed terminal count and minimality of p . At the root, repeated-cut regularity is $s^{-o(1)}\kappa$, so reducing post-energy by a fixed power would lose the terminal exponent by the full tail bound.

Removing incompatible rows. Construct the gates backwards. Suppose the j -gate is fixed and set $a = j - h$, $\delta = s^h$. At $a > 0$ make an ordinary preserving split; at $a = 0$ use the root state. The resulting states g_a have total energy at most $s^{-o(1)}E_a$ and regularity at most $s^{-o(1)}\kappa_a$. At j their row measure is at worst $s^{-o(1)}\kappa_a$ -regular by Lemma 3.7. Delete labels whose newly read mass is less than $\delta^D\kappa_a w_j$. Their total mass is at most

$$s^{-o(1)}\delta^D\kappa_a E_j / \kappa_j.$$

After synthesis they remain $s^{-o(1)}\kappa_a$ -regular. Their tail readout, relative to the required scale $E_j\sqrt{\kappa_j}$ at j , therefore loses at least $\delta^{D-3p/2-o(1)}$. Since $p < 3$, an absolute D makes the deletion harmless. At $j = 1$, use the output ratio bound directly.

Every retained descendant test projects to a starting test Q_λ^- with base width $O(r_j)$ in ℓ_a position units and normal width $O(\delta^{-1}f_j)$. Base backflow follows from the quadratic identity; normal backflow follows from Lemma 3.4, taking its truncation exponent smaller than $h/10$. Each such projected test contains every predecessor on the allowed correspondence chains. All label counts here are polynomial by their weight count and the polynomial lower bound for w_j .

Call a starting row ambiguous if it lies in two projected tests with $M > \delta^{-D_0}$. Lemma 3.11, its partner count, and the j -gate weight count bound its total mass by

$$s^{-o(1)}E_j(\kappa_a/\kappa_j)\delta^{-C}\delta^{\eta D_0}.$$

Choose $D_0 = D_0(\eta)$ large enough that this is smaller than E_a by a fixed power of δ . Restricted rows retain regularity after synthesis, so their tail is negligible by the model exponent. Remove them and also rows belonging to no projected retained test. The latter have negligible transfer to the j -gate. Both removals use one row projection followed by synthesis, not successive assumptions about disjointness of synthesized states.

Shear charts. Partition the remaining rows into charts G . First record a lattice Heisenberg base cell of width r_j . Choose one attached test in a fixed order and round its five shear data at that cell center: $\beta(T_a)$, K , P , and its two normal values. Round in normal units

$$f^* = \delta^{-D_*}f_j, \quad D_* > D_0 + 2.$$

The five data determine the polynomial by a triangular change of coordinates. Nonambiguity makes every attached test's data lie within a bounded number of these rounded values. Thus each j -label meets only boundedly many charts. In each chart, after subtracting the common rounded shear, base and normal support are bounded in (r_j, f^*, ℓ_a) units, and every attached j -test has bounded coefficients in those units.

Full chart regularity. The charts now have compatible geometry. To apply the model exponent to their synthesized states, we also need full regularity. The next peel gives regularity

on a bounded range of tests; the subsequent reanalysis, smoothing, and cutoff estimates extend it to the full test family.

Within each chart perform a truncated peel: normalized widths belong to $[\delta^{D_1}, 1]$, with the Planck floor retained, and normalized shear coefficients have absolute value at most δ^{-D_1} . Centers are unrestricted. The physical weight multiplier is $w^* = r_j^{3+\eta}(f^*)^{1-\eta}$. Keep one density and width class coherently across all charts; there are subpower many classes, so some class preserves the output. Let b_* be its density bound, or its floor if it is the remainder. Then $b_* \leq s^{-o(1)}\kappa_a$, the raw mass in each chart is $O(b_*w^*)$, and the synthesized chart energies satisfy $\sum_G e_G \leq s^{-o(1)}E_a$.

We justify applying the full model exponent with regularity b_* to each chart. Same-cut Gaussian reanalysis preserves its truncated regularity: normal shell enlargements can be covered by translates at unchanged widths and coefficients. Smooth the input and the entire subexperiment by independent base boosts, base position translations, and the two normal translations, each of size δ^l , where $l > D_* + 10$. Retain the smoothing parameter as an exact auxiliary index. These are exact symmetries. The smallest attached output accuracy is δ^{1+D_*} , so assigned masses persist on bounded enlarged tests. In normalized units the smoothed base and joint densities are at most

$$C\delta^{-5l}b_*w^*.$$

This follows by averaging translations, whose coordinate Jacobians are one, and using the raw chart mass bound. Base support is bounded; normal support outside $s^{-o(1)}$ is negligible by Gaussian tails.

Here are explicit choices showing that the fixed density cost disappears from the regularity estimate. Choose D_2 with $(1-\eta)D_2 > 5l + 1$. Equation (3.8) gives

$$\frac{\mu(Q)}{b_*w^*w(Q)} \leq C\delta^{-5l} \frac{\min(r^4, 1) \min(f^2, 1)}{r^{3+\eta}f^{1-\eta}}.$$

Thus any width below δ^{D_2} is harmless. Widths above one are handled by bounded base covering or, for the normal width, by dropping the normal conditions and covering the $s^{-o(1)}$ normal support with unit translates. For widths in $[\delta^{D_2}, 1]$, choose D_3 so that $2^{-3}D_3 > 5l + 4D_2 + 1$. Equation (3.9) makes tests with a coefficient above δ^{-D_3} harmless, even after dividing by their minimum weight δ^{4D_2} . All remaining tests pull back under each jitter to permitted truncated tests, provided $D_1 > D_2 + D_3 + 10$. For them the original truncated regularity bound is used directly and incurs no factor δ^{-5l} . This proves full regularity $s^{-o(1)}b_*w^*$ in normalized units. Tiny-energy chart pieces use Equation (3.10); the others satisfy the polynomial setup relative to their own energy.

Saturation and induction. The model exponent and the bounded chart overlap now bound the coherent cubic readout at j by

$$s^{-o(1)}\delta^{1/2-\Gamma}\sqrt{b_*}\sum_G e_G. \tag{3.17}$$

Indeed, for a label reached by at most C chart states, $\|\sum_{G=1}^C v_G\|^3 \leq C^2 \sum_G \|v_G\|^3$. The already established read-mass lower bound and weight count, by Hölder, give the lower bound

$$s^{o(1)}E_j\sqrt{\kappa_j} = s^{o(1)}\delta^{1/2-\Gamma}E_a\sqrt{\kappa_a}.$$

Consequently $b_* = \kappa_a s^{o(1)}$, and the floor remainder can be excluded. Use the heavy tests in the selected truncated level as the new documented labels. Their total weight is at most $s^{-o(1)} E_a / \kappa_a$. Their truncated widths and coefficients, together with chart compatibility, give the width bounds and Equation (M). The constants depend only on the choices above, hence only on η , not on the number of mesh steps. Any reachable j -test is attached to the actual supporting row of the new gate; no assertion about its mass in an earlier state is used.

This completes the backward induction. Finally insert ordinary regularization before the gates in forward order, preserving the terminal exponent at every step. Proposition 3.9 gives the desired benchmarks. At the final cut its prefix minimality and the repeated-cut regularity bound give the same conclusion; the root uses its given bounds. Additional finite row statistics may be flattened in this same forward order. Dyadic upper endpoints may be clipped at r_j , so repeated rounding does not create exponential drift in $r_i/r_j \leq 1$. Constants per step disappear before the mesh limit because every inner mesh is finite. $\square$

3.5 Extraction and the hybrid Planck floor

Lemma 3.12 (Group extraction). *Consider a fixed positive depth gap $[i, j]$ in a documented configuration, with ordinary pre-gate regularity at i . Partition its starting rows into groups, including their starting intervals, and synthesize each group separately. Suppose every used j -label is reachable from at most $s^{-o(1)}$ groups; all other transfers must be negligible by uniform packet locality. Then one can extract group subexperiments realizing Γ, p in the closure. Their reachable output weight satisfies*

$$\sum_{\lambda \text{ reachable from } G} w_\lambda \leq \frac{e_G}{\kappa_i} s^{-(p+d)(j-i)-o(1)}.$$

The same conclusion can be imposed simultaneously at finitely many intermediate documented depths with the corresponding overlap bound.

Proof. The group energies sum to at most $s^{-o(1)} E_i$, and each group has regularity $s^{-o(1)} \kappa_i$ after reanalysis. Hölder, gate count, and read mass give cubic output at least $s^{o(1)} E_j \sqrt{\kappa_j}$. The overlap assumption compares this to the sum of the group readouts with subpower loss. Each group readout is at most $s^{(1/2-\Gamma)(j-i)-o(1)} e_G \sqrt{\kappa_i}$. Thus near-extremal normalized groups must exist.

Summing reachable weights over groups costs at most $s^{-o(1)} E_j / \kappa_j$. Groups violating the displayed relative count with an extra factor $s^{-\varepsilon}$ have total energy at most

$$s^{(p+d)(j-i)+\varepsilon-o(1)} \kappa_i E_j / \kappa_j = s^{\varepsilon-o(1)} E_i.$$

Their readouts are negligible by the group upper estimate. Delete them, then extract a near-maximal remaining group and let $\varepsilon \downarrow 0$. Its terminal homogeneous class has exponent p by the same minimality and count argument as Proposition 3.9. Finitely many intermediate count deletions are handled in the same way.

Only polynomially many groups are relevant: there are polynomially many labels and at most subpower many groups reaching each. Remove globally unreachable groups as one complementary row projection. Negligible transfers are then estimated with polynomial group and test counts. The fixed-threshold argument following Equation (3.10) discards tiny energies before the uniform exponent is applied. A loss $s^{-\varepsilon'}$ from a later outer limit is allowed only when $\varepsilon'/(j-i) \rightarrow 0$; choose larger vanishing slack in the count deletions. These observations ensure that extraction produces genuine polynomial subexperiments in the closure, not a formal limit object. $\square$

Proposition 3.13 (Bottom Planck floor). *If the hybrid exponent satisfies $\Gamma_{\text{hyb}} > \Gamma_{\text{cl}}$ for the same weight, then every hybrid extremizing family at Γ, p has*

$$H/(sf_1^2) = s^{o(1)}. \quad (\text{P1})$$

At every fixed positive documented depth, also $H/(\ell_i f_i^2) = s^{o(1)}$.

Proof. Admissibility gives the upper bound one. Suppose the ratio is at most s^ξ for a fixed $\xi > 0$. Set $a = 1 - h$, with $h > 0$ sufficiently small relative to ξ, η , and $\delta = s^h$. Apply the deletion and chart partition in the proof of Proposition 3.10 on $[a, 1]$. Each output has bounded chart overlap; summed chart energy is at most $s^{-o(1)}E_a$, raw chart mass is at most $s^{-o(1)}\kappa_a w^*$, and regularity down to the root Planck floor is at most $s^{-o(1)}\kappa_a w^*$ in chart units. The normalized output normal width is δ^{D^*} , whereas the final packet velocity width is smaller by $s^{\xi/2}$.

Jitter as in the preceding proof. The averaged initial row measure can now be regarded as classical input regular down to width zero. Above the hybrid floor use hybrid regularity. Below that floor use the density estimate $s^{-o(1)}\kappa_a w^* \delta^{-5l}$ and Equation (3.8). The fixed positive gap ξ makes the tiny-width gain dominate δ^{-5l} when h is sufficiently small. All input cutoffs then hold classically for non-tiny chart pieces.

We spell out why arbitrary hybrid masks inside the segment do not invalidate the classical upper comparison. In chart units use a normal phase lattice of side $\zeta = \delta^{D^*+3}$. At a final time t , bin analyzed rows by $(N', D'_t - tN')$. Along the fixed stack, all packet drifts in these units are much smaller than ζ if h is small enough. Thus, at every fixed exact index, contraction and localization bound the output mass in one lattice cell by a constant times initial row mass in finitely many neighboring cells, plus negligible fiber energy. Only polynomially many cells matter.

Reduce the assigned mass in each cell by its common excess factor so that it fits this neighboring initial mass. The reduction loses only the negligible error after summing cells and integrating exact indices. Divide once more by the bounded neighbor-overlap constant and allocate the remaining assigned masses on those initial cells, separately for each final time. These are legitimate classical assignments: exact base indices have the same endpoint, and a neighbor cell arrives within $O(\zeta)$ in normal phase, less than the output positional tolerance δ^{1+D^*} . Hence it lies in a bounded enlargement of the original assigned test. Keep the jitter parameter among the exact indices throughout. This is an energy comparison, not a positive packet identity for a masked quantum evolution.

Applying the classical exponent chart by chart bounds the retained hybrid cubic readout by $s^{-o(1)}\delta^{1/2-\Gamma_{\text{cl}}}E_a\sqrt{\kappa_a}$. Saturation requires $s^{o(1)}\delta^{1/2-\Gamma_{\text{hyb}}}E_a\sqrt{\kappa_a}$, a contradiction. The prefix ending at any documented $i > 0$ also realizes Γ, p , by its read mass and count, so the same argument applies there. $\square$

3.6 Curvature costs and shape profiles

The matched gates control differences between successive shears. We now ask how large the shear coefficients themselves must remain among experiments with the extremal exponents Γ, p . Their least possible growth rates will distinguish the configurations treated by the lower-dimensional estimates from those requiring the entropy argument.

For a leaf label define

$$R_2 = \frac{r_1^2}{f_1}(|\beta(T_1)| + s|K|), \quad R_1 = \frac{r_1}{f_1}|\alpha + 2\beta(T_1)x - 2Kb|. \quad (3.18)$$

The quantity R_1 measures the first-order horizontal variation of the normal shear in units of f_1 . The remaining variation has parabolic order two: angular increments have size r_1 , horizontal position increments have size sr_1 , and the tilted vertical increment has size sr_1^2 . Its size in normal velocity units is bounded by a constant times R_2 . The same ratios apply to the normal position condition, whose width is sf_1 .

Among families at Γ, p , minimize in the closure the exponent $c = c_2$ of an upper bound $s^{-c-o(1)}$ for $\max(1, R_2)$ on contributing labels. If $c = 0$, also minimize the corresponding exponent c_1 for $\max(1, R_1)$. These are costs of the shear, not packet dimensions. Section 4 excludes simultaneous zero cost in the classical model when $\Gamma > \eta/2$, and in the hybrid model when $\Gamma > \Gamma_{\text{cl}}$. The positive-cost alternatives are reduced below to affine width profiles.

Lemma 3.14 (Finite costs and compact profiles). *The preceding minima have feasible finite values. Normalize $r_1 = f_1 = 1$. Along successively finer matched meshes one may extract Lipschitz limits*

$$u(i) = \lim \log_s r_i, \quad n(i) = \lim \log_s f_i, \quad b_t(i) = n(i) - tu(i) \quad (t = 1, 2). \quad (3.19)$$

Here u, n vanish at 1 and u is nonincreasing, hence nonnegative. In the strict-gap hybrid case, $n(i) = (1 - i)/2$. Along any connecting chain from i to j ,

$$\begin{aligned} |\beta_i(T_j) - \beta_j(T_j)| &\leq s^{\min_{[i,j]} b_2 - o(1)}, \\ |K_i - K_j| &\leq s^{\min_{k \in [i,j]} (b_2(k) - k) - o(1)}, \\ |P_i(F) - P_j(F)| &\leq s^{\min_{[i,j]} b_1 - o(1)}. \end{aligned} \quad (\text{A})$$

One may also impose the individual curvature bound

$$|\beta_i(T_i)| + \ell_i |K_i| \leq s^{\min(-c, \min_{[i,1]} b_2) - o(1)}. \quad (\text{B})$$

Proof. Apply the one-step chart partition at the root and Lemma 3.12. Subtraction of the full chart shear bounds all reached leaf coefficients by fixed powers of s^{-1} in leaf units. This proves finite feasibility for c . When approaching finite c , the same chart partition can be used without subtracting its quadratic terms. After a base boost and translation, the absolute quadratic coefficients are still bounded by fixed powers in leaf units: conversion from $s|K|$ to full-duration K costs at most s^{-1} . Subtracting only the linear part then gives finite feasibility for c_1 when needed. Once c_1 is minimized, that linear subtraction is unnecessary, since the leaf-center derivatives and the bounded quadratic coefficients already bound the linear term. Constant normal translations affect neither cost. Base boosts and translations preserve R_2 and the corresponding ray derivative P . Each output knows boundedly many charts, so all these extractions retain the relative count bound.

After $r_1 = f_1 = 1$, Proposition 3.10 gives a uniform Lipschitz bound for the interpolated width exponents and their endpoint values. Arzelà–Ascoli on finer meshes gives Equation (3.19). Equation (Pl) at each positive depth, compared with its value at depth one, yields $n(i) = (1 - i)/2$ in the strict hybrid case, and continuity handles zero. Telescoping Equation (M) proves Equation (A). For the first inequality evaluate each earlier affine coefficient at the later time T_j , which lies within its interval; the corresponding K error is multiplied by at most that interval's length. A finite mesh contributes only $O(L_* \text{ mesh}) + o(1)$ to the exponent. Remove labels on no full geometric chain, whose transfer is negligible.

For Equation (B), work first on a fixed fine mesh, with a deletion tolerance larger than several mesh-error constants. Suppose the part of the i -gate violating the bound could preserve the terminal exponent. Ordinary-regularize just before that gate. Its total raw mass is at most $s^{-o(1)}E_i$. At the next depth j , keep only labels that can continue to a leaf in the graph imposed by the two curvature comparisons in Equation (M). Their total read mass is at least $s^{o(1)}E_j$.

For any predecessor on the high part, the leaf bound and telescoping force its affine trajectory to satisfy

$$|\beta_i(T_1)| \leq s^{\min(-c, \min_{[i,1]} b_2) - \text{err}}$$

at some terminal time T_1 within the reached j -interval. The exponent err is smaller than the deletion tolerance. An affine function whose value or duration-scaled slope violates Equation (B) either never enters this range or enters it on an interval much shorter than ℓ_j . Thus each high input row predicts only $O(1)$ possible T_j values. For each T_j , restrict the input to the rows predicting it and apply contraction; all other transfers into the kept rows are negligible. Summing gives total read mass at most $s^{-o(1)}E_i$, contradicting $E_j/E_i = s^{-d(j-i)}$ and $d > 0$. Therefore the low part preserves the output. Make this deletion at every mesh depth, then pass to finer meshes. Strict tolerances and the ordered closure limits justify the same conclusion for limiting extremal configurations. $\square$

Proposition 3.15 (Shape constraints). *If $c > 0$, then*

$$b_2(i) \geq c(i-1),$$

with equality at every depth when $c \neq 1$. If $c = 0 < c_1$, then $b_1(i) = c_1(i-1)$, and the quadratic terms of every gate shear may be replaced by their linearization at the test center, with vanishing exponent enlargement of its normal widths.

Proof. Write $b = b_2$. At a positive depth i where $b(i) = \min_{[i,1]} b$, the prefix realizes Γ, p by read mass and count. Multiplying Equation (B) by $r_i^2/f_i = s^{-b(i)+o(1)}$, its normalized terminal curvature cost is at most $(b(i) + c)_+/i$. Minimality of c therefore implies

$$ci \leq (b(i) + c)_+.$$

Since $c > 0$ and $i > 0$, this yields $b(i) \geq c(i-1)$ at every positive future minimum. For any other depth, a later future minimum gives the same lower bound, because $c(i-1)$ increases with i ; continuity handles zero.

Comparison with root labels using Equations (A) and (B) gives

$$|K_i| \leq s^{\min(-c, \min_{k \in [0,i]} (b(k)-k)) - o(1)}. \quad (3.20)$$

Suppose first $0 < c < 1$ and at some interior i the shape inequality is strict. The lower comparison line for $b(k) - k$ is strictly decreasing. By continuity, Equation (3.20) improves $\ell_i |K_i| \leq s^{c(i-1)-o(1)}$ by a fixed power at this i . The minimum of b on $[i, 1]$ likewise strictly exceeds $c(i-1)$. Consequently, for some $\varepsilon > 0$, every connected leaf satisfies

$$|\beta_i(T_i) - \beta_1(T_1)| \leq s^{c(i-1)+\varepsilon}, \quad s|K_1| \leq s^{-o(1)}.$$

At the regularized i -gate, bin $\beta_i(T_i)$ at that precision and subtract the rounded constant quadratic shear. Every reached leaf knows only boundedly many bins. Group extraction gives an experiment

on $[i, 1]$ at Γ, p whose terminal curvature cost, after dividing exponents by $1 - i$, is strictly smaller than c . This contradicts its minimality.

For $c > 1$, a strict shape inequality at i instead gives

$$\min_{k \in [i, 1]} (b(k) - k) > c(i - 1) - i,$$

since its lower comparison line is strictly increasing. Equation (A) then compares the pair $(\beta(T_i), K)$ for the i -label and a leaf to precisions $s^{c(i-1)+\varepsilon}$ and $s^{c(i-1)-i+\varepsilon}$, respectively; extrapolating β to T_i costs at most ℓ_i times the K error. Bin this jointly predictable pair and subtract its polynomial trajectory. Group extraction again lowers c , a contradiction.

Now let $c = 0 < c_1$. A future minimum of b_1 is also a future minimum of $b_2 = b_1 - u$, since u is nonincreasing. At such a depth, Equation (B) makes the normalized prefix curvature cost zero. On a connecting chain the center derivatives at depths i and 1 differ by at most $s^{\underline{b}_{1,i}-o(1)}$, where $\underline{b}_{1,i} = \min_{[i,1]} b_1$. To see this, first compare them at the actual ray by Equation (A). Moving to the center at i costs at most the curvature bound times r_i ; its exponent is at least $\underline{b}_{1,i}$ because

$$\min(0, \min_{[i,1]} b_2) + u(i) \geq \underline{b}_{1,i}.$$

The leaf-center change has zero cost. Lexicographic minimality at positive future minima therefore gives $b_1(i) \geq c_1(i - 1)$, as in the curvature argument. At any strict interior point, bin and subtract the jointly predictable linear coefficient. The extracted tail keeps zero curvature cost while strictly lowering c_1 . Thus equality holds everywhere.

Both b_1 and $b_2 = b_1 - u$ are now nondecreasing. Equation (B) therefore bounds curvature at depth i by $s^{b_2(i)-o(1)}$. Taylor expansion at the base center leaves the coefficient $P_i(x, b)$ times the increments in X and B_{T_i} in the respective normal conditions. Every remaining term is quadratic in horizontal increments or linear in the tilted vertical increment, so has size at most $s^{-o(1)} f_i$ in velocity and $s^{-o(1)} \ell_i f_i$ in position. The existing row supports can consequently be kept. $\square$

3.7 Affine reduction

The preceding constraints need not make both width profiles affine. We obtain affine profiles by restricting to a short depth interval near a common differentiability point and normalizing that interval. The extraction must preserve the extremal exponents and minimal shear costs; otherwise the new profiles would not describe the configurations still to be excluded.

Proposition 3.16 (Affine extremal configurations). *If at least one relevant minimal cost is positive, one can pass to extremizing documented configurations with*

$$u(i) = J(1 - i), \quad J \geq 0, \quad \begin{cases} b_2(i) = q(i - 1), & c > 0, \\ b_1(i) = c_1(i - 1), & c = 0 < c_1. \end{cases} \quad (\text{F})$$

For $c \neq 1$, $q = c$. For $c = 1$, either $q = 1$ or $q < 1$; no nonnegative lower bound on q is assumed. In the latter option,

$$|\beta_1(T_1)| \leq s^{-1-o(1)}, \quad s|K_l| \leq s^{-o(1)} \quad \text{at every gate } l.$$

For $0 < c \leq 1$ and $q = c$, one has $\ell_l |K_l| \leq s^{b_2(l)-o(1)}$, and the own-time values $\beta_i(T_i), \beta_l(T_l)$ on forward connections differ by at most $s^{b_2(i)-o(1)}$. For $c > 1$, compare instead $\beta_i(T_i), K_i$

with $\beta_l(T_i), K_l$ at respective precisions $s^{b_2(i)-o(1)}, s^{b_2(i)-i-o(1)}$. The linear alternative retains its individual curvature bounds and the center-derivative comparison at precision $s^{b_1(i)-o(1)}$.

All relevant terminal minimal costs, gate counts, read-mass bounds, Equations (M) and (A), and the permission to insert ordinary regularization persist. Base support, significant normal support, and reachable shear parameters are bounded by fixed powers in every normalized inner experiment.

Proof. Choose an interior common differentiability point of the Lipschitz profiles and set $J = -u'$ and $q = b'_2$ when relevant. If $c = 1$ and the shape is not the equality line, there are points with $q < 1$: otherwise absolute continuity and the endpoint value would contradict the strict gap somewhere above $i - 1$. If the shape is the equality line, take $q = 1$. Take shrinking segments $[A, B]$ about this point, put $h = B - A$ and $\Delta = s^h$, and normalize duration and terminal widths to (ℓ_A, r_B, f_B) . For each fixed h , take the mesh fine enough and all preceding closure errors small enough that their exponents are $o(h)$. Differentiability then gives Equation (F) uniformly in the normalized depth. It remains to justify extracting actual experiments without losing the costs or compatibility.

At the ordinary-regularized A -gate partition rows by a containing lattice base chart in r_B, ℓ_A units, and boost and translate to its center. Every later connected label at depth $l \leq B$ knows only $s^{-o(h)}$ possible charts, since $r_l \leq s^{-o(h)} r_B$ and its base backflow is bounded in those units. Use the following additional bins, written temporarily in the old physical width exponents:

- (i) If $c > 1$, bin $(\beta_A(T_A), K_A)$ at widths $s^{b_2(A)}, s^{b_2(A)-A}$, and subtract the rounded polynomial shear. Equation (A), extrapolated to T_A , makes these bins known to every connected later label up to $s^{-o(h)}$ choices.
- (ii) If $0 < c < 1$, bin only $\beta_A(T_A)$ at width $s^{b_2(A)}$ and subtract that constant quadratic coefficient. The bounds for K in Proposition 3.15 make every later own-time coefficient predict the bin.
- (iii) If $c = 1$ and $q \leq 1$, bin the pair at widths $s^{b_2(B)+A-B}$ and $s^{b_2(B)-B}$, and subtract the rounded trajectory. On this segment $b_2(k) - k$ is affine nonincreasing up to $o(h)$, so Equation (A) again gives the required prediction.
- (iv) In the linear alternative subtract no curvature. Bin the A -label's center derivative at width $s^{b_1(A)}$ and subtract that linear shear; Proposition 3.15 supplies the joint prediction from every connected later label.

The subtractions are made after centering and include free constant terms. All residual curvature data have bounded-power size in Δ units. For example, in case (iii), the uncertainty of K contributes at most $s^{b_2(B)+A-B-o(h)}$ to the terminal quadratic coefficient, and at most $s^{b_2(B)-o(h)}$ to its terminal duration-scaled K term. This gives terminal cost at most one, and when $q < 1$ yields precisely $s|K_l| \leq s^{-o(1)}$ after normalization. Cases (i), (ii), and (iv) give the other displayed bounds directly. Their terminal cost is at most its prescribed minimum; minimality prevents a strict improvement.

When $c > 0$, also bin and subtract a linear shear at a sufficiently large-power tolerance in Δ units. This has no effect on the curvature costs. Indeed compare derivatives first on the connecting ray by Equation (A); $\min b_1$ differs from $b_1(B)$ by $O(h)$. The already bounded residual curvature permits recentering at the common base chart, still at a fixed-power cost. Every later label therefore predicts the additional bin up to $s^{-o(h)}$ choices. In the linear alternative

these derivatives are already bounded. Finally bin the actual residual normal phase at A at a sufficiently large-power tolerance and remove its constant velocity and position. Base backflow, bounded residual coefficients, and bounded packet drifts make each later reached label predict these bins with the same small multiplicity. The starting labels themselves also meet only that many groups.

Apply Lemma 3.12, including the relative count conditions at the finitely many intermediate depths. Transform the selected experiment by the indicated symmetries and remove labels unreachable from the selected group. Negligible errors are evaluated on each inner polynomial setup before its outer limits. Gate counts are now relative to the extracted experiment's own energy and inherited regularity. Its raw starting mass is at most that regularity times a fixed power of Δ^{-1} , by its bounded support; synthesized tails are negligible. Its energy is at least that regularity times an inverse fixed power for a near-extremal extracted group: otherwise contraction onto the normalized contributing output tests would make its normalized readout negligible by Equation (3.10). Thus all cutoff requirements hold after normalization.

If a documented root gate is needed, restore it on the extracted post-state. Classically keep the old row ownership. In the hybrid case reanalyze and retain rows having a nonzero original group row, in the same exact fiber, within $s^{-\varepsilon}$ packet widths. Use essential neighborhoods, defined by positive fiber mass in the relevant box, and assign one witnessing original label. Gaussian localization makes projection and synthesis approximate the identity on this input up to negligible error. Enlarge the root normal widths accordingly. Choose $\varepsilon = o(h)$ smaller than the truncation slack at the first mesh step. Every connection from the restored gate then extends geometrically to an original supporting row, so the original compatibility estimates remain valid with vanishing normalized errors.

The mesh in the normalized depth may now tend to zero, always after the inner scale limits. In relative count selections take vanishing Δ -exponent slack larger than the total overlap and extremality errors. This proves realization of Γ, p and the same terminal cost minima in the closure. All arguments concerned finite masks and finite meshes; no infinite product of operators has been taken. $\square$

Remark 3.17 (Forward preparation of later observations). One may ordinary-regularize all available gates on each finite affine mesh in forward order, interleaving any prescribed finite row statistics at that gate, before flattening the final readout. For any subsequently chosen fixed interior depth, a nearest available gate then already has the benchmarks in Equation (3.16). Its prefix and tail have positive limiting depth gaps, so Proposition 3.9 applies. This procedure neither asserts relative saturation on every shrinking adjacent mesh gap nor transfers profiles retroactively through new earlier masks.

3.8 Causal laws of the raw rows

Fix finitely many observed gates, with ordinary pre-gate regularization prepared in forward order. Let $\mathbb{P}^i$ be the probability measure on their retained pre-synthesis rows at depth i , including time, exact indices, and documented label, obtained by dividing raw row mass by its total. At depth one use the final assigned mass. Proposition 3.10 and pre-gate energy give

$$s^{o(1)} E_i \leq \text{normalizing mass} \leq s^{-o(1)} E_i. \quad (3.21)$$

Nearest mesh gates may represent requested depths, with vanishing exponent ambiguity. All these laws belong to actual finite inner setups.

Correspondence predecessors below are taken in the current starting-state essential support and through the pre-existing intermediate physical masks. We do not require membership in the original state's intermediate supports: restricting the input may undo cancellation at those cuts.

We next compare the mass of an event of later rows with the mass of its possible predecessors. If all predecessors of a later event U belong to an earlier row set S , locality permits us to synthesize only S . The cylinder exponent then bounds the output on U . This is the rarity estimate below; it does not require a hybrid output sample to have a sampled ancestral packet.

Lemma 3.18 (Rarity transport). *Let $i < j$. Let U be a measurable event of retained j -rows and S a measurable event of retained i -rows. Suppose every correspondence predecessor in the active essential support of a row in U belongs to S , uniformly at the selected truncation. Then*

$$\mathbb{P}^j(U) \leq s^{-o(1)} \mathbb{P}^i(S)^{2/3} + \text{negl.} \quad (\text{Sp})$$

Proof. Restrict synthesis at i to S . Its summed input energy is at most $s^{-o(1)} \mathbb{P}^i(S) E_i$, and its regularity remains at most $s^{-o(1)} \kappa_i$ in each interval. By locality, its output on U agrees with the original output up to negligible error: input outside the active essential support has zero state. The tail exponent bounds its cubic readout by

$$s^{(1/2-\Gamma)(j-i)-o(1)} \sqrt{\kappa_i} \mathbb{P}^i(S) E_i.$$

Hölder and the j -gate weight count give

$$\begin{aligned} \sum_{\lambda \text{ on } U} m_\lambda &\leq \left(\sum_{\lambda} m_\lambda^{3/2} w_\lambda^{-1/2} \right)^{2/3} \left(\sum_{\lambda} w_\lambda \right)^{1/3} \\ &\leq s^{-o(1)} E_j \mathbb{P}^i(S)^{2/3}. \end{aligned}$$

The scale powers cancel by $2d = 2\Gamma + p - 1$. Divide by Equation (3.21). Superpolynomially small input pieces are treated by Equation (3.10); polynomial weights and counts absorb the negligible operator errors. At final readout the restriction to U means partial assignment, so the same argument applies. $\square$

For the classical model a final sample of nonzero mass does have an exact-particle path through earlier pointwise masks, with ownership at the ancestor disjoint gates. We use this additional observation only in the classical case. Lemma 3.18 is the substitute that is valid in both models.

To apply this estimate when a row is marked by the existence of a successful completing path, we need measurable versions of the marked rows and their predecessor sets. Use a Borel inner version of the end event and a Borel outer version of all its geometric predecessors. The following lemma preserves the completion of every retained end row and the pointwise predecessor inclusion required by locality. Its proof is given at the end of this subsection.

Lemma 3.19 (Measurable completion events and causal restrictions). *Fix an inner scale, a finite stack of cuts, a finite query list, and a correspondence truncation.*

- (i) *The side and row data admit standard Borel codes in which the primitive correspondences, time ownership, physical masks, analyzed-state essential supports, and assignment supports have Borel versions. All energetically active paths occur in a Borel finite-path relation. Its coordinate projections, including completion events, are analytic and hence measurable in the completion of every Borel row law [5, Section 3, Theorem 2].*

- (ii) *For an analytic row event A and a finite Borel row law μ , there are Borel versions $A^- \subset A \subset A^+$ with $\mu(A^+ \setminus A^-) = 0$. For a pre-synthesis mask, μ is the full analyzed row-energy law $\|V_i g_i\|^2$ in the active intervals, before fractional assignments. An assigned law suffices only for a terminal readout. An inner version of a completion event retains a completion for every one of its rows.*
- (iii) *Let B be a Borel inner version of a marked end-row event, intersected with the current end-state positive-norm set H_j ; it has the same full analyzed end-row mass as the marked event. Let S consist of all predecessors of B under the unrestricted truncated geometric correspondence from the starting gate to the marked gate. This relation uses the current starting-state essential support and the pre-existing intermediate physical masks. It does not impose the success event that selected B again, or the original state's intermediate supports. The set S is analytic and has Borel versions*

$$S^- \subset S \subset S^+, \quad \mu(S^+ \setminus S^-) = 0$$

under the full analyzed starting-row energy law. Thus S^+ contains every geometric predecessor pointwise. Synthesis restricted to S^+ and S^- gives identical states, and the two restrictions have equal analyzed input energy. This assertion concerns the physical state; a completing witness may have zero conditional probability under a separate path law.

- (iv) *With finitely many possible witness bins, the marked end rows may first be disjointized by bin and given Borel inner versions in H_j . Each retained row still has a completion in its chosen bin. Form each bin's predecessor set by the unrestricted correspondence in (iii). If path concatenation places this set in a Borel allowed starting-bin set P_b , both predecessor versions may be chosen inside P_b . The original starting-bin partition and its overlap count are then preserved. Use the outer versions for pointwise locality and the inner versions for physical restrictions and input-energy accounting.*
- (v) *The versions may be chosen jointly in side and row coordinates, and simultaneously for any fixed finite or countable family of full analyzed and comparison laws. After a new restriction or a row law not dominated by those laws is formed, choose versions under that law before the next projection. Complements of analytic events are used only as completed-measurable events, or given Borel inner versions before entering another path relation. Conditional assertions obtained by disintegration hold for almost every side value; a uniform assertion requires a separate finite-grid or quantitative exceptional-side argument.*

Lemma 3.20 (Forward flattening of scores). *At finitely many gates in forward order, prescribe finitely many finite-valued measurable row names, each having polynomially many values. Conditional names may also condition on a common arbitrary exact field y . One may retain subpower choices of row restrictions, preserving the terminal exponent, so that the prescribed log probability scores have deterministic limiting values. Their conditional versions are evaluated under the actually retained law. Further restrictions of that same law of probability $s^{o(1)}$ preserve these deterministic values, outside events of power-small probability at every fixed positive score tolerance.*

Proof. Compute the probabilities under the gate's preselection law, and bin their nonnegative log_s scores. Scores beyond a sufficiently large power occur on power-small mass, by the finite-name capacity bound, also after conditioning on y . Their contribution cannot preserve the terminal

exponent: restrict to those rows and use the tail exponent, or Lemma 3.18. There remain subpower many bins for each fixed finite request. The coherent triangle inequality selects a class preserving the output. Such a class has probability $s^{o(1)}$, since a power-small fraction would again lose the output exponent. At final assignment one uses $(\sum_{b=1}^k m_b)^{3/2} \leq k^{1/2} \sum_{b=1}^k m_b^{3/2}$.

Here is the required change-of-law calculation. If $Q = P(\cdot | A)$ with $P(A) = s^{o(1)}$, then for any measurable name or conditioning field F ,

$$r_F = \frac{dQ_F}{dP_F} \leq P(A)^{-1}, \quad Q\{r_F < s^\varepsilon\} \leq s^\varepsilon.$$

The first inequality follows from $Q(B) \leq P(B)/P(A)$; the second follows by integrating r_F over the indicated event. Write D for one prescribed name and C for its finite-valued conditioning data. Apply these inequalities to (y, C, D) and (y, C) . Their ratio is the likelihood ratio for the conditional probability of D given (y, C) . Its $\log_s$ is arbitrarily small outside a power-small event. Thus the scores selected before conditioning have the same limiting values under the retained law. This is also Lemma 2.11.

Make later gate choices only after earlier ones have been realized. Countably many eventual requests mean increasing finite lists followed by subsequences; they do not mean infinitely many projections on one state. Dense-query extension is legitimate only when the corresponding names have explicitly verified small-list relations as the query parameters approach one another. $\square$

Lemma 3.21 (Transfer of conditional profiles). *Suppose names on the i - and j -row laws have the following finite-list relations on every connecting path. Given a common exact variable y , each conditioning name C_i determines at most $s^{-o(1)}$ possibilities for C_j , and conversely. Given y and a related pair of conditioning names, the observed names D_i, D_j likewise have bidirectional $s^{-o(1)}$ lists. If*

$$\log_s \mathbb{P}^i(D_i | y, C_i) = z + o(1)$$

outside power-small events at every fixed tolerance, then the same statement holds with i replaced by j . All lists must be measurable and uniformly valid for the correspondence geometry.

Proof. Fix $\varepsilon > 0$. At each (y, C_j) the set of D_j atoms of probability greater than $s^{z-\varepsilon}$ has cardinality at most $s^{-z+\varepsilon}$. Pulling this list back through the conditioning and observed-name lists gives at most $s^{-z+\varepsilon-o(1)}$ possible D_i values per (y, C_i) . Outside the exceptional predecessor set, each such value has conditional probability at most $s^{z-\varepsilon/4}$. Their union therefore has power-small $\mathbb{P}^i$ mass. Lemma 3.18 makes the corresponding large-atom event power-small at j .

For the other direction, the good D_i atoms of conditional probability at least $s^{z+\varepsilon/4}$ have at most $s^{-z-\varepsilon/4}$ values per (y, C_i) . Forward their observed-name lists and use the backward conditioning lists. The resulting set has at most $s^{-z-\varepsilon/4-o(1)}$ D_j values per (y, C_j) . Missing this set requires an exceptional predecessor and is power-small by Lemma 3.18. Within the set, atoms below $s^{z+\varepsilon}$ have total conditional probability at most $s^{3\varepsilon/4-o(1)}$ by counting. This proves the claim. The same proof within one law shows invariance under changes of discretization with the stated list bounds. $\square$

Lemma 3.22 (Time prediction). *Let $i < j \leq l$. Suppose a conditioning name at l has relay names at j and i such that an i -row gives at most $s^{-o(1)}$ possibilities for the relay at j , and that relay gives at most $s^{-o(1)}$ possibilities for the target conditioning name at l . The relations must hold on all connecting paths; an additional common exact variable y is permitted. Then any*

measurable list of at most $s^{-d(j-i)+\varepsilon}$ candidates for T_j per target conditioning datum has power-small probability under the l -law, for fixed $\varepsilon > 0$. Consequently the conditional log probability score of the time choice is at least $d(j-i) - o(1)$, outside power-small events at every fixed tolerance.

Proof. Mark j -rows whose relay predicts a target datum whose proposed time list contains their T_j . An i -row can reach marked rows at no more than $N = s^{-d(j-i)+\varepsilon-o(1)}$ possible times. For each such time restrict the input to its possible predecessor rows, and use contraction and locality. Summing squared norms over times costs at most N per input row, so marked mass at j is at most

$$s^{-o(1)}NE_i = E_js^{\varepsilon-o(1)}.$$

If $l > j$, apply Lemma 3.18 to transport this exceptional event; the final probability is at most $s^{2\varepsilon/3-o(1)}$. For $l = j$ divide directly by Equation (3.21). To obtain the conditional score claim, take the proposed list to be the time atoms larger than $s^{d(j-i)-\varepsilon}$; there are at most the stated number. Finite grids and removal of the resulting exceptional events give the corresponding prefix mass bounds. Availability of the relay at the tested prefix is essential to the input multiplicity estimate. $\square$

Lemma 3.23 (Recoverable phase names). *Let a finite phase-cell name be observed on a later law. Suppose that, for each name, all its predecessors at an ordinary-regularized gate i are covered by $s^{-o(1)}$ time/test pairs of weights at most $s^{-o(1)}w$. Its typical unconditional log probability score is at least*

$$\log_{1/s} \frac{E_i}{\kappa_i w} - o(1).$$

If also $w = w_is^{o(1)}$ and every documented i -label predicts at most $s^{-o(1)}$ names on all reachable rows, this lower bound is an equality in the limit.

Proof. Write $z = \log_{1/s}(E_i/(\kappa_i w))$. A list of at most $s^{-z+\varepsilon}$ names has predecessor mass at most $s^{\varepsilon-o(1)}E_i$ by regularity and the covering hypothesis. Lemma 3.18 makes its probability power-small on the later law. Apply this to the list of atoms larger than $s^{z-\varepsilon}$ to obtain the lower score bound.

For the reverse bound, the i -gate weight count and the prediction hypothesis cover all reachable later rows by at most $s^{-o(1)}E_i/(\kappa_i w_i) = s^{-z-o(1)}$ names. Any remaining rows have negligible probability by locality and Lemma 3.18. Within a list of that size, atoms smaller than $s^{z+\varepsilon}$ have total mass at most $s^{\varepsilon-o(1)}$. This proves equality. $\square$

The last three lemmas require actual geometric recoverability and finite measurable lists. Counting formal coordinates of a name, without proving these relations, does not establish their hypotheses.

Measurability of correspondence events. We finish by proving the measurable-version statement used in the causal arguments above.

Proof of Lemma 3.19. Only finitely many cuts and queries occur at this inner setup. Keep the exact-index spaces themselves in the row code, together with the Euclidean phase, time, and finite label coordinates; a standard Borel space admits an injective Borel realization in a Polish space. Thus equality of a preserved exact index is a Borel diagonal condition and distinct indices are never identified. Use the measurable separable-Hilbert coordinates of the direct integral

to represent the finitely many operator masks. Their countably many matrix coefficients have Borel versions under the reference measure; on the common exceptional null set set the mask to zero. This leaves the pointwise contraction and the multiplication operator unchanged. At every fixed cut and interval, take a Borel measurable-field representative of the analyzed state and let H_i be its Borel positive-norm set. The full analyzed row-energy law is concentrated on H_i . Changing the representative on a reference-null set, or inserting $\mathbf{1}_{H_i}$ before synthesis, leaves the physical state unchanged. In an i -to- j predecessor test use the current input state's H_i and the marked end event as vertex conditions. At intermediate cuts use the Borel masks and geometric correspondences, not the original state's support: a restriction at i can undo cancellation there. If a support condition for a particular propagated component is needed, recompute its Borel representative after that physical restriction. Thus ambient zero-state starting rows are not witnesses, without omitting possible intermediate transfer from the actual restricted input. Fractional assignment densities relative to the actual row measures likewise have Borel versions and give the same assigned measures. Finite time ownership, test inequalities, exact-index equality, and the truncated Gaussian correspondences in (3.7) are Borel in the code. Their finite intersection in the full path space is therefore Borel.

Projection of this Borel relation is analytic. Analytic and coanalytic sets are universally measurable, so completion of the active Borel law gives Borel inner and outer versions of equal mass. For finitely or countably many laws use a weighted sum of their normalizations to choose common versions. In particular, taking the inner version of the marked end rows preserves each retained row's literal witness. Intersect this inner version with the current end-state H_j , which is Borel and has full end-row energy. Taking the outer version of their predecessor set preserves inclusion of *all* geometric predecessors. One may also choose a Borel inner predecessor version $S^- \subset S$ of equal full analyzed input mass. The difference $S^+ \setminus S^-$ carries zero current pre-synthesis input state, so synthesis restricted to S^+ or S^- is identical. This last equality concerns the actual state, not an arbitrary probability law on completing paths; a witness may have zero conditional mass.

When finitely many witness bins are used, first form the analytic possible-end-row set for each bin. Lexicographically disjointize them, take Borel inner versions under the full analyzed end law, intersect each with the current end-state H_j , and then project the unrestricted starting-to-end geometric correspondence for each disjoint marked group. Take Borel inner and outer predecessor versions $S_b^- \subset S_b \subset S_b^+$ of equal full analyzed starting-row energy. If path concatenation places S_b in a Borel allowed starting-bin set P_b , intersect both versions with P_b ; the original starting-bin partition and its overlap count are unchanged. Use S_b^+ for pointwise locality and S_b^- for the physical state and input-energy accounting. Every kept end row still has a witness in its assigned bin. A later physical restriction uses its Borel version in a newly formed finite-path relation. If an approximate inner version is convenient, losing row energy at most ξ^2 changes the restricted input state by at most ξ in norm; the following contractions preserve that bound. Choose ξ smaller than every required power of s before the exponent limits, so polynomially many descendants and tests add only negligible errors. Finally, choose joint versions under the side-row law and disintegrate. This proves almost-everywhere, not pointwise, conditional assertions. $\square$

4 Lower-dimensional estimates

This section supplies lower-dimensional estimates and excludes extremal configurations in which both shear costs vanish. Its two external inputs are the wave-envelope estimate of Guth, Wang,

and Zhang and the planar Furstenberg theorem of Ren and Wang. We prove the Hilbert-valued, weighted, and probabilistic consequences used here.

The wave-envelope estimate gives a kinetic bound for the base rays. A separate scalar-channel estimate controls angular memory, which will also be used with the canonical names in Section 5. The kinetic bound excludes hybrid zero costs when the positive hybrid exponent is strictly larger than the classical exponent. The planar incidence input, also used in the later projection arguments, excludes classical zero costs when $\Gamma > \eta/2$. These exclusions leave the positive-cost configurations for Sections 5–7. The polynomial cutoff convention, the order of limits, and the meaning of negligible errors are those of the cylinder framework. In particular, the number of masks is fixed during each inner scale limit. Constants depending on that number are permitted; an ordinary power of the scale depending on that number is not.

4.1 The wave-envelope estimate and a kinetic bound

The scalar wave-envelope estimate below is [12, Theorem 1.3], in the smooth-cutoff form used here. The following proof derives its Hilbert-valued and weighted extensions.

Theorem 4.1 (Wave-envelope estimate and extensions). *Let $L \geq 2$, and let the Fourier transform of F be supported in an $O(L^{-1})$ -neighborhood of a fixed compact annular patch of the cone in $\mathbb{R}^3$. Use a smooth, boundedly overlapping decomposition into angular sectors ϑ of width $L^{-1/2}$. For each dyadic $L^{-1/2} \lesssim R \lesssim 1$, let τ range over a boundedly overlapping cover by angular caps of width R . Let $U_{\tau,L}$ be the associated envelope box, with dimensions L , LR , and LR^2 in the adapted ray frame, and tile space by its translates. Then, for every $\varepsilon > 0$,*

$$\|F\|_{L^4}^4 \lesssim_\varepsilon L^\varepsilon \sum_{L^{-1/2} \lesssim R \lesssim 1} \sum_{\tau: |\tau| \sim RU} \sum_{U|U_{\tau,L}} |U|^{-1} \left(\int_U \sum_{\vartheta \subset O(\tau)} \|F_\vartheta\|^2 \right)^2. \quad (\text{GWZ})$$

The assertion holds for functions taking values in a separable Hilbert space, with a constant independent of that space. Fixed thickness constants, bounded enlargements of tiles, and adapted rapidly decreasing tile weights are allowed.

Proof. For the scalar case, apply the cited Theorem 1.3. Here the enlarged cap in the displayed formula accommodates the bounded overlap of the smooth partition. For completeness, the Hilbert-valued extension introduces no dimension-dependent loss. In a finite-dimensional subspace apply the scalar theorem to $\langle F, G \rangle$, where G is a standard complex Gaussian vector. Its fourth moment is a fixed constant times $\|F\|^4$. Moreover,

$$\mathbb{E}(|\langle v, G \rangle|^2 |\langle w, G \rangle|^2) = \|v\|^2 \|w\|^2 + |\langle v, w \rangle|^2 \leq 2\|v\|^2 \|w\|^2.$$

Average the scalar inequality and use this bound inside each double integral on the right. Orthogonal finite-dimensional projections and monotone convergence give the separable case. The weighted versions follow by summing translates with rapidly decreasing coefficients; changing an adapted smooth cutoff amounts to convolution on its dual scale and gives the same summable enlargement. For a scalar formulation localized by a decaying spatial weight, see also [12, Theorem 1.5]. $\square$

We will use a fixed linear image and dilation of this theorem. At frequencies of size L , consider

$$\rho = \frac{\xi^2}{4\zeta}, \quad \zeta \asymp L, \quad \left| \frac{\xi}{2\zeta} \right| \lesssim 1,$$

with thickness $O(1)$. The change $(\rho, \xi, \zeta) \mapsto (\rho + \zeta, \rho - \zeta, \xi)$ identifies this quadratic cone with a circular cone. A cap centered at the parameter $x_0 = -\xi/(2\zeta)$ has physical envelope of bounded length in direction $(1, x_0, x_0^2)$ and transverse widths R, R^2 in the coordinates

$$b - x_0 t, \quad y - 2x_0 b + x_0^2 t.$$

Its volume is comparable to R^3 . These coordinates also specify the orientations in the following positive-measure consequence.

Lemma 4.2 (Kinetic estimate for parabolic rays). *Let $0 < \delta < 1/2$. Let μ be a finite positive measure on rays $z = (X, B, Y)$, of total mass E , supported where $|X| \lesssim 1$. Assume, for every center and $\delta^2 \lesssim R \lesssim 1$,*

$$\mu\{|X - x| \leq R, |B - b| \leq R, |Y - y - 2x(B - b)| \leq R^2\} \leq \kappa R^3.$$

For each of a δ -separated set of times in a fixed bounded interval, let D range over squares of side comparable to δ . Assume bounded overlap also after a fixed enlargement. With $Q_z(t) = (B + tX, Y + tX^2)$, one has

$$\delta^{-1} \sum_{t, D} \mu\{Q_z(t) \in D\}^2 \lesssim_\varepsilon \delta^{-\varepsilon} \kappa E. \quad (\text{kin})$$

Proof. Set $L = \delta^{-4}$ and form the $L^2(\mu)$ -valued function

$$\begin{aligned} F(t, b', y'; z) &= \delta^{-3} \chi(t) \phi\left(\frac{b' - B - tX}{\delta^2}\right) \\ &\quad \cdot \phi\left(\frac{y' - Y - tX^2 - 2X(b' - B - tX)}{\delta^4}\right) e^{iL(X^2 t - 2Xb' + y')}. \end{aligned}$$

Choose Schwartz functions χ, ϕ with sufficiently small compact Fourier support. Require χ to be bounded away from zero on the time interval under consideration and ϕ to be bounded away from zero near zero. The three packet coordinates show that

$$\zeta \asymp L, \quad -\frac{\xi}{2\zeta} = X + O(\delta^2), \quad \rho + X\xi + X^2\zeta = O(1).$$

Consequently $\rho - \xi^2/(4\zeta) = O(1)$, so Theorem 4.1 applies in the scaled coordinates just described. Spatial integration in the two packet coordinates gives $\|F\|_{L^2}^2 \lesssim E$.

A fine angular projection has a uniformly smooth multiplier in these rescaled frequency coordinates. Thus its packet components vanish unless X belongs to a bounded enlargement of that fine angular sector, and they retain arbitrary rapid decrease in the same spatial packet coordinates. We claim that for an R -cap and any of its tiles,

$$\int_U \sum_{\vartheta \subset O(\tau)} \|F_\vartheta\|^2 \lesssim \kappa R^3.$$

Only X within $O(R)$ of the cap center x_0 occurs. A ray meeting a tile near time t_0 backflows into horizontal width $O((1 + |t_0|)R)$ and tilted vertical width $O((1 + |t_0|)R^2)$. Indeed its tilted vertical velocity relative to the central ray is $(X - x_0)^2 = O(R^2)$. These enlarged boxes can be covered by a polynomial number in $1 + |t_0|$ of the boxes in the hypothesis. The time decay of χ

absorbs that polynomial. For packets not meeting the tile, sum spatial shells in packet units: their energy decreases faster than any prescribed power, whereas the number of regularity boxes needed in the shell grows only polynomially. Integrate the normalized spatial density of each packet first, and then integrate against μ . This proves the claim, including weighted tiles.

At a fixed cap scale the sum of these tile integrals is $O(E)$ by overlap. Since $|U| \asymp R^3$, Equation (GWZ) gives

$$\int \|F(t, b', y')\|^4 dt db' dy' \lesssim_\varepsilon \delta^{-\varepsilon} \kappa E.$$

For a square counted at time t , throughout a sufficiently short interval of length comparable to δ the packet cores of all counted rays lie in a fixed enlargement of the square. The integral of $\|F\|^2$ over that enlargement is therefore at least a constant times the counted mass. Cauchy–Schwarz in a spatial region of area $O(\delta^2)$, followed by integration over the short time interval, contributes at least $c\delta^{-1}\mu\{Q_z(t) \in D\}^2$. Bounded overlap in space and time proves Equation (kin). $\square$

4.2 The scalar channel

In this subsection the base variables, if present, are retained only as exact indices. The observable variables are the normal velocity and position. In the hybrid model the fiber indexed by $\sigma \in [1, 2]$ evolves by the Fourier multiplier $e^{-itHp^2/(2\sigma)}$, so that its velocity is $N = Hp/\sigma$. Write $d_\ell = (H/\ell)^{1/2}$ and $q_\ell = (H\ell)^{1/2}$. In the classical model $d_\ell = 0$.

Proposition 4.3 (Scalar-channel estimate). *Suppose the root analyzed measure has mass e and satisfies*

$$\mu_0\{|N - n| \leq f, |D - b| \leq f\} \leq b_0 f^{1-\eta} \quad (f > 0, f \geq d_1).$$

Here the subscript 1 means time length one. At all terminal intervals of length τ , make disjoint fractional assignments to tests of normal velocity width f_λ and current-position width τf_λ , where $f_\lambda \geq d_\tau$. The centers are constant on each test; no shear depending on an exact index is used in these tests. With any fixed finite number of admissible masks,

$$\sum_\lambda \frac{m_\lambda^2}{f_\lambda^{1-\eta}} \leq \tau^{-\eta} s^{-o(1)} b_0 e. \quad (\text{SC})$$

This holds along polynomial sequences for which τ^{-1} , b_0/e , e/b_0 , the phase density and support bounds, and H, H^{-1} when applicable are bounded by fixed powers of s^{-1} .

Proof. We first prove the free estimate. The density bound controls the mass of a small test by $s^{-C}ef^2$; for very large f use total mass. Consequently widths and reciprocal widths outside a sufficiently large polynomial range contribute an arbitrarily small power, by summing assigned mass. Split the remaining widths into $O(1 + \log(1/s))$ dyadic classes. Normal dilation reduces each class to $f = 1$; the change in the weight is canceled by the inverse change in the regularity constant. In the hybrid case admissibility now gives $H \leq \tau$. Each test is contained in boundedly many lattice cells of dimensions $1 \times \tau$. Splitting assignments between these cells and combining assignments in the same cell reduces their square sum to the square sum of the cell masses, up to an absolute constant.

For classical free evolution, express the latter square sum as a collision count against $\mu_0 \otimes \mu_0$. Pairs with velocity separation comparable to r , $\tau \leq r \lesssim 1$, can occupy a common cell at only $O(r^{-1})$ sample times. Such a collision also requires initial position separation $O(r)$. For each first ray the regularity hypothesis bounds the mass of the second rays in question by $Cb_0r^{1-\eta}$.

The contribution of this separation scale is at most $Cb_0e^{-\eta}$. Pairs separated by less than τ use at most τ^{-1} times and lie initially in a box of width $O(\tau)$; their contribution is at most $Cb_0e\tau^{-\eta}$. Sum the logarithmically many scales. Velocities in a common unit cell differ by only a fixed constant, so there are no additional separation scales.

We give the hybrid proof in detail. If $\tau = s^{o(1)}$, packet locality and contraction bound the mass of each output cell at one time by $s^{-o(1)}b_0$. Its input predecessors lie in a root regularity box of width $s^{-o(1)}$, since $d_\tau \leq 1$. Sum cell masses and the $\tau^{-1} = s^{-o(1)}$ times. Hence suppose τ has positive limiting depth, and use τ as the inner scale. All cutoff powers remain finite. Fix an arbitrarily small $\varepsilon > 0$. Every loss below is $\tau^{-C\varepsilon}$ with a fixed C ; tails outside $\tau^{-\varepsilon}$ -enlarged adapted neighborhoods are made negligible before ε is sent to zero.

Partition the input by smooth compactly supported unit Fourier-velocity caps with bounded overlap. Their energies e_{cap} sum to $O(e)$. The Gaussian final analyzer sees, in any unit output velocity window, only caps within distance $O(\tau^{-\varepsilon})$, modulo negligible operator norm. Thus Cauchy-Schwarz loses only $\tau^{-O(\varepsilon)}$ on reducing the square sum to the contributions of individual caps. Gaussian decay justifies summing this error over all relevant caps and cells; the polynomial support convention also permits restricting to polynomially many significant caps. Boost each cap to $|N| \lesssim 1$, and call its input g and its free evolution $u(t, x)$.

Plancherel in the analyzer's velocity variable identifies its spatial marginal with a Gaussian average of $\|u(T, \cdot)\|^2$ of width $q_\tau \leq \tau$. For any $t \in [T, T + \tau]$, the band's spatial mass on an interval transfers, in either time direction, into its $O(\tau^{1-\varepsilon})$ -neighborhood with negligible L^2 error. To verify this, insert a smooth Fourier plateau on the band. Away from velocities in its support, integration by parts in its propagation kernel gives a rapidly decreasing Schur tail. Its wavelength is $O(H) \leq O(\tau)$, which is smaller than the allowed neighborhood. The retained operator is uniformly bounded on L^2 .

At each t , enlarged spatial cells overlap $O(\tau^{-\varepsilon})$ times. Average the preceding comparison in each terminal interval and expand the square of a cell integral. Up to negligible errors and $\tau^{-O(\varepsilon)}$, the cell square sum is bounded by

$$\tau^{-1} \int_{|\Delta| \lesssim \tau^{1-\varepsilon}} \int_0^1 \int_{\mathbb{R}} \|u(t, x)\|^2 \|u(t, x + \Delta)\|^2 dx dt d\Delta. \quad (\text{pair})$$

Dropping the velocity cell after fixing a cap has only the already allowed multiplicity. A boost merely translates the spatial grids at each time. Errors can first be summed over polynomially many cells covering significant support; the remaining mass is negligible.

For the Fourier calculations, first fix two exact indices, with parameters σ, σ' , and view the integrand as the squared norm of the tensor product of their fields. We integrate the resulting inequalities over this pair of indices before applying input regularity. In particular, all energies in the regularity bounds below include integration over their exact indices. Decompose Fourier-velocity pairs into Whitney scales

$$r_0 = \max(\tau, \sqrt{H}) \lesssim r \lesssim 1.$$

At scale r , use products of smooth multipliers a_I, b_J on intervals of width $O(r)$, with center separation $O(r)$ and boundedly many partners per interval. Require separation $\gtrsim r$, except for the nearest pairs at r_0 . One explicit construction starts with smooth lattice partitions ψ_i^r , supported on $|N - ir| \leq r$, and the near-pair symbol $\sum_{|i-j| \leq C} \psi_i^r(N_1) \psi_j^r(N_2)$. Telescope these symbols through dyadic scales. Expand a difference using the partitions at both adjacent scales in both variables. If C is a sufficiently large fixed constant, every surviving tensor term is separated

by a constant times r and only boundedly many terms occur locally. The last near-pair symbol is the remainder at r_0 .

Insert a Schwartz time cutoff with compact Fourier support and modulus bounded below on $[0, 1]$. At fixed Δ and fixed indices, the spacetime Fourier supports of products at any one Whitney scale have bounded overlap. After multiplying frequency coordinates by H , these supports satisfy

$$\sigma N_1 + \sigma' N_2, \quad \sigma N_1^2 + \sigma' N_2^2 + O(H).$$

On fixing the first coordinate, the second is a positive quadratic in the transverse coordinate. At separation $r \geq \sqrt{H}$, its derivative has size comparable to r , and uncertainty $O(H)$ permits only boundedly many interval pairs of width r . In the nearest part, the positive coefficients in the first coordinate already give bounded overlap. Plancherel at each scale and Cauchy–Schwarz over the logarithmically many scales reduce Equation (pair) to the sum of the individual spacetime product integrals with the squared time cutoff.

Localize each initial factor $a_I g, b_J g$ further by $G(x/r - k)$, $k \in \mathbb{Z}$. Here G is Schwartz, has Fourier support in a sufficiently small neighborhood of zero, and $\sum_k G(v - k) = 1$. Such a function is obtained by choosing its Fourier transform supported near zero with the normalization prescribed by Poisson summation. The squared factors have bounded sum. Multiplication enlarges a velocity band by at most $cH/r \leq cr$, preserving the separation of separated pairs. Write $e_{I,k}$ for the energy of a localized factor, including its exact indices. Then

$$\sum_{I,k} e_{I,k} \lesssim e_{\text{cap}}, \quad e_{I,k} \lesssim \tau^{-O(\varepsilon)} b_0 r^{1-\eta} + \text{negl } e. \quad (\text{LC})$$

For the second bound, use the original boosted input, before cap selection. The localized operator needs only its analyzed rows with velocity within $O(\tau^{-\varepsilon} r)$ of I and position within $O(\tau^{-\varepsilon} r)$ of rk . On those rows apply the boundedness of synthesis, the Fourier multiplier, and the position multiplier, followed by input regularity. The root packet widths are $\sqrt{H} \leq r$. Excluded velocities give Gaussian frequency tails; excluded positions give Gaussian spatial tails and the Schwartz kernel tail of a multiplier whose spatial width is $O(H/r) \leq O(r)$. Applying Schur bounds after rescaling these kernels proves negligible norm for the discarded transfers. This proves Equation (LC) with constants independent of the exact-index spaces and Hilbert multiplicities.

Truncate the time integral to $|t| \leq \tau^{-\varepsilon}$. The time cutoff's Schwartz decrease, together with polynomial Bernstein bounds for the bandlimited fields, makes the error negligible. For a localized piece, discard the output outside

$$|x - rk - tN_I| \leq \tau^{-C\varepsilon} r,$$

where $C > 2$ is fixed large enough. To obtain the L^2 error bound, first discard the position multiplier's tail beyond $\tau^{-\varepsilon} r$, retaining a smooth propagation plateau equal to one on the original enlarged Fourier support. The nonstationary kernel tail then applies, since $r^2 \geq H$ and $|t| \leq \tau^{-\varepsilon}$. Tensor-product errors are negligible by Bernstein on the untruncated factors. These errors sum over polynomially many significant position indices; the remaining indices can be discarded together by the original support and tail bounds. On retained outputs a product with the shifted factor requires

$$|k - k'| \lesssim \tau^{-C'\varepsilon},$$

because $|N_I - N_J| \lesssim r$ and $|\Delta| \lesssim \tau^{1-\varepsilon} \lesssim \tau^{-\varepsilon} r$. At each point only $\tau^{-O(\varepsilon)}$ such localized terms occur, so Cauchy–Schwarz separates them at this loss.

For a separated localized pair, extend its spacetime integral to the whole plane. Bilinear Plancherel gives

$$\int_{\mathbb{R}^2} \|u_{I,k}(t, x)\|^2 \|u_{J,k'}(t, x + \Delta)\|^2 dx dt \lesssim r^{-1} e_{I,k} e_{J,k'}.$$

Indeed the map from momenta to their summed spatial and temporal frequencies has multiplicity at most two and Jacobian $|Hp_1/\sigma - Hp_2/\sigma'| = |N_1 - N_2| \gtrsim r$. The shift introduces only a phase. This calculation is componentwise valid for Hilbert-valued fields and then integrates over the independent exact indices. The shift interval cancels the prefactor τ^{-1} in Equation (pair), up to $\tau^{-O(\varepsilon)}$.

There are two possibilities for the nearest scale. If $r_0 = \sqrt{H}$, spatial Bernstein has bandwidth $O(r_0/H) = O(r_0^{-1})$; combine it with the other factor's conserved L^2 norm over the retained times. If $r_0 = \tau$, extend the shift integral to all of $\mathbb{R}$, after imposing the position-pair restriction just proved. The spatial and shift integrals then give the product of the two energies, and the surviving prefactor is $\tau^{-1} = r_0^{-1}$. Thus all cases are bounded by

$$\tau^{-O(\varepsilon)} r^{-1} e_{I,k} e_{J,k'}.$$

Each localized piece has only $\tau^{-O(\varepsilon)}$ allowable partners. Use Equation (LC) for one energy and sum the other to obtain $\tau^{-O(\varepsilon)} r^{-\eta} b_0 e_{\text{cap}}$ at each scale. Since $r \geq \tau$, summing scales and caps proves the free estimate after letting ε tend to zero. In particular this argument does not replace a finite-time spatial mass by an unrestricted spacetime L^4 norm when H is very small.

It remains to insert masks. Induct on their number. At the first cut of length ℓ carrying a mask, peel the raw analyzed rows into dyadic scalar-regularity levels. At threshold b , use the tests with widths $f, \ell f$ to remove disjoint heavy parts of mass greater than $b f^{1-\eta}$. The preceding threshold ensures that their union has regularity $O(b)$. Include an arbitrarily low polynomial floor remainder. If M is the row contraction, write it as the coherent sum $\sum_b M \mathbf{1}_{\text{level } b}$ before synthesis. The free estimate to this cut shows that its level's summed raw mass z_b satisfies

$$bz_b \lesssim s^{-o(1)} \ell^{-\eta} b_0 e.$$

The post-mask energies sum to at most z_b , and same-frame Gaussian locality gives regularity $O(b)$ after reanalysis. Apply the induction hypothesis separately on each descendant experiment, whose relative terminal time length is τ/ℓ , and sum its energies. The scale factors multiply as $\ell^{-\eta} (\tau/\ell)^{-\eta} = \tau^{-\eta}$. The floor has an arbitrarily small regularity constant; a trivial total energy bound makes its contribution negligible. Post-states too small to satisfy polynomial bounds relative to their own energy are instead estimated directly on the original polynomial terminal tests.

There are only logarithmically many levels. For fixed assignment multipliers the fourth root of the weighted mass-square sum is an ℓ^4 norm of Hilbert norms and hence a seminorm in the input state. The triangle inequality therefore sums the coherent levels at a subpower loss. Repeating this a fixed number of times completes the induction. Repeated cuts and subpower time gaps use the same-frame bound and the elementary estimate already proved, with subpower loss. $\square$

4.3 Memory of the base angle

Lemma 4.4 (Memory bound for documented labels). *Use an affine documented configuration as in Equation (F), with ordinary regularity imposed immediately before the relevant gates. Fix $0 < i = j - h < j \leq 1$, set $\delta = s^h$, and, separately for each starting interval, normalize time length*

to one and the ending widths to $r_j = f_j = 1$. In the quadratic case set $Q = q$ and $C = q - J$; in the linear case set $Q = 0$ and $C = c_1$. Suppose

$$L \geq 1, \quad 2L \geq 1 + \max(1, Q), \quad L \geq 1 + C. \quad (\text{ML})$$

Assign to each j -label a list of at most δ^{-R} cells of width δ^L for the normalized base angle X . Then

$$\mathbb{P}^j\{X \text{ belongs to its label's list}\} \leq s^{-o(1)} \delta^{(1-d+2\Gamma+\eta(L-1)-R)/2}. \quad (\text{mem})$$

The cell origins may depend on the label. The assertion of power rarity requires a strictly positive exponent after all outer errors.

In the classical model there is also a prediction version. Drop $L \geq 1 + C$ from Equation (ML). For each exact ancestor row at i , allow at most $s^{-o(1)}\delta^{-Z}$ candidate values for the ending derivative $P_j(F)$ on its true base ray, in the normalized units. Restrict the event to guesses accurate to $s^{-o(1)}\delta^{1-L}$. Then Equation (mem) holds with an additional $-Z$ in its exponent's numerator. In the linear case one may guess the enclosing test's linear coefficient instead.

Proof. Partition the exact base variables at the initial time into lattice boxes M of widths $(\delta^L, \delta^L, \delta^{2L})$, using the tilt of the center of each X -bin for the last coordinate. These variables do not mix under propagation. For a fixed listed angle cell, a terminal label permits initial horizontal and tilted vertical positions in intervals of width $O(\delta)$ each. To check this, backflow its terminal base footprint through a bounded time; variation within the angle bin contributes $O(\delta^L)$ horizontally and $O(\delta^{2L})$ vertically in the bin's tilt, while the terminal footprint has width $O(\delta)$ in both relevant directions. A change from the label's tilt to the bin's tilt has bounded coefficients. Therefore at most

$$s^{-o(1)}\delta^{2-3L-R}$$

base boxes M are relevant to any ending label and its whole list.

On the initial gate within M , bin both curvature coefficients, at its starting time and in segment units, to width $\delta^{-\max(1, Q)}$. Equation (A) shows that each ending label knows only $s^{-o(1)}$ possible bins, including the extrapolation of its quadratic coefficient back to the initial time. Subtract the shear of that rounded curvature trajectory. In the linear case omit this operation and use the enclosing linear tests supplied by the affine reduction. Residual curvatures of the two endpoint tests are now bounded by $s^{-o(1)}\delta^{-\max(1, Q)}$ throughout the segment.

The endpoint derivatives at a true ray differ by at most $s^{-o(1)}\delta^{-\max(0, C)}$. Recenter at the ray F_M through the base-box center; this adds at most $s^{-o(1)}\delta^{L-\max(1, Q)}$. Both quantities are bounded by $s^{-o(1)}\delta^{1-L}$, by Equation (ML). Bin the residual initial derivative at that accuracy and subtract the resulting linear shear. The ending label, M , and the curvature bins determine only $s^{-o(1)}$ possibilities for this last bin. The residual ending derivative at F_M is consequently at most $s^{-o(1)}\delta^{1-L}$.

For the prediction version, clip the lists to a polynomial parameter range large enough to contain all accurate guesses. Subtract the rounded curvature's derivative at the true ray from each guess, and bin the result to width δ^{1-L} . Make one copy of the classical ancestor row for each distinct resulting bin. This costs at most $s^{-o(1)}\delta^{-Z}$ in total energy. On the event of an accurate guess, recentering at F_M has the same error as before. The ending label and M again know only $s^{-o(1)}$ pertinent bins. If the guess uses the true derivative but the test is enclosed by a linear one, their coefficients differ by at most $s^{-o(1)}$, which fits the tolerance. Copying is used only in the classical model, where energy contributions add pointwise over exact indices.

After these subtractions, the part of each ending label over M lies in an ordinary scalar test of velocity width $s^{-o(1)}$ and position width $\delta s^{-o(1)}$. To verify the tighter position statement, Taylor-expand the two shear polynomials about F_M . Their horizontal linear terms are bounded by $\delta^{1-L}\delta^L = \delta$. The remaining terms are bounded by $\delta^{-\max(1,Q)}\delta^{2L} \leq \delta$. The tilted vertical coordinate has precisely the latter order, also after base free evolution. The velocity estimate follows from the same expansion with its less restrictive unit tolerance.

For each starting time and bin tuple, ordinary regularity before the gate gives scalar input regularity constant

$$s^{-o(1)}\kappa_i w_j \delta^{(3+\eta)L}.$$

Indeed a scalar regularity box pulled back by the subtracted common shear, together with the base box M , is a full cylinder test of base width δ^L . Restriction to the gate and same-frame reanalysis preserve this upper bound. Apply Proposition 4.3 to each such experiment, allowing all subsequent masks and retaining exact indices. The energies sum to at most $s^{-o(1)}E_i$, or to $s^{-o(1)}E_i\delta^{-Z}$ in the prediction version. The scalar cutoff conditions follow from the original density and phase cutoff bounds. As throughout, extremely small-energy pieces are bounded directly on the polynomial output tests.

Let $m_{\lambda,M}$ be the actual read mass of the tested event. Each pair (λ, M) sees only $s^{-o(1)}$ bin tuples. Other tuples have no allowed packet chain and contribute negligible operator norm. Cauchy–Schwarz in the contributing fields and Equation (SC) imply

$$\sum_{\lambda, M} m_{\lambda, M}^2 \leq s^{-o(1)}\kappa_i w_j E_i \delta^{(3+\eta)L-\eta-Z}.$$

There is no coherent loss over M , which is an exact base partition. In the classical prediction version use only accurate copied parts; they majorize the tested event pointwise and have the same small-list property.

The documented count and the base-box count give at most

$$s^{-o(1)} \frac{E_j}{\kappa_j w_j} \delta^{2-3L-R}$$

pairs. Apply Cauchy–Schwarz to their masses and divide by E_j^2 . Since $E_j/E_i = \delta^{-d+o(1)}$ and $\kappa_j/\kappa_i = \delta^{p+o(1)}$, the exponent is $d - p + 2 + \eta(L - 1) - R - Z$. Finally $d - p + 2 = 1 - d + 2\Gamma$, giving the assertion. $\square$

Corollary 4.5 (Memory when both costs vanish). *If $c_2 = c_1 = 0$, Equation (mem) holds with $j = 1$ and $L = 1$ for every fixed terminal depth gap, without assuming affine profiles.*

Proof. In normalized tail units the leaf curvature is at most $s^{-o(1)}\delta^{-1}$ throughout the segment. Its derivative on its own base footprint is at most $s^{-o(1)}$, by the two zero-cost bounds at the leaf. Recentring on a compatible base box of width δ changes the derivative by only $s^{-o(1)}$. Thus the Taylor bounds in the preceding proof hold directly, without curvature or derivative bins. The scalar-channel and counting arguments are unchanged. $\square$

4.4 The zero-cost hybrid case

Proposition 4.6 (Exclusion of a strictly nonclassical zero-cost case). *Suppose $\Gamma_{\text{hyb}} > 0$ and $\Gamma_{\text{hyb}} > \Gamma_{\text{cl}}$. The minimizing hybrid families cannot have $c_2 = c_1 = 0$.*

Proof. Write $\Gamma = \Gamma_{\text{hyb}}$ and assume both costs vanish. Normalize the leaf widths to one. The bottom Planck assertion (PI) gives $H \leq s$ and $H = s^{1+o(1)}$. At each leaf,

$$|\beta(T)| + s|K| \leq s^{-o(1)}, \quad |P(x, b)| \leq s^{-o(1)}.$$

Taylor expansion on its base footprint, whose horizontal and tilted vertical position widths are s , shows that its normal velocity and position lie in unsheared intervals of widths $s^{-o(1)}$ and $s^{1-o(1)}$ respectively.

Initial localization. Partition X into unit bins. Split the initial normal field by a smooth compactly supported unit Fourier-velocity partition of unity, and then by unit-width Schwartz position factors forming a partition of unity and having bounded Fourier support. Summed energies are $O(e)$. Every leaf sees only $s^{-o(1)}$ groups: its angle and normal velocity windows have subpower widths, and its position backflows into the corresponding subpower neighborhood of an initial position bin. This follows through the fixed stack of packet kernels since $d_s \leq 1$. Gaussian and Schwartz tails justify deleting all other transfers with negligible norm; first truncate to polynomially many significant input bins. Equivalently use arbitrary fixed-power enlargements and then send their exponents to zero.

By the triangle inequality on each assignment, it suffices to estimate the groups separately and sum their cube sums at subpower loss. Tiny-energy groups admit direct polynomial-test bounds. On a remaining group boost and translate so that $|X| \lesssim 1$, the normal Fourier velocities are bounded, and the initial normal position has negligible mass outside $|D| \leq s^{-o(1)}$. Denote its energy by e_g .

The group has input regularity $O(\kappa)$. The same assertion holds after replacing it by the direct sum of a smooth Fourier-velocity partition at any width $r \gg \sqrt{H}$. Here is the operator justification needed later. The conjugated operators in the root packet frame have rapidly decreasing Schur kernels in its normal packet units. For a vector of Fourier multipliers the ℓ^2 norm of its derivative of order k is $O(r^{-k})$. Rescale the Gaussian frequency-side inner product to root packet units and integrate by parts; since $r \gg \sqrt{H}$, this gives a bound for the whole vector kernel with summable tails. Unit position factors have the analogous space-side bound. At each exact base point these operators need only bounded enlargements, and then summable enlarged shells, of a normal test. Same-frame reanalysis therefore proves the claimed regularity for the sum measure. Integrating over the subpower normal support gives the base energy marginal regularity

$$\mu_{\text{base}}(\text{base test of width } R) \leq s^{-o(1)} \kappa R^3, \quad s^2 \lesssim R \lesssim 1.$$

We used $R^{3+\eta} \leq R^3$ and covered normal support by subpower many unit windows. All these weights are polynomial, so tails remain negligible.

After the boost, base leaf positions lie in squares of side $s^{1-o(1)}$. Pass from assignments to lattice position masses at subpower loss and drop angular restrictions. Let y denote all exact indices, $Y_d(T)$ the indices whose base position at T belongs to a square d of side s , and $u_y(T, x')$ the output field before final analysis. The unnormalized functional is bounded, up to subpower loss, by

$$\sum_{T,d} \int_{\mathbb{R}} \left(\int_{Y_d(T)} \|u_y(T, x')\|^2 dy \right)^{3/2} dx'. \quad (\text{HL})$$

Indeed the analyzer's spatial marginal is convolution by a probability Gaussian. Jensen removes that convolution in this upper bound, and Hölder on a normal bin of length s cancels the

functional's factor $s^{-1/2}$. We may replace the fields by approximations having negligible L^2 error at every time: the original assignment functional absorbs these errors by its polynomial parameters. We may likewise restrict the output to a polynomial range of x' when necessary.

A stopping scale and its envelopes. Fix $0 < \zeta < 1/4$, set $r \asymp s^\zeta$ dyadically, and choose comparable dyadic scales

$$\ell_0 = H/r^2, \quad B_* = H/r.$$

Then $1 \gg \ell_0 \gg B_* \gg s$ on sufficiently accurate outer approximations, and $d_{\ell_0} \asymp r$, $q_{\ell_0} \asymp B_*$. We will bound the expression in Equation (HL) by

$$s^{-o(1)}(r^{-D_0} + B_*^{-\Gamma})e_g\sqrt{\kappa},$$

where D_0 is absolute and independent of the number of masks. Separated velocity caps give the first term: a transverse crossing estimate will be combined with the kinetic bound. Nearby caps give the second term: we compare them with one experiment read at time length B_* , apply the exponent Γ to its prefixes, and use the kinetic bound on each remaining gap. Since $B_* = s^{1-\zeta+o(1)}$, both terms improve on the extremal exponent once $D_0\zeta < \Gamma$.

Split the group input coherently into $O(r^{-1})$ smooth velocity caps of width r centered at bounded lattice points θ_k . Call their outputs $u_{k,y}$. Let e_y be the original group fiber energy, so $\int e_y dy = e_g$ and each cap fiber has energy $O(e_y)$.

Add the cut ℓ_0 , including any masks already at that length. For a fixed stack of n stages use slack $s^{-\varepsilon}$, with $\varepsilon > 0$ arbitrarily small. At each mask restrict its rows to velocities within $C_n s^{-\varepsilon} \max(r, d_\ell)$ of the cap center, and post-compose with a smooth projection onto a slightly larger Fourier band. Choose successively increasing constants to include the preceding supports. The modified operators have bounded norm, and Gaussian frequency tails make their output differences negligible in L^2 , uniformly per fiber and branch. At the stopping interval I , write their states as $w_{I,k,y}$; their bandwidths in velocity are $O_n(s^{-\varepsilon}r)$. The fixed factors $s^{-C_n\varepsilon}$ below become subpower losses by taking the inner limit before $\varepsilon \downarrow 0$, even if the mask counts grow in a subsequent outer approximation.

For $T \in I$ the modified fields satisfy

$$\|u_{k,y}(T, x')\|^2 \lesssim_n \frac{s^{-C_n\varepsilon}}{B_*} \int_{|x'-z-(T-T_I)\theta_k| \leq s^{-C_n\varepsilon}B_*} \|w_{I,k,y}(z)\|^2 dz + \text{negl } e_y. \quad (\text{env})$$

To prove this, insert a smooth plateau on the stopping state's band before its first propagation. For duration at most ℓ_0 , its kernel has size $O(s^{-C_n\varepsilon}/B_*)$ and rapid tails beyond the displayed displacement. After the stop, $d_\ell \gtrsim r$ and $q_\ell \lesssim B_*$. The position kernel of a mask restricted to row velocities of width $O_n(s^{-\varepsilon}d_\ell)$ has absolute operator-valued bound $O_n(s^{-\varepsilon}/q_\ell)$ and Gaussian tails on scale q_ℓ : insert the two normalized analyzing packets, bound the row contraction by one, and integrate their product against the frame measure. Post-projection and bandlimited free evolution for a duration at most ℓ have Schwartz tails on the same scale, with slack. The effective uncertainty parameter is bounded in (q_ℓ, d_ℓ) units. Each kernel has integral bounded by $s^{-C_n\varepsilon}$. Compose after the first convolution and apply Cauchy–Schwarz against its absolute kernel. The resulting rapid tails can be bounded by negligible times e_y . This proves Equation (env), also for repeated cuts.

Expand the squared norm of $\sum_k u_{k,y}$ in Equation (HL). Pairs with centers within $s^{-2\varepsilon}r$ are bounded by $s^{-O(\varepsilon)} \sum_k \|u_{k,y}\|^2$. Call the remaining pairs high pairs, and write $V = |\theta_k - \theta_{k'}| \geq s^{-2\varepsilon}r$ for their separation. Only their sum will incur an ordinary power of r^{-1} .

The transverse crossing estimate. For normal fields f, g and $a > 0$, define the crossing energy

$$\mathcal{C}_a(f, g) = \int_{|x_1 - x_2| \leq a} \|f(x_1)\|^2 \|g(x_2)\|^2 dx_1 dx_2.$$

For each high pair and exact index,

$$s \sum_T \int_{\mathbb{R}} \|u_{k,y}(T, x')\|^2 \|u_{k',y}(T, x')\|^2 dx' \lesssim s^{-O_n(\varepsilon)} V^{-1} e_y^2 + \text{negl } e_y^2. \quad (\text{cross})$$

We prove the localization needed before the stopping cut, so that independent estimates on many short intervals do not introduce a scale loss. At each cut at or before the stop call the two modified band states $g_{1,I}, g_{2,I}$, and put $W_I = \ell_I V$. For sufficiently large fixed D , their stopping crossing energy satisfies

$$\sum_I \int_{|x_1 - x_2| \leq s^{-D\varepsilon} W_I} \|g_{1,I}(x_1)\|^2 \|g_{2,I}(x_2)\|^2 dx_1 dx_2 \lesssim s^{-O_{D,n}(\varepsilon)} e_y^2.$$

We establish this by pulling back successively from each cut to its parent, increasing the slack constant a fixed amount at each step. At the root total energy gives the assertion.

Work in a common Galilean frame where both band velocities are $O(V)$. They remain separated by a constant times V before the stop. For a parent I and its children J at the next cut write $W = \ell_I V$, $w = \ell_J V$, and

$$g_{\alpha,J} = A_{\alpha,J} U(T_J - T_I) g_{\alpha,I}.$$

The required one-step estimate is

$$\sum_{J \subset I} \mathcal{C}_{s^{-D\varepsilon} w}(g_{1,J}, g_{2,J}) \leq s^{-O_{D,n}(\varepsilon)} \mathcal{C}_{s^{-O_{D,n}(\varepsilon)} W}(g_{1,I}, g_{2,I}) + \text{negl } \|g_{1,I}\|_2^2 \|g_{2,I}\|_2^2.$$

Thus the child crossings must be charged to nearby pairs in the parent, with no factor for the number of children. We first control the position spread of the cut operators, then sum the freely evolving separated pairs by bilinear Plancherel.

Decompose each parent state into

$$g_{\alpha,I} = \sum_{b \in \mathbb{Z}} P_\alpha(\mathbf{1}_{[bW, (b+1)W]} g_{\alpha,I}),$$

where P_α is a smooth plateau of width comparable to V , equal to one on the input band and with the two enlarged supports still separated. At a child crossing, only pairs with $|b - b'| \leq s^{-O_D(\varepsilon)}$ contribute, up to negligible error. Indeed free band propagation has negligible tail beyond a slack multiple of W , and the child masks and projections have position width at most a slack multiple of w . The needed inequalities are

$$H/V \lesssim \ell_J V, \quad q_{\ell_J} \lesssim \ell_J V, \quad H/r \lesssim \ell_J V,$$

which follow from $\ell_J \geq \ell_0$ and $V \gg r$. Here is a quantitative tile argument for this step. For intervals $E_m = [mw, (m+1)w)$, put

$$B_{\alpha,J,t} = A_{\alpha,J} U(T_J - t) \tilde{P}_\alpha, \quad K_{mn} = \|\mathbf{1}_{E_m} B_{\alpha,J,t} \mathbf{1}_{E_n}\|_{2 \rightarrow 2},$$

where $\tilde{P}_\alpha$ is a slightly larger plateau equal to one on the support of P_α . Its two supports remain separated by a constant times V . Uniformly in J , $t \in J$, and the measurable row contractions, $\|B_{\alpha,J,t}\|_{2 \rightarrow 2} \leq C_n$, and there is $S_0 = s^{-C_n \varepsilon}$ such that, for every N and $R \geq 2S_0$,

$$\sup_m \sum_{|n-m| > R} K_{mn} + \sup_n \sum_{|m-n| > R} K_{mn} \leq C_{N,n} S_0^{C_n} (R/S_0)^{-N}. \quad (4.1)$$

We verify that this bound requires no derivatives of a mask. For the length- ℓ frame, Plancherel in packet velocity gives, for any position set E and $L > 0$,

$$\|\mathbf{1}_{\{D: \text{dist}(D,E) > Lq_\ell\}} V_\ell \mathbf{1}_E\|_{2 \rightarrow 2} \leq C e^{-cL^2}.$$

Indeed its squared norm is bounded by the supremum, for $x \in E$, of the Gaussian integral over those packet centers D . To bound a mask between separated position sets, split packet centers into those within one third of their separation from the input set and the complement. On the complement use this analysis bound; on the first part use its adjoint for synthesis into the output set. The row contraction preserves both sets of centers. Thus the off-diagonal mask norm has Gaussian decrease on scale q_ℓ , uniformly in the mask and Hilbert multiplicity.

The Fourier projections have Schwartz kernel tails on scale H/r . For $U(T_J - t)\tilde{P}_\alpha$, velocities are $O(V)$ and travel at most $O(w)$, while the kernel's wavelength is $O(H/V) \leq O(w)$. Outside a sufficiently large multiple of w , its phase derivative is bounded below by a constant times position displacement. Integration by parts gives arbitrarily rapid integral tails in displacement divided by w . All three widths are at most w by the displayed inequalities. Products of the finite number of such operators have the matrix tail bound (4.1): at large total tile separation one factor has comparably large separation, and convolution of their summable tails preserves arbitrary decay. The slack enlargements only replace w by $S_0 w$ and insert a fixed power of S_0 .

Choose $R = s^{-C'_{D,n} \varepsilon}$ with $C'_{D,n}$ larger than the constant in S_0 . Equation (4.1), with N arbitrarily large after fixing ε , allows deletion of all matrix blocks with $|m - n| > R$ in negligible operator norm. If B^R is the retained operator, Cauchy–Schwarz and the individual block bound $K_{mn} \leq C_n$ give the explicit local inequality

$$\|\mathbf{1}_{E_m} B^R f\|_2^2 \leq C_n (2R + 1) \sum_{|n-m| \leq R} \|\mathbf{1}_{E_n} f\|_2^2.$$

For $S \geq 1$, cover the crossing region by tile pairs $|m - m'| \leq S + 2$ and apply the local inequality to both factors. Each input pair occurs at most $(2R + 1)^2$ times. The triangle inequality between tile indices therefore gives

$$C_{Sw}(B_1 f_1, B_2 f_2) \leq C_n (2R + 1)^4 \mathcal{C}_{(S+2R+5)w}(f_1, f_2) + \text{negl} \|f_1\|_2^2 \|f_2\|_2^2. \quad (4.2)$$

The error estimate follows by viewing $C_a^{1/2}$ as the L^2 norm of a restricted tensor product and using the negligible operator-norm difference between B and B^R .

Apply the same matrix argument at tile width W to $A_{\alpha,J} U(T_J - T_I) P_\alpha$. All its movement and kernel widths are at most a slack multiple of W . After negligible deletion, on each output W -tile there are only $s^{-O_{D,n}(\varepsilon)}$ contributing parent pieces. Pointwise Cauchy–Schwarz for their sums in both factors bounds the child crossing by the sum of the crossings of individual parent piece pairs, with a slack factor. A contributing pair must have $|b - b'| \leq s^{-O_{D,n}(\varepsilon)}$: the output positions are within $s^{-D\varepsilon} w \leq s^{-D\varepsilon} W$ and both input tile indices are within a slack distance of their output tile. Dropping the output tile restrictions on each retained pair only increases its

crossing integral. This proves the earlier parent-pair restriction with its required restricted-region bookkeeping.

For a retained pair write $f_{\alpha,b} = P_\alpha(\mathbf{1}_{[bW,(b+1)W]}g_{\alpha,I})$ and $v_{\alpha,b}(t) = U(t - T_I)f_{\alpha,b}$. At every $t \in J$, $B_{\alpha,J,t}v_{\alpha,b}(t)$ is exactly the cut image of this piece. Use Equation (4.2) with $S = s^{-D\varepsilon}$, and write $L = s^{-O_{D,n}(\varepsilon)}$ for its enlarged radius and prefactor. All next-cut children have the same length ℓ_J and are disjoint. Hence averaging over each child gives

$$\begin{aligned} & \sum_{J \subset I} \mathcal{C}_{Sw}(A_{1,J}v_{1,b}(T_J), A_{2,J}v_{2,b'}(T_J)) \\ & \leq \frac{L}{\ell_J} \int_I \int_{|\Delta| \leq Lw} \int_{\mathbb{R}} \|v_{1,b}(t, x)\|^2 \|v_{2,b'}(t, x + \Delta)\|^2 dx d\Delta dt + \text{negl} \|f_{1,b}\|_2^2 \|f_{2,b'}\|_2^2 \\ & \leq C \frac{L^2 w}{\ell_J V} \|f_{1,b}\|_2^2 \|f_{2,b'}\|_2^2 + \text{negl} \|f_{1,b}\|_2^2 \|f_{2,b'}\|_2^2. \end{aligned} \quad (4.3)$$

The last line extends the time integral to $\mathbb{R}$ and uses bilinear Plancherel for the separated plateau supports, independently for each shift. Thus the only scale factor is $w/(\ell_J V) = 1$; there is no factor equal to the number of children.

Finally, $\|f_{\alpha,b}\|_2^2 \leq C \|\mathbf{1}_{[bW,(b+1)W]}g_{\alpha,I}\|_2^2$. If $|b - b'| \leq K_0 = s^{-O_{D,n}(\varepsilon)}$, every pair in these original tiles satisfies $|x_1 - x_2| \leq (K_0 + 2)W$. The tiles are disjoint, so

$$\sum_{|b-b'| \leq K_0} \|f_{1,b}\|_2^2 \|f_{2,b'}\|_2^2 \leq C \mathcal{C}_{(K_0+2)W}(g_{1,I}, g_{2,I}).$$

Combining this with Equation (4.3) proves the one-step pullback into the enlarged, restricted parent crossing energy. Finite induction proves the stopping estimate. Polynomially many intervals multiply only negligible errors, whose decay order was chosen after all cutoff powers were fixed.

From stopping energy to terminal crossings. Now insert Equation (env) for both outputs from a stopping interval. For fixed z_1, z_2 , their kernel cores overlap in a position interval of length at most $s^{-O_n(\varepsilon)}B_*$, and at only $s^{-O_n(\varepsilon)}B_*/(sV)$ terminal times. We used $B_* \gg s$ to absorb the possible extra endpoint time. They overlap only if $|z_1 - z_2| \leq s^{-O_n(\varepsilon)}\ell_0 V$. The two envelope prefactors, the position length, and s times the time count give $s^{-O_n(\varepsilon)}V^{-1}$. The stopping crossing bound proves Equation (cross). A constant additive tail from one envelope is integrated using the other's L^2 bound, or the spatial integral of its core, so unbounded position causes no extra error.

Summing the high pairs. At a given time put $H_y(x') = \|u_{k,y}(T, x')\| \|u_{k',y}(T, x')\|$, and define

$$M_{T,d} = \int_{Y_d(T)} e_y dy, \quad N_{T,d} = \int_{Y_d(T)} e_y^{-1} \int_{\mathbb{R}} H_y(x')^2 dx' dy,$$

with zero integrand when $e_y = 0$. Contraction gives $\int \int_{Y_d(T)} H_y dy dx' \lesssim M_{T,d}$, while Cauchy–Schwarz in y bounds its squared L^2 norm by $M_{T,d} N_{T,d}$. Interpolation, or Cauchy–Schwarz between these two integrals, therefore gives

$$\int_{\mathbb{R}} \left(\int_{Y_d(T)} H_y(x') dy \right)^{3/2} dx' \lesssim M_{T,d} N_{T,d}^{1/2}.$$

Lemma 4.2 applied to the original group's base marginal gives $\sum M_{T,d}^2 \leq s^{1-o(1)} \kappa e_g$. Equation (cross) gives $\sum N_{T,d} \lesssim s^{-1-O_n(\varepsilon)} V^{-1} e_g$. Cauchy-Schwarz in (T, d) bounds one high pair by $s^{-o(1)-O_n(\varepsilon)} V^{-1/2} e_g \sqrt{\kappa}$. There are $O(r^{-2})$ pairs, and the finite-sum inequality for the 3/2 power introduces only another fixed power of their number. Hence all high terms in Equation (HL) are bounded by

$$s^{-o(1)-O_n(\varepsilon)} r^{-D_0} e_g \sqrt{\kappa}$$

for an absolute D_0 , independent of the number of masks.

Summing the low contribution. Partition each stopping interval into intervals J_* of length B_* , and let $L_b = [bB_*, (b+1)B_*)$. Use one actual cylinder experiment whose root is the direct sum of the r -cap inputs, whose masks through the stopping cut act componentwise, and which thereafter evolves freely. For each J_* , let ν_{J_*} be this experiment's analyzed row measure at its left endpoint. All J_* and all position bins L_b thus belong to the same experiment; its root energy is $O(e_g)$ once, not the sum of copied energies over J_* . The analytical approximation operators used to prove the envelopes are not included in this experiment. For $T \in J_*$ and $x' \in L_b$,

$$\int_{Y_d(T)} \sum_k \|u_{k,y}(T, x')\|^2 dy \leq \frac{s^{-O_n(\varepsilon)}}{B_*} \nu_{J_*} \{y \in Y_d(T), |N| \leq C, \\ \text{dist}(D_{T_{J_*}}, L_b) \leq s^{-O_n(\varepsilon)} B_*\} + \text{negl } e_g.$$

To prove this, move the windows in Equation (env) from the stopping time to T_{J_*} . The inverse bandlimited transfer needs displacement uncertainty at most $\ell_0 O_n(s^{-\varepsilon} r) = O_n(s^{-\varepsilon} B_*)$, so local stopping energy is bounded by the freely continued energy on the displayed enlarged window. Pass to analyzed mass using its spatial Gaussian marginal, since $q_{B_*} \lesssim B_*$. Gaussian frequency tails allow restriction to $|N| \leq C$, because $d_{B_*} \asymp \sqrt{r}$ and cap centers are bounded. Take ε small relative to ζ and the fixed number of cuts. The modified stopping states differ negligibly from the actual direct-sum states. Cover enlarged windows by $s^{-O_n(\varepsilon)}$ ordinary B_* -bins, with the same overlap bound.

Integrate in x' ; a polynomial restriction is allowed as explained after Equation (HL). The low contribution is thus, up to slack and negligible errors, at most

$$B_*^{-1/2} \sum_{J_*, b} \sum_{T \in J_*, d} \left[\nu_{J_*} \{D_{T_{J_*}} \in L_b, |N| \leq C, y \in Y_d(T)\} \right]^{3/2}. \quad (\text{low})$$

Peel ordinary cylinder regularity levels of the positive measure ν_{J_*} , and let z_β be summed mass at threshold β . Within each level and each interval the regularity is $O(\beta)$. The heavy pieces are disjoint prefix assignments across the full J_* family, so the definition of Γ applied once to this actual direct-sum experiment gives

$$\sum_\beta z_\beta \sqrt{\beta} \lesssim s^{-o(1)} B_*^{1/2-\Gamma} e_g \sqrt{\kappa}.$$

Its input energy is $O(e_g)$ and its input regularity is $O(\kappa)$, by the direct-sum bound proved above. The scale B_* has fixed positive depth and the cutoff powers and number of masks are fixed first. An arbitrarily low floor contributes negligibly by total mass.

Split Equation (low) by these logarithmically many levels. For fixed J_*, b, β , restrict to its position and velocity window and take the base marginal. In units of duration B_* it has regularity

at most $C\beta R^{3+\eta} \leq C\beta R^3$ for $R \leq 1$: apply full cylinder regularity with normal width $O(1)$. This width is admissible because $d_{B_*} \asymp \sqrt{r}$. Apply Lemma 4.2 with $\delta \asymp s/B_*$. The sum of squares is at most $s^{-o(1)}\delta\beta$ times the marginal's mass, and the sum of masses is at most $O(\delta^{-1})$ times that mass. Cauchy–Schwarz bounds the corresponding 3/2 sum by $s^{-o(1)}\sqrt{\beta}$ times the marginal's mass. Summing the bins and levels and using the preceding prefix bound gives

$$s^{-o(1)-O_n(\varepsilon)} B_*^{-\Gamma} e_g \sqrt{\kappa}$$

for the low term. No ordinary power of the number of caps was used here.

Finally sum the original groups and let the inner slack losses vanish. Since $B_* = s^{1-\zeta+o(1)}$, the low term has exponent strictly below Γ . Choose $\zeta > 0$ so small that $D_0\zeta < \Gamma$; the high term also has exponent strictly below Γ . The outer zero-cost and Planck defects only contribute $s^{-o(1)}$. Their use in passing from labels to grids and groups has bounded multiplicity independent of the number of cuts; choosing the stopping scale from the actual H avoids accumulating its defect along the masks. The resulting strict improvement contradicts extremality and proves the proposition. $\square$

4.5 Planar incidences and classical labels of zero cost

We next exclude simultaneous zero quadratic and linear cost in the classical cylinder model. We first formulate the planar incidence input so that its application to conditional probability measures is explicit.

Theorem 4.7 (Discretized planar Furstenberg estimate). *Let $0 < a \leq 1$, $0 < b \leq 2$, and $\omega > 0$. There are $\sigma > 0$ and $\Delta_0 > 0$ with the following property. Work in fixed bounded coordinate charts for points and affine lines in the plane. Suppose that a probability measure ν on lines satisfies*

$$\nu(B(\ell, r)) \leq \Delta^{-\sigma} r^b \quad (\Delta \leq r \leq 1).$$

For every line in its support, suppose that a probability measure μ_ℓ on a set P_ℓ within distance $C\Delta$ of that line satisfies, uniformly in ℓ ,

$$\mu_\ell(B(x, r)) \leq \Delta^{-\sigma} r^a \quad (\Delta \leq r \leq 1).$$

If $0 < \Delta < \Delta_0$, then

$$\left| \bigcup_{\ell} P_\ell \right|_{\Delta} \geq \Delta^{-F(a,b)+\omega}, \quad F(a,b) = a + \min\{1, b, (a+b)/2\}. \quad (\text{RW})$$

Here $|\cdot|_{\Delta}$ denotes covering number; the constants may depend on the fixed charts and on C .

Proof. The separated-set version is the point–line dual of the discretized Furstenberg theorem of Ren and Wang [25, Theorem 4.1]. We explain the passage from probabilities, which is the only additional point needed here. Choose $a' < a$ and $b' < b$ sufficiently close to a, b for the prescribed final loss. Take the slack σ in the hypothesis smaller than both $\min(a - a', b - b')/2$ and one quarter of the slack allowed by the cited separated-set theorem at a', b' . Choose the additional sampling loss below another quarter of that slack. Replace line parameters by representatives of their Δ -grid cells. Points incident to a line remain within $O(\Delta)$ of its representative. On

the finite grid sample $\lceil \Delta^{-b'} \rceil$ line parameters independently from ν . For a dyadic ball of radius $r \geq \Delta$, the expected number of samples is at most

$$2\Delta^{-\sigma-b'} r^b \leq 2\Delta^{-\sigma} (r/\Delta)^{b'}.$$

For any prescribed small additional power loss, binomial tails bound all these counts simultaneously by that power times $(r/\Delta)^{b'}$. Indeed there are only polynomially many grid balls to consider, whereas the probability of exceeding a sufficiently large small-power multiple of the displayed expectation and 1 decays faster than every fixed power of Δ . This also bounds multiplicities in a single cell. The probability of one line grid cell is $O(\Delta^{b-\sigma})$, so the expected fraction of samples repeating an earlier cell is $O(\Delta^{b-b'-\sigma})$. Discard these repetitions at a small-power cardinality cost. Perform the same grid sampling on each retained line with exponent $a' < a$; its point-cell probabilities are $O(\Delta^{a-\sigma})$. The expected fraction of point repetitions is $O(\Delta^{a-a'-\sigma})$. The number of lines is polynomial, so the binomial count bounds hold simultaneously; by averaging, discard any small fraction of lines with excessive point repetitions. Equalize cardinalities at a further small-power cost. The resulting sets satisfy the nonconcentration and cardinality hypotheses of the separated-set theorem. Its conclusion, continuity of F , and choices of a', b' sufficiently close to a, b give (RW). Small changes of width and projection of the point supports onto representative lines change covering numbers only by bounded factors. All small losses are chosen before Δ tends to zero. $\square$

For a finite or countable discrete name U , side information D , and $0 < \rho < 1$, put

$$i_\rho(U \mid D) = \frac{-\log \mathbb{P}(U \mid D)}{\log(1/\rho)},$$

evaluated at the sampled value of U . Conditional probabilities are measurable versions on the probability spaces of the cylinder construction.

Lemma 4.8 (Likelihood and list comparisons). *For arbitrary additional side information D' and $\epsilon > 0$,*

$$i_\rho(U \mid D) \geq i_\rho(U \mid D, D') - \epsilon \tag{LP}$$

outside a set of probability at most ρ^ϵ . If $I(U; D' \mid D) = o(\log(1/\rho))$, the difference of these scores tends to zero in probability. More precisely, if the sampled U belongs on an event E to a list $\mathcal{L}(V, D)$ of at most ρ^{-b} names, then

$$\mathbb{P}(E \cap \{i_\rho(U \mid D) > i_\rho(V \mid D) + b + \epsilon\}) \leq \rho^\epsilon.$$

If V instead belongs to a list indexed by (U, D) , reverse U, V in this comparison. Thus a change of discrete names by such lists costs at most $b + \epsilon$ in the indicated direction. The exceptional probability is measured on E under the original law; no conditioning on E is implicit. Finally, restricting the law to an event of probability bounded below preserves any fixed finite collection of normalized scores up to an error tending to zero in probability under the restricted law.

Proof. Write $p(u \mid D)$ and $q(u \mid D, D')$ for the two conditional masses. Conditioning on (D, D') and summing over u gives

$$\mathbb{P}\{q(U \mid D, D') < \rho^\epsilon p(U \mid D)\} \leq \rho^\epsilon.$$

This is (LP). If $Z = \log(q(U \mid D, D')/p(U \mid D))$, the same argument gives $\mathbb{P}(Z < -t) \leq e^{-t}$ and hence $\mathbb{E}Z_- \leq 1$. Since $\mathbb{E}Z = I(U; D' \mid D)$, Markov's inequality applied to Z_+ proves the mutual-information assertion.

For lists, conditional on the known data, the total probability of names in a list of size L that each have conditional probability less than ρ^ϵ/L is at most ρ^ϵ . Combine this observation with the chain rule for conditional masses and the fact that $\mathbb{P}(U \mid D) \geq \mathbb{P}(U, V \mid D)$ at the sampled names. The argument still applies if only values attained on the specified event are listed.

Let E be the restricting event, with $\mathbb{P}(E) \geq c > 0$. The new conditional mass is

$$\mathbb{P}(U \mid D, E) = \mathbb{P}(U \mid D) \frac{\mathbb{P}(E \mid U, D)}{\mathbb{P}(E \mid D)}.$$

Both factors in the ratio are at most 1. Under the restricted law, the probability that either is smaller than ρ^ϵ is at most $2\rho^\epsilon/c$, by integrating the corresponding conditional probability of E . Thus the normalized logarithm of the ratio tends to zero in probability. This also proves the assertion with nonatomic side information. $\square$

Lemma 4.9 (Conditional planar incidence bootstrap). *Fix $a, g \in (0, 1]$ and $\xi > 0$. There are a finite collection of pairs $0 < z' \leq z \leq 1$ and a tolerance $\chi > 0$ with the following property. Let $(Q, \mathcal{L}, D)$ vary with $\rho \downarrow 0$, where Q and $\mathcal{L}$ lie in fixed bounded point and line charts and are incident up to $O(\rho)$. Subscripts denote grid names of width ρ^z . If, simultaneously on the chosen pairs, with probability tending to one,*

$$i_\rho(Q_{z'} \mid \mathcal{L}_z, D) \geq az' - \chi, \quad i_\rho(\mathcal{L}_{z'} \mid Q_z, D) \geq gz' - \chi, \quad (\text{cond})$$

then, with probability tending to one,

$$i_\rho(Q_1 \mid D) \geq a + \min\{1, a + g\} - \xi. \quad (\text{inc})$$

The same statement holds at any fixed output depth, with the target multiplied by that depth. The side information and conditional law may vary along the sequence.

Proof. We first justify one bootstrap step. Suppose the unconditional line scores, conditional only on D , are at least $yz' - \chi$ at a sufficiently fine finite grid of depths z' between 0 and a fixed $z > 0$. We claim that the point score at depth z is at least $(F(a, y) - \omega)z$ in probability, when the input tolerance and mesh are sufficiently small in terms of ω and z .

Otherwise pass to a subsequence on which violation has probability at least a fixed $c > 0$. A violating point belongs to a cell of conditional mass greater than $\rho^{(F(a, y) - \omega)z}$; for each D there are fewer than $\rho^{-(F(a, y) - \omega)z}$ such cells. Intersect this event with all the good input-score events. Choose a value of D for which this intersection has conditional probability bounded below, say by $c/2$. Retain fine line cells on which the conditional probability of the intersection is at least $c/4$. Their total retained witness mass is bounded below in terms of c . Restrict the original line probability to these witnesses and normalize. Any occupied coarser line cell contains a good witness, so its original mass satisfies the required bound $\rho^{yz' - \chi}$; restriction and normalization cost only a constant depending on c .

In each retained fine line cell, restrict its conditional point probability to witnesses and normalize. Each occupied coarser point cell contains a witness to the first inequality of (cond), so its mass is bounded by a constant times $\rho^{az' - \chi}$. Cover Euclidean balls by boundedly many grid cells. Between successive queried scales use the next larger queried cells; a depth mesh of

size ϑ costs at most $\rho^{-2\vartheta}$ in the ball estimates. At the coarsest end the trivial probability bound gives the same small-power loss. Thus, in units $\Delta = \rho^z$, the retained line and point measures meet Theorem 4.7 with exponents y, a and arbitrarily small nonconcentration slack. Replace each fine line cell by a representative line; its witnesses move by $O(\Delta)$. Equation (RW), with a loss less than ωz , contradicts the upper bound on the number of popular point cells. This proves the step. Point–line duality proves the symmetric step.

By (LP), the starting marginal bounds are $x = a$ and $y = g$, up to arbitrarily small losses. The two steps iterate

$$x \longleftarrow F(a, y), \quad y \longleftarrow F(g, x).$$

Use the preceding values on both right-hand sides, so the sequences are increasing and bounded. At their fixed point, $y > a$ and $x > g$. For example, $y \leq a$ would give $x = a + y \geq 2y$ and then $F(g, x) > y$, a contradiction. Consequently the fixed-point equations reduce to

$$x = a + \min\{1, (a + y)/2\}, \quad y = g + \min\{1, (g + x)/2\}.$$

This map is a contraction in the maximum norm. Its fixed point is $(2a + g, a + 2g)$ when $a + g \leq 1$, and $(1 + a, 1 + g)$ when $a + g \geq 1$. Thus finitely many iterations suffice to reach the strict target in (inc). Choose the finite grids and tolerances backwards from the final iteration. The argument for output depth z uses ρ^z as basic scale and is identical after this change of units. $\square$

We now return to a classical cylinder extremal configuration. Assume that both minimal shear costs vanish, normalize the terminal widths to $r_1 = f_1 = 1$, and retain the notation of the cylinder construction:

$$\Gamma = d + (1 - p)/2, \quad p \geq 1, \quad 0 < d \leq 1, \quad E_a = es^{-da}, \quad \kappa_a = \kappa s^{pa}.$$

The assigned terminal law is $\mathbb{P}^1$. All following statements use the ordered limits and finite meshes of that construction. In particular, ordinary regularizations are inserted in forward order, their total energies and regularity constants obey the displayed benchmarks up to $s^{-o(1)}$, the documented terminal count is at most $s^{-o(1)}E_1/\kappa_1$, and terminal assigned mass is at least $s^{o(1)}E_1$.

Let x_c, n_c be the unit-bin indices of the exact velocities X, N and set

$$A = (X - x_c, N - n_c), \quad V(A) = (A_1, A_1^2, A_2).$$

Use boosted spatial coordinates

$$(B_t - tx_c, Y_t - 2x_c B_t + tx_c^2, D_t - tn_c).$$

Their velocity is $V(A)$, and A lies in a fixed bounded square. For $0 \leq a \leq 1$, define the nested name

$$C(a) = (x_c, n_c, T_1^{(a)}, \text{position}(T_1)^{(a)}),$$

where the superscript denotes absolute dyadic bins of width comparable to s^a . Conditional on $C(a)$, refining to $C(b)$ uses at most $O(s^{-4(b-a)})$ names for $b \geq a$.

Lemma 4.10 (Geometric and probabilistic bounds for the boosted names). *Suppose $\Gamma > \eta/2$. The preceding configuration has the following properties, simultaneously on every fixed finite list of depth queries.*

1. A terminal label determines $C(1)$ to $s^{-o(1)}$ possibilities. After deterministic flattening of the relative scores of $C(a)$ given $C(0)$, their limiting profile $D(a)$ is nondecreasing and Lipschitz, with $D(0) = 0$ and $D(1) \leq p + d$.
2. For $0 < v < 1$ and $0 \leq a' \leq 1$, the typical score of $A^{(v)}$ given either its terminal label or $C(a')$ is at least $(3 - p)v$ in $\log(1/s)$ units. Every fixed strict deficit has power-small probability.
3. For fixed $0 < a < a + zh \leq 1$, the time name at width s^{a+zh} , conditional on the exact particle and $C(a)$, has typical score at least dzh in $\log(1/s)$ units, with the same meaning of strict deficit.
4. Fix $K = 3$, $a > 0$, and $h > 0$ with $a + Kh < 1$. For every $\epsilon > 0$, uniformly over measurable choices of an affine plane $\Pi_{C(a+Kh)} \subset \mathbb{R}^3$,

$$\mathbb{P}^1\{\text{dist}(V(A), \Pi_{C(a+Kh)}) \leq s^{\epsilon h}\} \leq s^{c+o(1)}$$

for some $c > 0$ depending on the fixed parameters and strict gap.

Proof. Zero shear cost and Taylor expansion on a terminal test imply unsheared velocity widths $s^{-o(1)}$ and spatial widths $s^{1-o(1)}$ after a bounded velocity boost. The possible unit boost bins therefore cost only $s^{-o(1)}$ per label. This proves the first assertion about $C(1)$. Conversely, for each fixed value c_0 of $C(0)$, take the union of the root predecessors of *all* terminal samples with $C(0) = c_0$. This whole union lies in only boundedly many root time/test pairs of bounded weight: A and terminal time are bounded, so exact backflow places the initial boosted positions in a bounded box. Consequently a list of at most $s^\tau e/\kappa$ values of $C(0)$ has predecessor mass at most $Cs^\tau e$, for every fixed $\tau > 0$. Rarity transfer (Sp) makes the corresponding terminal event power-small. Apply this to the list of atoms with probability greater than $s^{-\tau}\kappa/e$. This is the popular-list argument of Lemma 3.23, and gives the typical lower score $\log_{1/s}(e/\kappa) - o(1)$ for $C(0)$ without counting terminal times separately. The documented count and the label-to-name list give the typical upper score $\log_{1/s}(E_1/\kappa_1) + o(1)$ for $C(1)$. Their difference is $p + d$. Apply forward deterministic flattening at final readout on successively finer finite lists. The relative branching bound and Lemma 4.8 imply that all limiting increments lie between 0 and $4(b - a)$. Dense-list extension gives the asserted profile and endpoint bound.

For the angular assertion, ordinary-regularize at $j = 1 - v$. Fix a terminal label and an angular s^v -cell. In duration units of the j -interval, terminal spatial uncertainty is $s^{v-o(1)}$ in each of the three boosted spatial coordinates. Backflow through a duration at most 1 adds uncertainty $O(s^v)$ horizontally and in the normal position. After tilting by the angular-cell center, the velocity curvature contributes $O(s^{2v})$ in the last base coordinate, while terminal position uncertainty there is still $s^{v-o(1)}$. Consequently $s^{-v-o(1)}$ full tests with $r = f = s^{v-o(1)}$ cover all predecessors. Each has weight $s^{4v-o(1)}$, so their total mass is at most

$$s^{-o(1)}\kappa_j s^{3v} = s^{-o(1)}\kappa_1 s^{(3-p)v}.$$

Classical masks are pointwise contractions and do not enlarge this mass. Labels of read mass below $s^\tau\kappa_1$ have total probability at most $s^{\tau-o(1)}$ by the count. On the remaining labels the conditional angular atom bound follows, with arbitrarily small τ . A label predicts $C(a')$ with a subpower list. Testing popular angular atoms given $C(a')$, and pulling these lists back to labels, gives the same bound by Lemma 4.8.

For the time assertion apply Lemma 3.22, with the exact particle as common side information and $C(a)$ as target. At each of the cuts a and $a + zh$, use the current time and boosted-position bins at the common width s^a as relay names. Bounded boosted velocity moves position by only $O(s^a)$ throughout the depth- a interval. Thus the earlier row predicts the later relay with bounded ambiguity, and that relay predicts $C(a)$ with bounded ambiguity. The lemma and (Sp) exclude lists of fewer than $s^{-dzh+\tau}$ fine time names for any fixed $\tau > 0$. Testing the atoms that would violate the stated score gives the assertion. Approximating depths on the finite cut mesh costs only the ordered vanishing exponent errors.

It remains to prove plane avoidance. Write the plane condition as

$$|\alpha A_1 + \beta A_1^2 + \gamma A_2 + c_0| \lesssim s^{\epsilon h}, \quad \alpha^2 + \beta^2 + \gamma^2 = 1.$$

Choose a fixed $h' > 0$ sufficiently small compared with ϵh , with $i = 1 - h' > a + Kh$, and put $\Delta = s^{h'}$. Fix a small $\lambda > 0$. When $|\gamma| \leq \Delta^\lambda$, the boundedness of A_2 reduces the condition to a quadratic sublevel set in A_1 of thickness $O(\Delta^\lambda)$. At least one of $|\alpha|, |\beta|$ is bounded below. Such a sublevel set is covered by boundedly many intervals of length $O(\Delta^{\lambda/2})$: when the quadratic coefficient is bounded below this follows by factoring at the vertex, and otherwise the derivative is bounded below on the bounded interval, after a bounded partition. A terminal label knows only subpower many choices of the conditioning chart. Apply the zero-cost case of (mem) with $L = 1$ on a final depth gap $\lambda h'/3$. Its cell width is larger than these intervals, so a bounded list suffices for each chart. The exponent $1 - d + 2\Gamma$ is positive. This part has power-small probability.

Suppose now $|\gamma| > \Delta^\lambda$. Ordinary-regularize at i . A row at this cut knows only subpower many compatible boost and $C(a + Kh)$ charts, since its exact particle and current time start predict the coarser name. Make this subpower number of copies and restrict each to the selected plane slab. In its segment duration units subtract the velocity shear

$$N_{\text{sh}}(A_1) = -\gamma^{-1}(c_0 + \alpha A_1 + \beta A_1^2).$$

The corresponding spatial shear is one of the allowed cylinder shears. Its coefficients are $O(\Delta^{-\lambda})$ on any nonempty bounded chart. Choose h' small enough that $s^{\epsilon h} \Delta^{-\lambda} \leq \Delta$. The residual velocity therefore has magnitude $O(\Delta)$. Partition the residual initial normal position into intervals of length Δ . On each such slab the base marginal satisfies

$$\mu\{|X - x| \leq R, |B - b| \leq R, |Y - y - 2x(B - b)| \leq R^2\} \lesssim s^{-o(1)} \kappa_i R^{3+\eta} \Delta^{1-\eta}, \quad 0 < R \leq 1.$$

Indeed the specified set is contained in a full input test with normal width comparable to Δ . Since $R^{3+\eta} \leq R^3$, (kin) applies with constant $s^{-o(1)} \kappa_i \Delta^{1-\eta}$.

For each chart and initial normal slab, let e_* be its input mass and $m_{t,b}$ its base position-cell masses on the Δ time grid. Equation (kin) and the trivial time-copy bound give

$$\sum_{t,b} m_{t,b}^2 \lesssim s^{-o(1)} \kappa_i \Delta^{2-\eta} e_*, \quad \sum_{t,b} m_{t,b} \lesssim \Delta^{-1} e_*.$$

Cauchy-Schwarz therefore bounds their 3/2-sum by $s^{-o(1)} \sqrt{\kappa_i} \Delta^{(1-\eta)/2} e_*$. Subsequent classical contractions only decrease these masses. A terminal label meets at most $s^{-o(1)} \Delta^{-O(\lambda)}$ base position cells and initial normal slabs: its spatial diameter after the shear is $s^{-o(1)} \Delta^{1-O(\lambda)}$, and residual velocity is $O(\Delta)$ over the unit duration. Each label also sees only subpower many charts. Summing the preceding estimate and using the finite-list convexity inequality for each label yields

$$\sum_{\nu} m_{\nu}^{3/2} \lesssim s^{-o(1)} \Delta^{(1-\eta)/2 - O(\lambda)} E_i \sqrt{\kappa_i},$$

where m_ν are terminal masses restricted to the plane event. Their total is at least $s^{o(1)}\mathbb{P}^1(\text{event})E_1$. The terminal count and Hölder's inequality yield the reverse estimate

$$\sum_{\nu} m_{\nu}^{3/2} \geq s^{o(1)}\mathbb{P}^1(\text{event})^{3/2}E_1\sqrt{\kappa_1} = s^{o(1)}\mathbb{P}^1(\text{event})^{3/2}\Delta^{1/2-\Gamma}E_i\sqrt{\kappa_i}.$$

Thus the event probability to the power $3/2$ is at most $s^{-o(1)}\Delta^{\Gamma-\eta/2-O(\lambda)}$. Choose λ sufficiently small relative to $\Gamma - \eta/2$. This proves the fourth assertion, uniformly in the plane choice. Maximizing conditionally over a countable sufficiently fine parameter grid, and then enlarging the slab by a bounded factor, also shows that the conditional supremum of slab probabilities tends to zero for typical $C(a + Kh)$ names. Measurable approximate maximizers suffice. $\square$

Proposition 4.11 (Exclusion of classical zero costs). *A classical cylinder extremal configuration with $\Gamma > \eta/2$ cannot have both minimal shear costs equal to zero.*

Proof. Suppose such a configuration exists and use Lemma 4.10. Put $m = 3 - p$. The relation between p, d, Γ gives $m = 2 - 2d + 2\Gamma > 0$, while $p \geq 1$ gives $m \leq 2$. The profile supplied by that lemma satisfies $D(1) \leq p + d$. We will prove $D'(a) \geq 1 + 3d$ at almost every interior depth, contradicting $p + d = 1 + 3d - 2\Gamma$. At a short gap, conditional sampling and planar incidences will force information in a two-dimensional projection of the spacetime position. Recovering the two omitted coordinates from time uncertainty gives the required derivative bound.

We first identify depth parameters at which conditioning changes angular information negligibly on a short depth interval. For each dyadic $h > 0$, take subsequential limits, simultaneously on dense depth lists, of

$$f_h(a) = \lim \frac{H(A^{(Kh)} \mid C(a))}{\log(1/s)}.$$

These functions have values in $[0, 2Kh]$, are nonincreasing, and are Lipschitz with constant 4, by relative branching of C . The limiting normalized information equals $(f_h(a) - f_h(a + Kh))/h$. Integrating it over $0 < a < 1 - Kh$ gives at most $2K^2h$. Summing over dyadic h and applying Tonelli's theorem shows that, for almost every interior a ,

$$\frac{I(A^{(Kh)}; C(a + Kh) \mid C(a))}{h \log(1/s)} \rightarrow 0 \quad \text{as dyadic } h \downarrow 0$$

in the ordered limiting sense. Fix such an a at which D is also differentiable.

Write $\rho = s^h$, $C = C(a)$, and $C' = C(a + Kh)$. Subtract the lower endpoints of the time and position bins of C and divide by their common width. The finer name C' determines a representative $O \in \mathbb{R}^4$ with error $O(\rho^K)$ for

$$(0, W) + \theta e(A), \quad e(A) = (1, A_1, A_1^2, A_2).$$

Here W is determined by the exact particle and C , by evaluating position at the start of the time bin; all displayed variables are bounded. At each fixed finite collection of queried $z > 0$, Lemma 4.10 gives the lower scores

$$i_{\rho}(A_z \mid C') \geq mz - o(1), \quad i_{\rho}(\theta_z \mid C, \text{particle}) \geq dz - o(1)$$

in probability.

Draw M rows independently conditional on C' , with M any fixed integer, and put $L_i = A_i^{(Kh)}$, using bin representatives when evaluating $e_i = e(L_i)$. The rows share O and C . The time bound

survives conditioning on all pins. Indeed L_i is known from particle i , and conditional independence gives

$$I(\mathbf{L}_{\neq i}; \text{particle}_i, \theta_i \mid C) \leq I(\mathbf{L}_{\neq i}; C' \mid C) \leq \sum_{j \neq i} I(L_j; C' \mid C) = o(\log(1/\rho)).$$

For completeness, the second inequality follows by expanding conditional entropy: the pins have additive entropy given C' , whereas their entropy given C is at most the sum of their individual entropies. The first inequality is conditional data processing. The information bound also controls the information about θ_i conditional on its particle, by the mutual-information chain rule. Apply Lemma 4.8. Independently, the angular bounds conditional on C' survive conditioning on the other pins by the product law.

For every fixed small $\epsilon > 0$, with probability tending to one, all four-element subsets satisfy

$$|\det(e_i, e_j, e_k, e_\ell)| \geq \rho^{O(\epsilon)}, \quad |(L_i)_1 - (L_j)_1| \geq \rho^{O(\epsilon)} \quad (i \neq j).$$

To see this, sample the vectors successively. Uniform conditional plane avoidance from Lemma 4.10 excludes a slab about any plane containing the affine span of previously sampled V 's. The resulting lower bounds for successive distances multiply to a determinant bound. For fewer than three predecessors extend their span to a plane. Vertical planes give the second inequality. A union bound suffices because M is fixed. Rounding errors $O(\rho^K)$ are smaller than all these tolerances. The implied power constants are absolute.

For $i < j$, let π_{ij} be orthogonal projection to the quotient plane with kernel $\text{span}(e_i, e_j)$, using measurable orthonormal coordinates. Write $g_{ij}(A_k)$ for the projective direction of $\pi_{ij}e(A_k)$. Choose the finite query depths that will be needed in Lemma 4.9. Color each increasing triple $i < j < k$ by the vector of ϵ -quantized scores

$$i_\rho((g_{ij}(A_k))_z \mid C', \mathbf{L}_{\neq k})$$

at these depths. There are finitely many colors: scores outside $[0, 2]$ can be assigned one overflow color, which has probability tending to zero because a direction has $O(\rho^{-z})$ cells and $z \leq 1$. Take M sufficiently large that the finite Ramsey theorem [24, Theorem B, p. 267] provides four indices all of whose triples have the same color. Only this monochromatic four-set conclusion for a finite coloring of triples is used here.

We verify the recovery estimate used with this monochromatic set. For its largest index k , choose the smallest index i and the other two indices j, ℓ . A direction for $\pi_{ij}e(A_k)$ specifies the three-dimensional linear space $\text{span}(e_i, e_j, e(A_k))$; the other direction specifies $\text{span}(e_i, e_\ell, e(A_k))$. The determinant bound implies that these spaces meet transversely in $\text{span}(e_i, e(A_k))$, with inverse norms at most $\rho^{-O(\epsilon)}$. Thus two direction cells of width ρ^z constrain any bounded candidate with first coordinate 1 to within $O(\rho^{z-O(\epsilon)})$ of the affine line through $e_i, e(A_k)$. On this line, parameterized by $(1-t)e_i + te(A_k)$, the equation “third coordinate equals the square of the second coordinate” has residual

$$t(1-t)((A_k)_1 - (L_i)_1)^2.$$

The roots 0, 1 are simple with derivative bounded below by a power $\rho^{O(\epsilon)}$. Consequently the candidate angular parameter lies in a bounded number of balls of radius $\rho^{z-O(\epsilon)}$. This is a list of at most $\rho^{-O(\epsilon)}$ cells at the queried width.

It follows from the angular lower bound and list comparison that the joint score of these two directions is at least $mz - O(\epsilon) - o(1)$, on transverse samples outside a vanishing exceptional set.

The sum of their individual scores is at least the joint score minus $o(1)$ by (LP) and the chain rule. Their colors agree, so each individual score is at least $(m/2)z - O(\epsilon) - o(1)$, simultaneously at all query depths. Since there are only finitely many index choices, one fixed triple $i < j < k$ has these properties on an event of probability bounded below independently of ρ . Restrict to that event. By Lemma 4.8 all previously obtained finite score bounds remain valid in probability, with vanishing additional losses. One can further restrict to a fixed bounded slope chart after one of finitely many fixed rotations, retaining positive probability.

Set $D_* = (C, \mathbf{L}_{\neq k})$ and define

$$Q = \pi_{ij}O, \quad \mathcal{L} = \pi_{ij}(0, W_k) + \mathbb{R}\pi_{ij}e(A_k).$$

These form a point–line incidence to accuracy $O(\rho^K)$ in bounded charts. We check (cond) with $a = d$ and $g = m/2$. After adjoining particle k , the line is known. Its projected speed is at least $\rho^{O(\epsilon)}$, by transversality, so a point cell of width $\rho^{z'}$ permits at most $\rho^{-O(\epsilon)}$ time cells of that width. The conditional time bound and (LP) prove the first inequality of (cond). For the second, adjoin C' . The point name is then known, and a line cell specifies its direction to boundedly many direction cells. The score bound obtained from the monochromatic triple and (LP) prove the required line score. All tolerances can be made smaller than the tolerance in Lemma 4.9. Hence

$$i_\rho(Q_1 \mid D_*) \geq d + \min\{1, d + m/2\} - o(1)$$

with an arbitrarily small prescribed strict loss.

We must also recover the two coordinates removed by π_{ij} . Complete the quotient coordinates to an orthonormal basis using first the unit vector along the part of e_j perpendicular to e_i , and then the unit vector along e_i . Given particle j and D_* , the quotient point Q_1 has only boundedly many possibilities: its velocity component is annihilated up to $O(\rho^K)$, because L_j is known. The next coordinate has speed at least $\rho^{O(\epsilon)}$ and therefore reveals θ_j to a list of $\rho^{-O(\epsilon)}$ cells. Its score conditional on Q_1, D_* is at least $d - O(\epsilon)$ by the time bound, the bounded-list version of the chain rule, and (LP). For the last coordinate adjoin particle i . Both preceding coordinates are predictable with bounded ambiguity, whereas the last coordinate reveals θ_i with the same accuracy. This contributes another $d - O(\epsilon)$. The chain rule, bounded changes between orthonormal coordinate grids, and (LP) to remove the pins give

$$i_\rho(O_1 \mid C) \geq 3d + \min\{1, d + m/2\} - \text{arbitrarily small strict loss}.$$

The order of choices is as follows. First choose the target strict loss and the finite incidence iterations and query grids. Next choose the color accuracy, transversality tolerance, and input score tolerances sufficiently small. Then fix a finite Ramsey number M . Only after these choices let dyadic $h \downarrow 0$ at the selected a , taking all preceding scale and mesh limits first for each h . The stationary information loss then tends to zero, and every event restriction after this information comparison has probability bounded below by a constant depending on M , not on the inner scale.

Given C , the name O_1 and the refinement $C(a + h)$ determine each other with bounded ambiguity. Thus its limiting score in $\log(1/\rho)$ units is $(D(a + h) - D(a))/h$, which tends to $D'(a)$. Since $d + m/2 = 1 + \Gamma > 1$, the preceding inequality gives $D'(a) \geq 1 + 3d$ for almost every a . Integrating and using Lemma 4.10 gives

$$1 + 3d \leq D(1) \leq p + d = 1 + 3d - 2\Gamma,$$

a contradiction. This proves the proposition. $\square$

5 Canonical names and conditional observations

This section constructs the information variables used in the positive-cost cylinder argument. There are two distinct uses of probability. Canonical entropies concern the assigned law of a single experiment. Independent observations and the walks constructed below are auxiliary experiments, conditioned on a specified common side variable. We shall keep their laws separate.

Throughout, we work on an affine extremal configuration satisfying (F), with ordinary regularity imposed before the documented gates. Thus

$$r_1 = f_1 = 1, \quad p + d = 1 + 3d - 2\Gamma.$$

In the hybrid case the growth exponent is strictly larger than the classical comparison exponent, so that (Pl) is available. The limiting conventions of the framework apply: every depth gap, finite query list, and operator stack is fixed before the preceding approximation errors tend to zero. In particular, an error denoted by $o(h)$ at a tangent gap h is obtained by taking those earlier limits first.

5.1 The canonical coordinates

For a real number or vector z , let $[z]_b$ be its absolute dyadic cell at width comparable to s^b , or the fixed representative of that cell when a coordinate is required. The convention applies also to negative b . We also write $X^{(v)}$ for the angular cell $[X]_v$. Let t_a be the left endpoint of the depth- a ancestor of the terminal time. Approximate gates may be used, with their errors taken to zero before any subsequent gap is reduced.

In the classical model, y_0 denotes the exact particle, including all exact auxiliary indices. A positive-time label on its realized path is determined by y_0 and the corresponding interval start. The label at the final assignment is still sampled. In the hybrid model the base coordinates are exact; the sampled normal coordinates are extended backward as a classical ray only for the purpose of defining names. Comparisons of this artificial ray with physical rows will always use (Pl).

A canonical name is a finite record of a ray in a specified shear frame. It stores the rounded coefficients of that frame, then the base coordinates and the residual normal coordinates. For example, a linear shear with recorded coefficient L replaces normal velocity and position by $N - LX$ and $D_{t_a} - LB_{t_a}$. Their bins describe a cell in that known frame; they do not specify an exact particle. The displays below choose which curvature coefficients to retain, and the templates choose the precision of the entries.

Finite partitions and their conditional entropies across resolutions also underlie the local entropy methods of Hochman–Shmerkin [13, Section 4]. Here the records must additionally retain the time gate and the changing shear.

Definition 5.1 (Curvature displays). Fix one of the following displays on an entire parameter patch.

- (i) In the quadratic equality case, write

$$b(a) = b_2(a) = c(a - 1), \quad b_1(a) = (c - J)(a - 1), \quad c > 0,$$

and put $P^\diamond = P_1(F)$. If $c > 1$, let R_a, k_a be the rounded values of $\beta_1(t_a), K_1$, at depths $b(a), b(a) - a$, respectively, and extend R_a with slope k_a from t_a . If $c \leq 1$, round only $\beta^\diamond = \beta_1(T_1)$ at depth $b(a)$, and set $k_a = 0$.

- (ii) In the pure linear case $c_2 = 0 < c_1$, set $R_a = k_a = 0$, $b_1(a) = c_1(a - 1)$, and take $P^\diamond$ to be the linearized terminal coefficient at the center of its own base test. Earlier tests are replaced by their enclosing linear tests from (F), with vanishing exponent loss.
- (iii) In the strict case $c_2 = 1$, $q < 1$, fix an endpoint $D \in (0, 1]$; hybrid applications use $D = 1$. Use the encountered label λ_D , put $P^\diamond = P_D(F)$, and first record

$$\bar{k} = [K_D]_{b_2(D)-D}.$$

Subtract the common shear whose curvature trajectory is $(\bar{k}t, \bar{k})$, with zero other coefficients. All subsequent coordinates in this display are the residual coordinates. Set

$$b(a) = b_2(D) + a - D \quad (a \leq D),$$

round the residual $\beta_D(T_D)$ at depth $b(a)$ to obtain R_a , and set $k_a = 0$. The actual derivative exponent remains $b_1(a) = (q - J)(a - 1)$. For $D < 1$ the law remains the final classical assigned law; its encountered D -label supplies the data.

Write P for the specified derivative, after the global subtraction in case (iii).

The horizontal resolution is $v \geq 0$. We use one of the templates

template	$l(a, v)$	$w(a, v)$
quadratic	$b(a) + v$	$b(a) + 2v$
clipped	$\min\{b(a) + v, b_1(a)\}$	$l(a, v) + v$
pure linear	$b_1(a)$	$b_1(a) + v$

The clipped template is used only when $b'_1 > 0$. All derivatives of b , or of b_1 when it occurs as an active branch, are positive. An explicitly indicated enlargement parameter $g \geq 0$ changes only the normal phase widths from $w, a + w$ to $w - g, a + w - g$. Unless indicated otherwise, $g = 0$.

Definition 5.2 (Canonical name). The name $\mathcal{C}(a, v)$ records $\bar{k}$ when present, t_a, R_a, k_a , and

$$L = [P - 2R_aX + 2k_aB_{t_a}]_l.$$

It also records

$$[X]_v, \quad [B_{t_a}]_{a+v}, \quad [Y_{t_a} - 2[X]_vB_{t_a}]_{a+2v}.$$

Finally it records the residual normal velocity and position at t_a , at depths $w - g, a + w - g$, after subtracting the shear

$$\begin{aligned} N &\longmapsto N - LX - R_aX^2 - k_a(Y_{t_a} - 2XB_{t_a}), \\ D_{t_a} &\longmapsto D_{t_a} - LB_{t_a} - R_aY_{t_a} + k_aB_{t_a}^2. \end{aligned}$$

Here R_a in the displayed expressions is its value at t_a .

All names are used on significant polynomial support. The cutoff convention permits us to remove negligible mass and enlarge a fixed polynomial support on each inner setup. Thus finite names have the cardinality bounds needed for likelihood and entropy arguments. Exact side variables need not have finite range. Conditional kernels can be taken on the standard Borel codes of Lemma 3.19. Completing-path tests below are projections of the specified Borel finite-path relations. Before another path test uses a completed-measurable row restriction, we take its Borel version under that stage's active row law as in the lemma.

Lemma 5.3 (Refinement and comparison). *Fix a template and g . If both a and v increase, the finer canonical name determines the coarser one up to bounded lists. The relative branching exponent is bounded by a constant times the sum of the two depth changes. At approximated gates the lists may have size $s^{-o(1)}$. Names at nearby parameters admit mutual comparison lists with exponent tending to zero with the parameter displacement.*

Proof. The time cells are nested. For base coordinates, backflow uses $B_t = B_{t'} + (t - t')X$, $Y_t = Y_{t'} + (t - t')X^2$. After subtracting the tilt $2[X]_v B_t$, the unresolved contribution of the time displacement is quadratic in the unresolved angular increment. Its sizes are therefore s^{a+v} and s^{a+2v} . Changing the angular cell center produces its known linear correction and an error of the latter size. These facts give both the base lists and their relative branching bounds.

Compare two curvature displays in the coarse time units. Their differences have sizes $O(s^{b(a)})$ and $O(s^{b(a)-a})$. For a nonconstant display, extrapolation backward from the finer time preserves these bounds because $b' - 1 \geq 0$. Constant displays require no slope comparison. After both displays have been guessed, the unresolved part of their derivative difference has size at most $s^{b(a)+v}$, which is no larger than s^l . Thus the derivative cells have bounded comparison lists.

For completeness, the normal estimate uses the difference of the two shears, rather than either shear separately. Its derivative on the actual base ray is $O(s^l)$, since both displayed derivatives approximate P . Its curvature difference has size $O(s^{b(a)})$ in the coarse time units. On the unresolved base cell, Taylor's identity bounds its velocity uncertainty by

$$O(s^{l+v} + s^{b(a)+2v}) = O(s^w),$$

and its position uncertainty by s^a times this quantity. These estimates are stronger when the normal widths are enlarged. Shear subtraction commutes with backflow. This proves the lists for the residual coordinates even when the absolute shear or absolute position is large. The same calculations with unequal resolutions give at most $s^{-C(|\Delta a|+|\Delta v|)-o(1)}$ choices. Comparison through a common coarser pair proves the final assertion. $\square$

5.2 Entropy bounds and label prediction

We request deterministic final-readout profiles for the finite names used below. Let $\mathcal{H}(a, v)$ denote the limiting entropy of $\mathcal{C}(a, v)$, divided by $\log(1/s)$, and put

$$D_0 = \lim \log_{1/s}(e/\kappa), \quad W(a, v) = (3 + \eta)v + (1 - \eta)(w(a, v) - g).$$

Proposition 5.4 (Canonical entropy and endpoint recovery). *On its domain of use, the excess satisfies*

$$E(a, v) = \mathcal{H}(a, v) - D_0 - (p + d)a - W(a, v) \geq 0. \quad (\text{canon})$$

For hybrid names this assertion is used only when $g = 0, w < 0$, and at their terminal limits by continuity. The following equality and prediction statements also hold, with lists of size $s^{-o(1)}$.

(i) *For $D = 1, g = 0$, equality holds at $(1, 0)$. In the classical equality and pure linear cases with $b'_1 > 0$, equality holds at $v = u(a)$. In a classical strict family ending at D , it holds at $(D, u(D))$. At these equality points use the template with $l = b + v = b_1$, or the pure linear template.*

(ii) *In a strict family every row connected from $i < D$ to D , and its label, predicts $\bar{k}$. After its subtraction,*

$$|K_i| \leq s^{b_2(D)-D-o(1)}, \quad |\beta_D(T_D) - \beta_i(T_i)| \leq s^{b(i)-o(1)}.$$

(iii) In an equality or pure linear family, suppose $J > 0$, $a < j < 1$, $v = u(j) > 0$, and

$$l(a, v) \leq \min_{[j, 1]} b_1.$$

Then λ_j predicts $\mathcal{C}(a, v)$ on every completing path; in the hybrid case require $w < 0$.

(iv) In the classical cases with $b'_1 > 0$, fix $0 < a < j$. If $v - u(j) \geq 0$ is sufficiently small compared with $j - a$ and $l(a, v) \leq b_1(j)$, then $(\lambda_j, X^{(v)})$ predicts $\mathcal{C}(a, v)$. The strict version uses $j = D$. There λ_D alone suffices at $v = u(D)$; this last assertion also holds for hybrid $D = 1$ and $w < 0$, without the condition $b'_1 > 0$.

No equality is asserted at an arbitrary point where $\min_{[a, 1]} b_1 < b_1(a)$.

Proof. Fix a canonical name at depth a . On its ancestor rows the base coordinates lie in the displayed base cell; after the displayed shear the two normal coordinates lie in cells of widths s^{w-g} , s^{a+w-g} . Conversion between t_a and the actual gate time has the same precision because the residual velocity is known to that precision. Thus those rows are covered by a small list of regularity tests of weight s^W , in a small list of time intervals. Their benchmark mass is at most $s^{pa+W-o(1)}\kappa$. Division by the assigned-law benchmark $s^{-da}e$, and the sparse-list consequence of (Sp), exclude cells whose typical probability is larger than $(\kappa/e)s^{(p+d)a+W-o(1)}$. This is (canon). In the hybrid comparison the artificial ray differs from the ancestor row by at most $s^{-o(1)}$ in normal velocity and $s^{a-o(1)}$ in position, by (P1). Since the base is identical, all shears preserve this additive comparison. It fits the required widths when $w < 0$. The classical virtual endpoint uses the same sparse-list argument on the final law.

For the upper bounds at the stated equality points, the reached label determines the name to a small list. Curvature comparisons are those of (F). If $c \leq 1$, omitting the slope of the curvature trajectory costs at most the own curvature width in its own duration units. The derivative discrepancy along an actual ray is bounded by $s^{\min b_1 - o(1)}$ on the connecting interval. Once the curvature cuts are guessed, recentering over the label's base cell adds at most $s^{b_2(a)+u(a)-o(1)}$ to that derivative. The Taylor calculation in Lemma 5.3 then gives the normal widths. In a strict family the residual own slope is bounded by $s^{b_2(D)-D-o(1)}$, so its omission is harmless at D . The documented label count supplies the reverse entropy inequality.

Assertion (ii) follows directly from (A). The slope comparison on $[i, D]$ predicts the global bin and gives the first bound. Multiplying this residual slope bound by the earlier duration gives the artificial depth $b(i)$. The coefficient comparison itself is finer, since $\min_{[i, D]} b_2 > b(i)$, and gives the second bound.

For (iii) and (iv), first guess the earlier curvature cuts. Backflow of the later base test, with $X^{(v)}$ when required, gives the earlier base widths: its uncertainties $s^{j+u(j)}$ and $s^{j+2u(j)}$ fit when the right excess $v - u(j)$ is small compared with $j - a$. At zero excess the inequalities are weak and the allowed small lists suffice. At the later time the unresolved angular part of the relative derivative has size at most $s^{b(a)+v-o(1)}$; the unresolved position part of a curvature slope has the additional later-time factor. The remaining derivative discrepancy fits by the stated condition on b_1 . Taylor's identity now gives the normal comparison, using $n(j) \geq w(a, v)$, with strict slack for a small positive excess.

When a curvature slope was omitted, perform this comparison at T_j , not across the whole earlier duration. The omitted slope has size $s^{b_2(j)-j-o(1)}$; its contribution from unresolved own positions to the derivative is $s^{b_2(j)+u(j)-o(1)}$. It fits the later normal widths. The unresolved angular increment multiplies only the curvature *difference*, of size $s^{b(a)-o(1)}$. Having compared

velocity and position at T_j , backflow gives the earlier position assertion. These bounds depend only on the later label and the allowed common base region, so the lists are uniform over all completing paths of that label. $\square$

5.3 Deterministic profiles and their traces

The final-readout flattening is performed on growing finite lists of queries, after the ordinary gates have been prepared in forward order. Requests may include canonical names, angular bins, mixed time and coefficient bins, and exact side data such as y_0 . Each retained class has exponent-zero fraction by the count constraint and minimality of p ; subsequent final restrictions preserve the earlier deterministic profiles. A virtual endpoint chosen after preliminary profiles is fixed before requesting its additional profiles. Its exact side variable is common over that request; no interpolation across unrelated virtual labels is used. The same procedure applies to one intermediate label selected after the preliminary profiles: only the final readout is refined again.

For any fixed finite query list and any fixed positive tolerance, the profile assertions hold off power-small sets. Indeed the flattening fixes the preselection log probabilities; likelihood comparison bounds the change under an exponent-zero restriction, and list counting controls exceptionally small probabilities. Consequently such deterministic profiles persist under a later restriction of fraction $s^{o(h)}$, with errors vanishing in h -units after the earlier limits. This fact does not assert stability of an arbitrary mutual-information budget under that restriction.

Let

$$U(x; a, v) = \lim \frac{-\log \mathbb{P}(X^{(x)} \mid \mathcal{C}(a, v))}{\log(1/s)}$$

denote the deterministic angular score. It also equals the limiting conditional entropy. It is nonnegative, nondecreasing and 1-Lipschitz in x , and vanishes at $x = v$. Lemma 5.3 gives Lipschitz dependence on (a, v) . Its increments in x decrease when either conditioning depth increases.

Lemma 5.5 (Angular trace and time stationarity). *At almost every interior (a, v) , and at almost every v for each fixed admissible a , there is $m(a, v) \in [0, 1]$ such that*

$$U(v + p'h; a, v) = m(a, v)p'h + o(h) \quad (p' \geq 0). \quad (\text{ang})$$

The trace has a version nonincreasing in a at fixed v . Moreover, at almost every interior root, and at almost every time on an admissible fixed- v boundary interval,

$$U(v + Gh; a, v) - U(v + Gh; a + Gh, v) = o(h) \quad (\text{ng})$$

for every fixed G , as $h \downarrow 0$.

Proof. At a fixed a , take derivatives in x for a countable dense set of conditioning depths. Their values lie in $[0, 1]$, and are nonincreasing as the conditioning depth increases. Monotone envelopes therefore define the trace from conditioning depths below x . For $x \in [v, v + h]$, its derivative at conditioning depth v lies between that trace and the derivative at depth $x - \varepsilon$, provided $h < \varepsilon$. At common Lebesgue points, integration over $[v, v + h]$, followed by $h \downarrow 0$ and then $\varepsilon \downarrow 0$, makes the two bounds agree. This proves (ang). Countable envelopes and continuity recover the original profiles. Time monotonicity follows from the corresponding monotonicity of every difference quotient.

For fixed dyadic h , the function $a \mapsto U(v + Gh; a, v)$ is nonincreasing and bounded by Gh . The integral of its decrement over a translation of length Gh is consequently $O_G(h^2)$ on a compact interval. Divide by h and sum over dyadic h . The resulting summable nonnegative functions tend to zero almost everywhere. Nesting bounds an arbitrary smaller gap by a comparable dyadic gap. The same proof can be integrated in v , giving the interior statement. For a fraction- $s^{o(h)}$ final restriction, first transfer the deterministic scores. Chain rule and bounded relative name capacities then recover the normalized entropy difference; direct conditioning of the old information budget is not required. $\square$

Lemma 5.6 (Mixed coefficient trace). *Let V_* be an absolute scalar parameter and S a fixed side variable. Flatten*

$$H(x, z) = \lim H_s([V_*]_x \mid S, t_z).$$

The x -derivatives lie in $[0, 1]$ and decrease with z . On an affine graph $x = B(z)$ of slope $c' > 0$, there are almost-everywhere traces $0 \leq \nu_-(z) \leq \nu(z) \leq 1$. More precisely, at almost every a , fix offsets $\alpha_1 \leq \alpha_2$ and β for which the following queries are admissible, and put $I = [\alpha_1, \alpha_2]$. Then

$$\begin{aligned} & H(B(a) + \alpha_2 h, a + \beta h) - H(B(a) + \alpha_1 h, a + \beta h) \\ &= h(\nu_-(a)|I \cap (-\infty, c'\beta]| + \nu(a)|I \cap (c'\beta, \infty)|) + o(h). \end{aligned}$$

The coefficient-depth interval is split at $x = B(a + \beta h)$; the displayed lengths are in units of h . The larger trace is taken from earlier conditioning times at a fixed coefficient depth.

Proof. Use monotone envelopes of the derivatives for countably many time depths. At a point of the graph, squeeze the integrand on each side between its trace and its value with time moved by a fixed ε to that side. Lebesgue differentiation in x , the positive affine change of variable $x = B(z)$, and then $\varepsilon \downarrow 0$ prove the assertion. Joint and conditional scores agree with these deterministic entropy increments by chain rule and flattening. This argument uses no regularity of a coefficient as a function of the exact particle. $\square$

For later classical applications define $p_*(a)$ to be the supremum of depths with zero limiting conditional score for the *raw final* derivative $P_1(F)$, given y_0, t_a . Use the raw final linear coefficient in the pure linear case. Then p_* is nondecreasing; if $b'_1 > 0$, $p_*(a) \geq b_1(a)$, since the earlier owned label and (A) predict that accuracy. Below p_* , typical values admit lists of arbitrarily small exponent; strictly above it their deterministic score is positive. At a virtual endpoint with $b'_1 > 0$, raw $P_D(F)$ and $P_1(F)$ differ by at most $s^{b_1(D)-o(1)}$, so these conclusions transfer at strictly coarser query depths.

Proposition 5.7 (Memory consequences). *Put $k_0 = 1 - d$ and*

$$L_{\text{raw}} = \max\{1, (1 + Q)/2, 1 + C_{\text{rate}}\}, \quad (Q, C_{\text{rate}}) = \begin{cases} (q, q - J), & \text{curvature families,} \\ (0, c_1), & \text{pure linear.} \end{cases}$$

Whenever the left label comparison in Proposition 5.4(iii) applies at $v = u(j)$, the angular trace satisfies

$$m(a, v) \geq \frac{k_0 + 2\Gamma + \eta(L_{\text{raw}} - 1)}{L_{\text{raw}}}.$$

Fix a classical depth j , and let λ_j denote the random label encountered there on the final assigned law. Define $R_j(v) = \lim H_s(X^{(v)} \mid \lambda_j)$, with conditional entropy averaged over that label random variable. Then $R_j(u(j)) = 0$, and

$$\liminf_{z \downarrow 0} \frac{R_j(u(j) + z)}{z} \geq \frac{k_0 + 2\Gamma + \eta(L_{\text{raw}} - 1)}{L_{\text{raw}}}. \quad (\text{mr})$$

These lower bounds hold on the same flattened law at every tested boundary prefix whenever λ_j predicts the boundary name; they are not assertions after conditioning on each individual label value. Under the small right-excess comparison, $m(a, v) \geq R'_j(v)$ almost everywhere. At typical classical j with $C_{\text{rate}} > 0$, the following improvements are available:

- (i) If $p_*(j) > b_1(j)$, with the extra prediction persisting immediately to the left, replace L_{raw} by $\max\{1, (1 + Q)/2\}$.
- (ii) If $Q \leq 1$ and $0 < C_{\text{rate}} < 1$, an additional lower bound is $k_0 + 2\Gamma - C_{\text{rate}}\nu(j)$, where ν is the mixed trace for the raw derivative, side y_0 , and graph $B = b_1$.

Proof. Pull a proposed angular list back to λ_j using its canonical prediction list. Apply (mem) with time gap $h = z/L_{\text{raw}}$, then transfer the resulting rarity to the final law by (Sp). This proves the prefix statements and their derivative consequences. Conditioning additionally on $\lambda_j, X^{(v)}$ proves the right-excess inequality.

For the prediction variant of (mem) on $[j - h, j]$, only a list for the slope at depth $b_1(j)$ is needed, given y_0, t_{j-h} . If strict extra prediction persists, this list has arbitrarily small exponent in s^h . In case (ii), its cost is at most $C_{\text{rate}}\nu(j) + o(1)$: the score at $B(j - h)$ vanishes, and the remaining interval is above the graph at the earlier conditioning time. Apply Lemma 5.6. Restrict the final event to successful slope predictions, whose failure is negligible at a fixed strict tolerance. Their lists are then available on every tested completing path. The prediction memory bound and (Sp) prove both enhancements. $\square$

5.4 Common side variables and orientation marks

We now arrange for several observations to share a common phase, to the accuracy of the tangent queries. The side variable below specifies this common frame; independence will hold after conditioning on that variable.

Write $D_{\text{end}} = 1$ for the quadratic-equality and pure-linear displays, and $D_{\text{end}} = D$ for the strict-curvature display of Definition 5.1. We now work at (a, v) , with $a > 0$ strictly below the chosen endpoint, and set $\delta = s^h$. A quadratic branch has $l = b + v$ and rate $c' = b' > 0$. A linear branch has $l = b_1 < b + v$, or is pure linear, and has rate $c' = b'_1 > 0$. The following side constructions are used only on patches where their strict inequalities hold.

Definition 5.8 (Admissible side variable). Choose one of the following variables Z .

- (i) *Coarse quadratic side.* Suppose $l(a, v) < \min_{[a, D_{\text{end}}]} b_1$, and, in the hybrid case, $w(a, v) < 0$. Fix $S = \mathcal{C}(A, V)$ on the same unclipped branch, with $A < a$, $V > v$, and

$$V - v \gg (1 + c' + |q| + J)(a - A) > 0.$$

Retain all strict domain inequalities and put $Z = (S, t_a, R_a, k_a)$. When label recovery is needed for $c = 1$, also require $v, V < u(a)$ with slack.

- (ii) *Exact quadratic side, classical.* Let $S = (y_0, \bar{k}, [P^\diamond]_{x_*})$, omitting $\bar{k}$ when absent, with $x_* > l(a, v)$. The raw coefficient bin must have zero conditional exponent, or arbitrarily small-cost covering lists, given y_0, t_a . One may choose $x_* < p_*(a)$, and also $x_* < b_1(D)$ in a virtual comparison. Put $Z = (S, t_a, R_a, k_a)$.
- (iii) *Coarse pure linear side.* Take $S = \mathcal{C}(A, V)$, with $V - v \gg (1 + c')(a - A) > 0$, and put $Z = (S, t_a, L)$. Maintain $w < 0$ in the hybrid case.
- (iv) *Exact linear side, classical.* Put $Z = (y_0, t_a, \bar{k}, R_a, k_a, L)$, omitting unused entries. In a clipped branch require $v > b_1 - b$, and remain short of a virtual endpoint. The additional bins have zero exponent given y_0, t_a , by ownership and (A).

The anchors and static coefficient depths are fixed on a patch. Countably many such patches cover all the stated admissible roots.

Lemma 5.9 (Common phase). *Given an admissible side Z , choose a representative point of its support and let C_* be its root canonical name. Subtract the displayed root shear, center and boost in that root cell, and dilate duration, horizontal width, and normal width by s^a, s^v, s^w , respectively. In this common frame there is a bounded phase*

$$F = (X, B, Y, N, D'_0), \quad \text{with also } P \text{ on a quadratic branch,}$$

measurable in Z , which differs from the corresponding actual phase of every conditional observation by $O(s^\varepsilon)$ for some positive patch slack. In particular the error is smaller than every fixed δ -power when h is sufficiently small. The actual root name and C_ determine each other to bounded lists. The sampling scheme and the representative can be chosen independently of v on a smaller patch.*

Proof. For an exact side, the phase coordinates are fixed by the particle. In the quadratic case $x_* > l$ also resolves the residual derivative to finer precision than the root unit.

For a coarse side first subtract the S -shear. Its base uncertainties are s^V, s^{A+V}, s^{A+2V} , and its normal uncertainties are $s^{w(A,V)}, s^{A+w(A,V)}$; the same bounds hold after flow to t_a . At a representative base point the root and side derivatives differ by $O(s^{\min\{l(a,v), l(A,V)\}})$. Their curvature difference has size $O(s^{b(A)})$ in earlier-duration units. Changing to the fixed representative frame therefore adds velocity error at most

$$O(s^{\min\{l(a,v), l(A,V)\}+V} + s^{b(A)+2V}),$$

with an additional factor s^A for position; omit the second term in the pure linear case. The inequalities in Definition 5.8 put all these errors strictly below the root widths. The quadratic derivative itself varies by at most $O(s^{l(A,V)})$. The change of base tilt fits because $A+v+V > a+2v$.

An actual cell may differ from the representative cell at a grid boundary. Guess its boundedly many neighboring linear-cut and angular-center cells first. A linear-cut difference multiplies only unresolved base coordinates at their root widths when converting residual names; the resulting errors are root normal widths. The tilt conversion has the same property. This gives the two-directional lists without comparing normal coordinates in different frames directly. All fine phase comparisons above are in the single representative frame.

A measurable representative can be obtained by realizing the conditional kernel with a uniform seed and fixing a seed that is valid almost everywhere, by Fubini. The same choice can be made with the patch parameter under Lebesgue measure, and kept fixed as v varies. Only its real coordinates and cell data are used. Conditional observations still use the original kernel; no extra shared mark is conditioned upon. $\square$

For the chosen endpoint D_{end} , put $T_{\text{end}} = T_{D_{\text{end}}}$, the start time of its endpoint interval. On a quadratic branch put $\beta_{\text{obs}} = \beta_{D_{\text{end}}}(T_{D_{\text{end}}})$, in the coordinates of the chosen curvature display. Define the marks, before common-root dilation of the arguments on the right, by

$$\theta = \frac{T_{\text{end}} - t_a}{s^a}, \quad r = \begin{cases} \frac{\beta_{\text{obs}} - R_a(T_{\text{end}})}{s^{b(a)}}, & \text{quadratic,} \\ \frac{P - 2R_a X + 2k_a B_{t_a} - L}{s^l}, & \text{linear.} \end{cases} \quad (\text{xi})$$

For the equality case $c > 1$ also include $\mathfrak{k} = (K_1 - k_a)s^{a-b(a)}$. The endpoint is 1, or the fixed virtual D . Marks are bounded. Their joint law with Z is independent of v on the chosen patch; in particular the linear branch has $l = b_1(a)$, so its cut L is independent of v . Subscripts on marks below denote ordinary bounded-chart bins at δ -depths. Stored marks always have a specified finite cutoff depth, chosen before the inner scale limit.

Proposition 5.10 (Conditional angular dimension). *Conditional on Z , the time mark has lower surprise dp' at every tested positive prefix p' , to arbitrary strict tolerance. At a typical time root, the joint mark increment on $p' \rightarrow p' + t$, where $p' > 0$ and $t \ll p'$, has conditional surprise at least dt , given its parent and Z . More precisely:*

- (i) *If $c' \geq 1$, the time increment itself has this lower bound given the joint parent. In a linear branch with $c' > 1$, one may also condition on $r_{p'+t}$ when $p' + t < c'p'$.*
- (ii) *If $0 < c' < 1$, the mixed trace gives $\nu \in [0, 1]$ such that the r -increment, given $Z, \theta_{p'+t}, r_{p'}$, has exponent νt , while the time increment given $Z, \theta_{p'}, r_{p'+t}$ has exponent at least $(d - c'\nu)_+ t$. Their joint increment has exponent at least dt .*

In the classical exact linear case with $Q \leq 1$ and $0 < b'_1 < 1$, ν is the raw final derivative trace with side y_0 and graph b_1 ; this also applies strictly before a virtual endpoint with $b'_1 > 0$.

Proof. At the gate $i = a + p'h$, every predecessor on a tested completing path predicts the static side and the other root data to small lists. For a coarse side this uses exact base flow, the strict normal slack for hybrid backflow, and (A). For an exact predicted coefficient, use its arbitrarily small-cost list and discard its negligible final failure event. The row predicts $\theta_{p'}$, and it predicts r to depth p' if $c' \geq 1$, or depth $c'p'$ otherwise. In a linear branch the latter accuracy always holds. When $c > 1$, the predicted curvature trajectory is evaluated at the endpoint; its bounded normalized slope makes replacing that time by the gate time cost only $\delta^{p'-o(1)}$.

Here is the causal counting step. Use Lemma 3.19: take a Borel inner version of the prediction-success event, mark rows at a later gate j' that have a continuation to it, and retain a Borel inner version of that analytic mark under the full analyzed j' -row energy law, inside the current end-state support. For each of their times, restrict the i -inputs to a Borel outer version of the listed predecessor set. It includes all predecessors under the unrestricted truncated i -to- j' geometric correspondence, not just predecessors on successful completions, and has the same full analyzed input energy. A path from a complementary input to a marked row would concatenate with its completing path, and so would contradict the definition of that predecessor list. Correspondence locality therefore makes complementary transfer negligible. Ordinary contraction, summed over the listed next time cells, costs their multiplicity times the raw mass at i . Applying (Sp) onward to the readout prohibits lists of exponent strictly below $d(j' - i)$, including the small predictor multiplicities. Testing popular cells proves the stated time lower surprises. The same proof starts at $p' = 0$.

For $c' < 1$, use Lemma 5.6 with $V_* = \beta^\diamond$, $B = b$ for constant quadratic cuts, or with the raw slope and $B = b_1$ in a linear branch. The side is the specified static side. Its additional root data have zero exponent beyond $S, t_a, [V_*]_{B(a)}$. At the later time, $p' > c'(p' + t)$ for sufficiently small t/p' , so the r -increment lies above the graph and has rate ν . Adding the coefficient bins between $B(i)$ and the tested depth to the causal time bound loses at most $c'(\nu - \nu_-)t$: this is precisely the mixed coefficient information exposed by the new time bits. The r -increment before those bits has rate at least ν . Chain rule gives joint rate at least

$$d - c'(\nu - \nu_-) + \nu \geq d,$$

and also the separately stated time bound. Known shifts of bin origins and virtual raw coefficient comparisons cost small lists only. $\square$

5.5 Separation forced by a positive minimal cost

Proposition 5.11 (Angular separation). *In an equality quadratic family with $c > 0$, the conditional (θ, r) law for an admissible side avoids arbitrary affine lines with a positive power. At $c = 1$ this assertion requires the coarse side with the additional left label slack in Definition 5.8; it is not asserted for a general exact side. More precisely, for a line chosen measurably per Z , its δ^λ -neighborhood has total probability at most $\delta^{\varepsilon\lambda}$ in the exponent limits, for some $\varepsilon > 0$, each tested $\lambda > 0$, and sufficiently small h . The constants are uniform on bounded lists of positive λ .*

In a pure linear family, the analogous statement holds for predicting a value of r . In a clipped classical linear branch with $p_(a) = b_1(a)$, two conditionally independent marks satisfy $|r_i - r_j| \geq \delta^\lambda$ with probability tending to one for every fixed $\lambda > 0$; no uniform positive avoidance exponent is asserted in this case.*

Proof. For the clipped assertion, root additions to y_0, t_a have zero cost and the offset in (xi) is then known. Every strictly finer raw slope bin has positive deterministic score by the definition of p_* . After trimming exceptional heavy bins, conditional independent sampling gives the separation. In a virtual use keep the query below $b_1(D)$.

We prove the power statements by contradiction. Failure gives a predicted tube of width $s^{(1-o(1))z}$, where $z = \lambda h > 0$, carrying mass $s^{o(z)}$; all previous errors are taken to zero in z -units first. A nearly vertical line confines the time coordinate to a power-small interval and is excluded by Proposition 5.10. Hence, after an arbitrarily small loss, the line is the graph of an affine function with slope $s^{-o(z)}$ in mark coordinates. In physical units it predicts an affine trajectory β_* with

$$|\beta_1(T_1) - \beta_*(T_1)| \leq s^{b(a)+(1-o(1))z}.$$

Exact extra-prediction sides are restricted to their successful lists, taking their costs and failure tolerances sufficiently small in z -units.

We shall extract a segment $[i, j]$. By Lemma 3.19, take a Borel inner version of the retained final event, mark j -rows having a completion to it, and then take a Borel inner version of this analytic mark under the full analyzed j -row energy law, inside the current end-state support. Every retained marked row still has a completion. Their mass is at least $s^{o(z)}E_j$ by (Sp); the count bound gives a cube-sum lower bound $s^{o(z)}E_j\sqrt{\kappa_j}$. Group starting rows by the binned polynomial subtraction and their starting interval. Disjointize the finite possible-bin relations and retain their Borel inner versions, so each marked end row has a bin witnessed by a completion. Use the Borel outer predecessor set for that bin, formed from the unrestricted truncated i -to- j

geometric correspondence. It includes all its correspondence predecessors without changing full analyzed input energy; its Borel inner partner carries the same full analyzed state inside the original starting-bin partition. Each end label has $s^{-o(z)}$ chosen bins, as checked in the cases below. Path concatenation and locality then justify restricting synthesis to those starting sets. Input energy summed over groups is at most $s^{-o(z)}E_i$, each group's regularity is at most $s^{-o(z)}\kappa_i$, and the end counts gain only the same multiplicity. Hölder preserves the cube-sum lower bound over groups. After rescaling, these grouped experiments realize the preceding extremal parameters in the closure. Thus a fixed improvement in the minimal cost is impossible.

Suppose first $0 < c < 1$. Set $i = a + z/2$, $j = a + 3z/4$, and choose $0 < \omega < \min\{(1-c)/2, c/4\}$. Subtract the prediction at the i -time start, rounded at depth $b(i) + \omega z$. Starting rows know it from the side. Moving the prediction to the endpoint time costs at most $s^{b(a)+z/2-o(z)}$, and comparing the end label with the final coefficient costs $s^{b(j)-o(z)}$. Both are finer than the chosen bin, so the end label knows few bins. The residual end coefficient is at most $s^{b(i)+\omega z-o(z)}$, while the duration-scaled own slope already has size $s^{b(j)-o(z)}$. In segment units the curvature cost is at most $c - 4\omega + o(1) < c$.

For $c > 1$, set $i = a$, $j = a + \gamma z$, with $1/c < \gamma < 1$, and choose $\gamma < \tau < 1$. On the part where $s^j|K_1 - \beta'_*| > s^{b(a)+\tau z}$, the end trajectory has the same slope discrepancy to within a factor two, since $c\gamma > 1 > \tau$. Its meeting the predicted trajectory within the tube confines the terminal time, per end row and predicted line, to length $s^{j+(1-\tau-o(1))z}$. The causal time count excludes this part by a power. On the remainder, extrapolation to t_a gives coefficient and slope errors bounded by

$$s^{b(a)+(\tau-\gamma)z-o(z)}, \quad s^{b(a)-a+(\tau-\gamma)z-o(z)}.$$

Subtract the predicted pair at those base depths improved by ωz , with $0 < \omega < \tau - \gamma$. It is jointly list-recoverable, and both residual curvature terms at the end have cost strictly below c .

For $c = 1$, use $i = a$, $j = a + 2z$. The additional left slack makes λ_j predict the coarse side: its base width has $u(j) > V$, its derivative comparison fits $l(A, V)$, and its normal comparison fits by the same Taylor and hybrid-drift bounds as in Proposition 5.4. It also predicts the root curvature bins. Subtract the predicted pair at depths $b(a) + \omega z, b(a) - a + \omega z$, where $0 < \omega < 1/2$. The residual end coefficient is bounded by $s^{b(a)+\omega z-o(z)}$; the end-duration times the residual slope is bounded by $s^{b(j)-o(z)}$. Hence the end curvature cost is strictly below one. This case does not require the slope of the prediction to be inferred from the end curvature alone: it is predicted by the recovered side.

Finally, in the pure linear case choose $i = a$, $j = a + \gamma z$ with $c_1\gamma > 1$. A value prediction for r predicts the terminal linear coefficient at depth $b_1(a) + z - o(z)$. Subtract its bin at $b_1(a) + \omega z$, $0 < \omega < 1$. The end label knows the bin by its finer coefficient comparison. The curvature cost remains zero and the linear cost becomes at most $c_1 - \omega/\gamma + o(1)$, a contradiction.

For clarity, the recovery margins behind the group count are as follows. When $0 < c < 1$, the prediction-transport and end-label errors have depths $b(a) + z/2$ and $b(j) = b(a) + 3cz/4$, both finer than the subtraction depth $b(i) + \omega z = b(a) + (c/2 + \omega)z$ by the two restrictions on ω . When $c > 1$, both errors in the displayed value-slope pair exceed their subtraction depths by $(\tau - \gamma - \omega)z > 0$. At $c = 1$, the recovered coarse side supplies the slope bin to a small list, and the end coefficient has depth $b(j) = b(a) + 2z > b(a) + \omega z$. In the pure linear case the terminal coefficient has depth $b_1(j) = b_1(a) + c_1\gamma z > b_1(a) + \omega z$. Thus the pre-existing side and rounding lists leave only $s^{-o(z)}$ bins per fixed end label in each case. $\square$

5.6 Replica budgets and conditional fibers

Draw finitely many observations independently given Z , and store their marks $\Xi_i = (\theta_i, r_i)$, including $\mathfrak{k}_i$ when required, to fixed finite depths. Let $F = F(Z)$ and $C_* = C_*(Z)$ be the representative phase and name from Lemma 5.9. The cutoff for each stored mark is chosen sufficiently large for the finitely many ensuing queries; it is never replaced by an exact mark inside an information estimate without a new justification.

For the normalized phase and angle, $F_{\leq p'}$ and $X_{\leq p'}$ denote their coordinatewise dyadic names at δ -depth p' ; these names also determine their coarser dyadic bins.

Proposition 5.12 (Replica information budgets). *At almost every interior admissible root there are arbitrarily small tangent gaps with sufficiently accurate preceding experiments such that, for any fixed finite collection of queries,*

$$I_\delta(F_{\leq p'}; \Xi \mid C_*) = o(1). \quad (\text{vb})$$

Define a fiber read Q_i , at a sufficiently large finite depth relative to still finer stored marks, by binning

$$B + \theta_i X, \quad Y + \theta_i X^2, \quad D'_0 + \theta_i N, \quad \begin{cases} P - 2r_i X, & N - PX + r_i X^2, & \text{quadratic,} \\ N - r_i X, & & \text{linear.} \end{cases}$$

Then

$$\begin{aligned} I_\delta(X_{\leq p'}; Q_i, \Xi \mid C_*) &= o(1), \\ i_\delta(X_{\leq p'} \mid C_*) &= m(a, v)p' + o(1) \quad \text{in probability.} \end{aligned} \quad (\text{fib})$$

At a fixed boundary root with one observation, only the consequence of (ng) for adding that observation and its fiber read is asserted; neither (vb) nor an interior angular trace is asserted there.

Proof. The marks and their law before evaluation of the representative phase are independent of v on the patch. Their total entropy is $O(h \log(1/s))$. For sufficiently large fixed K , the pair $C_*(v), C_*(v + Kh)$ determines $F_{\leq p'}$ up to bounded lists. In a quadratic template the new derivative bin reads P to depth K . The new residual phase can be converted to the old shear frame using the known shear difference and finer base bins; every old coordinate is then known to depth at least K . In a linear template the linear cut is unchanged. Consequently telescoping the decreasing function $H_s(\Xi \mid C_*(v))$ over a translation of size Kh bounds the integral of $I_s(F_{\leq p'}; \Xi \mid C_*)$ by $O(h^2) + o(1)$. Here $I_s = hI_\delta$, since $\log(1/\delta) = h \log(1/s)$.

Integrate also in a on compact patches when needed. For fixed dyadic h , take lower limits of the sum of the finitely many nonnegative errors through the earlier approximations. Fatou preserves the integral bound. After division by h , the dyadic integrals are summable, so almost every root admits accurate experiments with all the required normalized errors tending to zero. Countably many finite cutoffs and tuple sizes are handled by a diagonal. This establishes existence of the stated tangent laws; it does not assert stationarity for arbitrary sampling schemes that change with v .

Consider next observation i and its actual child name $\mathcal{C}(a + Kh, v)$, with K larger than all required query depths, also after multiplication by c' and, for nonconstant quadratic cuts, by $c' - 1$. It predicts C_* and reads Ξ_i, Q_i to bounded lists. Here are the details for the last assertion.

The child time θ' approximates θ_i arbitrarily finely. Write $W' = B + \theta'X$, $V' = Y + \theta'X^2$. The difference between child and root shears, in the common frame, is

$$\begin{aligned} c_{\text{obs}} + \mathcal{L}X + \mathcal{R}X^2 + \mathfrak{b}(V' - 2XW'), \\ d_{\text{obs}} + \mathcal{L}W' + \mathcal{R}V' - \mathfrak{b}(W')^2. \end{aligned}$$

All coefficients and offsets are determined by the two names. In the quadratic case $\mathcal{R}, \mathfrak{b}$ approximate $r_i, \mathfrak{k}_i$, and $\mathcal{L} + 2\mathcal{R}X - 2\mathfrak{b}W'$ approximates P . Thus $\mathcal{L} - 2\mathfrak{b}W'$ reads $P - 2r_iX$; adding back the known $\mathfrak{b}V'$ to the residual velocity reads $N - PX + r_iX^2$. The base reads supply W', V' , and the residual position supplies $D'_0 + \theta'N$. For a linear branch the same derivative expression reads r_i , while $\mathcal{R}, \mathfrak{b}$ are negligible because $b + v > l$, or identically zero. It therefore reads $N - r_iX$. Changing the curvature basis in a clipped child costs s^{b+v} , smaller than every required $s^l \delta^{M'}$ by the fixed branch slack. Nonconstant quadratic cuts read r_i with error $O(\delta^K + \delta'^K)$ and the slope to depth $(c' - 1)K$. Large constant offsets are converted at the exact child-bin time before comparison. Finally the common phase error is negligible by Lemma 5.9.

Equation (ng) bounds the information that the child name gives about the fine angle, conditionally on the root. Thus adding (Q_i, Ξ_i) costs $o(1)$. To add the other marks, first include a sufficiently fine phase bin. Given it and Ξ_i , the fiber read has only bounded ambiguity. The remaining conditional phase information is bounded by (vb), using chain rule. This proves the first line of (fib). The second line is the deterministic angular profile and the root comparison from Lemma 5.9. $\square$

Corollary 5.13 (Stationary conditional walks). *One may resample the common side and its phase successively from their joint law conditional on C_*, Ξ, Q_i , optionally retaining an angular prefix bin. Each leg preserves the joint marked marginal. Fine angular increments of the new endpoint have negligible normalized information from the preceding history beyond that already present in C_* and the retained prefix. For a leg with $A = \Delta X$, its increments satisfy, up to arbitrarily fine specified errors,*

$$\begin{aligned} \Delta B = -\theta_i A, \quad \Delta Y = -\theta_i(2XA + A^2), \quad \Delta D'_0 = -\theta_i \Delta N, \\ \Delta P = 2r_i A, \quad \Delta N = PA + r_i A^2 \quad \text{(quadratic),} \quad \text{(leg)} \\ \Delta N = r_i A \quad \text{(linear).} \end{aligned}$$

For a finite walk without a retained prefix, the new angular draws have joint negligible information relative to independent draws given C_ , including relative to the starting phase and all marks.*

Proof. Conditional resampling preserves the conditioning data and their joint marginal. Given those data it is independent of the rest of the history. Apply (fib) to this unchanged marginal; chain rule shows that retaining an angular prefix only decreases the information budget for the remaining angle bits. Summing the finite number of budgets gives the joint assertion. The first three identities in (leg) follow from equality of the first three fiber reads. The derivative read gives $\Delta P = 2r_i A$. Substitution in the remaining quadratic read gives $\Delta N = PA + r_i A^2$; the linear read gives the last identity. Increase the finite cutoff to make all rounding errors smaller than the queried scales. $\square$

These walk assertions are information estimates, not total-variation assertions. In particular they do not permit retaining a set of merely subpower probability and reusing the old budget without a separate argument.

5.7 Random scores and diagonal traces

The walk estimates provide small information budgets, while the projection tests to come use scores of individual outcomes at several depths. We next justify passing from finite-name scores to local rates and specify which changes of probability law preserve those budgets.

Throughout this subsection, $\rho \downarrow 0$ and scores are divided by $L_\rho = \log(1/\rho)$. Side data D may take values in an arbitrary standard Borel space. All conditional laws are regular conditional laws. For a finite name A , write

$$i_\rho(A \mid D) = -L_\rho^{-1} \log \mathbb{P}(A = A(\omega) \mid D).$$

Every assertion about several names is made first for a fixed finite list of queries. Countable limits below are limits of these joint score laws; they do not assert a coupling of experiments at different scales.

Lemma 5.14 (Pathwise score limits). *Suppose a finite collection of name arrays A_q , with depths in a bounded interval, has the following properties. Coarser names are determined from finer names up to lists of size $\rho^{-o(1)}$, and, conditionally on A_q, D , the name $A_{q'}$, $q' \geq q$, has at most $\rho^{-K(q'-q)-o(1)}$ possible values. The same bounds hold for the joint arrays under consideration, and their total range sizes are at most a fixed power of ρ^{-1} . Then, after a subsequence, the scores at countably many fixed queries have a joint distributional limit with these properties:*

- (i) *Cumulative scores are nonnegative, nondecreasing, and Lipschitz in each named-variable refinement argument, with conditioning held fixed and with the corresponding branching constant. They extend from rational queries to real queries.*
- (ii) *The chain rule holds, and adding conditioning can only decrease a conditional score. In particular, for $r \leq r' \leq a \leq b$, whenever F_r is a nested conditioning array,*

$$\begin{aligned} i(A_b \mid F_{r'}, D) - i(A_a \mid F_{r'}, D) \\ \leq i(A_b \mid F_r, D) - i(A_a \mid F_r, D). \end{aligned} \tag{5.1}$$

Here and below i without a scale denotes a limiting random score.

- (iii) *Mutual determination with exponent-zero lists identifies the corresponding limiting scores. All these statements hold simultaneously almost surely for the countable family of queries.*

Proof. If a conditional distribution is supported on a list of size N , then the total probability of values of conditional probability less than ρ^ϵ/N is at most ρ^ϵ . Apply this observation to each refinement and each list comparison. It bounds the refinement score between zero and $K(q' - q) + \epsilon + o(1)$, except on a set of probability tending to zero. Bounded total range gives the same control on whole scores; values exceeding the range exponent by ϵ have probability at most ρ^ϵ , also conditionally on D . Thus every finite vector of scores is tight. A diagonal subsequence gives a consistent joint limit on the countable query set.

For exactly nested names the log-probability chain rule is an identity. For list nesting, insert the coarser name into the finer name. Its additional conditional score tends to zero by the preceding list bound, so the same identities survive in the limit. Equation (LP) gives the conditioning inequality at each fixed query. Apply it to the increment name with the coarse name already conditioned upon to obtain Equation (5.1). A countable union of null sets gives all inequalities simultaneously. The loss from refining a conditioning name F_r to $F_{r'}$ is nonnegative

and at most $i(F_{r'} \mid F_r, D)$, by the chain rule and nonnegativity of conditional scores. It is therefore at most $K(r' - r)$ with the appropriate branching constant. Thus the required mixed score functions are Lipschitz also in their conditioning depths. The Lipschitz estimates extend the limiting scores continuously from rational depths. Comparisons with neighboring rational depths also give the claimed limit at any additional fixed real query. The same reasoning applies when several arrays are refined together. $\square$

The following elementary trace statement is useful because differentiating a two-variable Lipschitz function on its diagonal is not justified by the usual almost-everywhere differentiability theorem alone.

Lemma 5.15 (Monotone diagonal trace). *Let $H(q, r)$ be continuous for $0 < r \leq q < T$, with $H(r, r) = 0$. Assume that $H(\cdot, r)$ is nondecreasing and K -Lipschitz, uniformly in r , and that*

$$H(b, r') - H(a, r') \leq H(b, r) - H(a, r) \quad (r \leq r' \leq a \leq b).$$

There is a measurable function $g : (0, T) \rightarrow [0, K]$ such that, for almost every p , uniformly over $p \leq u \leq v \leq p + t$,

$$H(v, u) = g(p)(v - u) + o_p(t) \quad (t \downarrow 0). \quad (5.2)$$

Proof. For rational r , choose the derivative $f_r(q) = \partial_q H(q, r)$ outside one common null set. Increment monotonicity gives $0 \leq f_{r'}(q) \leq f_r(q) \leq K$ for rational $r < r' < q$. Put

$$g(q) = \inf_{\substack{r \in \mathbb{Q} \\ 0 < r < q}} f_r(q).$$

For any fixed real u , differentiate the increment inequalities comparing $H(\cdot, u)$ with all rational conditioning depths. For almost every $q > u$ and each rational $r < u$, they imply

$$g(q) \leq \partial_q H(q, u) \leq f_r(q).$$

For the first inequality, compare with a rational depth between u and q . Absolute continuity therefore gives, for every $u < v$ and rational $r < u$,

$$\int_u^v g(q) dq \leq H(v, u) \leq \int_u^v f_r(q) dq. \quad (5.3)$$

Choose p to be a Lebesgue point of g and of every f_r defined in a neighborhood of p , and outside the common derivative null set. These conditions exclude only a null set. For fixed rational $r < p$, the lower and upper errors in Equation (5.2) are bounded, uniformly over the indicated triangle, by

$$\int_p^{p+t} |g(q) - g(p)| dq \quad \text{and} \quad \int_p^{p+t} |g(q) - g(p)| dq + \int_p^{p+t} (f_r(q) - g(q)) dq,$$

respectively. The first is $o(t)$. The limsup of the second divided by t is at most $f_r(p) - g(p)$. Taking the infimum over rational $r < p$ makes it zero and proves the assertion. $\square$

Corollary 5.16 (Prefixes with future conditioning). *Let F_q, U_q, V_q satisfy Lemma 5.14, and suppose F_q, D determines U_q, V_q to exponent-zero lists. For almost every fixed p , almost surely in the limiting score law, there are rates $g_U(p), g_V(p), g_{UV}(p)$ such that*

$$i(U_{p+t} \mid F_p, D) = g_U(p)t + o(t), \quad i((U, V)_{p+t} \mid F_p, D) = g_{UV}(p)t + o(t).$$

Moreover, for every fixed $0 \leq \lambda \leq 1$,

$$i(U_{p+\lambda t} \mid F_p, V_{p+t}, D) = \lambda(g_{UV}(p) - g_V(p))t + o(t). \quad (5.4)$$

The error can be taken uniform in λ . Tuples may replace V .

Proof. For $A = U, V, (U, V)$, apply Lemma 5.15 pathwise to $H_A(q, r) = i(A_q \mid F_r, D)$. Its diagonal vanishes by the list hypothesis, and its increment inequality is Equation (5.1). Put $u = p + \lambda t$ and $v = p + t$. The chain rule first gives

$$i(U_u \mid F_p, V_u, D) = H_{UV}(u, p) - H_V(u, p) = \lambda(g_{UV}(p) - g_V(p))t + o(t).$$

Write $i(A; B \mid C) = i(B \mid C) - i(B \mid A, C)$ for limiting conditional information. These quantities are nonnegative by Equation (LP). Since U_u is determined by F_u, D , the additional loss from the future part of V satisfies

$$\begin{aligned} 0 &\leq i(U_u; V_v \mid F_p, V_u, D) \\ &\leq i(F_u; V_v \mid F_p, V_u, D) \\ &= H_V(v, p) - H_V(u, p) - H_V(v, u) = o(t). \end{aligned}$$

The second inequality follows by expanding the information involving (U_u, F_u) in both orders and using nonnegativity. The last equality uses nesting and determination of F_p, V_u from F_u, D ; its $o(t)$ bound follows from the uniform triangular trace. This proves Equation (5.4). Finally, Fubini's theorem converts the pathwise almost-everywhere statement into an almost-sure statement at almost every fixed p . $\square$

Lemma 5.17 (Changes of probability law). *Let P_ρ, Q_ρ be probability laws with $\text{KL}(P_\rho \parallel Q_\rho) = o(L_\rho)$. At samples drawn from P_ρ , their scores for any fixed finite collection of names, including conditional scores with arbitrary common side data, differ by $o(1)$ in probability. If $P_\rho(E) \geq c > 0$, then $Q_\rho(E) \geq \rho^{o(1)}$.*

Restricting a law to an event of probability at least $c > 0$ preserves its finite-name scores up to $o(1)$ in probability. It also preserves any conditional mutual-information bound $o(L_\rho)$ between two variables given common side data. The latter assertion is not made for events whose probabilities merely have exponent zero.

Proof. Let $R = dP_\rho/dQ_\rho$. The elementary estimate $P_\rho(\log R < -u) \leq e^{-u}$ implies $\mathbb{E}_{P_\rho}(\log R)_- \leq 1$, and hence

$$\mathbb{E}_{P_\rho}(\log R)_+ \leq \text{KL}(P_\rho \parallel Q_\rho) + 1 = o(L_\rho).$$

Markov's inequality proves $\log R = o(L_\rho)$ in probability. Relative entropy decreases under measurable maps, so this also applies to the likelihood ratios for each joint name and its conditioning field. Subtracting those two log ratios proves the conditional-score assertion. Applying the log-sum inequality to E, E^c gives

$$\text{KL}(P_\rho \parallel Q_\rho) \geq P_\rho(E) \log \frac{1}{Q_\rho(E)} - \log 2,$$

which proves the event assertion.

For restriction to E , use $\text{KL}(P_\rho(\cdot \mid E) \parallel P_\rho) = \log(1/P_\rho(E))$. To verify the information assertion, let X, W, D denote the variables in question and set $Q = P_D P_{X|D} P_{W|D}$. With $c_\rho = P(E)$, the mixture identity for relative entropy gives

$$c_\rho \text{KL}(P_{XWD|E} \parallel Q) \leq I_P(X; W \mid D) + h(c_\rho),$$

where $h(c) = -c \log c - (1-c) \log(1-c) \leq \log 2$. This identity remains valid if E also depends on auxiliary variables: apply the mixture identity to the marginal laws of (X, W, D) . The relative entropy on the left is at least $I_{P(\cdot|E)}(X; W \mid D)$, by decomposition against the product of the restricted conditional marginals. Division by $c_\rho \geq c$ proves the result. $\square$

These results are used without selecting rare limiting profiles. A strict violation is tested at a fixed admissible p , on an event of positive probability in its random score law, using finitely many queries and slightly relaxed tolerances. The joint convergence of those scores transfers positive probability to sufficiently advanced experiments. Their information errors are then taken sufficiently small for this fixed event and layer, before the layer shrinks.

Remark 5.18 (Freezing coefficients). Suppose a local projection has smooth coefficients depending on the phase point, such as the contact differential $dY - 2X dB$. Fix their values to accuracy $O(\rho^{p_0})$ using a name already in the conditioning, where $0 < p_0 < p$. Inside an F_p cell, freezing changes an increment by $O(\rho^{p+p_0})$; Taylor expansion of the original coordinate contributes $O(\rho^{2p})$. Both are smaller than ρ^{p+t} when $0 < t < \min(p, p_0)$. The two descriptions therefore give bounded lists for one another at this layer, conditionally on the parent and the coefficient descriptions. The values at the parent representative must be translated exactly: they are known offsets and are not increment errors. The same argument applies to a mark coefficient frozen to accuracy ρ^ϵ when $t < \epsilon$. Any cost of adding this coefficient information is a separate information budget; the geometric comparison does not supply it.

For a Heisenberg recentring, the error left by changing horizontal cell representatives is bilinear in unresolved horizontal displacements. If their scale is ρ^a , this error is $O(\rho^{2a})$. Thus a vertical name at scale ρ^b is predicted up to bounded lists whenever $b \leq 2a$, giving the required nesting and branching in that wedge. The score calculus applies in its interior; it supplies no derivative trace across the boundary $b = 2a$.

5.8 A projection rule with conditional records

The record in the next proposition may contain more information than the direction it determines. In particular, different records with the same direction may carry different lists of projection values.

Proposition 5.19 (Conditional projection rule). *Let $0 \leq S \leq 2$, $0 < b \leq 1$, and $\Delta \downarrow 0$. Let Z be a bounded planar source, W a record determining a direction $g(W)$, and D common side data. Assume*

$$I(Z; W \mid D) = o(\log(1/\Delta)). \quad (5.5)$$

Suppose there is an event G of probability bounded below such that its unnormalized conditional marginals obey, for every needed ball and interval of radius $\Delta \leq r \leq 1$,

$$\begin{aligned} \mathbb{P}(G, Z \in B(z, r) \mid D) &\leq \Delta^{-o(1)} r^S, \\ \mathbb{P}(G, g(W) \in J(r) \mid D) &\leq \Delta^{-o(1)} r^b. \end{aligned} \quad (5.6)$$

For each D, W , let $\mathcal{I}(D, W)$ be a list of at most $\Delta^{-x-o(1)}$ intervals of length $O(\Delta)$, and suppose that on G the projection of Z in direction $g(W)$ belongs to their union, up to $O(\Delta)$ error. Then

$$x \geq \min\{S, (S + b)/2, 1\}. \quad (5.7)$$

The assertion is a strict-error statement: a fixed strict deficit is impossible if all preceding exponent losses and the information ratio are sufficiently small. Consequently a sufficiently fine finite grid of prefix radii suffices, and errors of size $\Delta^{1-o(1)}$ in directions, representatives, or output resolution are allowed. The same conclusion holds with $c_\Delta = \mathbb{P}(G)$ tending to zero, provided

$$\log(1/c_\Delta) = o(\log(1/\Delta)), \quad I(Z; W \mid D) = o(c_\Delta \log(1/\Delta)).$$

Proof. The case $S = 0$ follows from the nonempty output list on G , so assume $S > 0$. A finite choice of rotated projective charts allows restriction to a positive-probability part of G where projections have the form

$$\pi_u(z) = z_1 + uz_2,$$

with bounded u, z . Multiplying a projection by a bounded nonzero factor only changes interval constants. First restrict to G . Lemma 5.17 preserves the vanishing normalized information budget. If $c_D = \mathbb{P}(G \mid D)$ and $\mathbb{P}(G) \geq c$, then under this restricted law the parents with $c_D < c/2$ have total probability at most $1/2$. On the other parents, normalization of Equation (5.6) costs at most $2/c$. The conditional information is nonnegative and has vanishing average after normalization by $\log(1/\Delta)$. We can therefore choose a parent with these mass bounds and with conditional information $o(\log(1/\Delta))$.

Fix that parent. Write P for the restricted joint law and μ, ν for its source and record marginals. Let R be the relation that the projected source is in the record's output list. Since $P(R) = 1$, the log-sum inequality gives

$$\eta := (\mu \times \nu)(R) \geq \exp(-\text{KL}(P \parallel \mu \times \nu)) = \Delta^{o(1)}.$$

Thus product sampling still hits the lists, although a fixed positive product probability is not asserted.

Fix a small $\epsilon > 0$. For sufficiently advanced experiments, the marginal bounds have constants at most $\Delta^{-\epsilon}$ and $\eta \geq \Delta^\epsilon$. Draw $M = \lceil \Delta^{-b} \rceil$ independent full records $W_1, \dots, W_M$ from ν . With probability tending to one faster than any fixed power of Δ , their empirical direction law ν_M obeys

$$\nu_M(J(r)) \leq C\Delta^{-2\epsilon}r^b \quad (\Delta \leq r \leq 1). \quad (5.8)$$

Indeed, for each dyadic interval of radius r , the count is binomial with mean at most $M\Delta^{-\epsilon}r^b$; the asserted threshold is larger by a factor comparable to $\Delta^{-\epsilon}$ and is at least a constant multiple of $\Delta^{-2\epsilon}$. The binomial upper-tail bound is $\mathbb{P}(N \geq k) \leq (eMq/k)^k$ for a count with M trials and success probability q : sum over the k successful indices and use $\binom{M}{k} \leq (eM/k)^k$. This bound, followed by a union bound over $O(\Delta^{-1})$ dyadic intervals down to scale Δ , proves the claim. Every other interval is covered by a bounded number of these intervals at comparable radius.

The expected average relation density of this sample is η . Since the probability that Equation (5.8) fails is $o(\eta)$, there is a sample satisfying that equation and

$$\int \theta(z) d\mu(z) \geq \eta/2, \quad \theta(z) = M^{-1} \#\{j : (z, W_j) \in R\}.$$

Let $E = \{z : \theta(z) \geq \eta/4\}$. The bounds $0 \leq \theta \leq 1$ imply $\mu(E) \geq \eta/4$. The probability $\mu_E = \mu|_E/\mu(E)$ has planar ball masses at most $C\Delta^{-2\epsilon}r^S$. For each $z \in E$, put the uniform probability on its successful sampled records. Its direction masses are at most $C\Delta^{-3\epsilon}r^b$, by Equation (5.8) and $\theta(z) \geq \eta/4$.

Dualize: the source point z parametrizes the line

$$\ell_z = \{(u, z_1 + uz_2) : u \in \mathbb{R}\}.$$

The map from z to the line parameters is bi-Lipschitz on the bounded chart, so this family supports an S -Frostman probability with the preceding small power loss. On each ℓ_z , the successful records give a b -Frostman probability: the direction parameter and distance along the line are uniformly comparable. Yet all these points lie within $O(\Delta)$ of a union of at most

$$CM\Delta^{-x-\epsilon} \leq C\Delta^{-b-x-\epsilon}$$

grid cells, using the output lists separately for every sampled *record*. Repeated records or repeated directions cause no problem. Apply Equation (RW) with its point exponent b and line-family exponent S . For any fixed $\omega > 0$, choosing ϵ small enough gives the lower covering bound

$$\Delta^{-b-\min\{1, S, (S+b)/2\}+\omega}.$$

Comparison with the upper count excludes every strict deficit in Equation (5.7).

For completeness, suppose mass bounds are supplied only at radii Δ^{σ_j} on a grid with successive gaps at most γ , including the two endpoints. Enclose a ball of arbitrary radius by a bounded number of cells at the next coarser tested radius. The additional loss is at most $\Delta^{-2\gamma}$, since $S \leq 2$ and $b \leq 1$. Choose γ before the other small losses to fit the strict deficit. Finally, errors of size $\Delta^{1-\tau}$ are handled by rerunning the proof at that effective final resolution. The output count exponent becomes $x/(1-\tau)$, and the small power constants are rescaled by the same factor; the ball powers S, b are unchanged. The conclusion converges to the displayed one as $\tau \downarrow 0$.

For the last assertion, the restriction estimate in Lemma 5.17 has upper bound $I(Z; W | D)/c_\Delta + h(c_\Delta)/c_\Delta$. This is $o(\log(1/\Delta))$ under the stated assumptions, since $h(c)/c \leq \log(1/c) + 1$. Selecting parents with $c_D \geq c_\Delta/2$ loses at most half the restricted probability and adds only an exponent-zero normalization factor to the mass bounds. The rest of the proof is unchanged. $\square$

Remark 5.20 (Obtaining the hypotheses from scores). If at a sample the source prefix score at radius Δ^σ is at least $S\sigma - \epsilon$, retain only samples with this property. Every occupied source cell in the retained law then has original conditional mass at most $\Delta^{S\sigma-\epsilon}$; restriction can only reduce its unnormalized mass. Intersecting finitely many such conditions gives the source bound in Equation (5.6). The same argument applies to direction cells and to unnormalized direction bounds first proved under finer side data and then integrated to D .

Conversely, a projection score at most $x + \epsilon$, conditionally on the full record and D , places the projection in a list of at most $\Delta^{-x-\epsilon}$ popular cells: each listed cell has probability at least $\Delta^{x+\epsilon}$. Thus projection-score inequalities give precisely the lists required by Proposition 5.19. Bin the proposed rates on a positive-probability strict violation before making these finite tests. Corollary 5.16 supplies the needed linear prefix rates also when the source is conditioned on a future reading of another array. No bound on the total number of source cells is assumed.

5.9 Radial spreading and pin products

Projection tests whose directions come from two marks require information about the lines joining those marks. The estimates below turn the time marginal bounds into a dichotomy for tube masses, and provide a product-coordinate estimate for the low-exponent alternative. We begin with a finite-scale radial argument. It is useful to keep the measure unnormalized: no assertion of conditional independence will be made after restricting to a relation of pairs. All measures in this subsection are supported in a fixed bounded subset of $\mathbb{R}^2$. Constants may depend on that subset. We write $\mathbf{l}(x, y)$ for the unoriented line through distinct points x, y , and $N_R(\ell)$ for its R -neighborhood.

Lemma 5.21 (Finite radial exclusion). *Assume the discretized Furstenberg estimate (RW). For $0 < A < d \leq 1$ there are $\eta_0 > 0$ and $\Delta_0 > 0$ with the following property. There do not exist $0 < \Delta < \Delta_0$, a Borel measure μ of mass at most one, and a symmetric measurable relation E of distinct pairs such that*

$$\mu(B(x, R)) \leq \Delta^{-\eta_0} R^d \quad (\Delta \leq R \leq 1), \quad (5.9)$$

$$(\mu \times \mu)(E) \geq \Delta^{\eta_0}, \quad (5.10)$$

$$\begin{aligned} \mu(N_\Delta(\mathbf{l}(x, y))) &\geq \Delta^{A+\eta_0} && ((x, y) \in E), \\ \mu(N_R(\mathbf{l}(x, y))) &\leq \Delta^{-\eta_0} R^A && ((x, y) \in E, \Delta \leq R \leq 1). \end{aligned} \quad (\text{rad-assume})$$

Replacing fixed constants in the radii or bounds by other fixed constants does not change the conclusion, after reducing η_0 and Δ_0 .

Proof. Put

$$F(A, d) = A + \min\{1, d, (A + d)/2\}, \quad g = F(A, d) - 2A > 0.$$

Choose $\kappa, \epsilon_0 > 0$ so small that $\kappa + \epsilon_0 < g/4$ and $\kappa < (1 + A/d)/2$. We choose a separation parameter $\lambda > 0$ in the clustered case below. We prove the assertion with $\eta_0 = \eta$, where η is chosen last, sufficiently small relative to these parameters. The exact requirements on η are collected at the end of the proof.

Cover the bounded space of lines meeting the support by finitely many slope-intercept charts and partition each chart into Δ -squares. Assign each witness line to one such square c , and let T_c be a fixed $C\Delta$ -tube containing the Δ -neighborhoods of every line assigned to c . Define

$$J_c = \iint_{T_c \times T_c} \mathbf{1}_{\{|u-v| \geq \Delta^\kappa\}} d\mu(u) d\mu(v).$$

A pair at separation $r \geq \Delta^\kappa$ belongs to at most $C(1 + r^{-1}) \leq C\Delta^{-\kappa}$ such pairs of footprints. Indeed its allowable line directions occupy intervals of total length $O(\Delta/r)$, while the intercept has only $O(1)$ choices at width Δ for each direction cell. Consequently

$$\sum_c J_c \leq C\Delta^{-\kappa}.$$

Call a cell spread if $J_c \geq \Delta^{2A+\epsilon_0}$. There are at most

$$C\Delta^{-2A-\epsilon_0-\kappa} \quad (5.11)$$

spread cells.

For spread cells, this upper count will contradict the Furstenberg lower count $\Delta^{-F(A,d)+g/4}$. In a nonspread cell, the small mass of separated pairs instead forces most of the tube's mass into one ball of radius Δ^κ . We treat these two cases in that order.

Suppose first that spread cells account for at least half of the mass in (5.10). Pass to one line chart, losing a fixed factor, and discard pins whose partner mass in this relation is less than $\Delta^{2\eta}$. The retained pins have mass at least $\Delta^{2\eta}$. Their normalized distribution has ball bounds $\Delta^{-3\eta}R^d$, with a further fixed constant harmless. For each retained pin x , push its restricted partner measure to the line-parameter points $\mathfrak{l}(x, y)$ and normalize. This probability has ball bounds

$$C\Delta^{-3\eta}R^A, \quad \Delta \leq R \leq 1.$$

To check this, choose one witness line in a parameter ball which meets the support of the pushforward. Every partner belonging to that ball lies in a CR -neighborhood of this witness line, so (rad-assume) applies. Values $CR > 1$ are covered by the mass bound and a fixed constant.

Under planar duality the parameter points for a fixed pin lie on its dual line. In the chosen bounded chart the map from pins to dual-line parameters is bi-Lipschitz. Thus (RW) applies with point exponent A and line exponent d , giving at least $\Delta^{-F(A,d)+g/4}$ parameter cells. This contradicts (5.11), since $\epsilon_0 + \kappa < g/4$ and Δ is sufficiently small.

It remains to treat witness pairs assigned to nonspread cells. For any such occupied cell put $m_c = \mu(T_c)$; then $m_c \geq \Delta^{A+\eta}$. Averaging over $z \in T_c$ gives a point $z_c \in T_c$ such that

$$\mu(T_c \setminus B(z_c, \Delta^\kappa)) \leq J_c/m_c \leq \Delta^{A+\epsilon_0-\eta}.$$

Hence the cluster

$$K_c = T_c \cap B(z_c, \Delta^\kappa)$$

has mass at least $\frac{1}{2}\Delta^{A+\eta}$, and therefore at least $\Delta^{A+2\eta}$ when Δ is sufficiently small.

Choose

$$0 < \lambda < \kappa/10, \quad 10\lambda < \kappa(d - A).$$

Discard witness pairs at distance less than Δ^λ . Their total mass is at most $\Delta^{d\lambda-\eta}$, which is negligible compared with Δ^η . Since $\lambda < \kappa$, at least one endpoint of every remaining pair has distance at least $\frac{1}{4}\Delta^\lambda$ from its cluster. Symmetry permits orienting a relation of mass at least a fixed multiple of Δ^η so that its first endpoint x has this property. Retain pins whose partner mass is at least $\Delta^{2\eta}$; the set of retained pins again has mass at least $\Delta^{2\eta}$.

For each retained pin take a maximal separated collection of its witness directions, with angular separation

$$H = \Delta^{1-2\lambda}.$$

A ball of directions of radius CH accounts for partner mass at most $C\Delta^{-\eta}H^A \leq \Delta^{A(1-2\lambda)-2\eta}$, by the tube upper bound centered at one witnessing line. It follows that the collection has at least

$$\Delta^{-A+2A\lambda+4\eta} \tag{5.12}$$

members, with harmless rounding of this lower bound. Assign to each member the cluster of its witness cell. These clusters are disjoint: every point of such a cluster has distance $\gtrsim \Delta^\lambda$ from x , so its direction from x differs from the witness direction by $O(\Delta^{1-\lambda})$, which is much smaller than H .

Let E' consist of the new pairs (x, y) with y in one of these chosen clusters. The choices can be made measurably by a fixed finite grid and a deterministic greedy ordering. Using (5.12) and the cluster mass gives

$$(\mu \times \mu)(E') \geq \Delta^{2A\lambda+8\eta} \geq \Delta^{3\lambda}.$$

Retain with each new pair its original witness line ℓ_0 and cluster K . We have

$$K \subset B(y, C\Delta^\kappa) \cap N_{C\Delta}(\ell(x, y)), \quad \mu(K) \geq \Delta^{A+2\eta}. \quad (5.13)$$

Here $x \in \ell_0$, $y \in N_{C\Delta}(\ell_0)$, and $|x - y| \gtrsim \Delta^\lambda$. The angle of the two lines is $O(\Delta^{1-\lambda})$. On the cluster, whose diameter is $O(\Delta^\kappa)$, rotation about y costs at most $O(\Delta^{1+\kappa-\lambda}) = O(\Delta)$. This proves (5.13) without thickening its final tube by a power of Δ .

Set $q = (1 + A/d)/2$. By (5.9),

$$\mu(B(y, \Delta^q)) \leq \Delta^{(d+A)/2-\eta} = o(\Delta^{A+2\eta}).$$

Pigeonholing the dyadic annuli between radii Δ^q and $C\Delta^\kappa$ therefore assigns to every pair of E' a radius D in this range for which its cluster contributes at least $\Delta^{A+3\eta}$ mass in the annulus $D \leq |z - y| \leq 2D$. There are $O(\log(1/\Delta))$ choices of D . For one common D , the corresponding relation has mass at least $\Delta^{4\lambda}$. Choose y so that its set of pins has mass at least $\Delta^{5\lambda}$, and put

$$R = \Delta/D.$$

This pin submeasure has angular ball bounds

$$\mu\{x : (x, y) \in E', \angle(\ell(x, y), \ell) \leq CR\} \leq \Delta^{-2\eta} R^A \quad (5.14)$$

whenever the angular ball meets the retained pin set; increasing the exponent loss by η absorbs any additional fixed constant. In fact choose one pin in that ball and use its original witness line ℓ_0 . The new line differs from ℓ_0 by $O(\Delta^{1-\lambda})$, whereas $R \gtrsim \Delta^{1-\kappa}$ and $\lambda < \kappa$. Every pin in the angular ball is consequently in $N_{C'R}(\ell_0)$. The original upper bound in (rad-assume), not an upper bound asserted for a newly conditioned measure, proves (5.14).

A maximal collection of pin directions separated by a sufficiently large constant times R has at least $\Delta^{5\lambda+3\eta} R^{-A}$ members. Their $C\Delta$ -tubes in the annulus of radius D about y are disjoint. Each contains mass at least $\Delta^{A+3\eta}$ from its assigned cluster. Consequently

$$\mu(B(y, 2D)) \geq \Delta^{5\lambda+6\eta} R^{-A} \Delta^A = \Delta^{5\lambda+6\eta} D^A.$$

The ball upper bound, with its fixed-radius constant absorbed, is at most $\Delta^{-2\eta} D^d$. These inequalities imply

$$\Delta^{5\lambda+8\eta} \leq D^{d-A} \leq C\Delta^{\kappa(d-A)},$$

contrary to the choice of λ, η for small Δ .

To meet all the losses used above, after choosing $\kappa, \epsilon_0, \lambda$ in that order choose $\eta > 0$ with

$$100\eta < \min\{\lambda, d\lambda, \epsilon_0, d - A\}, \quad 10\lambda + 100\eta < \kappa(d - A),$$

and smaller than the loss allowed by (RW) with conclusion loss $g/4$. These are finitely many compatible requirements. Finally choose Δ_0 small enough to absorb the fixed constants and logarithmic factors in both cases. $\square$

Proposition 5.22 (Radial dichotomy). *Let $\xi_i = (\theta_i, r_i)$ and $\xi_j = (\theta_j, r_j)$ be conditionally independent, identically distributed planar marks given Z . Suppose their bounded-depth records obey the preceding score calculus, and their time marginals have prefix mass exponent at least $d > 0$ given Z , with $d \leq 1$. More precisely, at every prescribed finite collection of depths, discarding a set of vanishing average probability makes the unnormalized conditional time-interval bounds hold with exponent d and arbitrarily small losses. Assume (RW).*

Trim at the required depths to measures μ_Z dominated by the conditional mark laws. Pass to a joint subsequential limit of the needed scores and of

$$e(t) = -\log_{1/\rho} \mu_Z(N_{\rho^t}(\mathbb{I}(\xi_i, \xi_j))), \quad t > 0 \text{ rational.}$$

Then, almost surely,

$$e_* := \liminf_{t \downarrow 0} \frac{e(t)}{t} \in \{0\} \cup [d, \infty]. \quad (\text{rad})$$

If, in addition, the conditional tube supremum satisfies a uniform affine-line avoidance estimate of some positive exponent in these depth units, then $e_ \geq d$ almost surely.*

Proof. Discard at each tested depth the time bins which violate the asserted mass bound. Their average mass tends to zero; finitely many such trimmings have the same property. Each time interval is covered by a bounded number of comparable bins, so the trimmed time marginal, and hence every planar ball, has the required unnormalized bound. Increasing finite grids can be chosen diagonally. Only grids fixed before the ρ -limit are used.

The displayed tube masses give legitimate limiting random variables. Indeed an endpoint ball of a sufficiently smaller constant radius lies in the joining tube. Among $O(\rho^{-2t})$ planar cells, the total trimmed mass of those having mass less than $\rho^{2t+\alpha}$ is $O(\rho^\alpha)$. The probability that either endpoint was trimmed away tends to zero. Thus the exponents can be clipped on exceptional sets and jointly subsequenced at countably many rational queries. Nearby depth queries handle constant changes of width.

If $0 < e_* < d$ on a set of positive probability, choose an essential value $A \in (0, d)$. Given the tolerance in Lemma 5.21, restrict to $|e_* - A| < \epsilon$, with ϵ sufficiently small. At each such sample the lower inequalities $e(u) \geq (A - 2\epsilon)u$ hold for every sufficiently small rational u , while $e(t) \leq (A + 2\epsilon)t$ holds at arbitrarily small rational t . A countable union therefore supplies a fixed small t and a positive-probability event on which these inequalities hold at t and at every point of a prescribed finite grid σt , $0 < \sigma \leq 1$.

Choose that grid with mesh small relative to the tolerance of Lemma 5.21, including a first point so small that the trivial mass bound fills the gap to radius one. Monotonicity fills all intervening radii. Transfer the strict inequalities to the approximating laws and put $\Delta = \rho^t$. The time bounds give (5.9); the tube bounds give (rad-assume) with any prescribed sufficiently small loss. The event has fixed positive probability at this stage. Since both marks were sampled independently given Z , its surviving pair mass is the average of the corresponding $\mu_Z \times \mu_Z$ relation masses. One can consequently choose Z with pair mass bounded below by a fixed positive constant, in particular by Δ^{η_0} for small Δ . The event and its transpose may be combined to make the relation symmetric. This contradicts Lemma 5.21.

A uniform positive tube-avoidance exponent gives $e_* > 0$. The dichotomy then gives the last assertion. $\square$

The radial dichotomy bounds tubes under the original conditional mark law. In the interior projection tests one also fixes the parent cell of a moving mark and auxiliary angular data. The

next lemma carries the tube bounds to directions from that parent, keeping the source law and the full pin record explicit.

Lemma 5.23 (Parent-centered directions and comparison laws). *Let P be a law of a side variable Z , finitely many full records $\mathbf{V}$, and a finite auxiliary name A . Suppose that the record marginal is conditionally iid given $Z = z$, with law ν_z . Each record has a bounded planar projection $\pi_z(V_i)$; write $\mu_z = (\pi_z)_\# \nu_z$ for its law. The full record may retain additional entries, including a stored curvature slope. Let S be a function of Z , and set*

$$R(dZ, d\mathbf{V}, dA) = P(dZ, d\mathbf{V})P(dA | S).$$

Assume $\text{KL}(P||R) = o(\log(1/\delta))$. Choose distinct records V_x, V_y , and write $x = \pi_z(V_x)$, $y = \pi_z(V_y)$. Let K be the depth- b parent cell of y , with representative c_K , where $b > 0$. For parents of positive mass put $\mu_z^K = \mu_z|_K / \mu_z(K)$. The common data are

$$D = (Z, K, A, \text{the held full records other than } V_x, V_y).$$

Fix $t > 0$ sufficiently small compared with b , and a finite grid of radii $r = \delta^{\sigma t}$, $0 < \sigma \leq 1$. Let T_z be a measurable trimming set in the full record space and put $\mu_z^T = (\pi_z)_\#(\nu_z|_{T_z})$. In particular the trimming may depend only on the planar projection. Let $W_y = \psi_D(\pi_z(V_y))$ be a finite Borel source name in the tested prefix family. Under R , its conditional law given D is the image of μ_z^K under ψ_D ; in applications it is a bin of the rescaled child increment of the moving mark in K . On an event G , suppose that $V_x \in T_z$, that $|x - y| \geq \delta^\chi$ with $b - \chi > t$, and that the original witness lines satisfy the tested bounds

$$\mu_z^T(N_{Cr}(\mathbf{l}(x, y))) \leq C\delta^{-\varepsilon t}r^{d-\varepsilon}.$$

Here the fixed enlargement constants needed below are included among the finite tests. Then, with $g_K(x)$ the direction from c_K to x ,

$$R(G, g_K(x) \in J(r) | D) \leq C\delta^{-\varepsilon t}r^{d-\varepsilon} \quad (5.15)$$

for every tested direction interval $J(r)$. A fixed swap or sign change of its coordinate representation preserves the assertion.

After deleting a set of P -probability tending to zero, the corresponding unnormalized conditional bound under P holds with factor $C\delta^{-3\varepsilon t}$. Source prefix bounds for W_y supplied by the original conditional law $\mu_z(\cdot | K)$, with their own source trimming when needed, transfer in the same way. Moreover,

$$I_P(W_y; V_x | D) \leq \text{KL}(P||R) = o(\log(1/\delta^t)).$$

All assertions fix b, t , the finite grid, and the tested positive-probability event before sending the information error to zero. No lower bound for $\mu_z(K)$ is required.

Proof. Since S is known from Z , conditioning on A does not change the full record kernel under R . Conditional iid sampling also allows the other full records to be held. On parents of positive probability, the exact conditional pair law is consequently

$$R(dV_x, dV_y | D) = \nu_z(dV_x)\nu_z^K(dV_y), \quad \nu_z^K = \frac{\nu_z|_{\pi_z^{-1}(K)}}{\mu_z(K)}.$$

Its moving planar marginal is $\mu_z^K = (\pi_z)_\# \nu_z^K = \mu_z|_K / \mu_z(K)$. This factorization holds before selecting G . Take the jointly Borel inner version of G from Lemma 3.19 under the sum of the P - and R -laws on the full record tuple. It preserves their masses and the pointwise witness bounds; all conditional assertions here are for almost every common datum. Its direction marginal on G is

$$\begin{aligned} & \int \mathbf{1}_{J(r)}(g_K(\pi_z(V_x))) \nu_z(dV_x) \int \mathbf{1}_G(D, V_x, V_y) \nu_z^K(dV_y) \\ & \leq \mu_z^T(\mathcal{W}_{D,J}), \end{aligned}$$

where $\mathcal{W}_{D,J}$ is the set of planar projections of surviving pin records in that direction interval having some full-record witness V_y with projection in K . The inner probability is at most one. In particular the bound does not contain a factor $\mu_z(K)^{-1}$.

If $\mathcal{W}_{D,J}$ is nonempty, choose one full-record witness and denote its planar projections by (x_0, y_0) . Since y_0 is within $O(\delta^b)$ of c_K and the pair is separated, the directions of $\mathbf{l}(x_0, y_0)$ and $\mathbf{l}(x_0, c_K)$ differ by $O(\delta^{b-x}) = o(r)$. Every participating pin lies in a fixed bounded region and makes angle $O(r)$ with that witness direction about c_K . It is therefore in $N_{Cr}(\mathbf{l}(x_0, y_0))$. The original trimmed tube bound proves Equation (5.15). Notice that the source law is still μ_z^K ; it has not been replaced by the conditional law of the trimmed measure.

For the change of law use the full finite tuple carrying the event, the common data, and all tested source and pin names. Its relative entropy is at most the stated budget by data processing. Write $f = dP/dR$ on this tuple and $f_D = dP_D/dR_D$. The likelihood estimates in Lemma 5.17 show, for fixed $\varepsilon, t > 0$, that outside a set of vanishing P -probability,

$$f \leq \delta^{-\varepsilon t}, \quad f_D \geq \delta^{\varepsilon t}.$$

The conditional likelihood on the retained event is at most $\delta^{-2\varepsilon t}$. Integrating it over either the source or the pin proves the claimed unnormalized marginal bounds under P . Intersecting further with a popular-output event only decreases these masses.

Finally, decomposition of relative entropy conditional on D , followed by its decomposition against the product kernel $\nu_z \times \nu_z^K$, bounds $I_P(V_y; V_x \mid D)$ by $\text{KL}(P \parallel R)$. The name W_y is a function of V_y, D , so data processing gives the asserted bound on $I_P(W_y; V_x \mid D)$, also with the full pin record retained in the projection lists. The scale conversion is valid because t is fixed. If a strict violation is retained on an event of probability $c(t) > 0$, take the earlier information error negligible compared with $c(t)t \log(1/\delta)$ before invoking the positive-event restriction rule. This uses no uniform assertion as $t \downarrow 0$. $\square$

For the linear-fiber estimates, the projection direction is parametrized by the scalar $(\theta_j - \theta_i)(r_j - r_i)$. We next control this scalar on witness sets whose joining tubes have unusually large mass, then combine that estimate with the high radial alternative.

Lemma 5.24 (A low-exponent pin estimate). *Fix $0 < d \leq 1$. There is a constant C_d with the following property, for every fixed $\epsilon > 0$ and all sufficiently small $\Delta > 0$ depending on ϵ . Let μ have mass at most one and time marginal*

$$\mu\{(\theta, r) : \theta \in I\} \leq \Delta^{-\epsilon} |I|^d \quad (\Delta \leq |I| \leq 1).$$

Fix a pin $x = (\theta_x, r_x)$ in the same fixed bounded set, and let W_x be any subrelation of points $y = (\theta_y, r_y)$ for which

$$|\theta_y - \theta_x|, |r_y - r_x| \geq \Delta^\epsilon, \quad \mu(N_\Delta(\mathbf{l}(x, y))) \geq \Delta^\epsilon.$$

For $f_x(y) = (\theta_y - \theta_x)(r_y - r_x)$, every interval I of length Δ^σ , $0 \leq \sigma \leq 1$, satisfies

$$\mu\{y \in W_x : f_x(y) \in I\} \leq C_d \Delta^{d\sigma - C_d \epsilon}. \quad (5.16)$$

Finite grids of prefix hypotheses suffice, with their mesh added to the exponent loss.

Proof. It suffices to treat $3\epsilon/d < 1$: by increasing C_d , the conclusion is immediate from the total mass bound for larger ϵ . The time bound also bounds planar balls. Put $a = \Delta^{3\epsilon/d}$. A ball of radius a about x has mass at most $C\Delta^{2\epsilon}$, so at least half the mass of each witness tube is outside that ball, for small Δ . Choose a maximal family of witness directions separated by $C\Delta/a$. Outside $B(x, a)$ their Δ -tubes are disjoint when C is sufficiently large. The number of chosen directions is at most $C\Delta^{-\epsilon}$. Maximality places all witness points within $O(\Delta/a)$ of one of the representative lines.

Each representative is a witness, so its slope m satisfies $c\Delta^\epsilon \leq |m| \leq C\Delta^{-\epsilon}$. On its enlarged tube,

$$f_x(y) = m(\theta_y - \theta_x)^2 + O(\Delta^{1-C_d'\epsilon}).$$

For the points under consideration, $|\theta_y - \theta_x| \geq \Delta^\epsilon$. On each of the two possible signs of this difference, the inverse of the quadratic has derivative at most $C\Delta^{-2\epsilon}$. Thus $f_x(y) \in I$ restricts θ_y to at most two intervals of total length $C\Delta^{\sigma-C_d''\epsilon}$. Apply the time-marginal bound and sum over at most $C\Delta^{-\epsilon}$ representative lines. This gives (5.16); intervals longer than one are handled by the total mass bound. Rounding the queried lengths to a finite grid merely adds its mesh to the loss. $\square$

Proposition 5.25 (Pin-product alternatives). *In addition to the hypotheses of Proposition 5.22, assume the following. The score calculus and the preceding conditional projection rule apply to all fixed finite queries used below. At almost every parent depth p , the joint mark increment, conditional on Z and its own parent $\xi_{j,p}$, has prefix mass exponent at least d on every fixed finite grid of sufficiently small layers $p \rightarrow p+t$. Finally, for every $\chi > 0$,*

$$\mathbb{P}\{|\theta_i - \theta_j| \geq \rho^\chi, |r_i - r_j| \geq \rho^\chi\} \longrightarrow 1.$$

Put $f = (\theta_j - \theta_i)(r_j - r_i)$ and $\mathcal{G}_+ = \{e_* \geq d\}$, $\mathcal{G}_0 = \{e_* = 0\}$. Then the limiting cumulative scores satisfy

$$i(f_p \mid Z, \xi_i) \geq dp \quad \text{on } \mathcal{G}_+ \quad (5.17)$$

at the tested depths. Here ξ_i denotes a record stored to a sufficiently large fixed depth, chosen before the ρ -limit.

On $\mathcal{G}_0$ the following local substitute holds. Any strict violation of a differentiable layer rate on a set of positive probability can be tested at a fixed small layer t on a positive-probability subset where $e(t) < \epsilon t$. With $\Delta = \rho^t$, the unnormalized witness sublaw of f , given Z, ξ_i , then has prefix interval bounds of exponent d with loss $O_d(\epsilon)$, as in (5.16). The assertion is for a fixed selected layer before earlier approximation and information errors are sent to zero.

Proof. We first record precisely how a high radial bound produces a direction bound. On any finite grid of sufficiently small depths $u = \sigma t$, a high witness satisfies

$$\mu_Z(N_{\rho^u}(I(\xi_i, \xi_j))) \leq \rho^{(d-\epsilon)u},$$

with a further arbitrarily small absolute loss when transferring to the approximating laws. For fixed Z, ξ_i , an angular bin which meets the witness set has all its witnessing partners inside a

constant enlargement of one witness tube. This gives the same bound for the unnormalized partner-direction sublaw. One may instead fix Z and a j -parent bin at depth $p \gg t$. Use its representative as the center. After discarding pairs closer than a sufficiently small fixed power, the change of center is negligible at every queried angular width. An angular bin with a witness again places all its participating pins inside an enlargement of that witness tube. Thus the pin-direction sublaw, unnormalized and with this parent held, has the same exponent d . These are bounds on a restricted measure, not claims that the two marks remain independent after the witness restriction.

The cumulative score of f given Z, ξ_i is Lipschitz, starts at zero, and is differentiable at almost every depth. If (5.17) fails on a set of positive probability, Fubini's theorem and the fundamental theorem of calculus provide a fixed differentiability depth $p > 0$ and a positive-probability high set where the incremental rate is strictly less than d . Choose $t \ll p$ small enough to test this strict deficit, together with a finite prefix grid on which the preceding high tube bounds hold.

Adding $\xi_{j,p}$ to the conditioning can only lower the tested limiting incremental score, by the score calculus. The coarse product is known to bounded lists from this parent and the pin. After translating and dilating the j -parent, the product increment is, up to $O(\rho^p)$ in dilated units, the linear form with coefficient vector

$$(r_{j,\text{ctr}} - r_i, \theta_{j,\text{ctr}} - \theta_i).$$

Its norm is bounded below by an arbitrarily small power with probability tending to one. Since $t \ll p$, both its quadratic remainder and the finite-record errors fit the tested resolution, with arbitrarily small exponent loss. The source increment has prefix exponent at least d by hypothesis. The gradient directions, given $Z, \xi_{j,p}$, have unnormalized exponent at least d on the high relation: they are the joining directions just estimated, with the two coordinates interchanged. Before restriction, the pin is independent of the j -increment given this parent, since the two marks were independent given Z . The conditional projection rule applies and gives projected rate at least $\min\{d, (d+d)/2, 1\} = d$, contradicting the strict deficit. Integrating the almost-everywhere rate bound proves (5.17). Popular-cell trimming converts it to the corresponding unnormalized interval-mass bounds whenever needed.

On $\mathcal{G}_0$, the definition of a pointwise liminf supplies arbitrarily small rational t with $e(t) < \epsilon t$ at each sample. A strict violation of a differentiable limiting rate persists at every sufficiently small layer at that sample. A countable union therefore selects one fixed such t on a positive-probability violating set. Impose the two scalar separations at threshold $\rho^{\epsilon t}$; their exceptional probability tends to zero in the earlier limits. The trimmed time marginal has the required bounds at the finite queried depths. At $\Delta = \rho^t$, the condition on $e(t)$ gives the tube lower bound of Lemma 5.24, with an arbitrarily small additional loss. That lemma proves the asserted interval bounds at all the necessary prefixes.

Finally, these conditional bounds can be averaged over Z and independently sampled auxiliary records when the parent conditioning keeps the other pin fixed and permits that mixture. A different conditioning requires its own justification. In particular the selected low-set event may have a small positive probability depending on t . One first fixes this t and its finite queries, and only then makes the earlier information errors negligible relative to that probability. Nothing here permits transferring an information budget through a restriction whose probability is merely a power of the fine scale. $\square$

6 Interior phase inequalities

This section derives the local inequalities used in the cylinder argument. Its input is the tangent construction of the preceding section, including the information estimates (vb) and (fib), the angular profile (ang), and the projection calculus. We keep these inputs explicit: an artificial quadratic branch does not acquire an angular spreading hypothesis merely by having the same phase coordinates as an equality branch.

The output is the list of admissible time–width directions in Proposition 6.12. We first express changes of canonical entropy through local phase-reading rates; the remaining arguments bound those rates and the quadratic width defect.

6.1 Tangent data and contact readings

Fix a differentiability point (a, v) in an interior patch for which the replica construction is valid. Write

$$c = c', \quad m = m(a, v), \quad k_0 = 1 - d, \quad \delta = s^h,$$

where $0 < d \leq 1$, $c > 0$, and $0 \leq m \leq 1$. On a linear branch $c = b'_1$. The calculations below have $g = 0$. All auxiliary cutoffs and all finite lists of depth queries are chosen before the inner scale limits. Errors in information, divided by $\log(1/\delta)$, tend to zero before a microlayer length tends to zero. Statements with an arbitrarily small loss mean that this order is respected.

Definition 6.1 (Admissible interior tangent data). The common root is C_* , the side is Z , and the phase $F = F(Z)$ is either

$$F = (X, B, P, Y, N, D) \quad \text{in the quadratic case,} \quad F = (X, B, Y, N, D) \quad \text{in the linear case.}$$

Here D denotes the normalized coordinate called D'_0 in the tangent construction. Conditional on Z , finitely many observations are independent with their prescribed common distribution. Their stored marks are $\Xi_i = (\theta_i, r_i)$, together with $\mathfrak{k}_i$ in the nonconstant quadratic case. Marks are stored to a sufficiently large finite depth. Set $\mathcal{D} = (C_*, \Xi)$.

We use the following properties of this construction.

- (i) The information estimates (vb) and (fib) hold for every fixed finite set of phase, angular, and mark queries. In particular the angular score at depth p is $mp + o(1)$.
- (ii) The nested phase and projection names satisfy the score calculus: limiting scores have nonnegative Lipschitz increments; the chain rule, conditioning inequalities, and comparisons by negligible lists hold at actual samples; local mixed-prefix scores are linear at almost every layer. The projection rule stated in the preceding section applies with its specified finite-prefix Frostman hypotheses.
- (iii) The time mark has dimension at least d . Given a joint mark parent at depth $p > 0$, the joint mark increment on $p \rightarrow p + t$ has score at least dt , up to strict losses. For $c \geq 1$ the time increment itself has this lower bound. For a linear branch with $c > 1$, this remains true after conditioning on r_{p+t} , provided $p + t < cp$. If $0 < c < 1$, the mixed mark profile supplies a number $\nu \in [0, 1]$ with

$$\begin{aligned} i_\delta(r_{p+t} \mid Z, \theta_{p+t}, r_p) &= \nu t + o(t), \\ i_\delta(\theta_{p+t} \mid Z, \theta_p, r_{p+t}) &\geq (d - c\nu)_+ t - o(t). \end{aligned}$$

These assertions are made on the finite queries allowed by that profile; t is chosen small enough relative to p .

- (iv) Resampling the side and phase conditionally on (C_*, Ξ, Q_i) preserves the marked marginal. On a resampling leg, with $A = \Delta X$, the increments obey (leg) to any fixed tested accuracy. The finite-walk likelihood comparison supplied by (fib) remains available after conditioning on a common angular prefix when one is used.

In the quadratic case, *radial spreading* means that the high alternative of (rad) has $e_* \geq d$ almost surely for the replicas. For conclusions involving G_i when $c > 1$, we also use the stated uniform avoidance of affine lines. In the linear case, *crossing* means that both $|\theta_i - \theta_j|$ and $|r_i - r_j|$ exceed δ^χ with probability tending to one for each fixed $\chi > 0$.

Subpower separation in this section always means the last type of statement, with each positive exponent fixed before the inner limits. It permits inversion with an arbitrarily small exponent loss. It does not give a scale-independent lower bound.

Let F_p be the isotropic phase name at depth p . On a layer $p \rightarrow p + t$, $t \ll p$, use the contact forms

$$H = dY - 2X dB, \quad J = dN - P dX, \quad K = dD - P dB \quad (6.1)$$

in the quadratic case; only H is used in the linear case. These are linear readings of increments, not differentials of the corresponding products. For example, J is not $d(N - PX)$.

For nested names, freeze X, P at a fixed positive depth $p_0 < p$. After translation at the known parent F_p , changing this freeze changes an increment by $O(\delta^{p+p_0})$. Genuine quadratic remainders are $O(\delta^{2p})$. Both errors are negligible on a layer with $t < p_0$ and $t < p$. Mark coefficients may be frozen in the same way, at a fixed positive depth greater than t , keeping the known translation at the parent.

Put $W_i = dB + \theta_i dX$. The readings needed below are

$$\begin{aligned} x_i &= K + \theta_i J - r_i H, & \pi_i &= (W_i, H, x_i), & G_i &= (L_i, J - \mathfrak{k}_i H, K - \beta_i H), \\ L_i &= dP - 2\beta_i dX + 2\mathfrak{k}_i dB, & \beta_i &= r_i - \theta_i \mathfrak{k}_i, & & \text{quadratic,} \\ x_i &= dD + \theta_i dN - r_i W_i, & \pi_i &= (W_i, H, x_i), & G_i &= (dN - r_i dX, dD - r_i dB), & \text{linear.} \end{aligned} \quad (6.2)$$

The quadratic G_i is used on the equality branch with $c > 1$.

Write $|U|(p)$ for the limiting random rate of the score of U at depth $p + t$, conditional on $F_p, \mathcal{D}$. Set

$$|U | V| = |(U, V)| - |V|, \quad \alpha(p) = |F|(p), \quad w(p) = \alpha(p) - m.$$

The conditioning in $|U | V|$ includes the child reading of V . Rates are defined simultaneously at typical layers for any finite or countable collection of readings. Relative capacities and the exponential bound for tiny cell probabilities identify their expectations with entropy derivatives.

A bar denotes a common subsequential expected logarithmic Cesàro mean at zero. Thus one averages against dp/p over intervals whose logarithmic lengths tend to infinity and whose lower endpoints tend to zero. Constant dilations have the same mean. We take one subsequence for the finite list of averages used in an argument. Exchangeability gives equal expected means for different row indices.

6.2 The cost of time and width

Proposition 6.2 (Differential cost). *For admissible interior tangent data,*

$$\partial_a \mathcal{H} \geq d + \min(1, c)(\bar{w} + \overline{|x_i|}) + |1 - c| \begin{cases} \overline{|\pi_i|}, & c < 1, \\ \overline{|G_i|}, & c > 1, \end{cases} \quad (\text{time-rate})$$

where the last term is omitted at $c = 1$. In the linear case, $\partial_v \mathcal{H} \leq \bar{\alpha} + 1$.

In the quadratic case define, for $p \leq z \leq 2p$, the Heisenberg name $F_{p,z}$ by reading X, B, P to depth p and

$$Y - 2[X]_p B, \quad N - [P]_p X, \quad D - [P]_p B$$

to depth z . Let

$$e(\lambda) = \frac{H_\delta(F_{2\lambda} \mid \mathcal{D}) - H_\delta(F_{\lambda, 2\lambda} \mid \mathcal{D})}{\lambda} \quad (6.3)$$

in the limiting entropy law. Then

$$\partial_v \mathcal{H} = 2\bar{\alpha} - \bar{e}. \quad (\text{width-rate})$$

Proof. Let O be the actual child name at $(a + \lambda h, v)$. Given C_* , its limiting entropy divided by λ is $\partial_a \mathcal{H}$. Even after giving a sufficiently fine phase name, the time bits cost at least $d\lambda$: reveal Z and apply the conditional time-mark bound. The remaining entropy is bounded below by a sum of

$$I_\delta(F_{p+t}; O \mid C_*, F_p) \quad (6.4)$$

over consecutive layers. The chain rule makes this a telescoping sum of information about phase.

We will lower-bound the phase information by integrating the expected values of the following local rates over the indicated intervals:

phase-depth interval	local rate
$(0, \min\{1, c\}\lambda)$	w
$(c\lambda, \lambda), \quad c < 1$	$ \pi_i $
$(\lambda, c\lambda), \quad c > 1$	$ G_i $
$(\max\{1, c\}\lambda, (1+c)\lambda)$	$ x_i $.

The middle interval is absent when $c = 1$. The first rate comes from recovering the phase after paying for its angle; the other rates come from readings recoverable from the output.

For $p + t < \min(1, c)\lambda$, the output together with C_* and X_{p+t} reads F_{p+t} up to negligible lists. Its positions give the two base coordinates at the child time; undoing its displayed shear gives the common-frame coordinates and, in the quadratic case, the residual derivative. The extra angular increment has conditional cost at most mt by (ang). Consequently Equation (6.4) pays the expected w -rate on this interval.

For the remaining intervals, we must identify the projected readings and show that their coefficient information is paid for. Let

$$\theta' = (t_{a+\lambda h} - t_a)/s^a, \quad W' = B + \theta' X, \quad V' = Y + \theta' X^2.$$

The child-time error is $O(\delta^\lambda)$. The converted base positions read W', V' to depth λ . In common parent units, the child shear difference has known constant offsets and coefficients $\mathcal{L}, \mathcal{R}, \mathbf{b}$ at time θ' . Its residual velocity and position are displayed to depths $c\lambda$ and $(1+c)\lambda$. The derivative

$$\mathcal{L} + 2\mathcal{R}X - 2\mathbf{b}W'$$

approximates P in the quadratic case, or r_i in the linear case, to depth $c\lambda$.

In a linear branch the curvature coefficients vanish, or are negligible at every tested fixed power by the branch slack. For constant quadratic cuts, $\mathcal{R} = r_i + O(\delta^{c\lambda})$ and $\mathbf{b} = 0$. For the quadratic equality branch with $c > 1$,

$$|\mathbf{b} - \mathbf{t}_i| + |\mathcal{R} - \theta' \mathbf{b} - \beta_i| \lesssim \delta^{(c-1)\lambda}. \quad (6.5)$$

This is the own-time coefficient comparison, extrapolated back to normalized time zero after subtraction of the root trajectory. All its nonconstant normalized coefficients are bounded; constant offsets are retained exactly in the translations.

In the quadratic case the own-time algebra can be checked without an implicit change of origin. Put

$$\mathbf{a} = \mathcal{R} - \theta' \mathbf{b}, \quad \epsilon_0 = P - \mathcal{L} - 2\mathbf{a}X + 2\mathbf{b}B.$$

The derivative comparison gives $|\epsilon_0| \lesssim \delta^{c\lambda}$. With actual displayed coefficients held fixed, the residual velocity has gradient

$$J - \mathbf{b}H + \epsilon_0 dX,$$

and the residual position has gradient

$$K + \theta' J - \mathcal{R}H + \epsilon_0(dB + \theta' dX).$$

Subtracting θ' times the former from the latter gives $K - \mathbf{a}H + \epsilon_0 dB$. These identities are evaluated at the actual phase and follow directly by differentiating the shear polynomials; the known constant offsets have zero variation. At a parent representative the derivative error is $O(\delta^p + \delta^{c\lambda})$. Multiplication by the parent width gives $O(\delta^{2p}) + O(\delta^{p+c\lambda})$. The second term is negligible on a microlayer with $t < c\lambda$, and fits a binary layer whenever $p \leq c\lambda$. The derivative display itself has gradient $dP - 2\mathbf{a}dX + 2\mathbf{b}dB$. Equation (6.5) therefore gives the three claimed G_i readings with the stated precision.

In the linear case the shear is linear, with $|r_i - \mathcal{L}| \lesssim \delta^{c\lambda}$. The displayed residual velocity and the backflowed residual position have respective gradients

$$dN - \mathcal{L} dX, \quad dD - \mathcal{L} dB.$$

They differ from the two components of G_i by $(r_i - \mathcal{L})dX$ and $(r_i - \mathcal{L})dB$. Before backflow the position gradient is $dD + \theta' dN - \mathcal{L}(dB + \theta' dX)$, which gives x_i with the additional $O(\delta^\lambda)$ time-coefficient error.

Center each displayed quantity by its value at F_p , evaluated with its actual displayed coefficients and offsets. Taylor's formula has error $O(\delta^{2p})$. The residual-position gradient converges, with a positive power error on an interior layer, to x_i . Below depth λ , the base-position gradients additionally give W_i, H . If $c > 1$, subtract θ' times the displayed velocity from the position. In the quadratic case the two resulting gradients give $K - \beta_i H$ and $J - \mathbf{t}_i H$, and the derivative equation gives L_i . In the linear case they give the two components of G_i . Thus the intermediate interval gives π_i when $c < 1$ and G_i when $c > 1$, while the final interval always gives x_i .

For completeness, the projected entropy is indeed paid by Equation (6.4), even though its coefficients may depend on phase. Freeze the needed angular and curvature coefficients into a name Ω at a depth q with

$$t < q < \begin{cases} \min(p, \lambda, c\lambda), & \text{linear or constant quadratic cuts,} \\ \min(p, \lambda, (c-1)\lambda), & \text{nonconstant quadratic cuts, } c > 1. \end{cases}$$

Shrink the microlayer first so that such a fixed q exists. On a compact interior interval of depths, the preceding output comparisons recover Ω from O, C_*, F_p up to negligible lists. The centered reading U is also determined, up to such lists, by both $(F_{p+t}, C_*, F_p, \Omega)$ and (O, C_*, F_p, Ω) . For finite tested names A, B, Ω, U and arbitrary conditioning data C , the chain rule and the two reconstruction bounds give

$$\begin{aligned} I_\delta(A; B \mid C) &\geq H_\delta(U \mid C, \Omega) - H_\delta(\Omega \mid B, C) \\ &\quad - H_\delta(U \mid A, C, \Omega) - H_\delta(U \mid B, C, \Omega). \end{aligned}$$

To obtain this inequality, first adjoin Ω to B , at cost at most $H_\delta(\Omega \mid B, C)$, retain the conditional information given Ω , and apply the chain rule to the two reconstructions of U . Use $A = F_{p+t}$, $B = O$, and $C = (C_*, F_p)$. All three error terms tend to zero at the fixed layer, before division by t . Conditioning further on the stored marks lowers this entropy. Translation at the parent and the coefficient-freezing comparison then identify the relevant contact-reading rate.

Tile compact subintervals of the three depth ranges by smaller layers. Bounded convergence for the score derivatives and nonnegativity for the information sums permit passage to their integrals. Truncate near the endpoints, then remove the truncations using the uniform relative capacities. Finally average in λ logarithmically. Dilation invariance of this mean gives Equation (time-rate).

For width variation, increasing v by λh in a linear branch reads at most F_λ and one additional λ -depth scalar in vertical position. This proves the linear upper bound. In a quadratic branch the same change refines X, B, P by λ and the three recentered coordinates by 2λ . In changing the linear shear, the new binned coefficient differs from the true normalized P by $O(\delta^\lambda)$. After translation, its effect on unresolved horizontal widths is $O(\delta^{2\lambda})$. Hence the new width name and $F_{\lambda, 2\lambda}$ determine one another up to bounded lists, given C_* . Equation (vb) allows marked conditioning in these finite-depth entropy comparisons. Since F_0 has zero normalized entropy, logarithmic averaging of Equation (6.3) proves Equation (width-rate). $\square$

6.3 Angular modes and projection amplification

Lemma 6.3 (Angular and commutator modes). *The angular fiber direction is*

$$\ell_i = \begin{cases} (1, -\theta_i, 2r_i; 0, 0, 0), & (dX, dB, dP; H, J, K) \quad \text{quadratic}, \\ (1, -\theta_i, r_i, -\theta_i r_i; 0), & (dX, dB, dN, dD; H) \quad \text{linear}. \end{cases} \quad (6.6)$$

Any nondegenerate scalar component in this direction has conditional rate at least m after adjoining finitely many simultaneous readings that annihilate ℓ_i . A conditional angular-coordinate rate is at most m .

In the quadratic case, if $m > 0$, the center also has the mode

$$b_{ij} = (\theta_i - \theta_j, r_i - r_j, \theta_i r_j - \theta_j r_i) \quad \text{in } (H, J, K). \quad (6.7)$$

A nondegenerate component of this mode has residual rate at least m after conditioning on center readings that annihilate it. Both x_i and x_j annihilate b_{ij} .

Proof. Adjoin Q_i, X_p to the conditioning. Along this fiber the phase is determined by X to every fixed tested accuracy. Differentiating the fiber equations gives Equation (6.6). Their quadratic remainder on an angular parent cell is $O(\delta^{2p})$. Thus F_p and the annihilating child readings have negligible list cost with the enlarged conditioning. A nondegenerate component recovers X_{p+t} up

to negligible lists. For a subpower lower bound on its derivative, fix the separation exponent much smaller than the strict loss in the test, apply one-dimensional inversion, and then let that exponent decrease. Equation (fib) gives the residual rate m . The reverse upper bound for angular rates follows from (ang) and conditioning.

For the center mode, walk successively along fibers i, j , with endpoints F_0, F_1, F_2 and angular coordinates X_0, X_1, X_2 . Adjoin the starting side, very fine X_2 bits, and $X_{1,p}$. The finite-walk likelihood comparison and (fib) leave rate m for $X_{1,p+t}$. The separation $|X_1 - X_2| \geq \delta^\chi$ holds with probability tending to one for every fixed $\chi > 0$: conditional on the previous history, a new angular prefix has positive score at each positive depth, so it cannot lie in a bounded list of prescribed bins with nonvanishing probability.

Apply (leg) on the two legs and differentiate with X_2 fixed. The resulting center derivative at F_2 is

$$2(X_2 - X_1)b_{ij} dX_1.$$

In particular its last component is $2(X_2 - X_1)(\theta_i r_j - \theta_j r_i) dX_1$. The same parent-cell Taylor and inversion argument applies. The endpoint has the original marked marginal, so the result is a statement about its original conditional rates. Substitution in the formula for x_i or x_j verifies the last assertion. $\square$

Lemma 6.4 (Projection amplification). *Let a planar pair have joint local rate S , possibly after conditioning on common finer readings. Suppose that a scalar projection of rate x leaves a residual rate at least $M > 0$. Suppose also that its direction has Frostman exponent d on the finite prefixes required by the projection rule, and that its full varying record has negligible information with the planar source. Then*

$$x \geq \min(1, d + M).$$

Proof. The chain rule gives $S \geq x + M$. The projection rule gives $x \geq \min(1, S, (S + d)/2)$. If $x < 1$, then $S > x$, so the only remaining possibility is $2x \geq S + d \geq x + M + d$. This gives $x \geq M + d$.

For random rates these deductions are applied to a positive-probability set with a strict deficit. Bin the rates to sufficiently short fixed intervals, use mixed-prefix linearity to obtain the finite-prefix source bounds, and use popular projection cells to obtain the output lists. The information error vanishes at this fixed layer. The projection rule excludes the deficit. This procedure also allows the high-set radial witnesses and the low-set pin-product witnesses of the preceding section; it never conditions on an unspecified subpower-probability event. $\square$

Proposition 6.5 (Basic phase rates). *Put $l_* = \min(1, d + m)$. For quadratic data with $m > 0$ and radial spreading, write $C_z = (H, J, K)$. Then*

$$|W_i | C_z| \geq l_*, \quad |C_z| - |x_i| \geq m + l_*, \quad \alpha - |C_z| \geq 2m + l_*. \quad (\text{modes})$$

When the quadratic G_i is used and line avoidance holds,

$$|G_i| \geq 2m + |x_i|, \quad 2\mathbb{E}|G_i| \geq \mathbb{E}\alpha - 1. \quad (\text{Gq})$$

For linear data with $m > 0$ and crossing,

$$|W_i | H| \geq l_*, \quad |x_i| \geq l_*, \quad |G_i| \geq m + l_*, \quad 2\mathbb{E}|G_i| \geq \mathbb{E}\alpha - 1. \quad (\text{modes-lin})$$

For linear data, without a positivity assumption on m ,

$$w - |\pi_i| \leq 1. \quad (\text{miss})$$

Proof. In all uses of Lemma 6.4, the planar source and any common child conditioning depend only on phase and the held records. Equation (vb), followed by the conditional chain rule, permits removal of the varying record. Its directional sublaw bounds, initially conditional on Z and the held records, integrate to the tested conditioning. This verifies the information requirement without asserting independence after a witness restriction.

For W_i , use the source (X, B) , conditional on the center in the quadratic case and on H in the linear case. The direction parameter is θ_i , with exponent d , and ℓ_i gives residual m .

For the quadratic center inequality, condition on x_i to child depth and use (H, J) . It determines all center coordinates with x_i . Vary j and project with kernel the image of b_{ij} . The direction is a nonsingular linear transform of the joining-line direction in mark space. Radial spreading and Lemma 6.3 give a projected rate at least l_* and residual m . Their sum gives $|C_z| - |x_i| \geq m + l_*$.

For the remaining horizontal rate, condition on the whole center and quotient (dX, dB, dP) by ℓ_i . One may use the quotient coordinates $(W_i, dP - 2r_i dX)$. The missing coordinate has rate at least m . On the quotient, vary j and project with kernel the image of ℓ_j . Its direction is another nonsingular transform of the joining line, and its residual is m . Thus the quotient rate is at least $m + l_*$, proving the third bound in (modes). Time separation suffices for the nondegenerate coordinate comparisons.

Next, x_i is a combination of the two center components of quadratic G_i . Given x_i , the form $J - \mathfrak{k}_i H$ sees b_{ij} . Given the center, L_i sees ℓ_j . Both tests are nondegenerate because

$$|r_j - r_i - \mathfrak{k}_i(\theta_j - \theta_i)|$$

has subpower separation by line avoidance. They contribute two residual copies of m . The pair G_i, G_j determines all three center coordinates and two independent horizontal combinations. Indeed the two coefficient pairs $(\beta_i, \mathfrak{k}_i)$ and $(\beta_j, \mathfrak{k}_j)$ have subpower separation by the same line test. One scalar coordinate remains, with rate at most one. Subadditivity and exchangeability prove the second bound in (Gq).

For the linear x_i bound, use the planar source (dX, x_i) and put $f = (\theta_i - \theta_j)(r_j - r_i)$. Both projections x_i and $x_i - f dX$ have residual rate exactly m : ℓ_i and ℓ_j , respectively, give the lower bound, and the angular upper bound gives equality. Consequently their projected rates agree. Vary j in the second projection and use the pin-product direction alternatives with Lemma 6.4. This proves $|x_i| \geq l_*$. For G_i , its image of ℓ_j is nondegenerate by crossing and has direction θ_j . The same argument gives a projected rate l_* and residual m . Two G readings determine every coordinate except Y , so the last expected inequality follows.

Finally (π_i, X) misses only one additional scalar coordinate by rank. In this comparison all inversions have bounded coefficients. Its missing angular rate is at most m , and the other scalar costs at most one. Thus $\alpha \leq |\pi_i| + m + 1$, which is (miss). $\square$

Lemma 6.6 (A many-leg vertical rate). *For linear admissible tangent data with $m > 0$,*

$$|H| \geq l_*. \quad (\text{H-walk})$$

For each strict-error application a sufficiently large finite number of replicas suffices.

Proof. Walk $2n$ distinct legs. Adjoin the starting side, very fine even angular coordinates, including X_{2n} , and the odd angular parents at depth p . These data predict the endpoint parent

to negligible lists. Summing the B, Y formulas in (leg), and translating at that parent, makes the endpoint H reading equivalent to

$$S_n = \sum_{k=1}^n (\theta_{2k} - \theta_{2k-1}) (X_{2k-1}^2 - 2X_{2n}X_{2k-1}).$$

The coefficient of H may first be frozen at the known fine endpoint or at the prescribed fixed positive prefix; the increment comparison gives the same rate.

Let s_k be the rate of the partial sum with these data, and set $s_0 = 0$. The source consisting of S_{k-1} and the new odd quadratic scalar has joint prefix rate at least $s_{k-1} + m$. To see the second contribution, reveal the other fine endpoint bits, which predict the previous sum. The derivative of the new scalar is $2(X_{2k-1} - X_{2n})$, with subpower separation, and (fib) gives its remaining angular rate m .

For the projection, omit the current pair of records. The difference of their time marks has exponent d . Here is the required conditional comparison. If X_j^* are fixed finite-depth angular names fine enough for all these tests, the walk rule gives

$$D_{\text{KL}} \left(\mathbb{P}_{Z_0, \Xi, (X_j^*)_j} \left\| \mathbb{P}_{Z_0, \Xi} \otimes_{C^*} \prod_j \mathbb{P}_{X^*|C^*} \right. \right) = o(\log(1/\delta)). \quad (6.8)$$

This follows by the conditional chain rule in chronological order: the next endpoint is sampled from its conditional fiber kernel, the marked marginal is stationary, and its angular information cost is (fib). The initial angular coordinate is included in Z_0 and need not be independent of it.

The reference law in Equation (6.8) factorizes the omitted record and the source angular bits conditionally on the remaining data. The source parents are predicted to negligible lists by the odd parents and the retained data. Conditional chain rules give the same negligible information budget after these parents are revealed. Conditional likelihoods at actual samples agree up to vanishing exponents. Trim excessive likelihood ratios and time-ball failures with any fixed small power slack. The unchanged start-side marked marginal supplies the unnormalized direction bounds, which integrate to the source conditioning. This verifies the projection-rule hypotheses at the actual law; no assertion merely typical for the reference law is transferred without a likelihood bound.

The projection rule therefore gives

$$s_k \geq \min \left(1, s_{k-1} + m, \frac{s_{k-1} + m + d}{2} \right).$$

Iterating this monotone map from zero converges to $\min(1, m + d)$: below that value each of its three entries exceeds the current value, and its least positive fixed point is $\min(1, m + d)$. Adding the auxiliary data can only lower the original endpoint score, whose parent was already predicted. Given any strict desired error, choose a finite n making the iterate sufficiently close to the limit, and only then choose the common replica experiment and cutoffs. $\square$

6.4 Orientation motions

The preceding estimates varied the angular coordinate with a mark held fixed. We now vary the mark in the equations relating two phases with shared fiber data. Holding one phase and the target angle fixed determines the resulting motion of the target phase. These motions make the time and orientation innovations available as further phase readings.

Lemma 6.7 (Mark-motion comparison). *Assume $m > 0$. Fix a record j and resample one or two sides R_s independently from the conditional kernel with the same $Q_j, \mathcal{D}$ as a target F . Put $A_s = X_F - X_{R_s}$. Each A_s , and $A_1 - A_2$ when two draws are used, has subpower separation. With R_s, X_F held to sufficiently fine tested precision, variation of the mark $\xi_j = (\theta_j, r_j)$ has differential*

$$\begin{aligned} dX &= 0, & dB &= -A d\theta_j, & H &= A^2 d\theta_j, & \text{both cases,} \\ dP &= 2A dr_j, & J &= A^2 dr_j, & K &= A^2(r_j d\theta_j - \theta_j dr_j), & \text{quadratic,} \\ dN &= A dr_j, & dD &= -r_j A d\theta_j - \theta_j A dr_j, & & \text{linear.} \end{aligned} \quad (\text{trans})$$

Here $A = A_s$, and the forms are frozen at the target. In particular x_j annihilates the full two-component motion. These motions give conditional lower-score tests with the mark innovation bounds of Definition 6.1, when the held readings annihilate the tested innovation.

Proof. The target and the resampled sides are exchangeable conditionally on the shared fiber data. Their angular information bounds give the stated separations. Holding R_s and X_F fixes A . In the quadratic case the leg formulas give

$$\begin{aligned} B_F &= B_R - \theta_j A, & Y_F &= Y_R - \theta_j(2X_R A + A^2), \\ P_F &= P_R + 2r_j A, & N_F &= N_R + P_R A + r_j A^2, \\ D_F &= D_R - \theta_j(P_R A + r_j A^2). \end{aligned}$$

Differentiating and substituting $X_F = X_R + A$ and $P_F = P_R + 2r_j A$ into the contact forms gives Equation (trans). In the linear case use $N_F = N_R + r_j A$ and $D_F = D_R - \theta_j r_j A$. Direct substitution proves the assertion about x_j .

The conditioning order is essential. First freeze every coefficient involving the moving mark at a depth below p and above t . With the full record known, translate at F_p to compare the original and frozen readings. Equation (vb) then removes the remaining fine record from the list test. Only after this removal do we adjoin Z_{R_s} and the motion data. At a mark parent of depth p , these data predict F_p and every held annihilating child reading up to negligible lists. An informative component recovers the scalar innovation by inversion, with arbitrarily small power loss. Its Taylor error is $O(\delta^{2p})$. Holding the other mark scalar to depth $p + t$, when permitted by its innovation law, adds only an error of that order of accuracy.

Auxiliary angular bits do not invalidate removal of the moving record. For example, for two draws and fixed sufficiently fine names,

$$I_\delta((X_{R_1}^*, X_{R_2}^*); Z_F, \Xi \mid C_*) = o(1). \quad (6.9)$$

Indeed let $\mathcal{B} = (Q_j, \mathcal{D})$ and $Y_s = X_{R_s}^*$. The variables Y_1, Y_2 are conditionally independent given $\mathcal{B}$, and their joint conditional law depends on (Z_F, Ξ) only through $\mathcal{B}$. Since $\mathcal{B}$ is determined by these latter data, the conditional chain rule gives

$$\begin{aligned} I_\delta((Y_1, Y_2); Z_F, \Xi \mid C_*) &= I_\delta((Y_1, Y_2); \mathcal{B} \mid C_*) \\ &\leq \sum_{s=1}^2 I_\delta(Y_s; \mathcal{B} \mid C_*) = o(1). \end{aligned}$$

The inequality subtracts the nonnegative term $I_\delta(Y_1; Y_2 \mid C_*)$; the last equality is (fib) for the stationary marginals. The target bits X_F^* may be included in the finite phase budget in (vb). At R_s , its own angular bits are known, and the other angular bits satisfy the corresponding

version of Equation (6.9). Finally, (Z_{R_s}, Ξ) has the original marked marginal, so the replicas are conditionally independent there and the stated mark innovation law applies. Conditional chain rules permit the specified mark parents and child scalar bins. This proves the score comparison without asserting independence conditional on all the exact motion data simultaneously. $\square$

Proposition 6.8 (Orientation rates). *For quadratic data with $m > 0$ and radial spreading,*

$$|x_i| \geq d. \quad (\text{cross-q})$$

If also $c < 1$, set $a_ = (d - cv)_+$ and $n_* = \min(1, d + \nu)$. Then*

$$|x_i| \geq n_*, \quad |H \mid x_i| \geq a_*, \quad |C_z| - |x_i| \geq a_* + \nu, \quad |\pi_i| \geq l_* + a_* + |x_i|. \quad (\text{mix-q})$$

For linear data with crossing, $m > 0$, $c < 1$, and $\nu > 0$,

$$|x_i| \geq n_*, \quad |\pi_i| \geq 2l_* + n_*, \quad w - |\pi_i| \geq \nu. \quad (\text{mix-l})$$

The many-leg term in this assertion is obtained with an arbitrary strict small loss using finitely many replicas.

Proof. For the quadratic x_i bound, move a record $j \neq i$. Its gradient in mark space is

$$A^2(r_j - r_i, \theta_i - \theta_j).$$

Keep Z_R , the moving mark parent $\xi_{j,p}$, the needed target angular bits, and other records except i, j . Use i as the varying projection record. The input mark increment has dimension at least d . By the parent-representative version of radial spreading, the gradient directions over i have unnormalized Frostman exponent d on the high witness relation. Swapping and signing coordinates does not change that assertion. Pair separation makes the gradient magnitude nondegenerate up to subpower loss.

To verify the output and independence hypotheses, start from a proposed low-score list for x_i conditional on $F_p, \mathcal{D}_{-j}$. The removal of record j is allowed by (vb). Given its parent and the motion data, F_p has negligible lists. Taylor expansion at the phase predicted by the mark parent, followed by dilation by δ^{-p} , turns this into a list for the stated linear projection. The law of the angular bits has negligible likelihood cost relative to retaining (Z_R, Ξ) and sampling those bits independently given C_* . Under this comparison kernel the planar source and the projection record are independent given the test data. Trim likelihood and direction failures with fixed small power slack; the marked marginal is unchanged. Thus the source and direction dimensions are both at least d , and the projection rule gives $|x_i| \geq d$. Precisely, Lemma 5.23 applies with side Z_R , auxiliary name X_F^* , $S = C_*$, moving mark j at parent depth $b = p$, and pin record i . It supplies both the unnormalized parent-direction bound and its conditional likelihood transfer, without dividing the pin bound by the moving parent's probability.

For the mixed quadratic bounds, move i itself. Hold r_i to child depth. The H component reads the time innovation, of rate at least a_* , and x_i annihilates it. Next hold θ_i to child depth. The J component reads the r_i innovation, of rate ν , while both H and x_i may be held. The chain rule gives the two center bounds. If $\nu > 0$, use the planar pair (J, K) conditional on H and the projection $K + \theta_i J$. Holding θ_i to child depth, this projection and H annihilate the r_i innovation, while J reads it. Remove the full i record from the planar source by (vb), then use the time direction and Lemma 6.4 with residual ν . The resulting projected rate is n_* . Restoring

the record and H identifies it with x_i . When $\nu = 0$ use (cross-q). Finally W_i , even conditional on the whole center, costs at least l_* by (modes).

In the linear case condition on (X, B, H) to child depth. The planar source is (dN, dD) and the projection is $dD + \theta_i dN$. With θ_i held to child depth, the r_i innovation has rate ν , is annihilated by this projection, and is read by dN . Lemma 6.4 gives rate n_* , also for x_i with this finer conditioning. Add $|W_i \mid H| \geq l_*$ and $|H| \geq l_*$ from (H-walk). This proves the first two claims. Given π_i , the angular mode gives rate m for X ; after those readings are held, dN still reads the r_i innovation with residual ν . Consequently $\alpha \geq |\pi_i| + m + \nu$. $\square$

Proposition 6.9 (Two time innovations). *For linear admissible data with $m > 0$ and crossing,*

$$\begin{aligned} |\pi_i| &\geq 2d + l_*, & c < 1, \nu = 0, \\ w - |x_i| &\geq 2d, & c = 1, \\ w - |G_i| &\geq 2d, & c > 1. \end{aligned} \tag{double}$$

Proof. In the first case move $j \neq i$, holding r_j to child depth. The time innovation still has rate at least d . The reading

$$V = dD + \theta_i dN - r_j W_i$$

annihilates that motion. It is a projection of G_j with direction θ_i . The image of ℓ_i in G_j is nondegenerate by crossing and is annihilated by V . Lemma 6.4 therefore gives $|V| \geq l_*$.

For $c = 1$ move i and set $V = x_i$. Do not hold the refined r_i ; the joint-parent time innovation still has rate d , and x_i annihilates the full motion. For $c > 1$, move i , set $V = G_i$, and hold r_i to child depth. The allowed depth condition $p + t < cp$ gives the time rate d , and G_i annihilates this remaining motion.

In all cases use two conditional draws R_1, R_2 as in Lemma 6.7. The motion of (W_i, H) has components $(-A_s, A_s^2)d\theta$. After adding the auxiliary angular bits, first test $H + A_1 W_i$ with the second draw. Its coefficient is $A_2(A_2 - A_1)$, with subpower separation, and V is held. Next hold both V and $H + A_1 W_i$, and test W_i with the first draw. The held combination annihilates this motion and the coefficient of W_i is $-A_1$. The chain rule gives $|(W_i, H) \mid V| \geq 2d$. The conditioning comparison in Lemma 6.7 justifies each test before dropping the auxiliary bits.

In the first case (W_i, H, V) and π_i are equivalent with the marked data, since their last components differ by $(r_i - r_j)W_i$. In the other two cases the held readings also annihilate ℓ_i , leaving angular rate m . These observations prove all three inequalities. $\square$

6.5 Quadratic width and binary time layers

Two estimates remain before we can collect the admissible directions. The first bounds the entropy cost of refining the horizontal width in a quadratic phase. The second uses binary time layers to improve the time bound at large quadratic cost.

Proposition 6.10 (Horizontal width). *For quadratic admissible tangent data, $\bar{e} \geq 2m$. If $m > 0$ and radial spreading holds,*

$$\bar{e} \geq 2m + d. \tag{e-q}$$

Proof. Fix $p < z < 2p$ and refine the horizontal depth from p to $p + t$, leaving z fixed. Given $F_{p,z}$, this increment is equivalent to the new horizontal bits: recentering contributes only $O(\delta^{2p})$. For every angular mode, $Q_i, X_p, \mathcal{D}$ predict $F_{p,z}$ up to negligible lists. Indeed the contact-center derivative on the fiber is $O(\delta^p)$ on the parent, and its variation is $O(\delta^{2p})$.

Two time-separated angular modes are linearly independent. First test a horizontal scalar annihilating ℓ_i and seeing ℓ_j ; then test a scalar seeing ℓ_i . These tests remain valid after holding the scalar annihilating both modes. Lemma 6.3 therefore gives horizontal increment rate at least $2m$. Integrating in p from λ to 2λ with $z = 2\lambda$ fixed yields $e(\lambda) \geq 2m$ in the averaged sense. Boundary endpoints are handled by the Lipschitz continuity of the array on the wedge.

For the additional rate, a horizontal scalar annihilating both modes is

$$U = (\theta_i - \theta_j)(dP - 2r_i dX) - 2(r_j - r_i)(dB + \theta_i dX). \quad (6.10)$$

Set $q_0 = z - p$ and $u = p - q_0 > 0$. Resample R on the i fiber while sharing X_{q_0} . Then $A = X_F - X_R$ is $O(\delta^{q_0})$ and, for every fixed $\chi > 0$, at least $\delta^{q_0+\chi}$ in magnitude with probability tending to one. This follows from the conditional version of (fib). The marked marginal is stationary, and the target angular bits beyond the common prefix retain negligible information with Z_R, Ξ given C_*, X_{q_0} . To see the exact comparison, set $\mathcal{S} = (C_*, X_{q_0})$ and $\mathcal{B} = (Q_i, \mathcal{D}, X_{q_0})$. The resampling kernel is the conditional side law given $\mathcal{B}$. The conditional Markov property and (fib) give

$$\begin{aligned} I_\delta(X_F^*; Z_R, \Xi \mid \mathcal{S}) &\leq I_\delta(X_F^*; Q_i, \mathcal{D} \mid C_*, X_{q_0}) \\ &\leq I_\delta(X_F^*; Q_i, \mathcal{D} \mid C_*) = o(1). \end{aligned}$$

The second inequality uses that X_{q_0} is a prefix of X_F^* . Thus the comparison law keeps (Z_R, Ξ) unchanged and samples X_F^* independently with its conditional law given $\mathcal{S}$. The common prefix is already determined by Z_R .

Start the moving mark at depth u . Given Z_R , fine X_F bits, and $\xi_{i,u}$, the name $F_{p,z}$ has negligible lists. Horizontal uncertainty is $O(A\delta^u) = O(\delta^p)$. Guess the boundedly many corresponding horizontal bins. With these bins fixed, the recentered uncertainty is

$$O(A^2\delta^u) + O(\delta^{2p}) = O(\delta^z).$$

The mixed quadratic mark remainder is $O(A^2\delta^{2u})$, which is smaller.

Freeze the i coefficients in Equation (6.10) at their u -parent representatives. Indeed its expanded form is

$$U = (\theta_i - \theta_j)dP - 2(r_j - r_i)dB - 2(\theta_i r_j - \theta_j r_i)dX.$$

Each coefficient changes by $O(\delta^u)$. On a horizontal parent its centered increment has size $O(\delta^p)$, so the change is $O(\delta^{p+u})$, negligible at depth $p+t$ when $t < u$. Absolute shifts at the parent are retained during this comparison with the full record. Remove the remaining fine i mark by (vb) before adjoining motion data. With X_F fixed, the resulting mark-plane projection is

$$2A((r_j - r_{i,u})d\theta_i + (\theta_{i,u} - \theta_j)dr_i).$$

After dilation of the mark parent, a low-rate list at depth $p+t$ becomes a projection list at depth t , up to arbitrarily small power losses. Its source has joint dimension at least d . More explicitly, the common test data are $Z_R, X_F^*, \xi_{i,u}$ and the held records other than i, j . The source is the child increment of ξ_i and the projection record is j . Under the comparison law they are independent given these data: the marks are independent given Z_R , and X_F^* is independent of them given $\mathcal{S}$. Conditional chain rules retain a vanishing information error at each fixed u, t . The original low-score list can be unioned over the bounded parent possibilities described above, giving a list for each j record. The mark innovation bounds apply at $Z_R, \xi_{i,u}$. Radial

spreading at this same parent gives the unnormalized direction bound of exponent d over j on the high witness relation. Finite-prefix likelihood trimming, with fixed small power slack, verifies the projection hypotheses at the actual samples. This is Lemma 5.23 with side Z_R , auxiliary name X_F^* , $S = (C_*, X_{q_0})$, moving mark i at parent depth $b = u$, and pin record j . The required separation exponent can be chosen so that $u - \chi > t$, since $t \ll u$. Thus U has incremental rate at least d in the strict-error sense.

This last argument does not assume homogeneous horizontal increments under the finer center conditioning. More explicitly, an expected increment rate strictly below $2m + d$ along a sequence $t \downarrow 0$ would, by bounded relative capacities and the two residual angular rates, give a positive-probability strict deficit for the U list. Radial witnesses provide the required finite-prefix bounds with probability tending to one as t decreases. Select one sufficiently small fixed t , take the tangent information errors to zero first, and apply the projection rule to exclude this deficit. Integration over the open wedge, followed by Lipschitz endpoint control, proves (e-q). $\square$

Proposition 6.11 (Binary time estimate). *Assume quadratic admissible data, $m > 0$, $c > 3$, and line avoidance. At a binary layer $p \rightarrow 2p$, freeze contact coefficients at F_p , and let*

$$b_{\text{miss}}(p) = p^{-1} H_\delta(F_{2p} \mid F_{p,2p}, \mathcal{D}, (L_i, L_j)_{2p})$$

in the limiting entropy law. Then $0 \leq b_{\text{miss}} \leq 1$ and $e(p) \geq b_{\text{miss}}(p) + 2m$. Put $r_0 = c 2^{-\lfloor \log_2 c \rfloor} \in [1, 2)$. For the time lower bound, the term $(c - 1)|G_i|$ in (time-rate) may be replaced by

$$\frac{1}{2} \left[(c - r_0)(\bar{\alpha} - \bar{b}_{\text{miss}}) + (r_0 - 1)(\bar{\alpha} - 1) \right]. \quad (\text{bin})$$

Proof. The center name and two independent horizontal forms leave only one scalar coordinate; line avoidance supplies subpower singular-value bounds. Relative capacities therefore give $0 \leq b_{\text{miss}} \leq 1$. The joint increment of L_i, L_j given $F_{p,2p}$ costs at least $2mp$. Indeed L_i annihilates ℓ_i exactly and L_j sees it, and conversely. Given $Q_i, X_p, \mathcal{D}$, the center bins to depth $2p$ have bounded lists by the fiber Taylor estimate. The horizontal invariance is linear, so these two comparisons apply on the whole binary layer. The conditional chain rule yields $e(p) \geq b_{\text{miss}}(p) + 2m$.

On $(\lambda, r_0\lambda)$ use the microlayer estimate and $2\mathbb{E}|G_i| \geq \mathbb{E}\alpha - 1$. Tile $[r_0\lambda, c\lambda]$ by binary intervals $[p, 2p]$, where $p = c2^{-j}\lambda$. On each such interval,

$$2p \leq c\lambda, \quad p \leq (c - 1)\lambda.$$

The output comparison in Proposition 6.2 then reads G_i to depth $2p$ in the contact basis at F_p . The Taylor remainder is $O(\delta^{2p})$, and the coefficient error in (6.5) times the parent width is no larger. Freeze the true $(\beta_i, \mathfrak{k}_i)$ at depth p as output-predicted auxiliary data before adding marked conditioning. Subtract the velocity using the actual child bin time θ' ; this avoids a λ -depth error from an incorrect terminal-time backflow.

The pair G_i, G_j at this accuracy recovers the three recentered center readings and L_i, L_j , up to subpower losses. Hence symmetry and subadditivity lower-bound the information from a single output by half the isotropic binary increment minus $pb_{\text{miss}}(p)/2$. Logarithmic averaging of that isotropic increment divided by p gives $\bar{\alpha}$: integrate its expected derivative on $[p, 2p]$ and use dilation invariance. The binary interval lengths sum to $(c - r_0)\lambda$, and the initial interval has length $(r_0 - 1)\lambda$. Their sum proves (bin). $\square$

6.6 Interior action bounds

We now combine the phase rates into directions of decrease for the canonical excess. Put $t = 1 - a$. Along a path with $dv/dt = k \geq 0$, the inequality below gives $dE(1 - t, v(t))/dt \leq -2\Gamma$ wherever it applies. We call $k = 0$ freezing: the horizontal resolution is held fixed. Recall that $c = c'$ is the rate of the active template branch, so $c = b'_1$ in the linear cases, including a clipped branch.

Proposition 6.12 (Admissible local actions). *On a valid branch of (canon), the actions listed below satisfy*

$$(\partial_a - k\partial_v)E \geq 2\Gamma. \quad (\text{surplus})$$

Except where explicitly allowed, assume $m > 0$, radial spreading in the quadratic case, crossing in the linear case, and line avoidance for the quadratic $c > 1$ estimates.

- (i) *Quadratic freezing: $k = 0$ is admissible if $m \geq k_0/(1 + c)$ for $0 < c < 1$, or $m \geq k_0/2$ for $c \geq 1$.*
- (ii) *Quadratic motion: for $c \geq 1$, $k = (c + 1)/4$ is admissible.*
- (iii) *At quadratic cost $c = 1$, without radial spreading and allowing $m = 0$, $k = 1/2$ is admissible when $d \leq 1/4$.*
- (iv) *Linear freezing: $k = 0$ is admissible if $m \geq k_0$. For $c < 1$, it suffices more generally that $m \geq k_0 - cv$, with $m > 0$ still required.*
- (v) *Linear motion with $0 < m \leq k_0$: for $c \geq 1$, use $k = (1 + c)/2$; for $c < 1$ and $m \geq k_0/(1 + c)$, use $k = c$.*
- (vi) *Linear motion with $2d \leq c < 1$: $k = 1$ is admissible for every $m \in [0, 1]$, including zero. Crossing is needed only when $m > 0$.*

All assertions are stable under sufficiently small errors in their hypotheses and inequalities, with the corresponding additive error in (surplus).

Proof. Write x, π, G, w, α, e for the expected barred rates. Equations (time-rate), (width-rate), and (canon) reduce the claim to $S \geq 0$, where

$$\begin{aligned} T &= \min(c, 1)(w + x) + (1 - c)_+\pi + (c - 1)_+G, \\ S &= T - 1 - 2d - c + k \begin{cases} 5 - 2\alpha + e, & \text{quadratic,} \\ 3 - \alpha, & \text{linear.} \end{cases} \end{aligned} \quad (6.11)$$

For $c > 3$ in the quadratic case we may replace the last term in T by (bin). The omitted favorable η term is ηc in the linear case and $\eta(c - k)$ in the quadratic case. It is nonnegative for the stated speeds.

Quadratic freezing. Suppose first $c < 1$. Equations (modes) and (mix-q) give

$$T \geq l_* + (1 + c)n_* + (1 - c)a_* + cm + c \max(m + l_*, a_* + \nu). \quad (6.12)$$

If $\nu \leq k_0$, then $n_* = d + \nu$, $a_* \geq d - c\nu$, and the last maximum is at least $m + l_*$. Thus

$$T \geq (1 + c)l_* + 2d + 2cm + (1 + c^2)\nu \geq (1 + c)l_* + 2d + 2cm.$$

The desired bound follows from $2cm \geq (1+c)(1-l_*)$. If $m \geq k_0$ this is immediate. Otherwise it is equivalent to $(1+3c)m \geq (1+c)k_0$, which follows from $m \geq k_0/(1+c)$ and $1+3c \geq (1+c)^2$ for $c \leq 1$. If $\nu \geq k_0$, then $n_* = 1$ and the maximum is at least $a_* + \nu$. Equation (6.12) gives

$$T \geq 1 + c + l_* + cm + a_* + c\nu \geq 1 + c + 2d,$$

because $l_* \geq d$ and $a_* + c\nu \geq d$.

For $c \geq 1$, the center and horizontal bounds give $w \geq 2m + 2l_* + x$ and $G \geq 2m + x$. Hence

$$T \geq 2cm + 2l_* + (1+c)d.$$

If $m \leq k_0$, subtracting $1 + 2d + c$ gives $(1+c)(2m - k_0) \geq 0$. If $m \geq k_0$, substitute $l_* = 1$ and use monotonicity in m .

Quadratic motion. For $1 \leq c \leq 3$, use $2G \geq \alpha - 1$, $x \geq d$, $e \geq 2m + d$, and $k = (c+1)/4$. Cancellation of the α terms gives

$$4S \geq (3-c)(1-d) + 2(c-1)m \geq 0.$$

For $c > 3$, set $b = \bar{b}_{\text{miss}} \in [0, 1]$. The two width bounds imply $e \geq 2m + \max(b, d)$. Substitution of (bin) gives

$$4S \geq (c+1)(1 + \max(b, d) + 2m) - 2(c-r_0)b - 2(r_0-1) - 4m - 4d.$$

For $b \leq d$ the right side decreases in b , so its minimum occurs at d . For $b \geq d$ it is affine, so only $b = d, 1$ need be checked. The respective values are

$$(c+3-2r_0)(1-d) + 2(c-1)m, \quad 4(1-d) + 2(c-1)m,$$

both nonnegative since $r_0 < 2$.

At $c = 1$, using only $e \geq 2m$ and $x \geq 0$ gives

$$S = \frac{1}{2} - 2d - m + x + \frac{1}{2}e \geq \frac{1}{2} - 2d.$$

This proves the third assertion, including $m = 0$.

Linear freezing. If $c < 1$ and $\nu > 0$, Equation (mix-1) gives $T \geq 2l_* + (1+c)n_* + c\nu$. Its excess over $1 + 2d + c$ is

$$2 \min(m, k_0) - (1+c)k_0 + (1+c) \min(k_0, \nu) + c\nu.$$

For $\nu \geq k_0$ this is nonnegative. For $\nu \leq k_0$, if $m \geq k_0$ it is also immediate. Otherwise $m \geq k_0 - c\nu$ lower-bounds it by $(1-c)k_0 + \nu \geq 0$. If $\nu = 0$, angular invariance gives $w \geq \pi$, so $T \geq \pi + cx \geq 2d + (1+c)l_*$. The threshold gives $l_* = 1$. For $c = 1$ use $w \geq x + 2d$ and $x \geq l_* = 1$. For $c > 1$ use $w \geq 2d + G$, $G \geq m + l_*$, and $x \geq l_*$. These yield $T \geq 2d + c(m+1) + 1 \geq 1 + 2d + c$.

Linear motion. For $c \geq 1$, put $k = (1+c)/2$ and use $2G \geq \alpha - 1$. Cancellation gives $S \geq 1 + x - 2d - m$. Since $m \leq k_0$, $l_* = d + m$; the bound $x \geq l_*$ leaves $S \geq k_0$. For $c < 1$, put $k = c$. The α terms cancel directly, leaving

$$S = -cm + cx + (1-c)\pi + 2c - 1 - 2d.$$

Use $x \geq l_*$ and $\pi \geq 2d + l_*$. The latter follows from (double) if $\nu = 0$; if $\nu > 0$, (mix-1) implies it because $l_* + n_* \geq 2d$. Thus

$$S \geq (2c-1)k_0 + (1-c)m \geq \frac{2c^2}{1+c}k_0 \geq 0.$$

Finally take $k = 1$ and $2d \leq c < 1$. Equation (miss) gives

$$S \geq 1 - 2d - m + cx.$$

When $m > 0$, crossing and ℓ_j give $x \geq m$; when $m = 0$ the same inequality is trivial. Since $m \leq 1$, this lower bound is at least $c - 2d \geq 0$.

Every calculation involves finitely many linear inequalities with bounded coefficients on a compact parameter range. Strict-error versions therefore give the claimed stability; changing the speed costs the corresponding Lipschitz error. $\square$

7 Completion of the cylinder estimate

We now finish the proof of Theorem 3.6. The canonical excess E is nonnegative and vanishes at the equality points specified in Proposition 5.4. Proposition 6.12 supplies directions in which it decreases when followed backwards in time. The final contradiction comes from a path starting at an equality boundary along which this decrease remains valid. The remaining difficulties are the unit-cost quadratic boundary and intervals on which the angular trace vanishes.

All entropy limits in this section use the ordered limits of the canonical construction. In particular, a finite list of macroscopic queries is fixed before the configuration, mesh, and scale limits; a tangent gap is subsequently allowed to tend to zero.

7.1 A line estimate at unit quadratic cost

At unit quadratic cost the needed fixed-width time estimate is $2 + 3d$. On an equality boundary it contradicts the canonical entropy upper bound; in the strict-curvature case it gives the missing lower bound for $\partial_a E$. The next lemma isolates this estimate when the orientation marks lie close to a line determined by the side data. Its hypotheses record time uncertainty, angular uncertainty along a fiber, and a mismatch between the target line and the line obtained by resampling with the same fiber label. Its conclusion concerns the entropy of one observation output.

Lemma 7.1 (Fixed-width line estimate). *Let $\delta \rightarrow 0$. Let C be a common root name and let side data $\mathcal{Z}$ determine C , a quadratic phase $F = (X, B, P, Y, N, D)$, and coefficients (β_F, κ_F) . The normalized phase, coefficients, and marks have size $\delta^{-o(1)}$. Conditional on $\mathcal{Z}$, draw two independent observations with the same conditional law, whose marks satisfy*

$$r_i = \beta_F + \kappa_F \theta_i + O(\delta^{10-o(1)}). \quad (\text{s-line})$$

Suppose the following hypotheses hold, with arbitrary fixed strict resolution slack and with exceptional probabilities tending to zero.

- (i) *The conditional score of $\theta_{i,p+t}$ given $\mathcal{Z}, \theta_{i,p}$ is at least dt , for the finitely many queries used below through depth 3, including queries starting at 0, where $d > 0$.*
- (ii) *Each observation carries a fiber label S_i such that (C, S_i) determines its marks and the quadratic entries of Q_i in the fiber formulas to single-valued error $\delta^{10-o(1)}$.*
- (iii) *Each observation has an output O_i with these recovery properties. The pair (C, O_i) reads θ_i to depth 1, and together with X_{p+t} reads F_{p+t} whenever $0 < p < p+t < 1$, up to lists of*

vanishing exponent. If $1 < p < p + t < 2$ and $t \ll \varepsilon, p$, it also predicts $(\theta_i, r_i)_\varepsilon$ and, using these coefficients and F_p , reads

$$K + \theta_{i,\varepsilon} J - r_{i,\varepsilon} H$$

to depth $p + t$. Here H, J, K are the contact forms, evaluated in a fixed positive-depth prefix frame and translated at F_p .

(iv) For each i individually,

$$I_\delta(X_6; S_i \mid C) = o(1), \quad i_\delta(X_p \mid C) = M(p) + o(1)$$

in probability at the tested prefixes, where M is deterministic, $M(0) = 0$, and $M(p) \geq g_* p$ for some $g_* > 0$.

(v) Draw R independently from the posterior law of the side and observation given (C, S_i) . Its line satisfies

$$|\kappa_R - \kappa_F| \geq \delta^\chi \quad \text{with probability tending to one, for every fixed } \chi > 0. \quad (\text{mis})$$

Then

$$H_\delta(O_i \mid C) \geq d + g_* + 2 \min(1, 2d) + d + \min(1, d + g_*) - o(1). \quad (\text{line-time})$$

In particular, $g_* \geq 1 - 2d$ implies the lower bound $2 + 3d - o(1)$. The finite accuracy constants 6 and 10 may be increased.

Proof. The angular capacities make M Lipschitz. Put $m(p) = M'(p)$ at its Lebesgue points. Fix the tested observation i , its posterior draw R , and the other target observation j . Write $A = X_F - X_R$ and, for a small fixed $\varepsilon > 0$, condition on

$$\Omega = (\beta_F, \kappa_F, \theta_i, r_i, \theta_j, r_j, A)_\varepsilon.$$

Its normalized entropy given C is $O(\varepsilon)$. In the contact frame of the current prefix define

$$\begin{aligned} L &= dP - 2\beta_{F,\varepsilon} dX + 2\kappa_{F,\varepsilon} dB, \\ J' &= J - \kappa_{F,\varepsilon} H, \quad K' = K - \beta_{F,\varepsilon} H, \\ G &= (L, J', K'), \quad x = K' + \theta_{i,\varepsilon} J', \quad y = J' - \tfrac{1}{2} A_\varepsilon L. \end{aligned}$$

Unless specified otherwise, rates are conditional on (C, Ω, F_p) ; an additional held reading is held to child depth $p + t$. First restrict p to compact subintervals of $(0, 1)$ and $(1, 2)$. After ε and a positive contact-prefix depth are fixed, take t smaller than these depths and than p .

We first account for the cost of Ω . For each of the nested refinement families

$$\begin{aligned} X \text{ beyond } p & \mid C, S_i, X_p \quad \text{or } C, S_j, X_p, \\ \tilde{\theta} \text{ beyond } p & \mid C, \mathcal{Z}_R, X_{F,6}, \tilde{\theta}_p, \\ F \text{ beyond } p & \mid C, F_p, \Omega_0, \end{aligned} \quad (7.1)$$

where Ω_0 is any fixed subtuple needed below, the integrated expected information loss on adding Ω is at most $H_\delta(\Omega \mid C) = O(\varepsilon)$. This is the conditional chain rule applied along the refinement family. The cumulative losses have nonnegative Lipschitz limits, so their derivatives have the same integral bound.

Before this addition the first family has rate $m(p)$ by stationarity. For the second, the posterior draw has the original side-and-observation marginal. Conditional independence given (C, S_i) and the first stationarity hypothesis give

$$I_\delta(X_{F,6}; \mathcal{Z}_R, \text{observation at } R \mid C) = o(1).$$

Its time innovations therefore retain the lower rate d , by (LP). The same information comparison and $M(z) \geq g_* z$ show that $|A| \geq \delta^\chi$ with probability tending to one for each fixed $\chi > 0$: the contrary event puts X_F in a bounded list of depth- χ bins predicted by X_R .

These budgets justify all subsequent rate errors as follows. Fix a strict proposed deficit and a positive lower bound for its probability. Let $\varepsilon \downarrow 0$, discard layer points where one of the finitely many loss densities is too large, and then choose typical remaining points for the pathwise and mixed-prefix differentiations. At each such point let $t \downarrow 0$. The expected loss divided by t tends to zero along this procedure; the likelihood inequalities (LP) turn it into a vanishing score error on all but a vanishing probability. The same argument gives a vanishing product-likelihood cost, measured relative to $t \log(1/\delta)$, for each planar projection test. Conditioning additionally on a child reading which does not use the removed subtuple costs no more: the chain rule bounds the remaining information by that of the full source, with the negligible list entropy of that reading added. All rates have bounded capacities, so discarded probabilities and layer sets contribute vanishing errors to the integrals. This procedure requires no uniform differentiability in ε .

The output chain rule now gives

$$H_\delta(O_i \mid C) \geq d + \int_0^1 (\alpha_0(p) - m(p)) dp + \int_1^2 \mathbb{E}|x|(p) dp - o(1), \quad (\text{L-chain})$$

where α_0 is the expected isotropic phase rate before adding Ω . Indeed, conditional on fine phase, the time contribution is at least d after conditioning further on $\mathcal{Z}$. Below depth 1, adding the angular bits recovers the phase bits and costs at most their rate m . Between depths 1 and 2, first add the coarse mark coefficients predicted by the output and then add Ω . Translate the displayed reading at F_p and use (s-line) to obtain x . Tiling compact interior intervals and taking limits proves (L-chain); bounded capacities allow the omitted endpoints to tend to 0, 1, 2.

Both angular fiber modes ℓ_i, ℓ_j annihilate G to the required accuracy. They give

$$|F| \geq |G| + 2m(p) - o(1). \quad (\text{L-mode})$$

For completeness, add S_i, X_p for the first mode, or S_j, X_p for the second. The fiber equations predict the parent and all annihilating readings; replacing line coefficients by their depth- ε representatives introduces error $O(\delta^{p+\varepsilon-o(1)})$. First use $dB + \theta_{i,\varepsilon} dX$, which sees ℓ_j , and then dX , which sees ℓ_i . The two times are separated by every fixed power threshold, by conditional independent sampling and their time-score bound. Each inversion thus costs an arbitrarily small power, and the first budgets in (7.1) supply the two residual rates $m(p)$.

Next vary $\tilde{\theta}$ at R , holding $\mathcal{Z}_R$ and $X_{F,6}$ and starting inside a depth- p time parent. The shared fiber label identifies the target mark with $(\tilde{\theta}, \beta_R + \kappa_R \tilde{\theta})$ to the fine prescribed accuracy. Equations (leg) and (trans) give the response in G as

$$(\kappa_R - \kappa_F)(2A, A^2, -\theta_i A^2) d\tilde{\theta},$$

up to the coarse-coefficient error. Here $\beta_R - \beta_F = -\theta_i(\kappa_R - \kappa_F)$ to the same accuracy. This response annihilates x, y , whereas its J' and L components are separated from zero by arbitrary

power slack, using (mis) and the separation of A . The parent is predicted by the fiber formulas. The remaining Taylor error is $O(\delta^{2p-o(1)})$, which is finer than δ^{p+t} . The contact forms are frozen; no derivative of the line field is taken. Consequently the time innovation, with the second budget in (7.1), gives

$$|J' \mid x| \geq d - o(1), \quad |L \mid x, y| \geq d - o(1). \quad (\text{L-res})$$

Apply the conditional projection rule of Proposition 5.19 twice. For the source (J', K') , initially retain only the line coefficients from Ω , and vary the time slope $\theta_{i,\varepsilon}$. Conditional on $\mathcal{L}$, the trimmed direction sublaw has exponent d ; integrating this unnormalized bound supplies the required direction bound at the source parent. The last budgets in (7.1) justify reinstating the remaining records for the score and list tests. If S is the joint source rate, (L-res) gives $S \geq |x| + d - o(1)$, and the projection bound $|x| \geq \min(1, S, (S + d)/2) - o(1)$ therefore gives $|x| \geq \min(1, 2d) - o(1)$.

For the second application condition on x to child accuracy, retain the target entries of Ω , and vary A_ε in the projection y of (L, J') . Here is the exact information comparison for this conditioning. Write T for the target subtuple of Ω , $B_* = A_\varepsilon$, and $D_* = (C, F_p, T)$. Let W_* be the child bin of the frozen reading x , and let Z_* be the planar child source (L, J') . Both are determined to negligible lists by a sufficiently fine relative child phase name V_* and D_* , without using B_* . The name V_* may be taken at depth $p + (1 + \xi)t$ with arbitrarily small fixed $\xi > 0$; the extra capacity is $O(\xi t)$. The conditional chain rule gives

$$I_\delta(Z_*; B_* \mid D_*, W_*) \leq I_\delta(V_*; B_* \mid D_*) + H_\delta(Z_*, W_* \mid D_*, V_*), \quad (7.2)$$

$$I_\delta(V_*; B_* \mid C, F_p, T) = I_\delta(V_*; B_* \mid C, T) - I_\delta(F_p; B_* \mid C, T) + o(1). \quad (7.3)$$

For the first inequality, adjoin V_* to the source, and use $I(V_*; B_* \mid D_*, W_*) \leq I(V_*; B_* \mid D_*) + H(W_* \mid D_*, V_*)$. The second equality uses nested phase bins, with the negligible list error if their grids are only list-compatible. The cumulative mutual information on the right is nondecreasing and bounded by $H_\delta(B_* \mid C, T) = O(\varepsilon)$. Consequently its loss density is one of the integrated budgets already treated above. Divide (7.2) by t , use that budget at the retained layer points, and then let the strict resolution slack ξ tend to zero. This proves the needed vanishing source-record information after the child conditioning. In particular, W_* uses the frozen target coefficients and no A ; coefficient replacement is by a known translation at F_p and an error $O(\delta^{p+\varepsilon-o(1)})$, not by an A -dependent reading.

For the direction bound, let the full target data include its side and its two observations. The names D_*, W_* are functions of these data. Conditional on the full target data, R still has the posterior kernel given (C, S_i) . Stationarity and $M(z) \geq g_* z$ trim its angular law to unnormalized ball bounds of exponent g_* . More explicitly, for each required fixed depth and strict tolerance retain only posterior angular cells whose conditional mass given (C, S_i) is at most the corresponding $\delta^{g_* z - \text{tolerance}}$. Their union has probability tending to one by the stationary scores, and their unnormalized mass in any constant-multiple depth- z ball has the same bound up to a fixed neighbor factor, for every conditioning value. Translation and reflection give the same bound for A for every full-target value. Integrating these restricted kernels over the full target conditional on (D_*, W_*) preserves the bound. No normalization of the retained sublaw is used in this step. The projection uses layers $t \ll \varepsilon$, so rounding the direction is harmless. The last budget, now with mixed-prefix linearity for the source conditioned on x , again permits the full record in the test. Its residual beyond y is at least $d - o(1)$. Thus

$$|x| \geq \min(1, 2d) - o(1), \quad |G| - |x| \geq d + \min(1, d + g_*) - o(1). \quad (7.4)$$

Each assertion means that a fixed strict deficit of positive probability contradicts the corresponding finite projection test with the loss order specified above.

Finally $\alpha_0 \geq \mathbb{E}|F|$, and $\int_0^1 m(p) dp = M(1) \geq g_*$. Substitute (L-mode) and (7.4) into (L-chain), and then send all the budget and endpoint errors to zero. This gives (line-time). To check the last claim, if $d \leq 1/2$, use $g_* \geq 1 - 2d$ and $\min(1, d + g_*) \geq 1 - d$; if $d \geq 1/2$, use $g_* > 0$ and $\min(1, d + g_*) \geq d$. In either case the stated lower bound is at least $2 + 3d$. $\square$

7.2 Boundary applications of the line estimate

The line estimate has two uses. At the stationary equality boundary it forces the radial spreading needed by the interior estimates. In the strict-curvature case it gives a fixed-width time bound directly. Both applications must verify the posterior-line mismatch in (mis).

Lemma 7.2 (The stationary equality boundary). *In the classical equality configuration $q = c_2 = 1$, $J = 0$, the exact-side quadratic replica construction has the radial spreading property at almost every strong time a , where $p_*(a) > b_1(a)$. This holds on interior patches $v > 0$ with a common static depth $b(a) + v < x_* < p_*(a)$.*

Proof. Here $u = 0$ and $b_1 = b_2 = a - 1$. Use the unclipped quadratic template and request the canonical profiles also on the fixed boundary $v = 0$, where $E(a, 0) = 0$. Countably many depth queries suffice. Choose a typical strong a at which the prediction slack persists immediately to the left and (ng) holds on that boundary for all fixed accuracy demands. The reached label λ_a knows the boundary name. The strengthened version of (mem) with $L = 1$, ending at a , together with (Sp), therefore gives angular prefix scores with lower slope $k_0 + 2\Gamma$, to arbitrary strict slack. The comparisons can use nearby actual gates with their earlier errors already tending to zero. If this slope exceeds the angular capacity there is already a contradiction; otherwise it supplies a positive g_* with $g_* \geq 1 - 2d$.

Fix the exact side $Z = (y_0, [P]_{x_*}, t_a, R_a)$. It is the same side, with the same marks, at $v = 0$ and on the tested interior patch. Suppose the required uniform power avoidance fails. Then, along $z = \lambda h \downarrow 0$, a line predicted from Z has tube mass $s^{o(z)}$ at normalized width s^z . This formulation allows λ subsequently to decrease, as required in the high-set radial estimate (rad). A vertical line, or a slope of size at least $s^{-\chi z}$, restricts θ on the bounded mark support to an interval of positive power width; the conditional time-list bound excludes such mass. Hence, with subpower normalized coefficients, the remaining line event is

$$|\beta_1(T_1) - \beta_*(T_1)| \leq s^{b(a) + (1 - o(1))z}, \quad \beta_*(T) = R_a + s^{b(a)}(\beta_F + \kappa_F(T - t_a)s^{-a}).$$

Include successful lists of size $s^{-o(z)}$ for $[P]_{x_*}$ given (y_0, t_a) , using strong prediction and the deterministic profiles. Put $h' = z/400$, $\delta = s^{h'}$, and restrict the single final law to this event. All fixed strict time-score failures were power-small in these units before restriction, by the causal time-list count; likelihood comparison therefore preserves the time-score hypotheses.

Use $\mathcal{Z} = Z$, $C = C_*(a, 0)$, and the representative phase at $v = 0$. The predicted line satisfies (s-line). Let S_i contain depth-12 marks and representative quadratic fiber entries. They are read to negligible lists from $(C, \mathcal{C}(a + 40h', 0))$. Actual and representative roots are mutually predicted to negligible lists: guess neighboring derivative bins if necessary and transport normal centers with the base-cell accuracies; the true derivative varies by only $O(s^{x_*})$ on this side. The deterministic scores persist under the $s^{o(h')}$ restriction. Transferring those scores first, and then using the entropy chain rule, transfers (ng) to $I_\delta(X_6; S_i | C) = o(1)$ with the preceding angular

lower profile. This does not transfer an arbitrary small mutual information bound directly through a rare-event restriction. Further subsequences supply M . The output $O_i = \mathcal{C}(a + h', 0)$ has the recovery properties of Lemma 7.1, by the displayed-shear estimates underlying (time-rate). Boundary equality in (canon) gives

$$H_\delta(O_i | C) \leq 2 + 3d - 2\Gamma - \eta + o(1).$$

All profile errors are taken to be $o(h')$ before z decreases.

It remains to verify (mis). Its failure at a fixed small $\chi > 0$ gives, by independent posterior sampling and selection of a most popular interval, a prediction of a δ^χ interval for κ_F from (C, S_i) with nonvanishing success. Mark an end gate near $j = a + 60h'$ using the Borel inner completion mark of Lemma 3.19, inside the current end-state support, and apply group extraction on $[a, j]$. The successful completions have mass $s^{o(h')}$. The label λ_j knows the canonical names through $a + 40h'$ at width $0 = u(j)$, and hence C, S_i , to negligible lists. Starting rows know the lists for Z along these paths, including the successful derivative lists.

Choose $0 < \omega < \chi$ and bin the value and slope of the predicted trajectory at t_a at widths

$$s^{b(a)+\omega h'}, \quad s^{b(a)-a+\omega h'}.$$

Include the queried t_a in the list. The slope bin is predicted at the end from (C, S_i) . Once it is chosen, the value bin is read to small lists from $\beta_j(T_j)$: comparison with $\beta_1(T_1)$ and transport from T_j to T_1 cost $s^{b(j)-o(h')}$, and the line-tube error is still smaller. Extrapolation of the binned slope back to t_a costs only its chosen value width. Disjointize the finite possible-bin relations as in Lemma 3.19. Each retained marked end row has a witnessing completion in its chosen bin; path concatenation puts all its predecessors in the corresponding Borel outer starting list. This predecessor projection uses the unrestricted truncated $[a, j]$ geometric correspondence, not only successful completions. Its Borel inner partner carries the same full analyzed pre-synthesis input state in the original bin partition. After the common subtraction, the end curvature value is bounded by $s^{b(a)+\omega h'-o(h')}$, and its duration times the residual slope is at most $s^{b(j)-o(h')}$. As $b(j) - b(a) = 60h'$, the curvature cost on this segment is strictly less than 1. Group extraction contradicts its minimality. Fixed list slacks are chosen first to dominate mesh and closure errors and then sent to zero in h' units. Only the curvature cost is improved.

Thus (mis) holds. Lemma 7.1 contradicts the displayed boundary upper bound. This proves uniform line avoidance and hence the stated radial spreading. The restriction was made on the deterministic-profile single law before the new line experiment, so no stationarity claim on a subpower part of an already constructed many-replica law is needed. $\square$

Proposition 7.3 (Strict affine boundary). *Suppose $c_2 = 1$ and $q < 1$, with no lower bound on q . Use the artificial quadratic template ending at D , and set $v_0 = u(D)$; in the hybrid case take $D = 1$. If the boundary prefix memory supplies $g_* > 0$ with $g_* \geq 1 - 2d$, then at almost every $a < D$*

$$\partial_a \mathcal{H}(a, v_0) \geq 2 + 3d, \quad \partial_a E(a, v_0) \geq 2\Gamma.$$

The same conclusion holds for the clipped template on this boundary.

Proof. The boundary is an open quadratic branch strictly before D :

$$l(a, v_0) = b_1(D) + a - D < \min_{[a, D]} b_1, \quad w(a, v_0) = n(D) + a - D.$$

In hybrid applications this also supplies strict normal slack. The derivative to be proved is at fixed v_0 , so interior stationarity in width is unnecessary.

We first specify the forward profiles needed in the hybrid case. At each fixed rational $0 < j < 1$, preflatten the actual read rows $\mathbf{P}^j$ in chronological order, before the final deterministic profiles. Make finitely many requests and then diagonalize. On these rows use a version $\mathcal{C}^{[j]}(a, v)$, $a < j$, with the same exponents and endpoint constant as the artificial template, but with the j -label and its coordinates as the actual data. Namely, set $\zeta = b_2(D) - D$, bin $\bar{k}_j = [K_j]_\zeta$, subtract that trajectory, and use the residual $\beta'_j(T_j), P_j$ for the curvature and derivative inputs. Here the prime denotes residual coordinates. The queries include the roots at $v_0 = 0$ and their angular refinements. In the classical case any fixed $a < j < D$ is allowed: evaluate these names via the encountered label λ_j on the final read law, with no change to an earlier gate.

On connections from j to D , the names $\mathcal{C}^{[j]}$ and $\mathcal{C}$ have bidirectional lists of vanishing exponent, also at nearby times below j . To verify this, their global slope bins differ by at most $s^{\zeta-o(1)}$, by (A). Compare the total displayed curvature trajectories, including the global terms, at t_a . The discrepancy at T_D has accuracy $\min_{[j,D]} b_2 > b(a)$; the difference of global bins is multiplied by $O(s^a)$, and the residual j -slope by $O(s^j)$. The resulting error is $s^{b(a)-o(1)}$, and the slope error is $s^{\zeta-o(1)}$. The raw derivative discrepancy fits the displayed derivative accuracy by (A). On the shared base cell, the shear-difference Taylor formulas then give the normal widths. Hybrid fake drift fits by the strict slack and (Pl). These comparisons do not estimate the large global terms separately at absolute time. In hybrid, the forward inequality (Sp) transfers the already flattened j -row scores to the final law; classically the comparisons are pathwise. Thus the needed scores on the two systems agree, with power-small strict-tolerance exceptions and persistence under restrictions of vanishing exponent. No earlier gate is modified after computing later profiles.

This comparison concerns the canonical names and their angular refinements. It does not identify the new fine line-coefficient side data at j with analogous data at D . Those finer data are introduced on the j -law below, and are predicted from earlier gates ending at j . In particular the coarse bound $K_j - K_D = O(s^{\zeta-o(1)})$ is not used as a depth-30 prediction of their normalized slopes.

Fix a typical $a < D$ for the boundary memory, differentiability, and (ng). Choose $a < j < D$, rational in hybrid, and then $\delta = s^h$ with h small compared with these fixed gaps. For the latent side choose $A_0 < a$, $V > v_0$ with the patch slack of the coarse-side construction, and include

$$\mathcal{C}^{[j]}(A_0, V), \quad t_a, \quad R_a^{[j]} = [\beta'_j(T_j)]_{b(a)}, \quad \bar{k}_j.$$

Add depth-30 bins, in δ units, of the bounded quantities

$$(\beta'_j(t_a) - R_a^{[j]})s^{-b(a)}, \quad (K_j - \bar{k}_j)s^{-\zeta}.$$

Use their representatives for (β_F, κ_F) and choose the representative root and phase on this refined side. With

$$\theta = (T_j - t_a)s^{-a}, \quad r = (\beta'_j(T_j) - R_a^{[j]})s^{-b(a)},$$

Equation (s-line) follows because $b(a) = a + \zeta$.

The conditional time-score bound follows from the causal time-list count. Indeed a row at any nearby prefix gate between a and $a + O(h) < j$, on a path to the tested event, predicts the side to small lists. Put $\Delta_K = b_2(j) - j - \zeta = (1 - q)(D - j) > 0$ and choose $100h < \Delta_K$, with h

also smaller by a fixed large factor than all static-cell, derivative, and normal-drift slacks. On a connection from a gate $i \in [a, a + O(h)]$ to j , Equation (A) gives

$$K_j - K_i = O(s^{\zeta + \Delta_K - o(1)}), \quad \beta_j(t_a) - \beta_i(t_a) = O(s^{b(a) + \Delta_K - o(1)}).$$

For the second bound use $\min_{[i,j]} b_2 \geq a + b_2(j) - j$ and multiply the slope discrepancy by $O(s^a)$ when extrapolating to t_a . Thus both normalized errors are finer than the new depth-30 bins. The inequality

$$b_2(j) - j > \zeta$$

controls both $K_j - K_i$ and the extrapolated curvature discrepancy at t_a more accurately than every new tangent bin, by (A). First subtract the guessed global bin. The residual j -slope has size s^ζ , so the T_j -based root curvature cut also has a small list. Earlier static phase and derivative data are predictable using $l(A_0, V) < \min b_1$, exact base flow, and the hybrid normal slack. The expanded lists at the prefix gate therefore exclude too-popular time bins, with power-small exceptions at each fixed strict tolerance. In classical final-path tests the exact particle index is retained throughout.

Let S_i consist of depth-12 marks and quadratic fiber entries, and put $O_i = \mathcal{C}^{[j]}(a + h, v_0)$. The pair $(C, \mathcal{C}^{[j]}(a + 40h, v_0))$ reads S_i to negligible lists. Actual-root comparisons and the constant-cut displayed-shear calculations give the output recoveries in Lemma 7.1. The deterministic profile transfer just proved transfers (ng) to its stationarity hypothesis, with the prefix lower bound from final memory; moreover

$$H_\delta(O_i \mid C) \leq \partial_a \mathcal{H}(a, v_0) + o(1).$$

These statements persist under a vanishing-exponent fraction restriction of the j -row law.

We prove the remaining mismatch property while keeping a, j, D, h fixed. Suppose a δ^χ interval for κ_F is predicted from (C, S_i) , where $0 < \chi < 1$ is fixed and small. Choose $0 < \omega < \chi$ and group on $[j, D]$ by bins of K_j and $\beta_j(T_j)$ of widths

$$s^{\zeta + \omega h}, \quad s^{b(j) + \omega h}.$$

Starting rows know these bins. The end label knows the final canonical names at the used times below j at width v_0 , hence their j -versions, C, S_i , and $\bar{k}_j$, to small lists. It therefore predicts the slope bins on success paths. Given one such slope, it guesses the value bin at T_j from $\beta_D(T_D)$: Equation (A) and time transport apply because $\min_{[j,D]} b_2 > b(j)$. Include the ordinary time and start-rounding choices. After subtraction the residual end value is bounded by $s^{b(j) + \omega h - o(1)}$, and the residual slope times the terminal duration by $s^{b_2(D) - o(1)}$. Since $b_2(D) - b(j) = D - j$, the curvature cost on $[j, D]$ is at most $1 - \omega h / (D - j)$, a fixed strict improvement.

Classically, failure of (mis) yields a prediction with nonvanishing final mass. Use the Borel inner success and completion marks of Lemma 3.19 at D , inside the current end-state support; Equation (Sp) and the count bounds supply the group lower bound, using outer hulls of unrestricted geometric predecessors. Exact particle and time starts fix the earlier gate ownership, so these are actual path groups. Group extraction contradicts minimality of the curvature cost.

In hybrid the working law is $\mathbf{P}^j$, and a prediction-success event there might have little terminal mass. To address this, discard before j -synthesis all rows whose depth- χ slope bin has conditional mass at least s^τ given (C, S_i) , with $\tau > 0$ sufficiently small. These maps use the single observation on that row and their Borel versions under the full analyzed j -row energy law as in Lemma 3.19. For each conditioning value there are at most $s^{-\tau}$ discarded bins. We use a

closure argument because τ is not yet a vanishing exponent. Hold a, j, D, h, χ, ω fixed, and put $\gamma_0 = \omega h / (D - j) > 0$. If arbitrarily small τ allowed the discarded synthesis to approach terminal saturation, choose $\tau_n \downarrow 0$ and genuine approximating experiments whose terminal deficit and prior mesh, profile, and regularization losses tend to zero faster than τ_n . For each n group on the same fixed interval $[j, D]$ by the preceding construction. Each reachable end label has the expanded list of groups. Finite-overlap inequalities have normalized loss $O(\tau_n / (D - j))$, while other list and approximation losses can be made $o(\tau_n) / (D - j)$. Regularity is retained before propagation, grouped energies sum to the j -energy budget, and counts have the same overlap loss. The quantitative loss form of group extraction selects genuine experiments whose extremality and count losses tend to zero in the $[j, D]$ units. Their curvature costs are at most $1 - \gamma_0 + o(1)$, since h, j, D, ω remain fixed. They realize the original extremal exponents in the closure with cost at most $1 - \gamma_0/2$ eventually, contradicting minimality. This applies the vanishing-overlap clause along τ_n ; it does not apply a zero-loss extraction statement at fixed τ . Consequently some sufficiently small fixed $\tau > 0$ leaves a strict positive terminal-exponent deficit on the discarded part. The triangle inequality for the terminal functional shows that the complementary synthesis preserves the terminal exponent. Its pre-synthesis row fraction is consequently $s^{o(1)}$.

Restrict only the j -row probability law to this complement. On retained rows the original conditional slope-bin mass is below s^τ . Except on a vanishing restricted fraction, conditioning normalization given (C, S_i) multiplies mass by at most $s^{-\tau/2}$: the excluded conditioning cells have original retained mass at most $s^{\tau/2}$, whereas the total retained mass is $s^{o(1)}$. Neighboring-bin counting therefore makes posterior collisions at the tested precision vanish. The same fixed representative maps, time bounds, and deterministic scores are preserved by this restriction. For finitely many separation queries repeat the exclusion with previous errors sufficiently small; diagonal limits give (mis) for every fixed threshold. This is a restriction of the j -law for the entropy test, not a claim about the final law of a modified stack. The strict curvature improvement is needed only for fixed h .

Lemma 7.1 now gives $\partial_a \mathcal{H}(a, v_0) \geq 2 + 3d$. On this fixed-width quadratic branch, $\partial_a W = 1 - \eta$ and $p + d = 1 + 3d - 2\Gamma$ in (canon); hence $\partial_a E \geq 2\Gamma + \eta \geq 2\Gamma$. Finally, the inequalities used above hold for every $q < 1$: $b_2(j) - j - \zeta = (q - 1)(j - D) > 0$, and $\min_{[j, D]} b_2 > b_2(D) + j - D$ whether q is positive or negative. $\square$

7.3 Finite motion when the angular trace may vanish

Throughout the remaining argument write $k_0 = 1 - d$ and $p + d = 1 + 3d - 2\Gamma$, where $0 < d \leq 1$. Motion is in backwards time $t = 1 - a$, at speed $k = dv/dt \geq 0$. Thus (surplus) makes E decrease at rate at least 2Γ . On a fixed compact template domain the limiting profiles are Lipschitz with a fixed constant, uniformly for small normal enlargements $g > 0$. The two normal-bin list costs give

$$|E^g - E| \leq Cg. \quad (\text{enl})$$

Original and enlarged names may have deterministic profiles on the same law: first request a countable sequence $g \downarrow 0$ and dense depth lists; later finite requests are readout refinements.

Lemma 7.4 (Finite off-gap motion). *Consider either an unclipped classical quadratic equality branch with $b' = c > 3$, $b'_1 > 0$, and uniform strong-prediction slack $p_*(a) > b(a) + v$, or a classical linear branch with rate $c \geq 1$. The latter may be the right part of a clipped equality template, or a pure linear template. Fix a small $g > 0$ and let $h > 0$ be sufficiently small compared with g*

and the prediction slack. Set $\delta = s^h$. Let $\mathcal{K}$ be a sufficiently fine uniform finite grid in $[c, c+1]$, including its endpoints, and set

$$k_R = k/2 \quad \text{in the quadratic case}, \quad k_R = k \quad \text{in the linear case}.$$

In particular $k_R \geq 1$. For a fixed K_* sufficiently large compared with $c+1$, define

$$C = \mathcal{C}^g(a, v), \quad O = \mathcal{C}^g(a+h, v), \quad C^k = \mathcal{C}^g(a, v+k_R h), \quad M_X = \lim H_s(X^{v+K_*h} \mid C).$$

Provided the names stay on the specified branch, some $k \in \mathcal{K}$ satisfies

$$E^g(a+h, v) - E^g(a, v+k_R h) \geq \Gamma h - M_X. \quad (\text{step})$$

Proof. Use the actual C -shear at t_a . Center the residual normal velocity and starting position in their C -bins, and divide by s^{w-g} , s^{a+w-g} , respectively. The bounded pair $V = (V_n, V_d)$ is a function of (y_0, C) . The output and root know $\theta' = (t_{a+h} - t_a)/s^a$. Given C , let S_k contain

- (i) X^{v+K_*h} ;
- (ii) B_{t_a} to depth $a+v+k_R h$ and $Y_{t_a} - 2[X]_{v+K_*h} B_{t_a}$ to depth $a+2v+2k_R h$;
- (iii) in the quadratic case, $P - 2R_a X + 2k_a B_{t_a}$ to depth $l(a, v) + k_R h$, in the root curvature cuts.

These data refine with k , up to zero-cost lists. Request the finite deterministic profiles of (S_k, V_z) , $z, k \in \mathcal{K}$, and their needed joints, without altering the earlier profiles. Write $D(z \mid k) = \lim H_\delta(V_z \mid C, S_k)$.

Two geometric comparisons will be used:

$$\begin{aligned} (O, C) \text{ predict } (V_n)_c \text{ and } (V_d + \theta' V_n)_{c+1}, \\ (C, S_k, V_k) \text{ predict } C^k, \quad \lim H_s(S_k \mid O, C) \leq M_X + h(3k_R - 2). \end{aligned} \quad (7.5)$$

Here and below all prediction lists have vanishing exponent at fixed h . To verify the first line, the child basis minus the root basis on the actual ray has derivative discrepancy $O(s^l)$ and curvature discrepancy $O(s^b)$ in s^a duration units. In curvature cases $b+v \geq l$. Evaluating this difference at the known root base-bin center instead costs only $O(s^w)$ in normal velocity and $O(s^{a+w})$ in position, including the vertical base tilt. Since $(c+1)h \ll g$, these errors fit both child accuracies in the enlarged normal units. Evaluate all known offsets at the exact bin times. Thus the predictions are single centers with the required radii, not merely unspecified lists. For C^k , the S_k data supply its refined base and derivative bins. The same shear-difference comparison bounds the normal error within its requested flat widths.

For the entropy bound in (7.5), first give X^{v+K_*h} . Backflow from (O, C) reads both the horizontal and tilted vertical positions at normalized depth at least 1. In the tilted coordinate the velocity is $X^2 - 2[X]_{v+K_*h} X$, whose error about the refined center is quadratic. The two extra position costs are therefore $(k_R - 1)h$ and $(2k_R - 1)h$. In the quadratic case the output displays derivative refinement ch , which suffices because $k_R \leq c$. Changing back to the root curvature basis costs $O(s^b)$ in earlier-time units; multiplying by the unresolved base errors in the guessed position bins costs at most $O(s^{l+k_R h})$. No further derivative entropy is required.

Conditionally on (y_0, C, S_k) , time bins at δ -depth $z \in [0, 1]$ have typical score at least dz . Indeed (C, S_k) is predicted to small lists from (y_0, t_a) on all paths outside the profile exceptions.

Root curvature predictions follow from (A), (F), and cut ownership. The quadratic derivative query is strictly below $p_*(a)$; in a clipped linear branch it uses only $l = b_1(a)$, and in pure linear form it uses the center slope comparison. Exact classical flow predicts the phase entries. Expanding any popular time list by these predicted lists and applying the causal time count through $a + zh$ gives power-small failures at each fixed strict tolerance. Quantizing the child bin-start time changes only bounded factors. Likewise, deterministic profiles and the neighboring-bin likelihood bound give conditional V -ball mass at most $\delta^{D(z|k)-o(1)}$ about typical samples, for $z \in \mathcal{K}$. These are bounds for the unrestricted conditional mass of those balls. The trimmed time law has unnormalized interval bounds of exponent d .

We claim

$$H_\delta(O \mid C, S_k) \geq d + \min_{z \in \mathcal{K}} \{D(z \mid k) + d(c + 1 - z)\} - O(\text{mesh } \mathcal{K}). \quad (\text{sh})$$

The first term is supplied by $H_\delta(O \mid y_0, C, S_k)$. For the information with y_0 , draw an independent reference y'_0 given (C, S_k) , and let V' be its normal pair. If it is admissible for (O, C) , the first comparison in (7.5) forces agreement in V_n to $O(\delta^c)$ and in $V_d + \theta' V_n$ to $O(\delta^{c+1})$. Hence $|V' - V| \lesssim \delta^c$. If its distance lies between consecutive grid widths $\delta^{z+\mu}$ and $O(\delta^z)$, where $\mu = \text{mesh } \mathcal{K}$, the velocity difference is comparable to the full difference unless the latter is already at final width. Consequently the time projection confines θ' to an interval of length $O(\delta^{c+1-z-\mu})$. At the final width no time gain is charged.

Average reference admissibility over actual samples satisfying both typicality bounds. Given the actual (y_0, C, S_k) , use the conditional ball bound for V' and, for each fixed V' , the trimmed time interval bound. Summing the finitely many annuli bounds this average by the power in (sh) minus its first d , with strict tolerance and the $O(\mu)$ loss. Markov's inequality makes the reference admissibility probability at most that power with slightly more tolerance for most actual (O, C, S_k) . The actual posterior given (O, C, S_k) is supported on the admissible set. The relative entropy bound for a measure supported on a reference set of mass q , namely $\log(1/q)$, therefore gives the information term in (sh). All trimming was on the actual sample in this averaging argument; no exceptional-set transfer to the independent product law is used.

For $z_2 > z_1$, the increment $D(z_2 \mid k) - D(z_1 \mid k)$ is nonincreasing in k , because S_k refines. Thus the largest minimizer $z(k)$ in (sh) is nondecreasing. A nondecreasing self-map of a finite ordered set has a fixed point: iterate from its least element. At such a point $z(k) = k$, use (7.5) and refinement of C by O to obtain

$$\lim \{H_\delta(O) - H_\delta(C^k)\} \geq d + d(c + 1 - k) + 2 - 3k_R - M_X/h - O(\mu).$$

The weight exponent difference, later minus moved, divided by h , is $-(3 + \eta)k_R + (1 - \eta)(c - k)$. Subtracting this and $p + d$ as in (canon) gives

$$\frac{E^g(a + h, v) - E^g(a, v + k_R h)}{h} \geq (1 - d)(1 + k - c) + 2\Gamma + \eta(k_R + c - k) - M_X/h - O(\mu).$$

Both extra displayed terms are nonnegative: $k \in [c, c + 1]$, $d \leq 1$, and $k_R + c - k \geq 0$ in the stated branches. Choose μ small enough to retain Γ . The finite refinement proof can be applied at any macroscopic path node to the same profiles E^g, M_X ; it does not select microscopic configurations adaptively. This proves (step). $\square$

7.4 The off-gap path contradiction

The finite step loses only the angular information M_X . We compare that loss with angular scores at later backward times, so that its accumulated contribution vanishes across widths where the trace is zero. This lets us join finite steps to the differential motions from the interior estimates.

Write $E_t^g(v) = E^g(1 - t, v)$, $E_t(v) = E_t^0(v)$, and $m_t(v) = m(1 - t, v)$. Denote the cost in (step) with earlier root $(1 - t - h, v)$ by $M_X(t, v; h, g)$. Thus a permitted speed k satisfies

$$E_t^g(v) - E_{t+h}^g(v + kh) \geq \Gamma h - M_X(t, v; h, g).$$

On a fixed compact canonical domain there is a constant A_0 such that, if $t' > t + h + A_0 g$ and $0 \leq v - h \leq y \leq v$, then

$$M_X(t, v; h, g) \leq U(y + (K_* + 1)h; 1 - t', y). \quad (\text{shift})$$

Indeed the enlarged name at $(1 - t - h, v)$ predicts the ordinary name at $(1 - t', y)$. Nesting handles time, curvature, derivative, and base data. The positive forward rate of the normal-width exponent absorbs g in this time shift; backflow in the fine shear costs its enlarged velocity width and that width times the longer duration, and the shear-difference estimate on the base cell fits the ordinary coarse widths. This remains valid across a clipping switch. Finally $y + (K_* + 1)h \geq v + K_* h$, so conditioning and refinement prove (shift). For each fixed rational t' the angular profile properties give

$$F_h(t', y) := h^{-1}U(y + (K_* + 1)h; 1 - t', y) \longrightarrow (K_* + 1)m_{t'}(y) \quad (7.6)$$

for almost every y , with $0 \leq F_h \leq K_* + 1$.

Lemma 7.5 (Off-gap path contradiction). *Suppose that on an open time interval and in a sufficiently narrow right neighborhood of $v = Jt$ the following hold. The profiles E, E^g are nonnegative and uniformly Lipschitz on the compact domains used, satisfy (enl), and $E_t(Jt) = 0$. Equations (step) and (shift) apply, with bounded step speeds whose minimum is at least 1 and strictly greater than J ; step admissibility is uniform on compact subsets for h/g small enough. The angular rate has its nondecreasing-in- t version. There is a speed $k_+ \geq 1$, $k_+ > J$, for which (surplus) holds almost everywhere where $m_t(v) > 0$. Alternatively, that motion need only hold for $0 < m_t(v) \leq k_0$ provided the vertical estimate $\partial_t E_t(v) \leq -2\Gamma$ holds almost everywhere where $m_t(v) > k_0$, throughout the constant-width continuations used, including after crossing the seam. The normal-width exponent has positive forward rate on all branches needed for (shift). These conditions are incompatible with $\Gamma > 0$.*

Proof. Fix compact domains and all Lipschitz and speed bounds before choosing density scales. Let C_0 denote constants depending only on these bounds and K_* . Outside a null set of widths set

$$\sigma(v) = \inf\{t \in \mathbb{Q} : m_t(v) > 0\},$$

with the terminal endpoint convention if the set is empty. Take rational queries in a containing interval and truncate this threshold to its closure if necessary. Monotonicity implies positivity at almost every relevant point above σ , whereas $m_{t'}(v) = 0$ at each rational $t' < \sigma(v)$ outside its null set of exceptional widths.

We first justify integration on a prescribed positive segment. Fix a slab $[s, T]$ and a measurable width set P where $m_t(v) > 0$ for almost every t in that slab. Consider a proposed segment of speed k_+ , duration ℓ , and width interval W . At its initial point one has either

$$E_s(v) \geq 2\Gamma\ell + E_{s+\ell}(v + k_+\ell) - C_0|W \setminus P|, \quad (7.7)$$

$$\text{or} \quad E_s(v) \geq 2\Gamma(T - s) - C_0|W \setminus P|. \quad (7.8)$$

To see this without assuming the proposed line is generic, perturb its initial width. For almost every perturbation, Fubini's theorem and the Lipschitz chain rule give the moving estimate almost everywhere and valid vertical estimates at almost every encountered width. Time spent outside P is bounded by $|W \setminus P|/k_+$, with an error tending to zero under the perturbation. If freezing is allowed and the high-rate set $m > k_0$ on that line has positive time measure, choose a regular point arbitrarily close to its first essential occurrence. The measure of earlier high-rate times can be made arbitrarily small. Integrate the moving estimate to that point, using the Lipschitz bound on those earlier high-rate times, and then freeze to T . Monotonicity retains $m > k_0$ on the vertical continuation, and endpoint nonnegativity proves (7.8). If there is no positive-measure high-rate set, integrate to the moving endpoint to get (7.7). Let the earlier-high-time tolerance and then the perturbation tend to zero. A subsequence either always freezes or always reaches the proposed endpoint, and Lipschitz continuity gives the corresponding alternative at the original starting point.

Suppose $J > 0$ and put $T(v) = v/J$ on an interior seam-width interval. It is covered by

$$Z_0 = \{\sigma(v) \geq T(v)\}, \quad \{\sigma(v) \leq T(v) - 1/n\}, \quad n \geq 1.$$

One of these sets has a density point in that interval. If $v_* = Jt_*$ is a density point of $P = \{\sigma(v) \leq T(v) - \Delta\}$, $\Delta > 0$, fix a small $\varepsilon > 0$ and start at $(t_*, v_* + \varepsilon r)$. Propose a positive segment of duration r . For r small, every visited width in P has threshold below t_* , while density gives $|W \setminus P| = o(r)$. The segment stays to the right of the seam. Either (7.7) or (7.8), with $T = t_* + r$, implies

$$E_{t_*}(v_* + \varepsilon r) \geq 2\Gamma r - o(r).$$

Seam equality bounds the left side by $C_0\varepsilon r$, a contradiction after first choosing ε small and then r small.

Now let $v_* = Jt_*$ be a density point of Z_0 . Start at the same width and make steps of duration h until less than h remains in $[t_*, t_* + r]$. At a step starting at (t_i, v_i) , the speed bound gives $v_i - Jt_i \geq \varepsilon r$. For h small compared with εr , all $y \in I_i = [v_i - h, v_i]$ satisfy

$$T(y) - (t_i + h) \geq b_0 := \frac{\varepsilon r}{2J}.$$

These intervals have disjoint interiors, since every step increases width by at least h . Choose a fixed finite rational grid meeting every interval $(t + b_0/4, t + b_0/2)$ for $t \in [t_*, t_* + r]$. Choose g with $A_0g < b_0/4$. For each step take a grid point t'_i in this interval for $t = t_i + h$. It is admissible in (shift) for the entire I_i . On $I_i \cap Z_0$ it satisfies $t'_i < T(y) \leq \sigma(y)$, and hence $m_{t'_i}(y) = 0$. Integrating (shift) divided by h over I_i gives

$$M_X(t_i, v_i; h, g) \leq \int_{I_i} F_h(t'_i, y) dy.$$

On the good portions this integrand is dominated by

$$\sum_{t' \text{ in the fixed grid}} \mathbf{1}_{\{t' < \sigma(y)\}} F_h(t', y),$$

which tends to zero almost everywhere and is bounded. Notice that only grid times below the threshold are included. All visited widths lie in a window W_r of length $O(r)$ based at v_* . Disjointness and bounded convergence give

$$\sum_i M_X(t_i, v_i; h, g) \leq C_0 |W_r \setminus Z_0| + o_{h \rightarrow 0}(1).$$

First choose r using density so that the first term is small compared with Γr , then $g \ll \varepsilon r$, and only then let $h \rightarrow 0$. Telescoping (step), using (enl) and endpoint nonnegativity, gives

$$\Gamma(r - h) \leq C_0 \varepsilon r + C_0 g + C_0 |W_r \setminus Z_0| + o_{h \rightarrow 0}(1),$$

which is impossible with these choices.

Suppose finally $J = 0$. Fix $\varepsilon > 0$ and a compact positive-length interval I_0 of starting times. Let L_0 exceed the speed bound by a fixed margin. Fubini's theorem gives

$$\int_{I_0} |\{0 < v < L_0 r : \sigma(v) \in [t - r, t + 2r]\}| dt \leq 3L_0 r^2.$$

For r sufficiently small, there is therefore $t_* \in I_0$ with transition-set length at most εr . In $(0, L_0 r)$ put

$$P = \{\sigma(v) < t_* - r\}, \quad Z_0 = \{\sigma(v) > t_* + 2r\}.$$

The remaining set has length at most εr . Widths in P have positive trace throughout the slab $[t_*, t_* + r]$. Choose a rational $t' \in (t_* + 5r/4, t_* + 7r/4)$; the trace vanishes at that time for almost every width in Z_0 . Approximate P to symmetric-difference error εr by a finite union P' of intervals, with N endpoints. Now choose g with $A_0 g < r/8$ and $C_0(N + 1)g \leq \varepsilon r$, and subsequently let $h \rightarrow 0$ with $h \ll g$ and $h < \varepsilon r/2$.

Start at $(t_*, \varepsilon r)$. Outside P' make an off-gap step; inside P' apply the positive-segment comparison up to its next right endpoint or the final time. Stop immediately if the terminal alternative (7.8) occurs. Consistent half-open endpoint conventions prevent zero-length transitions. Width increases before any terminal freeze, so there are at most $N + 1$ positive segments. Changes between E and E^g cost at most $C_0(N + 1)g$. Their positive-segment width intervals are disjoint and lie in P' , so the total bad-width charge is at most $C_0 |P' \setminus P| \leq C_0 \varepsilon r$.

For off-gap steps the intervals $I_i = [v_i - h, v_i]$ again have disjoint interiors. Since their right endpoints are outside P' , their intersections with P' lie within distance h of its endpoints. Consequently

$$\left| \left(\bigcup_i I_i \right) \cap P \right| \leq \varepsilon r + 2Nh.$$

The one fixed t' is admissible in (shift) for every step. On Z_0 , Equation (7.6) and bounded convergence give zero limiting integral; on the transition set and the parts of P just estimated use $F_h \leq K_* + 1$. Hence

$$\sum_i M_X(t_i, v_i; h, g) \leq C_0 \varepsilon r + C_0 N h + o_{h \rightarrow 0}(1).$$

Each off-gap step pays Γ times its duration and each positive segment pays 2Γ , subject to the listed errors. A terminal freeze pays to the final time; otherwise less than h remains unused. Initial seam equality, initial displacement, and final nonnegativity yield

$$\Gamma(r - h) \leq C_0 \varepsilon r + C_0(N + 1)g + C_0 N h + o_{h \rightarrow 0}(1).$$

Let $h \rightarrow 0$ with earlier choices fixed. The choice of g gives a contradiction for ε sufficiently small, completing the proof. $\square$

7.5 Exhaustion of the affine configurations

Proof of Theorem 3.6. First suppose a classical extremal configuration has $\Gamma > \eta/2$. After it is excluded, suppose a hybrid configuration has $\Gamma > \eta/2$. It then has the strict gap $\Gamma > \max(\Gamma_{\text{cl}}, \eta/2)$ required for (Pl). The affine reduction (F) and the zero-cost arguments leave precisely the positive-cost cases treated below. All uses of (surplus) are on valid branches with the indicated crossing or spreading hypotheses.

We describe once the integration conventions. Almost-everywhere estimates on affine rays are integrated on generic arbitrarily nearby rays and passed to the desired one by Lipschitz continuity. A speed with surplus 2Γ may be decreased slightly while retaining surplus at least Γ . Rays from the terminal corner can start at a small positive backward time, with initial E tending to zero. If a high-rate linear portion is encountered, the freezing alternative in the proof of Lemma 7.5 applies.

We also require a short-slab observation. Suppose $b'_1 > 0$ and choose a typical j with $E(j, u(j)) = 0$. Work in a short slab ending at j , omitting a small fixed fraction next to its final edge. For $v > u(j)$ sufficiently close to $u(j)$, the pair (λ_j, X^v) recovers $\mathcal{C}(a, v)$ to small lists throughout the shortened slab. In the unclipped quadratic case, choose it so that $b(a) + v < b_1(j)$. If R_j is the angular profile given λ_j , conditioning gives

$$m(a, v) \geq R'_j(v) \quad \text{for almost every such } (a, v). \quad (7.9)$$

An integrated lower slope for R_j through $u(j)$ supplies positive-measure sets of arbitrarily close widths with derivative at least that slope minus any prescribed strict error. Otherwise integration would contradict the lower slope. A bound above capacity is already impossible. Choose these widths generically for all almost-everywhere assertions. If the supplied derivative exceeds a freezing threshold, (surplus) holds throughout the slab, except for any stipulated small time-density error. Its positive integral contradicts nonnegativity: the terminal width perturbation and the omitted edge cost arbitrarily little by Lipschitz continuity and $E(j, u(j)) = 0$. When a threshold depends on the mixed trace ν , choose j to be a Lebesgue point of that trace. The strict margin at j then survives off a time set of arbitrarily small relative length. All extra j -profile queries are readout refinements after j has been selected.

Finally, if $p_* > b_1$ on a positive-measure set of interior times, monotonicity of p_* and continuity of b_1 give a smaller open interval with uniform strict prediction slack. Otherwise $p_* = b_1$ at almost every time under consideration. We call these the strong and weak alternatives, respectively.

Equality curvature: $q = c_2 > 0$. Put $c = c_2$ and $C_1 = c - J = b'_1$. If $J > 0$, consider left quadratic points $0 < v < u(a)$ with $l < \min_{[a, 1]} b_1$ and, in hybrid, $w < 0$. Write $v = u(j)$ for a later $j < 1$. Left memory gives

$$m \geq \frac{k_0 + 2\Gamma + \eta(L_{\text{raw}} - 1)}{L_{\text{raw}}} > 0, \quad L_{\text{raw}} = \max\{1, (1 + c)/2, 1 + c - J\}. \quad (7.10)$$

The coarse-side construction and avoidance supply radial spreading, including the extra left-slack version when $c = 1$. If $0 < c \leq 1$, then $L_{\text{raw}} \leq 1 + c$, so (7.10) exceeds the quadratic freezing threshold $k_0/(1 + c)$. On every compact interior time slab all sufficiently small $v > 0$ meet the hypotheses. Integrate at those widths, let $v \downarrow 0$, and then let the trimmed endpoint tend to 1. The equality $E(1, 0) = 0$ contradicts the resulting positive drop. The same argument applies for $c > 1$ when $L_{\text{raw}} \leq 2$, using the threshold $k_0/2$.

If $c > 1$ and $J \geq (c+1)/4$, use a ray from the terminal corner at a speed just below $(c+1)/4$. It stays in the left region, has $2k < c$, and satisfies the derivative prediction condition; if b_1 decreases forward, $l < 0$ suffices. The positive quadratic motion in (surplus) is contradictory. This covers hybrid equality configurations because (Pl) gives $J = (c+1/2)/2$ there.

The remaining configurations are classical and have $b'_1 > 0$: either $J = 0$, or $c > 1$ and $0 < J < (c+1)/4$, with $C_1 > 1$. Indeed if $C_1 \leq 1$ in the latter range, then $L_{\text{raw}} \leq 2$, a case already excluded. In a strong interval use the unclipped quadratic template near the seam with exact side and prediction slack. Avoidance supplies radial spreading; at $c = 1, J = 0$ this is supplied by Lemma 7.2. If $c \leq 3$, the strengthened memory bound (mr) uses $L = \max\{1, (1+c)/2\}$. For $c < 1$ it exceeds $k_0/(1+c)$, and for $1 \leq c \leq 3$ it exceeds $k_0/2$. Equation (7.9) therefore gives the short-slab freezing contradiction. Shrink the slab to retain $l < p_*$ throughout. If $c > 3$, apply Lemma 7.5: its positive speed is $k_+ = (c+1)/4 > J$, and the finite-step speeds have minimum $c/2 > J$. They are all at least 1, the normal-width exponent increases forward, and no high-rate switch is needed.

In the weak alternative use the clipped template. Its right branch $v > u(a)$ is linear of rate C_1 , with exact construction and crossing at almost every weak time. If $c < 1$, necessarily $J = 0$ here and $C_1 = c$. At a typical mixed-trace time j , use both memory inequalities (mr). Together with (7.9), their strict margin gives positive m above $k_0 - cv(a)$ outside an arbitrarily small relative set of times. The low-rate linear freezing bound contradicts the short-slab observation. In all other weak cases $C_1 \geq 1$. For $0 < m \leq k_0$ use $k_+ = (1+C_1)/2$, and finite-step speeds have minimum C_1 ; both are strictly greater than J in the remaining range. For $m > k_0$, freeze on the right by the linear bound. A continued freeze crossing left uses the quadratic freezing bound, the same monotone clipped-template trace, coarse side, and line avoidance. The forward width rates are C_1 on the right and c on the left, both positive. Lemma 7.5 applies and gives a contradiction.

Pure linear cost: $c_2 = 0 < c_1$. Write $c = c_1$. The linear construction has crossing. If classical and $c < 1$, use both inequalities (mr) at a typical mixed-trace time and the short-slab argument just given; there is now no side-of-seam restriction. In hybrid, (Pl) gives $J = c + 1/2$. Use a ray from the corner at a speed slightly below c for $c < 1$, and slightly below $(1+c)/2$ for $c \geq 1$. It lies in $v < u(a)$, $w < 0$, and left memory gives

$$m \geq \frac{k_0 + 2\Gamma + \eta c}{1 + c} > 0.$$

For $c < 1$ this exceeds the threshold $k_0/(1+c)$ for linear motion at speed c ; for $c \geq 1$ use the stated positive linear motion while $m \leq k_0$. If $m > k_0$ on a positive time set, switch at a regular point to freezing. Its hypotheses remain valid backwards. These motions contradict nonnegativity and terminal equality. The same argument, with exact side, works classically for $c \geq 1$ and $J \geq (1+c)/2$. For the remaining classical case $c \geq 1, J < (1+c)/2$, use Lemma 7.5 on the pure linear template: $k_+ = (1+c)/2 > J$, finite-step speeds are at least $c > J$, and any high-rate freeze stays in a linear branch.

Strict curvature: $q < 1 = c_2$. Put $C_1 = q - J < 1$ and use the artificial template ending at D . At $v_0 = u(D)$ memory supplies the lower slope

$$\frac{k_0 + 2\Gamma + \eta(L_{\text{raw}} - 1)}{L_{\text{raw}}}, \quad L_{\text{raw}} = 1 + \max(0, C_1).$$

If $C_1 \leq 0$, then $L_{\text{raw}} = 1$, so this is positive and at least $1 - 2d$. This explicitly includes every negative q allowed by the affine reduction. If $C_1 > 0$, the same conclusion follows whenever $d \geq C_1/(1 + 2C_1)$, since $(1 - d)/(1 + C_1) \geq 1 - 2d$ is exactly that inequality. Take $D = 1$, integrate Proposition 7.3, and contradict $E(D, v_0) = 0$.

It remains that $C_1 > 0$ and $d < C_1/(1 + 2C_1)$. If $C_1 \leq 1/2$, then $d < 1/4$. From the terminal corner $D = 1$ take a quadratic ray of speed slightly below $1/2$. It satisfies $l < \min_{[a, D]} b_1$ and $w < 0$ strictly in the interior: along a speed $k < 1/2$ ray these are $-(1 - k)(1 - a) < -C_1(1 - a)$ and $-(1 - 2k)(1 - a) < 0$. The coarse side is valid. The unit-cost quadratic motion bound without spreading, which also permits $m = 0$, gives the contradiction. All hybrid strict configurations have been covered: (P1) gives $J = (q + 1/2)/2$, so $C_1 = q/2 - 1/4 < 1/4$.

Finally suppose $1/2 < C_1 < 1$. The configuration is classical, and $2d < C_1$. If the raw final derivative has a strong interval, choose a typical D in it, retaining prediction slack immediately to its left. The strengthened memory estimate (mr) has $L = 1$ for this boundary comparison via the encountered λ_D on final paths. Proposition 7.3 again contradicts $E(D, u(D)) = 0$. Otherwise take the clipped artificial template with $D = 1$ and move from the corner at speed 1, using generic small perturbations. This is strictly right of the clipping switch $v = b_1(a) - b(a) = (1 - C_1)(1 - a)$. The classical linear branch has crossing at almost every weak time. Its arbitrary- m motion bound for $2d < C_1 < 1$, including $m = 0$, gives the contradiction without requiring a positive angular trace.

Every affine positive-cost configuration has now been excluded. Together with the zero-cost arguments this proves the classical bound, and subsequently the hybrid bound with its required strict separation, as claimed. $\square$

8 Round cones and the reduction of thin shapes

The round model has two transverse directions. Its regularity tests include neighborhoods of null hyperquadrics. A thin neighborhood of one such hyperquadric becomes a cylinder after a conformal change of coordinates and a rescaling in one transverse direction. The purpose of this Section is to justify this reduction, including its effect on wave packets and regularity.

8.1 The round experiment

Definition 8.1 (Round experiment). Write a spacetime point as $(t, b, y) \in \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}$, and put

$$q(t, b, y) = ty - |b|^2, \quad L_A = \partial_t + A \cdot \partial_b + |A|^2 \partial_y.$$

The classical trajectories are $(B + tA, Y + t|A|^2)$, with arbitrary positive phase energy of total mass e . The initial phase support has finite-power size, and its density relative to Lebesgue measure is at most e times a finite power of the final scale. These bounds are uniform for the families in a fixed setup. Passive indices, when present, range over standard Borel spaces; separable Hilbert multiplicity is unrestricted. In the wave experiment the initial datum belongs to $L^2(\mathbb{R}^3; \mathcal{H})$, where $\mathcal{H}$ is any separable Hilbert space, and has squared norm e . Its evolution is

$$\widehat{U(t)v}(\xi, -\rho) = e^{-it|\xi|^2/(4\rho)} \widehat{v}(\xi, -\rho).$$

The parameter $H > 0$ and its reciprocal have finite-power size, and $H\rho$ belongs to a fixed compact subinterval of $(0, \infty)$. At length ℓ set

$$d_\ell = (H/\ell)^{1/2}, \quad q_\ell = (H\ell)^{1/2}.$$

The analyzing operator V_ℓ applies, for each $A \in \mathbb{R}^2$, the multiplier

$$\chi(H\rho)\psi((\xi/(2\rho) - A)/d_\ell), \quad \int_{\mathbb{R}^2} |\psi|^2 = 1,$$

and then takes the inverse spatial Fourier transform. Its row measure is $d_\ell^{-2} dA db dy$. Here ψ is smooth and compactly supported, and χ is smooth, supported in a positive band, and bounded by one. A *plateau analysis* has $\chi = 1$ on the incoming spectrum. Such an analysis is isometric; every analysis is contractive. Masks act pointwise by contractions on the rows and are followed by synthesis V_ℓ^* . After synthesis, subsequent plateau cutoffs may use a fixed band enlargement. Only finitely many cuts at fixed depths are used on any inner sequence. Initial row energy outside a finite-power phase box is smaller than every power of the final scale, relative to e .

For a classical experiment put $d_\ell = 0$. In both versions, μ_I denotes the row energy at the left endpoint of the interval I . The following normalization specifies the tests without depending on a choice of chart. For a test at time T , length ℓ , and width $r > 0$, use coordinates

$$\begin{aligned} u &= (t - T)/\ell, & x &= (b - b_* - \ell u A_*)/(\ell r), \\ z &= (y - y_* - 2A_* \cdot (b - b_*) + \ell u |A_*|^2)/(\ell r^2), & V &= (A - A_*)/r. \end{aligned}$$

The normalized rays again have velocities $(V, |V|^2)$.

Definition 8.2 (Round regularity). A round test imposes $|x|, |z|, |V| \leq 1$ at $u = 0$ and has $f = 1$. A hyperquadric test additionally imposes

$$|F| \leq f, \quad |G| \leq f, \quad 0 < f \leq 1,$$

where

$$F = c + du + \alpha \cdot x + \beta z + k(uz - |x|^2), \quad G = L_V F,$$

and

$$|\alpha|^2 - 4\beta d + 4kc = 1, \quad \max(|c|, |d|, |\alpha|, |\beta|, |k|) \leq 100.$$

Every test must satisfy $rf \geq d_\ell$. Its weight is $w = r^4 f^{1-\eta}$, where $0 < \eta < 1$ is fixed. Regularity with constant κ means $\mu_I(E) \leq \kappa w(E)$ for every test E . Initial regularity is required, and e/κ and κ/e have finite-power size. At a final cut one may assign measures dominated in sum by the row energy, with each assignment supported on its own test. Writing their masses as m_λ , the normalized functional is

$$\frac{s^{-1/2}}{e\sqrt{\kappa}} \sum_{\lambda} m_\lambda^{3/2} w_\lambda^{-1/2}.$$

Direct differentiation gives

$$L_V G = 0, \quad L_V g = 0, \quad g = \partial_V G = \alpha + 2(\beta + ku)V - 2kx, \quad |g|^2 = 1 + 4(\beta + ku)G - 4kF. \quad (8.1)$$

Translations, boosts with their indicated vertical tilts, rotations, and the preceding parabolic changes of scale preserve the class of experiments. The parameter in normalized coordinates is $H/(\ell r^2)$, and the change in regularity is exactly reciprocal to the change in weights. Multiplying the datum by the square root of the spatial Jacobian preserves its energy.

Lemma 8.3 (Packet locality). *For $\ell' \leq \ell$ and $|u| \lesssim \ell$, the kernel of $V_{\ell'} U(u) V_{\ell}^*$ vanishes unless $|A' - A| \lesssim d_{\ell'}$. On this angular support its absolute value is at most*

$$\frac{C_N}{q_{\ell}^2 H} \left(1 + \frac{|b' - b - uA|}{q_{\ell}} + \frac{|y' - y - u|A|^2 - 2A \cdot (b' - b - uA)|}{H} \right)^{-N}. \quad (\text{cone-kernel})$$

Consequently, after any fixed finite stack, retained correspondence paths from a row at cut i to a row at cut j satisfy

$$\begin{aligned} |A_j - A| &\leq s^{-O(\varepsilon)} d_{\ell_j}, \\ |b_j - b^p| &\leq s^{-O(\varepsilon)} q_{\ell_i}, \\ |y_j - y^p - 2A \cdot (b_j - b^p)| &\leq s^{-O(\varepsilon)} H. \end{aligned} \quad (\text{acc})$$

Here (b^p, y^p) is the prediction along the earlier row's ray. Removing all kernel pairs outside these correspondences changes the operators by an arbitrarily small power error, for every prescribed $\varepsilon > 0$.

Proof. The two angular symbols have disjoint supports unless the angular assertion holds. On their overlap use frequency coordinates $q_{\ell}(\xi - 2\rho A)$ and $H\rho$. The amplitude is compactly supported and smooth with uniformly bounded derivatives. The quadratic part of the propagation phase has the same property because $uH/q_{\ell}^2 = O(1)$. The remaining linear phases are precisely the two position differences in Equation (cone-kernel). Integration by parts gives that bound, including its normalization $1/(q_{\ell}^2 H)$. The row measures and the inverses of the packet scales have finite-power size. Schur's test therefore makes the omitted operator norm smaller than any prescribed power by choosing N last.

For a path through a finite sequence of cuts, an angular change at length ℓ_m acts for at most $O(\ell_m)$ additional time. It contributes $O(q_{\ell_m})$ in transverse position and $O(\ell_m d_{\ell_m}^2) = O(H)$ to the tilt-corrected vertical position. Replacing an intermediate tilt by the original tilt costs at most $O(d_{\ell_m} q_{\ell_m}) = O(H)$, with the same dilation factors. Summing finitely many such bounds proves Equation (acc). Successive truncation and contraction also show that an output region with no retained predecessor path has negligible energy. This last assertion uses operator norms and does not posit a positive transition kernel for the wave. $\square$

Lemma 8.4 (Enlarged tests). *Suppose $P \geq 2$. Enlarge the normalized round bounds by fixed powers of P , allow coefficients bounded by fixed powers of P with discriminant one, and replace both sublevels by $P^C f$. The resulting set has mass at most $\kappa r^4 f^{1-\eta} P^{C'}$, provided $r f \geq d_{\ell}$. The exponent C' depends only on the stated enlargement exponents. Restriction, synthesis, and plateau reanalysis at one cut consequently preserve regularity with an arbitrarily small exponent loss.*

Proof. If f exceeds a sufficiently small inverse power of P , cover by round tests; their number and the reciprocal of $f^{1-\eta}$ are polynomial in P . Otherwise Equation (8.1) gives $|g| \asymp 1$ on the set. Subdivide its normalized phase box into parabolic boxes of transverse width $a = P^{-D}$, centered on actual witnesses. After centering at a witness, boosting to its ray, applying the parabolic change of scale, and dividing the polynomial by a , its transverse linear coefficient is g at that witness. Its vertical and quadratic coefficients are $O(aP^C)$; its constant and central-ray derivative are $O(P^C f/a)$. Choose D first and the smallness threshold for f second. All coefficients then satisfy the bounds in Definition 8.2, after bounded further subdivisions. The discriminant remains

one. The new relative thickness can be taken as $P^{C''}f/a$; thus the absolute uncertainty width is at least rf . There are polynomially many boxes, and their total test cost has the asserted bound.

For reanalysis, Equation (cone-kernel) puts all relevant predecessors of a test inside such an enlargement with $P = s^{-O(\varepsilon)}$. Apply the preceding covering to their union and use contraction on that union. Its complement contributes an arbitrary power error. Finally let ε tend to zero after the inner scale limit. $\square$

All subsequent polynomial truncations are legitimate under Definition 8.1. Here is the useful quantitative reason. For a thin test, Equation (8.1) gives a velocity sublevel of normalized area $O(f)$ at each allowed position. A finite-power phase density therefore bounds mass divided by ew by $s^{-C}r^2f^\eta$. In the classical compact-support case, slicing instead in the original velocity coordinates gives the additional bound $s^{-C}r^{-3}f^\eta$ for large r . In the wave case $rf \geq d_\ell$ excludes arbitrarily thin tests at a fixed large width, and the first bound together with total mass excludes extreme widths. Thus, above any fixed power floor, widths, their reciprocals, and centers can be restricted to finite-power ranges. Summed assignment mass, rather than the number of labels, controls the discarded small assignments. Error estimates after subdivision use the original energy as reference; components too small to normalize are bounded directly by contraction and the lower weight cutoff.

Lemma 8.5 (Single-time estimate). *Relative to regularity at a parent cut, the energy in a test of widths (r, f) at any one descendant of relative length δ satisfies*

$$m \leq s^{-o(1)} \kappa r^4 f^{1-\eta} \delta. \quad (\text{single})$$

The same conclusion holds through any fixed finite stack.

Proof. Normalize angular units by r , but keep the parent time and position units. At the target time θ , the test has $|b|, |y| \leq \delta$, $|A| \leq 1$, and, in the non-round case,

$$F = \alpha \cdot b + \beta y + c\delta + d(t - \theta) + k((t - \theta)y - |b|^2), \quad |k| = K \leq 100/\delta,$$

with $|F| \leq \delta f$, $|G| \leq f$. The remaining coefficients are bounded, and the discriminant is one. Write $P = s^{-O(\varepsilon)}$. The uncertainty condition gives $d_1 \leq f\sqrt{\delta}$, and retained predecessors satisfy

$$|A - A_*| \leq Pf, \quad |b - b_*| \leq Pd_1, \quad |y - y_* - 2A \cdot (b - b_*)| \leq Pd_1^2.$$

Expanding first in velocity and then in position shows $|G| \leq P^C f$ and $|F| \leq P^C f$ on the predecessor rays at the parent time. The potentially largest quadratic error is $Kd_1^2 \lesssim f^2$.

Cover angles by caps of width $a \gtrsim \max(d_1, \delta)$, $a \leq 1$. Their number is at most

$$P^C a^{-2} (K\delta + f + a).$$

Indeed the relevant angles obey $|d + \alpha \cdot A + \beta|A|^2| \lesssim P^C (K\delta + f)$. When this sublevel and its a -enlargement are small, its gradient is bounded below by the discriminant identity; slicing the quadratic gives the claimed area bound. Otherwise the full angular area suffices. For a round test use $K = 0$ and $f = 1$ in these counts.

Within a cap centered at A_0 , only P^C transverse position cells of width a are needed. The required length in $y - 2A_0 \cdot b$ is at most the following quantity, with $(f + a)/K = \infty$ when $K = 0$:

$$P^C \left(a^2 + \min\{\delta, (f + a)/K\} \right).$$

This follows by writing $G = d + \alpha \cdot A_0 + \beta |A_0|^2 + k(y - 2A_0 \cdot b)$; the error is $P^C(a + Kd_1a)$. Divide the displayed length by a^2 to count vertical cells. Backflow preserves the resulting parent round boxes up to bounded enlargement.

Put $a_0 = \min(\sqrt{\delta}, 1/K)$, with the same convention when $K = 0$. If $f \leq a_0$, take $a = a_0$ and retain the hyperquadric condition. Lemma 8.4, applied after dividing the recentered polynomial by a , bounds the cost of each box by $P^C \kappa r^4 a^{3+\eta} f^{1-\eta}$. For $K \leq \delta^{-1/2}$ the total cost, divided by $\kappa r^4 f^{1-\eta}$, is at most $P^C \delta^{1+\eta/2}$; for $K > \delta^{-1/2}$ it is at most $P^C \delta K^{-\eta}$. Both are $O(P^C \delta)$.

If $f > a_0$, take $a = \max(a_0, d_1)$ and use only round bounds. Their total cost divided by $P^C \kappa r^4$ is

$$(K\delta + f + a) \left(a^2 + \min\{\delta, (f + a)/K\} \right) \lesssim \delta f.$$

For $K \leq \delta^{-1/2}$ this uses $a = \sqrt{\delta}$ and $f > \sqrt{\delta}$. For larger K , use $a \lesssim f$, $a^2 \lesssim K^{-2} + \delta f^2$, and $K \lesssim \delta^{-1}$. Since $f \leq f^{1-\eta}$, this proves the desired bound. Contraction on the covered predecessors, followed by Lemma 8.3, handles the wave and finite-stack versions. Letting ε decrease gives Equation (single). $\square$

8.2 Free gaps and regularity splitting

For each version, let Γ be the supremum of upper exponents of the normalized functional for free experiments, with the closure convention of the framework. Widths and parameters have fixed finite-power ranges on each inner sequence; exterior limits of their exponents are allowed. Equation (single) and total mass over $O(s^{-1})$ cuts imply $\Gamma \leq 1$, so this upper exponent is finite.

Proposition 8.6 (Composition of free gaps). *Inserting finitely many admissible pointwise row contractions followed by synthesis, as in Definition 8.1, at fixed depths does not increase Γ . Repeated operations at the same cut are permitted. Reconstruction uses plateau analyses, while an old non-plateau terminal analysis may be retained for coherent recombination.*

Proof. At each new cut peel off successes from tests at descending dyadic regularity thresholds b , from a high power multiple of the incoming regularity down to a low power multiple. The top threshold exceeds a crude polynomial regularity bound. At threshold b , the entering remainder is $O(b)$ -regular because the preceding threshold $2b$ has been exhausted. A success has mass exceeding bw . All success portions at this level are disjoint submeasures of that entering remainder, so their union is also $O(b)$ -regular, even if the peeling returns to previously used tests. Synthesize this union as one state per interval and threshold, not as one state per success. The polynomial weight cutoff makes the process finite; the number of threshold levels is logarithmic. Synthesis and Lemma 8.4 give post-states with regularity $s^{-o(1)}b$. If z is the raw success mass summed over intervals, the sum of post-state energies is at most z .

At depth $a \in (0, 1)$ the raw prefix cube sum is at least $z\sqrt{b}$. The free bound on the preceding segment gives

$$s^{-a/2} z \sqrt{b} \leq s^{-a\Gamma - o(1)} e \sqrt{\kappa}.$$

The free bound on the remaining segment therefore bounds its final cube sum by $s^{1/2 - \Gamma - o(1)} e \sqrt{\kappa}$. Summing the logarithmically many levels costs zero exponent. The low-floor remainder is bounded using its tiny regularity and the total cut energy. Applying the same argument on each parent proves the relative statement for two interior cuts and, by finite induction, the full assertion.

At a repeated cut, Lemma 8.4 supplies the necessary regularity without a positive-length gap. At the last cut, split before the terminal operators and use Lemma 8.5 together with total

mass. For coherent recombination freeze the allocation densities $a_\lambda \geq 0$ on the output rows, with $\sum_\lambda a_\lambda \leq 1$ at each cut. If T_λ is the corresponding output map, then

$$v \mapsto \left(\sum_\lambda w_\lambda^{-1/2} \|\sqrt{a_\lambda} T_\lambda v\|_2^3 \right)^{1/3}$$

is a seminorm by the triangle inequalities in L^2 and weighted ℓ^3 . The old terminal cutoff can consequently be retained throughout. $\square$

Proposition 8.7 (Saturation at inserted cuts). *Suppose $\Gamma > 0$. In the outer closure of homogeneous free exponent data, let p_0 be the minimal mass exponent at upper exponent Γ , and let d be the associated total-mass exponent. Then*

$$\Gamma = d + (1 - p_0)/2, \quad p_0 \geq 1, \quad 0 < d \leq 1.$$

Fix any finite depth list. After realizing these data with arbitrarily small exterior errors, every regularity splitting preserving the terminal functional has limiting dominant regularity and summed energy described by

$$\kappa_a = \kappa s^{p_0 a + o(1)}, \quad E_a = e s^{-d a + o(1)}. \quad (\text{CE})$$

In particular, an arbitrarily short fixed final gap may be taken free, with a regularly split incoming state, when classifying terminal shapes. Equation (CE) denotes identities of exponent data in this outer closure; it does not assert that one inner sequence attains the closure extrema exactly.

Proof. Homogenization gives $m_\lambda/(\kappa w_\lambda) = s^{p_0 + o(1)}$ and hence the identity for Γ . Equation (single) gives $p_0 \geq 1$, while total energy at $O(s^{-1})$ cuts gives $d \leq 1$; positivity of Γ gives $d > 0$. The retained terminal tests have total weight at most $s^{-(p_0 + d) + o(1)} e/\kappa$.

For completeness, minimality must be applied to free gaps, not simply asserted for masked experiments. Start with any saturated finite stack having terminal homogeneous exponent p' . Insert ordinary regularity splits immediately before each positive-depth group of operators, including terminal operators when necessary. Between successive splits, use the post-synthesis state as input and the next raw peeling successes as output. This is a free experiment. Every bound in the composition proof must be sharp in exponent, since otherwise the terminal functional loses a fixed power. Thus using full raw mass instead of retained mass, or inherited regularity instead of sharp regularity, also loses no power. The regularity exponent increment on each free gap is at least p_0 times its depth length: sum the uniform free bounds over starting intervals and select a near-maximal one, then apply the definition of p_0 to its homogeneous raw successes. The same argument applies to a last free gap ending in the retained final assignments. If terminal operators occur at that depth, their output satisfies $m_\lambda \leq s^{-o(1)} b_{\text{end}} w_\lambda$ by inherited regularity; the final homogeneous exponent is therefore at least the accumulated regularity exponent across the preceding free gaps.

To account for changes in masses under coherent splitting, denote the new final homogeneous exponent by p'' , and put $d'' = \Gamma + (p'' - 1)/2$. Free-gap minimality gives $p'' \geq p_0$. The retained labels still have total weight at most $s^{-(p' + d') + o(1)} e/\kappa$, where $d' = \Gamma + (p' - 1)/2$. For their new saturated masses the same total weight has exponent $p'' + d''$. Hence $p'' + d'' \leq p' + d'$, so $p'' \leq p'$. This proves $p' \geq p_0$ without identifying the masses before and after splitting. Starting with $p' = p_0$ forces equality on every free gap. The sharpness of the energy sums in Proposition 8.6 then gives both identities in Equation (CE). A split originally made without all these isolating

splits satisfies the same conclusion by applying the just-proved inequality $p' \geq p_0$ separately to its prefix and suffix. To implement this in the outer closure, fix the finite depth list first and realize the extremal data to arbitrary exterior accuracy. Single-time bounds, total energy, and sharpness of the composed estimates bound all added intermediate exponents. Extract convergent subsequences of these finitely many exponents and then let the exterior error vanish. The preceding inequalities pass to their limits and give Equation (CE). These exterior errors are distinct from inner $o(1)$ errors. Therefore a further fixed final depth can be inserted as close to the endpoint as desired, leaving its continuation free. $\square$

Proposition 8.8 (Thin endings reduce to cylinders). *Let $\Gamma_{\text{cl}}^{\text{cyl}}$ and $\Gamma_{\text{hyb}}^{\text{cyl}}$ be the classical and hybrid cylinder exponents. A saturated round experiment with terminal thickness of positive limiting exponent satisfies*

$$\Gamma \leq \Gamma_{\text{cl}}^{\text{cyl}} \quad \text{in the classical case,} \quad \Gamma \leq \max(\Gamma_{\text{cl}}^{\text{cyl}}, \Gamma_{\text{hyb}}^{\text{cyl}}) \quad \text{in the wave case.}$$

The conclusion includes infinite limiting thickness exponents.

The proof has three reductions. First group the thin ending tests around one hyperquadric. Then flatten that hyperquadric and transfer both input regularity and terminal assignment capacities through its conformal chart. Finally freeze the very fine base variables while retaining the normal quantum evolution; the factors introduced by the normal rescaling will cancel.

8.3 Conformal charts and thin families

We now work on a free gap of relative length $\delta = s^h$, where $h > 0$ is fixed before taking the scale limit. If the ending thickness has strictly positive limiting exponent, h can be chosen so small that

$$f \leq \delta^Z \tag{8.2}$$

for any prescribed fixed Z . Normalize by the common ending angular width and the parent time length. Ending weights are then comparable to $f^{1-\eta}$, and $d_\delta \lesssim f$.

Packet backprojections let us split into bounded parent phase charts: use unit angular cells, tilted unit position cells, and Lemma 8.3. A label meets only boundedly many such source cells, their raw energy budgets sum to at most e , and Lemma 8.4 preserves input regularity. Select a near-maximal chart by the uniform free bound. At positive saturation we may retain homogeneous labels satisfying

$$m_\lambda \geq \delta^4 \kappa f^{1-\eta}, \quad \#\{\lambda\} \lesssim \frac{\delta^{-5}e}{\kappa f^{1-\eta}}. \tag{8.3}$$

Indeed $\Gamma > 0$ implies that the homogeneous mass exponent of this free gap is less than 3, and the second bound follows by summing cut energy. Comparable widths produce the same conclusions.

In the bounded parent chart, write the ending hyperquadrics as

$$F_\lambda = c_\lambda + d_\lambda t + \alpha_\lambda \cdot b + \beta_\lambda y + k_\lambda q, \quad n_\lambda = (c_\lambda, d_\lambda, \alpha_\lambda, \beta_\lambda, k_\lambda).$$

Their discriminant is one, $|n_\lambda| \lesssim \delta^{-1}$, and at their readout time θ one has $|\beta_\lambda + \theta k_\lambda| \lesssim 1$. Predecessors lie in bounded sets S_λ satisfying $|F_\lambda(0)|, |L_A F_\lambda| \leq \delta^{-1}f$, after harmless small power enlargements.

Lemma 8.9 (Flattening a null hyperquadric). *Let $|n| \leq P$, $P \geq 2$, with discriminant one. Near any bounded ray on which $|F(0)| + |L_A F|$ is smaller than a sufficiently high inverse power of P , there is a rational conformal chart carrying $F = 0$ to $b'_2 = 0$. A polynomial number of such charts cover the relevant rays. On buffered subcharts, both chart directions have derivative bounds $C_j P^{Cj+C}$, time along a ray increases at a rate comparable to one, and parabolic phase boxes of width $R \leq 1$ map into polynomial enlargements of boxes of the same width. The induced linear map on hyperquadric coefficients preserves the discriminant and has norm and inverse norm bounded by powers of P .*

Proof. Center on the given ray. Equation (8.1) makes its transverse coefficient nonzero. Move its initial point in that direction to set $F(0) = 0$, and then move its velocity in that direction to set $L_A F = 0$. The implicit scalar equations have derivatives bounded below; the changes are polynomial in P times the original errors. Center and boost on this exact ray, then rotate so that $F = b_2 + \beta y + kq$. First make the linear q -isometry

$$u = b_2 + \beta y, \quad t_{\text{new}} = t + 2\beta b_2 + \beta^2 y,$$

retaining b_1, y . In these coordinates put $v = (0, 0, k, 0)$ and

$$x' = \frac{x + vq(x)}{\mathcal{D}(x)}, \quad \mathcal{D}(x) = 1 + 2\langle v, x \rangle + q(v)q(x). \quad (\text{conf})$$

Here $\langle \cdot, \cdot \rangle$ is the polar form of q . Then $b'_2 = F/\mathcal{D}$, the inverse uses $-v$, and $q(x') = q(x)/\mathcal{D}(x)$. Differentiation gives metric scale $\mathcal{D}^{-2}$. The time axis is fixed and has no poles, so inverse-polynomial neighborhoods of its buffered segment have all the stated derivative and time properties.

Division by $\mathcal{D}$ carries a polynomial $c + lx + k'q(x)$ to

$$c(1 - 2\langle v, x' \rangle + q(v)q(x')) + l(x' - vq(x')) + k'q(x'),$$

which proves linearity, invertibility, and preservation of the discriminant. Null lines map to null lines, as is also immediate from the fractional-linear restriction of Equation (conf) to a null line. Their polar covector $dy - 2A \cdot db + |A|^2 dt$ is preserved up to a controlled nonzero scalar. On a phase box this controls the vertical displacement to first order; Taylor's remainder is $O(P^C R^2)$. Transverse and angular displacements are $O(P^C R)$. Backflow to a common time preserves these conclusions. Subdivide the bounded phase region into sufficiently small polynomial cells around actual witnesses to obtain the stated covering. Boxes larger than the chart buffer use a crude polynomial covering instead. $\square$

Lemma 8.10 (Overlap of thin predecessors). *For $K = \max(1, \min_{\pm} |n_{\lambda} \pm n_{\lambda'}|/f)$,*

$$\mu_0(S_{\lambda} \cap S_{\lambda'}) \leq \kappa \delta^{-C-o(1)} f^{1-\eta}/K. \quad (\text{CI})$$

The fixed exponent C is independent of Z in Equation (8.2).

Proof. Flatten F_{λ} by Lemma 8.9, with P a fixed power of $1/\delta$. At time zero, $|A'_2|, |B'_2| \leq \delta^{-C} f$. Dropping these small coordinates from the second hyperquadric leaves, in $X = A'_1, B = B'_1, Y = Y'$, the two conditions

$$|c' + \alpha' B + \beta' Y - k' B^2| \leq \delta^{-C} f, \quad |d' + \alpha' X + \beta' X^2 + k'(Y - 2XB)| \leq \delta^{-C} f.$$

The five remaining coefficients are at most $\delta^{-C}Kf$. If $K > \delta^{-C}$, at least one of α', β', k' has size at least $\delta^C Kf$: otherwise nonempty intersection controls the two constants, while discriminant one forces the omitted transverse coefficient to be close to 1 or -1 , contradicting the definition of K and invertibility of the coefficient map.

Divide by f and apply the cylinder covering calculation Equation (I). It gives base cells of widths $R \gtrsim 1/M$, where $\delta^C K \leq M \leq \delta^{-C} K$, with $\sum R^{3+\eta} \leq \delta^{-C-o(1)}/K$. For $K \leq \delta^{-C}$ use a unit covering. Since $M \leq \delta^{-C}/f$, the extra second-coordinate tilts fit inside polynomial enlargements of full round cells. Pull them back by the chart. In each cell, the first shell and Lemma 8.4 give cost $\kappa \delta^{-C} R^{3+\eta} f^{1-\eta}$, using relative thickness f/R . If $R < f$, the ratio f/R is polynomially bounded and round tests give the same conclusion with polynomial loss. Their uncertainty widths may be enlarged by that same factor. Summing proves the result. $\square$

For a fixed label, partners at dyadic coefficient distance K number at most $\delta^{-C} K^{1-\eta}$. Their predecessor union lies in the first shell enlarged to thickness $\delta^{-C} Kf$. Regularity and contraction bound assignable mass at each time by $\kappa \delta^{-C} (Kf)^{1-\eta}$; sum over times and use Equation (8.3). Equation (CI) and the total label count now give

$$\mu_0\{\text{a pair with } K > \delta^{-D}\} \leq e \delta^{-C} \sum_{K > \delta^{-D}} K^{-\eta}.$$

Choose D so that this is a positive power smaller than e . Restriction and synthesis preserve regularity; the uniform free bound makes the removed state's final contribution negligible at saturation.

Choose $D_1 > D + 5$ and put

$$f_* = \delta^{-D_1} f. \quad (8.4)$$

Round one attached coefficient vector, up to sign, on an f_* lattice to partition the remaining rows into groups. Each label connects to boundedly many groups. Choose an actual discriminant-one representative F_L in each group. All connected coefficients differ from it by $O(f_*)$, and all initial rows satisfy $|F_L(0)| + |L_A F_L| \lesssim f_*$. With raw budgets e_L one has

$$\sum_L e_L \leq e, \quad e_L \leq \kappa \delta^{-C} f_*^{1-\eta}. \quad (8.5)$$

Large quadratic coefficients cannot carry saturation. Indeed, if $|k_L| > \delta^{-b}$, connected times satisfy $|\beta_L + k_L \theta| \lesssim 1$ and lie in boundedly many intervals of length $\delta^{b/2}$. Split at that length. On the prefix the single-time estimate and the bounded number of active intervals give exponent zero, while the suffix has exponent at most Γ . Composition therefore gains $\delta^{b\Gamma/2}$ against a saturated gap. Sum this bound using Equation (8.5) before subdividing the other groups. We may thus retain only $|k_L| \leq \delta^{-b}$; all other coefficients are then $O(1 + |k_L|)$, by centering at a connected label and using its coefficient bounds. Lemma 8.9 partitions each retained group into at most δ^{-Cb} buffered charts. Their norm and overlap costs are arbitrarily small powers when b is sufficiently small.

8.4 Transporting a thin wave through a chart

Fix one low-coefficient buffered chart, with raw budget still denoted e_L , and write $P = \delta^{-Cb}$. Normalize the transformed time range to $[-1, 3]$ and round transformed terminal event times to the δ -lattice. At the ideal cylinder input, physical packet length 4 gives its normalized root

analyzer; replacing 1 by 4 below only changes fixed constants. Shrink the phase patch by polynomial factors of P so that throughout a slightly longer interval all relevant rays have valid inverse charts, positive time derivative comparable to one, and inverse-polynomial spatial buffers. Pulled incident covectors then have $\rho' > 0$, $\rho' \asymp \rho$, and transformed angle in a small cone about the time axis. Choose a larger safe cone with another polynomial buffer. These choices are possible by Lemma 8.9; all order- j cutoff derivatives are bounded by $C_j P^{Cj+C}$.

The algebraic covariance alone does not identify the transformed L^2 -energy. We obtain that comparison from packet kernels.

Lemma 8.11 (Conformal wave transport). *Let u be the free cone wave synthesized from the retained patch. There is an exact free cone wave u' in the transformed variables, with a fixed positive radial band. At packet length $\ell = 1$ or δ , write $p \sim_\ell p'$ when the old row p and transformed row p' are close to the mapped and backflowed ray in angle, transverse position, and tilt-corrected vertical position, with respective widths $\delta^{-\gamma} d_\ell$, $\delta^{-\gamma} q_\ell$, and $\delta^{-\gamma} H$; for terminal rows, include the corresponding chart event time rounded to the δ -lattice. Put $N_\ell(E') = \{p : p \sim_\ell p' \text{ for some } p' \in E'\}$. Let μ_{in} be the retained raw old row-energy measure at the parent input, and let μ'_{in} be the transformed analyzed row-energy measure at time -1 . For every prescribed $\gamma > 0$ and $N > 0$, if b and the auxiliary localization losses are sufficiently small and Z sufficiently large, the following hold:*

(i) *for every Borel set E' of transformed input rows,*

$$\mu'_{\text{in}}(E') \leq \delta^{-\gamma} \mu_{\text{in}}(N_1(E')) + C_N \delta^N e_L;$$

(ii) *for any finite old terminal fractional assignments of masses m_λ , there are fractional assignments to u' at the rounded times, each supported on rows associated with its old label, whose fractions sum to at most one on every transformed row. If m'_λ is the mass aggregated over the rounded times for label λ , then*

$$m_\lambda \leq \delta^{-\gamma} m'_\lambda + \epsilon_\lambda, \quad \sum_\lambda \epsilon_\lambda \leq C_N \delta^N e_L.$$

The predecessor sets are measured in the completion of μ_{in} .

Proof. Equation and spectral localization. The classical conformal covariance of the four-dimensional wave equation [1, Section 2] takes the following form in our coordinates. Write $\square = 4\partial_t \partial_y - \Delta_b$. For Equation (conf), put $w = \mathcal{D}^{-1}$. Direct differentiation gives

$$\square \mathcal{D} = 8q(v), \quad q^{-1}(d\mathcal{D}, d\mathcal{D}) = 4q(v)\mathcal{D}, \quad \square w = 0.$$

Similarly $\square(wx') = 0$, using

$$q^{-1}(d\mathcal{D}, d(x + vq(x))) = 2v\mathcal{D} + 2q(v)(x + vq(x)).$$

The differential metric identity therefore gives

$$\square(wF \circ \Phi) = w^3(\square F) \circ \Phi. \tag{8.6}$$

Apply this to the inverse chart and multiply by a spatial cutoff equal to one on a buffered neighborhood of all transformed rays. Denote the result by Ψ .

Off the original packet flow by an inverse-polynomial buffer, every fixed derivative of u is $O_N(H^N \sqrt{e_L})$, for arbitrary N , with integrable weights at infinity. To verify this, express u as initial packet synthesis and use Equation (cone-kernel) at length one. Cauchy–Schwarz in the bounded initial row region costs only a fixed power of H^{-1} ; a fixed derivative costs another fixed power. The packet width $d_1 = \sqrt{H}$ is smaller than the buffer by a power when Z is large. Arbitrary integration order then absorbs these costs. Thus cutoff derivatives occur only where u has this decay, and Equation (8.6) proves the same error estimate for $\square\Psi$ in every fixed spatial Sobolev norm.

Choose a smooth safe-cone and radial-band projector $\mathcal{P}$ that equals one on all incident pulled covectors throughout the time interval, and evolve $\mathcal{P}\Psi(-1)$ exactly to define u' . Uniformly for every $t' \in [-1, 3]$, $\|(1 - \mathcal{P})\Psi(t')\|_{H^j} = O_N(H^N \sqrt{e_L})$ for every fixed j, N . Indeed insert the old compact-band Fourier representation of u . Outside the plateau the spatial phase gradient is separated from the output frequency ζ' by $cP^{-C}(H^{-1} + |\zeta'|)$, while its order- j derivatives are at most $C_j P^{Cj+C} H^{-1}$. Repeated integration by parts gives arbitrary powers of H , including integrable decay for $|\zeta'| \gg H^{-1}$. The exponents of P are linear in the total differentiation order, so the required choice of Z is independent of that order. On the retained band the forced equation is first order in t' after division by $-4i\rho'$; unitarity and integration in time show

$$\sup_{-1 \leq t' \leq 3} \|u'(t') - \Psi(t')\|_{H^j} = O_N(H^N \sqrt{e_L})$$

for every fixed j, N . Sobolev embedding also gives the pointwise comparison needed below. Smooth band projection and bounded-time propagation leave negligible analyzed energy outside a bounded spatial region and the enlarged safe angular cone, by the same phase argument.

Packet association. Put $Q_\ell = q_\ell^2 H$. At input, substitute the actual old synthesis into the weighted pullback and then analyze $u'(-1)$. At output, synthesize $u'(T')$, propagate to the image event, pull back, and analyze on the old slice. The preceding approximation justifies both substitutions with arbitrary power errors. Each resulting kernel has absolute value at most P^C/Q_ℓ : the analyzing kernel has size Q_ℓ^{-1} and integrable Schwartz profile in tilted position units (q_ℓ, H) , and a propagated source atom is bounded by C/Q_ℓ from its Fourier support.

Choose small $\varepsilon_2 \gg \varepsilon_1 \gg b$ and first restrict integration to the target packet core enlarged by $\delta^{-\varepsilon_1}$. Its complement has arbitrary decay by the Schwartz bound. If the source angle differs from the image target angle by more than $\delta^{-\varepsilon_2} d_\ell$, express the source atom by plane waves. In a transverse variable $(b - b_p)/q_\ell$, holding $y - 2A_p \cdot b$ fixed, the phase derivative has magnitude at least $cP^{-C}\delta^{-\varepsilon_2}$. Its variation over the retained core is at most $P^C\delta^{-C\varepsilon_1}$, because $q_\ell^2/H = \ell \leq 1$. An order- j higher derivative is bounded by $C_j P^{Cj+C} q_\ell^j/H$. Apply the integration-by-parts operator $(i\partial_v \phi)^{-1} \partial_v$ in a direction nonstationary at the center. The product rule bounds the resulting terms by

$$\frac{P^{C_0}}{Q_\ell} \delta^{N(\varepsilon_2 - C\varepsilon_1 - Cb) - C_0}$$

times finite-power normalization factors, for arbitrary N . Take $\varepsilon_2 > C\varepsilon_1 + Cb$. The inner choice of N then absorbs all fixed polynomial row-volume costs, even when the exponent of H^{-1} is very large. This proves angular localization without any loss proportional to that exponent.

Once angles match, use the propagated source packet envelope for spatial localization. The relevant propagation duration is at most $\ell \delta^{-O(\varepsilon_1 + b)}$. The chart sends the target core into the associated source core: the polar covector cancels first-order vertical variation and the remaining quadratic error is controlled by $q_\ell^2 \leq H$. Position-induced angle variation is controlled by

$q_\ell \leq d_\ell$. Backflow through the allowed time discrepancy adds transverse error $O(\ell d_\ell) = O(q_\ell)$ and corrected vertical error $O(\ell d_\ell^2) = O(H)$, with the same small-power enlargements. Applying the inverse chart proves the column version of this assertion, including after event-time rounding.

Transfer of assignments. For fixed source and target time indices, row and column association volumes in normalized row measure are therefore at most $Q_\ell \delta^{-O(\varepsilon_2)}$. Schur's test with the kernel bound proves the input energy comparison. For output assignments, let $h_\lambda(p)$ be their old fractions, including the time partition. Pointwise Cauchy–Schwarz bounds an old assigned mass by a small-power multiple of transformed energy with fractions

$$Q_\ell^{-1} \int_{p \sim p'} h_\lambda(p) d_\ell^{-2} dA_p db_p dy_p.$$

For a fixed p', T' these fractions sum to at most a small inverse power. The column volume gives this for each old time. To count those times, inverse-map the source center event at T' and call its old time t_0 . A mapped associated old event differs from that center by $O(\delta)$ in time, from rounding, and by at most $\delta^{-O(\varepsilon_2)}(q_\delta + H + \delta)$ in ordinary spatial coordinates; the last term accounts for bounded velocity during rounding. Since $q_\delta = \delta d_\delta \lesssim \delta$ and $H = \delta d_\delta^2 \lesssim \delta$, the inverse chart gives $|t_{\text{old}} - t_0| \leq \delta^{1-O(\varepsilon_2)}$. Old cuts have spacing δ , so only $\delta^{-O(\varepsilon_2)}$ indices occur. This bound is independent of which label assigned the event to that rounded time. Divide all fractions by the resulting bound to legitimate the new assignments. A fixed old label uses only polynomially many in P rounded times, because its position cell has diameter $O(\delta)$; Hölder's inequality bounds this aggregation cost by another small power. Choosing the losses so their combined exponent is below γ proves both conclusions. $\square$

Lemma 8.12 (Allocation from cell capacities). *Let x_i and y_j be energies in two finite measurable cell partitions, and let $A, D \geq 1$ and $\epsilon_i \geq 0$. Suppose*

$$x_i \leq A \sum_{j \in \mathcal{N}(i)} y_j + \epsilon_i,$$

and every source cell j belongs to at most D sets $\mathcal{N}(i)$. Assignments $m_{\lambda i}$ with $\sum_\lambda m_{\lambda i} \leq x_i$ can, after removing total mass at most $\sum_i \epsilon_i$, be transferred to the y -cells with every retained label losing at most a factor AD . Contributions from different cells may be aggregated back into the same label without an additional factor.

Proof. Reduce the assignments proportionally in each target cell until their sum is at most AY_i , where $Y_i = \sum_{j \in \mathcal{N}(i)} y_j$. The removed mass is at most ϵ_i . For $Y_i > 0$, allocate $m_{\lambda i} y_j / (AD Y_i)$ to cell $j \in \mathcal{N}(i)$. Its total over labels and incident i is at most y_j . Summing over j gives exactly $m_{\lambda i} / (AD)$, so aggregation over i preserves the stated label mass. Cells with $Y_i = 0$ have no retained assignments. $\square$

8.5 Regularity after flattening

In transformed coordinates use

$$X = A'_1, \quad B = b'_1, \quad Y = y', \quad N = A'_2/f_*, \quad D = b'_2/f_*, \quad H_{\text{flat}} = H/f_*^2.$$

Exact ray velocities in (B, Y, D) are $(X, X^2 + f_*^2 N^2, N)$. Initial and final normal coordinates satisfy $|D|, |N| \leq \delta^{-C\gamma}$, modulo dispensable tails; this follows from F_L, G_L , conformal flattening, and Lemma 8.11. The ideal cylinder replaces $X^2 + f_*^2 N^2$ by X^2 .

For waves, let $(\xi, -\rho, p_N)$ be the Fourier variables dual to (b'_1, y', D) , and put $\sigma = 2H\rho$. Evolution over elapsed time t_0 factors exactly as

$$U_b = e^{-it_0\xi^2/(4\rho)}, \quad U_\sigma = e^{-it_0H_{\text{flat}}p_N^2/(2\sigma)}. \quad (8.7)$$

Thus the normal evolution has parameter H_{flat} even when H is much smaller than the base observation scales. The comparison below replaces the base evolution by exact cylinder indices while retaining this normal propagator whenever its quantum scale is not negligible.

Lemma 8.13 (Input and terminal weights). *Fix a finite precision L_0 . Consider ideal input cylinder tests with base and normal widths $R, F_0 \in [\delta^{L_0}, 1]$, and shear coefficients bounded by δ^{-L_0} . Suppose $f_*F_0 \gtrsim d_1$. Their associated old input rows have mass at most*

$$\delta^{-C\gamma} \kappa f_*^{1-\eta} R^{3+\eta} F_0^{1-\eta}. \quad (\text{lift})$$

At output, split each connected old assignment by chart and rounded time. Each resulting assignment fits an ideal cylinder test of weight at most

$$\delta^{-C\gamma} (f/f_*)^{1-\eta}. \quad (\text{endlift})$$

The assertions remain true under sufficiently small fixed-power base errors and normal errors of size $\delta^{-C\gamma} \max(\delta^S, d_\ell/f_)$ in velocity and ℓ times this in position, where $\ell = 1$ or δ and S is sufficiently large. The constants are independent of Z once Z is sufficiently large relative to these fixed precisions.*

Proof. For an input cylinder test, subtract f_* times its position-shear polynomial from b'_2 , replacing the base quadratic by $q' = t'y' - |b'|^2$. This produces a full hyperquadric. On matched rays, its position and null-derivative sublevels have size at most $\delta^{-C\gamma} f_* F_0$. Terms introduced by the second-coordinate square and by $f_*^2 N^2$ are bounded by a fixed power times f_*^2 ; Equation (8.2), with Z sufficiently large, makes them smaller than the stated sublevels. Its discriminant is close to one, so normalize it to one. The full round box has width R : the absolute second-coordinate widths are much smaller than R . Pull back and backflow to old time zero. Lemma 8.9 costs only powers of P . Center at a witness, shrink by R , and divide the polynomial by R . Its coefficients are bounded by small inverse powers; the constant and time coefficients use the two sublevels. Lemma 8.4 with relative thickness f_*F_0/R now gives

$$\delta^{-C\gamma} \kappa R^4 (f_*F_0/R)^{1-\eta},$$

which is Equation (lift). The permitted errors and packet uncertainty are absorbed by its displayed enlargement.

At output, each connected polynomial differs after flattening from b'_2 by coefficients $O(P^C f_*)$. At its image event its position sublevel is $O(P^C \delta f)$ and its derivative sublevel is $O(P^C f)$. Propagation to the rounded time preserves these bounds. Normalize the b'_2 coefficient to one, divide by f_* , and drop the $O(f_*^2)$ terms just discussed. The resulting cylinder normal width is $\delta^{-C\gamma} f/f_*$, with shear coefficients bounded by $\delta^{-C\gamma}$. Its base width is $\delta^{-C\gamma}$: the old position cell and time rounding give positions within $\delta^{1-C\gamma}$, and angles stay bounded. This proves Equation (endlift). These tests depend only on the old label, chart, and rounded time, so the cell allocation below can aggregate back into their original labels. All allowed comparison errors fit their enlarged bounds. Since $d_\delta \lesssim f$, the hybrid uncertainty condition also holds. $\square$

Completion of input regularity. In each ideal input constructed below, include an average over independent exact base and normal translations and boosts of size δ^l , with l fixed larger than the terminal precision requirements. Retain the copies as passive indices and enlarge terminal tests. The exact cylinder symmetries preserve the shear family. This smoothing extends the moderate-test comparison to full regularity: uniformly in shear parameters,

$$\mu(\text{test}) \leq \delta^{-C(l)} e_L \min(R^4, 1) \min(F_0^2, 1).$$

In the hybrid case the normal smoothing uses Weyl translations and the same estimate holds conditionally at each base point. Together with Equation (8.5), this proves the required regularity outside a finite-power width range, by taking its cutoff exponent large depending on l, η . Bounded base support handles $R > 1$; used normal coordinates can be covered by unit zero-shear boxes for $F_0 > 1$, at small-power cost. Polynomial sublevel slicing against the jitter density discards tests whose coefficients exceed a sufficiently large finite power. The cylinder truncation argument therefore reduces full regularity to Equation (lift), with L_0 chosen after these cutoffs. Compact base support and negligible normal tails are retained in the constructions below.

Additive errors. All cell comparisons allow a prescribed error $\delta^J e_L$. Equation (8.5) and the finite-power relative weights allow J to be fixed so that these errors are negligible in the final cube sum. When several labels share a cell, remove its excess once, proportionally over labels, as in Lemma 8.12; do not sum an error separately over labels. Components of extremely small relative energy use the direct contraction bound instead of a normalized exponent estimate.

8.6 An almost classical base

Classically, use the transformed rays at time -1 as ideal cylinder input, then evolve with the ideal velocities. The omitted vertical displacement is $O(f_*^2 \delta^{-C\gamma})$, smaller than every fixed comparison precision once Z is large. Lemma 8.13, the jitter, and the energy budget therefore give the desired cylinder comparison directly.

For waves, first fix a large M depending on all comparison precisions. If $H_{\text{flat}} < \delta^M$, use the transformed initial analyzed energy as a positive measure on ideal rays. Partition phase into cells of a fixed power size, smaller than the comparison tolerances but larger than every rescaled packet width. Equation (cone-kernel) bounds each backflowed output cell by neighboring input capacities, with bounded overlap and negligible error. Lemma 8.12 transfers assignments while retaining their labels. Equation (lift) applies because $d_1/f_* \ll \delta^{L_0}$. Thus this regime reduces to the classical cylinder as well.

It remains to consider

$$\delta^M \leq H_{\text{flat}} \lesssim \delta(f/f_*)^2. \quad (8.8)$$

The base parameter $H = f_*^2 H_{\text{flat}}$ is still arbitrarily small relative to every fixed comparison precision.

Lemma 8.14 (Classicalizing the base). *In the range Equation (8.8), the transformed wave can be replaced, for the input and terminal comparisons of Lemma 8.13, by a fixed finite direct sum of ideal hybrid cylinder states. Their summed energy is at most $\delta^{-C\gamma} e_L$, each regularity is at most $\delta^{-C\gamma} \kappa f_*^{1-\eta}$, and their assignments satisfy Equation (endlift), with arbitrarily small total additive errors. The exponent $C\gamma$ is independent of the finite-power exponent of H^{-1} .*

Proof. Construction of the hybrid states. Use L^2 -normalized coordinates (b'_1, y', D) for $g = u'(-1)$, with Fourier variables and propagators as in Equation (8.7). Put $X = \xi/(2\rho)$ for the base Fourier

velocity. Smoothly restrict to $|H_{\text{flat}}p_N| \leq \delta^{-C\gamma}$, with slack. The removed norm is arbitrarily small relative to $\sqrt{e_L}$: integrate the initial analyzed angular tail and apply Plancherel. The spatial normal tail is likewise negligible by packet synthesis; the smooth frequency cutoff preserves it with slack. Let Π be a further real cutoff equal to one on the retained support, so $\Pi g = g$. All base positions are bounded up to negligible tails.

Choose observation cells with base sides δ^{S_1} and normal velocity and position sides

$$\nu_\ell = \max(\delta^{S_1}, \sqrt{H_{\text{flat}}/\ell}), \quad \ell\nu_\ell, \quad \ell = 1, \delta.$$

Here S_1 exceeds the required comparison precision. There are polynomially many cells. Allow a per-cell error $\delta^{J_1}e_L$, with J_1 large compared with their count exponent.

Choose $K \gg M + S_1 + J_1$. Split the base first by smooth compact square partitions in (X, σ) and then by smooth compact square partitions in (b'_1, y') , both of side δ^K . The spectral partition is a tensor product with a common σ -lattice. Write

$$W_j = M_j^{\text{pos}} P_j^{\text{freq}}, \quad (X_j, \sigma_j, B_j, Y_j)$$

for its operators and cell representatives. Spectral partition kernels have spatial width $O(H\delta^{-CK})$, so restriction to bounded base centers loses negligible energy when Z is large. Before that restriction, $\sum_j W_j^* W_j = \text{Id}$ exactly; afterwards the identity holds on g up to negligible error. The column map W is contractive. Give fiber j the normal input $W_j g$, taking its base spatial variable as an auxiliary Hilbert index. Evolve its representative classically, and evolve its normal datum with U_{σ_j} . This defines one direct-sum state for both input and output measurements. Base partitions commute with normal cutoffs, so its normal tails satisfy the required bounds; Gaussian packet tails allow the stated slack.

To meet the cylinder convention $1 \leq \sigma \leq 2$, partition these passive fibers into finitely many fixed bands $a \leq \sigma_j \leq 2a$. For every fiber in that band use, from the outset, cylinder parameters $(\sigma_j/a, H_{\text{flat}}/a)$ and its corresponding Gaussian analyzer. The propagator is unchanged because H_{flat}/σ_j is unchanged. The Gaussian position width is now $\sqrt{H_{\text{flat}}\ell/a}$; its angular width also differs from the displayed comparison scales only by a fixed factor. Thus all kernel and cell estimates below hold with these analyzers, with constants depending on the fixed bands. At the final cylinder estimate, energy budgets add, each band inherits the same regularity bound, and aggregation over the finitely many bands costs a fixed factor. The band count and the factors a are fixed on the inner setup, so there is no scale-exponent loss.

Cross-analysis and radial freezing. Let the exact and ideal analyzing maps at a measured time be

$$E = V_\ell^{\text{cone}} U_b U_\sigma, \quad I = \{V_\ell^{j, \text{flat}} U_{\sigma_j} W_j\}_j.$$

Use a plateau cone analyzer, so $E^* E = \text{Id}$ on the retained band. Flat analysis is isometric, giving $I^* I = \sum W_j^* W_j$. For $C_0 = E \Pi I^*$ we therefore have

$$C_0 I g = E g + O(\delta^{J_2} \sqrt{e_L}), \quad C_0^* E g = I g,$$

where J_2 can be prescribed before the localization order is chosen. The reference energy is the raw budget e_L ; no lower bound for $\|g\|^2/e_L$ is asserted. All three operators are contractive. Moreover C_0 differs in norm by at most $C\delta^{K-M-C\gamma}$ from

$$V_\ell^{\text{cone}} U_b \Pi \sum_j W_j^* (V_\ell^{j, \text{flat}})^*. \quad (\text{cap-cross})$$

To prove this, commute U_{σ_j} through W_j^* , which operates only in the base variables. On the outer spectral support of W_j^* , $|\sigma - \sigma_j| \lesssim \delta^K$. On Π , $H_{\text{flat}} p_N^2 \lesssim \delta^{-M-C\gamma}$, so

$$|U_{\sigma} U_{\sigma_j}^* - 1| \lesssim \delta^{K-M-C\gamma}.$$

Group the indices with common σ_j before summing: within each group use contraction of W^* ; between groups use bounded spectral overlap of their outer projections. This proves the norm estimate without a factor from the number of spatial or angular bins.

Off-diagonal bounds. We prove that Equation (cap-cross) has negligible norm between an observation cell and the complement of its enlarged neighbors. Compress to bounded base coordinates and $\delta^{-C\gamma}$ -bounded normal coordinates; the removed energy of Eg and Ig is negligible by the established localization and flat packet propagation. Angular mismatch in X is excluded by the outer spectral bin and cone symbol. The latter selects $H_{\text{flat}} p_N / \sigma$ within width $\sqrt{H_{\text{flat}} / \ell}$, whereas the flat synthesis atom has a Gaussian Fourier profile centered at $H_{\text{flat}} p_N / \sigma_j = N_j^{\text{row}}$ with the same width up to fixed factors. The difference of σ and σ_j is much smaller than observation precision. Normal velocity mismatch outside enlarged neighbors thus has arbitrary Gaussian gains.

For position comparison, the inverse Fourier kernel from auxiliary base point (b_0, y_0) and normal center D_0 has phase

$$\xi(b'_1 - b_0) - \rho(y' - y_0) - t_0 \xi^2 / (4\rho) + p_N(D - D_0).$$

The point (b_0, y_0) is in its spatial bin. A base discrepancy from $(B_j + t_0 X_j, Y_j + t_0 X_j^2)$ beyond the neighboring observation cells therefore gives a phase gradient of size at least $c\delta^{S_1}$ in a base frequency direction. In coordinates $(H\xi, H\rho)$ the large parameter is H^{-1} . Higher derivatives of the factored phase are bounded, and an order- n symbol derivative costs at most

$$C_n \max(\delta^{-K}, \delta^{-C\gamma} d_\ell^{-1})^n.$$

This includes differentiation through the normal angular dependence on ρ . Differentiating the reciprocal phase gradient also costs powers of δ^{-S_1} ; these are absorbed by the δ^{-K} term because $K > S_1$. Each integration by parts therefore gains a power of

$$H\delta^{-S_1} \max(\delta^{-K}, \delta^{-C\gamma} d_\ell^{-1}),$$

which is a positive power of δ when Z is sufficiently large, since $Hd_\ell^{-1} \leq \sqrt{H}$. Its positive exponent can be fixed before the integration order. Arbitrary iteration absorbs the finite-power normalizations and integration volumes.

For normal-position mismatch use frequency $H_{\text{flat}} p_N$. Symbol and Gaussian derivatives cost at most $C_n(\ell/H_{\text{flat}})^{n/2}$, with integrable Gaussian remainders. The linear phase derivative has size $|D - D_0|/H_{\text{flat}}$, greater by a small inverse power outside the enlarged normal cell. Repeated integration by parts supplies arbitrary powers of that enlargement. Gaussian tails do the same for normal velocity mismatch. Crude integration and Schur's test convert these pointwise bounds into the claimed off-diagonal norm estimate: the row measures, auxiliary spatial supports, packet normalizations, and bin counts all have fixed finite-power size on the inner sequence. Choose the order after these powers. The geometric smallness conditions are unchanged, because derivative exponents grow at most linearly in order.

Capacity and regularity comparisons. Contraction of C_0 and its reverse, the freezing error, and the off-diagonal bounds give each cell energy at most a bounded multiple of the opposite neighboring

capacities plus $\delta^{J_1} e_L$. A cell belongs to only $\delta^{-C\gamma}$ enlarged neighbor sets. Apply Lemma 8.12 and aggregate into the original terminal labels; there is no loss from the total number of observation cells. Their neighbors fit the errors in Lemma 8.13. At input the reverse comparison, summed over moderate test cells, proves Equation (lift). Choose J_1 above the cell count and K above the required freezing precision. The jitter extension then proves full cylinder regularity. This gives the asserted energy, regularity, and terminal assignments for the same finite direct-sum state at both measurements. $\square$

Proof of Proposition 8.8. Suppose there is a strict gap from the indicated cylinder exponents. Proposition 8.7 supplies a short free final gap with regular incoming states. Choose its fixed depth length so that Equation (8.2) holds at every required precision. The preceding reductions discard contributions that cannot saturate, then decompose into low-coefficient buffered patches with summed raw energy at most a small inverse power times e .

For each patch, Lemmas 8.11, 8.13, and 8.14, or the classical construction, give an ideal cylinder input of energy at most $\delta^{-C\gamma} e_L$, regularity at most $\delta^{-C\gamma} \kappa f_*^{1-\eta}$, and terminal tests of weight at most $\delta^{-C\gamma} (f/f_*)^{1-\eta}$. Transferred masses lose at most a small power. The cylinder estimate thus gives, for either relevant cylinder exponent Γ_c ,

$$\sum_{\lambda} \frac{m_{\lambda}^{3/2}}{f^{(1-\eta)/2}} \leq \delta^{1/2-\Gamma_c-C\gamma-o(1)} e_L \sqrt{\kappa}.$$

The factor $f_*^{(1-\eta)/2}$ from input regularity cancels its reciprocal from the terminal weights. Sum over patches and use the raw energy budgets and bounded label connections. Choose all small-power losses below the strict exponent gap. The resulting bound contradicts saturation of the free round gap.

Here is the order of choices, needed to ensure uniformity. Fix the group threshold D_1 , the jitter precision, and the moderate-test cutoffs. Next fix observation-cell and additive-error precisions, the normal threshold M , and the base partition precision K . Choose geometric and localization loss exponents small relative to the strict gap, then take Z large enough for every fixed precision. Finally choose the positive final depth length h sufficiently small. After fixing the inner sequence and its finite-power bounds, choose the integration orders; these orders remain fixed as the inner scale tends to zero. Their derivative costs have linear exponents, as proved above, so they do not alter the earlier small-loss choices. A positive infinite thickness exponent also satisfies every prescribed finite Z , proving that case as well. $\square$

9 The round-cone exponent

We retain the round-cone experiments, their regularity tests, and their free growth exponents from the preceding section. We write $\Gamma_{\text{cl,round}}$ and $\Gamma_{\text{wav,round}}$ for the classical and wave exponents, respectively. The allowance in the following theorem is deliberately larger than the losses in its proof.

Theorem 9.1 (Round-cone bound). *For every $0 < \eta < 1$, the round-cone experiments satisfy*

$$\Gamma_{\text{cl,round}} \leq 100\eta, \quad \Gamma_{\text{wav,round}} \leq 100\eta.$$

The same upper exponent bounds hold with a fixed finite list of admissible row contractions, with arbitrary separable Hilbert multiplicity, and with the finite-power support and error conventions of the round-cone model.

The assertion concerning contractions follows from Proposition 8.6. We prove the free assertions, first classically and then for waves. In a contradiction argument, Γ denotes a saturated exponent with

$$\Gamma > 100\eta, \quad \Gamma > \Gamma_{\text{cl,round}} \quad \text{in the wave case.}$$

As in the preceding section, take the least homogeneous mass exponent p in the free closure at Γ , and put

$$\Gamma = d + \frac{1-p}{2}, \quad p \geq 1, \quad 0 < d \leq 1. \quad (9.1)$$

At a prescribed depth a , the energy and regularity benchmarks are

$$E_a = es^{-da+o(1)}, \quad \kappa_a = \kappa s^{pa+o(1)}. \quad (9.2)$$

Throughout this section a finite list of depths is fixed before the inner scale limit. An assertion with $s^{o(1)}$ means its strict-power version with arbitrarily small prescribed error, after the earlier extremal and homogeneous approximations have been taken sufficiently accurate. We never construct an infinite stack of operators.

The thin-shape reduction leaves round labels. In the wave case we place them at the uncertainty scale. We then exclude concentration of velocities near a plane and use compatible names and stationarity to obtain an outgoing entropy bound. Direct integration closes the classical argument; for waves, the bound supplies the two differential inequalities used in the final descent.

9.1 Round labels occur at the uncertainty scale

Proposition 8.8, together with Theorem 3.6, shows that every homogeneous ending family which saturates the assumed exponent has $f = s^{o(1)}$. This statement also applies to a raw peeling family read before synthesis: insert a short positive free gap immediately before that readout and apply Equation (CE) on the gap.

Lemma 9.2 (Volume of a classical round test). *Normalize the parent length to one and let ν be a classical phase measure supported in a fixed bounded box of (b, y, A) , with Lebesgue density at most L_0 . Uniformly in the centers and coefficients of every admissible round or hyperquadric test,*

$$\frac{\nu(\text{test})}{r^4 f^{1-\eta}} \leq CL_0 \min(r^2, r^{-3}) f^\eta. \quad (9.3)$$

The constant depends only on the support box and the fixed test coefficient bound.

Proof. Use the normalized test coordinates (x, z, V) at its initial time. The phase Jacobian is r^6 : horizontal position contributes r^2 , vertical position contributes r^2 , and velocity contributes r^2 . For a hyperquadric test, F is independent of V , and

$$g = \partial_V G = \alpha + 2\beta V - 2kx, \quad |g|^2 = 1 + 4\beta G - 4kF.$$

The coefficient bound is 100. On the two sublevels, if $f \leq 1/1600$, the identity gives $|g|^2 \geq 1/2$. At least one of its two components then has magnitude at least $1/2$. Partition the angular sublevel according to such a component. Hold the other angular coordinate fixed. The function G in the remaining coordinate is a quadratic, so the portions where its derivative has magnitude at least $1/2$ consist of at most two monotonic intervals. On each interval the preimage of $[-f, f]$

has length at most $4f$. Integrating over the bounded other coordinate gives normalized angular area $O(f)$, uniformly at each fixed (x, z) satisfying $|F| \leq f$. Integration over $|x|, |z| \leq 1$ bounds the physical phase volume by Cr^6f .

For a second bound, hold the physical position (b, y) fixed and integrate only over the fixed bounded support box in A . On the same sublevel, $\partial_A G = g/r$, so the identical quadratic slicing argument gives angular area at most Cr^6f ; the other physical angular coordinate ranges in a fixed bounded interval. Integration over the compact physical position support yields phase volume at most Cr^6f . These arguments do not restrict the test centers: only the points satisfying its normalized round conditions and lying in the physical support are being counted.

For $f \geq 1/1600$, the crude phase volume is at most $C \min(r^6, 1)$, which is bounded by $C \min(r^6, r)f$ after enlarging the constant. This also handles round tests, for which $f = 1$. Combining the two volume bounds, multiplying by L_0 , and dividing by the weight proves Equation (9.3). $\square$

Proposition 9.3 (Angular uncertainty at a saturated wave cut). *In the wave contradiction, a homogeneous saturating family at any positive-depth cut j satisfies*

$$\frac{r_j}{d_{\ell_j}} = s^{o(1)}. \quad (\text{round-Pl})$$

After normalizing the terminal angular width to $r_1 = 1$, it follows that

$$H = s^{1+o(1)}, \quad u(a) = \frac{1-a}{2}, \quad r_a = s^{u(a)+o(1)}. \quad (\text{round-u})$$

Proof. The uncertainty condition and $f = s^{o(1)}$ give the lower bound in Equation (round-Pl). Suppose the ratio is large by a fixed power. Choose an entering free gap of ratio $\Delta = s^{h+o(1)}$, with $h > 0$ smaller than that power and smaller than the available gap. Normalize the parent time and position lengths to one and the ending angular width to one. By decreasing h , we can arrange $d_\Delta \leq \Delta^M$ for any fixed M . In these units write κ_* for the parent regularity.

Partition the entering plateau rows into bounded phase boxes, each with a bounded angular range after a boost. Equation (acc) shows that an ending label receives contributions from at most a subpower number of these boxes. Coherent splitting therefore has only a subpower cost. The raw mass e_g of one box satisfies $e_g \leq s^{-o(1)}\kappa_*$ by round regularity, and its synthesized field retains the same regularity upper bound.

Regard the raw row measure in this box as a positive measure of exact rays. Average it over independent smooth position translations and velocity boosts, of size Δ^2 , using the accompanying exact vertical tilt. More explicitly, at the initial time the transformation with parameters $(u, v, a) \in \mathbb{R}^2 \times \mathbb{R} \times \mathbb{R}^2$ is

$$(b, y, A) \mapsto (b + u, y + v + 2a \cdot (b + u), A + a).$$

At subsequent times the velocity boost additionally changes b by ta and y by $t|a|^2$, so it sends exact rays to exact rays. For each original phase point, the displayed map from (u, v, a) to the new phase point has Jacobian one. Smooth parameter densities at scale Δ^2 therefore give a jittered density at most $C\Delta^{-10}e_g$, with bounded support. These symmetries preserve admissible test bounds exactly, not just up to an enlargement. Transform the test center and its central velocity by the same symmetry. Substitution in the normalized coordinates leaves (x, z, V) unchanged, including their time extension. Thus r, f , the normalized polynomial coefficients,

their discriminant, and their bound of 100 are unchanged. The uncertainty parameter d_1 is also unchanged. Pullback of a test with $rf \geq d_1$ is consequently an admissible input test of the same weight; averaging its bound preserves regularity. For a test below that uncertainty threshold, Lemma 9.2 gives

$$\frac{\mu(\text{test})}{r^4 f^{1-\eta}} \leq C \Delta^{-C} e_g \min(r^2, r^{-3}) f^\eta.$$

If $rf < d_1 \leq \Delta^M$, the final factor is at most $\Delta^{M\eta}$: for $r \leq \Delta^M$ use r^2 ; for $\Delta^M < r \leq 1$ use $f \leq \Delta^M/r$; and for $r \geq 1$ use $r^{-3} f^\eta \leq \Delta^{M\eta} r^{-3-\eta}$. Choosing $M > C/\eta$ therefore gives admissible classical regularity $s^{-o(1)} \kappa_*$. This choice depends on η and the fixed jitter construction, not on the original polynomial clipping powers.

At each ending time, group the wave assignments into spatial cubes of side Δ . Equation (cone-kernel), applied to the complement of the ray preimage of an enlarged cube, and contraction on that preimage imply that its analyzed wave mass is bounded by the raw positive ray mass of the enlarged cube, up to negligible error. Here the packet position errors are much smaller than Δ . Jitter moves a ray by $O(\Delta^2)$ over the bounded time interval, so all its jittered copies belong to one further enlargement. These enlarged cubes have bounded overlap, have bounded angular width, and are round tests of bounded weight. Splitting a label among cubes and combining assignments in the same cube cost only $s^{-o(1)}$.

Consequently the classical round-cone bound controls the ending wave cube sum on this gap, relative to $e_g \sqrt{\kappa_*}$, with exponent $\Gamma_{\text{cl,round}} + o(1)$. The kernel errors are first made smaller than all required powers on the clipped inner sequence; components of negligible energy are estimated directly on the polynomial family of output tests. Summing the raw component budgets contradicts saturation of a gap whose exponent is strictly larger. This proves Equation (round-Pl). Substituting $d_\ell = \sqrt{H/\ell}$, first at the terminal cut and then at depth a , proves Equation (round-u). $\square$

9.2 A short step with velocities near a plane

We establish the estimate that will remove planar concentration from conditional velocity laws. Normalize a free parent interval to length one. The selected angular spectrum is bounded after a boost; ending round labels have angular width one and time mesh $\Delta = s^{h'+o(1)}$. Their weights are $s^{o(1)}$ and will be omitted in this subsection. Let e_g and κ_* be the input energy budget and regularity. In the wave case assume $H = \Delta s^{o(1)}$. A plane slab in velocity space has the form

$$|d_0 + \alpha \cdot A + \beta |A|^2| \lesssim \rho^\varepsilon, \quad |\alpha|^2 + \beta^2 = 1, \quad \rho = s^h, \quad (\text{plane})$$

where $h, \varepsilon > 0$ are fixed. If the slab meets the bounded angular range, then $|d_0| \lesssim 1$. Classical input rays are supported in this slab. For waves the spatial spectrum is supported within $s^{-o(1)} \sqrt{\Delta}$ in velocity of its bounded portion, in the prescribed positive compact radial bands.

Lemma 9.4 (Spatial readout). *For a free wave with this bounded angular spectrum,*

$$\Delta^{-1/2} \sum_\lambda m_\lambda^{3/2} \lesssim s^{-o(1)} \int_{\mathcal{U}} \|u(t, b, y)\|^3 dt db dy + \text{negligible error}. \quad (\text{space-3})$$

Here $\mathcal{U}$ can contain the bounded time interval and all space. If the contributing input rows are localized in bounded phase boxes, only their subpower enlarged spacetime region is needed.

Proof. Group the ending assignments at each cut T in ordinary spatial Δ -cubes. A label meets only $s^{-o(1)}$ such cubes. The analyzer at scale Δ and propagation for $|t - T| \leq \Delta$ have spatial kernel tails at scale $\Delta s^{-o(1)}$: rescale their compact smooth symbols, using $H = \Delta s^{o(1)}$ and bounded angular support, and integrate by parts. Thus a cube mass m_D is at most the spatial square energy in an enlarged cube D^+ at each $t \in [T, T + \Delta]$, plus negligible error. On D^+ , Holder's inequality gives

$$m_D^{3/2} \lesssim s^{-o(1)} \Delta^{3/2} \int_{D^+} \|u(t, b, y)\|^3 db dy.$$

Average over this time interval, multiply by $\Delta^{-1/2}$, and sum over the bounded-overlap spacetime cubes. The same argument with polynomial clipping and kernel truncation proves the localized statement. All errors can be made negligible relative to an original reference energy before estimating components of very small energy. $\square$

Proposition 9.5 (Plane-slab estimates). *There is a universal constant C with the following properties. Let $\lambda > 0$ be sufficiently small relative to ε . If $|\alpha|^2 - 4\beta d_0 \lesssim \rho^{2\lambda}$, one may choose $0 < h' < \lambda h$, inside any prescribed available positive gap, so that*

$$\Delta^{-1/2} \sum_{\lambda} m_{\lambda}^{3/2} \leq s^{-o(1)} \sqrt{\kappa_*} e_g. \quad (\text{tiny-cap})$$

In the complementary case choose a fixed $0 < h' \leq \varepsilon h/20$ inside the available gap, then take λ sufficiently small relative to h'/h . With $P = \rho^{-\lambda}$,

$$\Delta^{-1/2} \sum_{\lambda} m_{\lambda}^{3/2} \lesssim s^{-o(1)} P^C \Delta^{-4\eta} \sqrt{\kappa_*} e_g. \quad (\text{sec})$$

The estimates hold classically even after further positive restrictions or raywise contractions. For waves they concern the free continuation of the specified input state.

Proof. Put $D_0 = |\alpha|^2 - 4\beta d_0$. At an approximate zero of the polynomial in Equation (plane), its squared angular gradient is $D_0 + O(\rho^\varepsilon)$. If $D_0 \lesssim \rho^{2\lambda}$ and the slab is nonempty, $|\beta|$ is bounded below: otherwise $|\alpha|$ is bounded below and D_0 is bounded below. Completing the square shows that all slab velocities lie in a ball of radius $O(\rho^\lambda)$. Our choice $h' < \lambda h$ makes that radius $O(\Delta)$.

Classically, after boosting at its center, the preimage of an ending spatial Δ -cube has angular and horizontal widths $O(\Delta)$ and tilted vertical width $O(\Delta)$. It is covered by $O(\Delta^{-1})$ round phase boxes of radius $O(\Delta)$, of total weight $O(\Delta^3)$. Hence $\max m_{\lambda} \leq s^{-o(1)} \kappa_* \Delta^3$. Since the total assigned mass over all times is at most $s^{-o(1)} \Delta^{-1} e_g$, Equation (tiny-cap) follows from $\sum m^{3/2} \leq (\max m)^{1/2} \sum m$.

In the wave case the boosted horizontal and vertical frequency widths are at most $s^{-o(1)} \Delta^{-1/2}$ and $s^{-o(1)} \Delta^{-1}$. A spatial box of dimensions $(\sqrt{\Delta}, \sqrt{\Delta}, \Delta)$ at a bounded time can receive energy only from the correspondingly enlarged parent phase box, up to kernel tails. Round regularity at radius $\sqrt{\Delta}$ and contraction bound this energy by $s^{-o(1)} \kappa_* \Delta^2$. Reproduce the field with a smooth band kernel. Cauchy–Schwarz against that kernel, partitioning its tails into enlargements of these boxes, gives

$$\|u(t, b, y)\|^2 \lesssim s^{-o(1)} \kappa_*.$$

An arbitrarily small fixed-power enlargement followed by arbitrary kernel decay handles the tails on each inner sequence. Combine this bound with energy conservation and Lemma 9.4 to obtain Equation (tiny-cap).

We turn to $D_0 \gtrsim \rho^{2\lambda} = P^{-2}$. The exact section has angular gradient at least a constant multiple of P^{-1} , and each approximate solution is within $O(P^C \rho^\varepsilon)$ of it. Cover it by P^C buffered patches. Boost an exact direction in each patch to zero, rotate, and divide the plane equation by its nonzero transverse derivative. The section then reads $A_2 + \tilde{\beta}|A|^2 = 0$, with $|\tilde{\beta}| \lesssim P$. Apply the linear change

$$b'_2 = b_2 + \tilde{\beta}y, \quad t' = t + 2\tilde{\beta}b_2 + \tilde{\beta}^2y, \quad b'_1 = b_1, \quad y' = y. \quad (\text{L})$$

This preserves $ty - |b|^2$. On a ray set

$$D_A = 1 + 2\tilde{\beta}A_2 + \tilde{\beta}^2|A|^2.$$

The transformed direction satisfies

$$\frac{dt'}{dt} = D_A, \quad A'_1 = \frac{A_1}{D_A}, \quad A'_2 = \frac{A_2 + \tilde{\beta}|A|^2}{D_A}, \quad |A'|^2 = \frac{|A|^2}{D_A}.$$

Choose the patch to have sufficiently small inverse-polynomial radius so that D_A is bounded above and below. Its derivatives and those of the inverse chart have bounds $P^{C(n+1)}$ at order n . Our choices of h' and then λ ensure that $|A'_2| \leq \Delta^8$ classically and $|A'_2| \leq P^C \sqrt{\Delta}$ for waves. The patches have buffers larger than the finitely many admitted spectral broadenings.

Partition also into bounded input position charts. Each output label knows only $s^{-o(1)}P^C$ of these input charts, by Equation (acc). Restrict the parent rows and synthesize in the wave case. The sum of raw energy budgets is bounded by the original budget times this overlap; inherited regularity and spectral support have the stated bounds. Thus it suffices to work in one chart, where all relevant transformed coordinates and times have size at most P^C .

We first record its input regularity. At a fixed transformed time, test (A'_1, b'_1, y') in widths (R, R, R^2) with the usual tilt by twice the base angular center, and test (A'_2, b'_2) in widths (W, W) , with the angular center of A'_2 equal to zero. For $0 < R \lesssim 1$ and $W \leq P^C R$, the parent mass on the predicted rays satisfying these conditions is at most

$$P^C \kappa_* R^{3+\eta} W^{1-\eta}. \quad (\text{slice-reg})$$

For waves the widths are enlarged, if necessary, so that R, W are at least the parent uncertainty width. To verify the bound, pull back the round box and backflow to the parent time. The null covector $dy - 2A \cdot db + |A|^2 dt$ is preserved up to a controlled scalar; Taylor's formula therefore gives horizontal errors $P^C R$ and tilted vertical errors $P^C R^2$. The pullback of $b'_2 - b'_{2,*}$ is the polynomial $b_2 + \tilde{\beta}y - b'_{2,*}$, of discriminant one. Its value and ray derivative at the parent time are bounded by $P^C W$, because their values at the tested time have that bound and the propagation time is at most P^C . In angular R -units, divide this polynomial by R . Lemma 8.4 then applies with thickness $P^C W/R$ and gives $P^C \kappa_* R^4 (W/R)^{1-\eta}$. If $W > R$, the round bound alone is enough. Boxes larger than a chart buffer are covered by P^C boxes, and all coefficient bounds are still polynomial in P . This proves Equation (slice-reg).

For the classical estimate, put $W = \Delta^8$ and partition at $t' = 0$ into W -slabs of b'_2 . In one slab, project onto the base rays with velocity (X, X^2) , $X = A'_1$. Omitting $(A'_2)^2$ from the vertical velocity changes positions by at most $P^C W^2$, negligible at all scales R^2 with $R \gtrsim \Delta^2$. Equation (slice-reg) implies the hypothesis of Equation (kin) with constant $P^C \kappa_* W^{1-\eta}$, since $R^\eta \leq 1$. The affine change of spacetime compares old and new Δ -cells with P^C multiplicities. A ray meeting a new spacetime cell meets an enlarged spatial cell at its nearest grid time. Apply

Equation (kin) on the P^C required unit time intervals and use Cauchy–Schwarz with total counted mass at most $P^C \Delta^{-1}$ times the slab mass. The sum of $3/2$ powers in this slab is at most $s^{-o(1)} P^C \sqrt{\kappa_*} W^{1-\eta}$ times its mass. A full spatial cell meets at most $P^C \Delta/W$ slabs; convexity costs the square root of this number. After multiplication by $\Delta^{-1/2}$ the remaining factor is $W^{-\eta/2} = \Delta^{-4\eta}$. This proves Equation (sec) classically.

For waves the change in Equation (L) produces an exact transformed wave u' . The polar covector transforms with $\rho' = \rho D_A$. On the buffered patch this is a smooth invertible change of the spatial cone coordinates, with Jacobian and inverse bounded by powers of P and $\rho' H$ in a fixed positive band. Plancherel gives transformed-time energy at most $P^C e_g$. The same bound holds when only a subset of parent rows is synthesized: insert a common smooth spectral projection on the larger patch before making the change of variables.

Choose $W = P^{C_0} \sqrt{\Delta}$ so that $|A'_2| \leq W$. Cover the required transformed time range by intervals of length $L = P^{-C_2}$, where C_2 is large compared with the fixed bandwidth powers. At fixed b'_2 , multiply by a band-limited Schwartz time cutoff $\chi((t' - t'_0)/L)$. The resulting function of (t', b'_1, y') lies within thickness $O(L^{-1})$ of the base cone: the additional temporal frequency has size $\rho' |A'_2|^2 \leq P^C$. After scaling lengths by L , apply Equation (GWZ) at frequency L/H . The fine angular windows may depend on $\xi'_1/(2\rho')$ alone, and hence commute with restriction in b'_2 and the time cutoff. Comparable adapted windows give the same bound by convolution with rapidly decaying dual-sector kernels and a summable covering by translated tiles. At angular width R , where $(H/L)^{1/2} \lesssim R \lesssim 1$, the tile volume is $|U| \asymp L^3 R^3$.

The fine-cap square energy on such a tile, at fixed b'_2 , obeys

$$\int_U \sum_{\vartheta \subset O(\tau)} \|u'_\vartheta\|^2 \leq s^{-o(1)} P^C \Delta^{-\eta/2} \kappa_* |U|. \quad (\text{tile})$$

Indeed the second spatial bandwidth is $B_{\text{freq}} = P^C/\sqrt{\Delta}$. Reproduction and Cauchy–Schwarz in b'_2 reduce the left side to $C B_{\text{freq}}$ times the corresponding energy integrated over an enlarged W -slab. At each remaining tile time, Equation (slice-reg) bounds the energy of all the relevant fine caps by $P^C \kappa_* R^{3+\eta} W^{1-\eta}$. They have a common predecessor set of parent rows; after restriction to it, synthesis, the spectral change, and Plancherel with bounded overlap give this bound without a fine-cap counting factor. Integration over the tile time length $O(L)$ gives, after division by $|U|$,

$$P^C \kappa_* \Delta^{-1/2} W^{1-\eta} R^\eta \leq P^C \kappa_* \Delta^{-\eta/2}.$$

Powers of L^{-1} have been included in P^C .

For completeness, the predecessor restriction used here follows directly from the transformed packet kernel. If p is a parent row, its starting point becomes (t'_p, b'_p, y'_p) , and the phase is

$$(b' - b'_p) \cdot \xi' - (y' - y'_p) \rho' - (t' - t'_p) |\xi'|^2 / (4\rho').$$

The duration is at most P^C , and angular support lies within $P^C \sqrt{H}$ of its transformed direction A'_p . In variables $H\rho'$ and $\sqrt{H}(\xi' - 2\rho' A'_p)$ the amplitudes and their derivatives have bounds $P^{C(n+1)}$. The fine-cap cutoff obeys the same bounds because its width is at least $(H/L)^{1/2} \geq \sqrt{H}$. Integration by parts restricts the predicted positions to errors $P^C \sqrt{H}$ horizontally and $P^C H$ after tilt. These lie inside the enlarged conditions of Equation (slice-reg). Choose the enlargement power larger than the derivative power per order; increasing the order then gives arbitrary decay. Schur bounds on polynomially clipped rows, including the polynomial number of caps, turn this into negligible operator error. The reproducing kernel in b'_2 has arbitrary decay outside the

W -slab, because $B_{\text{freq}}W$ can be made larger than a fixed power of P . The time cutoff supplies the same decay outside an enlarged time range. Orders are chosen after the inner polynomial support and energy ratios are fixed.

Use Equation (tile) for one factor in each square on the right side of Equation (GWZ). Summing the remaining factor over tiles and integrating in b'_2 costs only the total spacetime square energy. Sum the P^C time windows to obtain

$$\int_{|t'| \leq P^{C_1}} \|u'\|^4 \lesssim s^{-o(1)} P^C \Delta^{-\eta/2} \kappa_* e_g.$$

The arbitrarily small exponent loss in Equation (GWZ) is included in $s^{-o(1)}$. Cauchy–Schwarz against the spacetime square energy gives an L^3 bound with factor $P^C \Delta^{-\eta/4} \sqrt{\kappa_*} e_g$. Changing coordinates back and using Lemma 9.4 proves Equation (sec), with room in its stated loss.

All nonnegligible patch, Jacobian, window, and overlap losses used a fixed power of P . The kernel orders affect only the negligible remainders. Thus the exponent C can be fixed before choosing λ, h, h' . In applications choose the large-discriminant h'/h first, then λ , and finally the tiny-cap step. $\square$

9.3 Finite configurations and compatible names

Let $\ell_a \sim s^a$ and $R_v \sim s^v$ be dyadically rounded. The time T_a is the left endpoint of the ancestor interval at depth a . Denote the lower corner of the dyadic angular R_v -cell by A_v . Define the phase name

$$C(a, v) = \left(T_a, A_v, [b(T_a)]_{\ell_a R_v}, [y(T_a) - 2A_v \cdot b(T_a)]_{\ell_a R_v^2} \right). \quad (\text{grid})$$

The brackets mean coordinatewise grid cells. For a wave row measured at a later cut, the positions in this definition are obtained by backflow along that row's own ray. This definition makes no assertion that a ray is unchanged by synthesis. The wave domain is $0 < a < 1$, $0 \leq v \leq u(a)$; classically we use $v = 0$.

Lemma 9.6 (Selected laws and rarity transfer). *For any fixed finite rational list of cuts and names, one can choose the saturated configuration in forward order so that the following properties hold to arbitrarily small exponent errors. At every requested subterminal cut a , the selected raw mass and the post-synthesis energy both have size E_a . The raw rows have intervalwise regularity at most $s^{-o(1)} \kappa_a$ and are carried on peeling labels with*

$$\sum_{\lambda_a} w_{\lambda_a} \leq s^{-o(1)} E_a / \kappa_a. \quad (9.4)$$

Write $\mathbf{P}^a$ for their normalized raw row law. If a portion tested at a later selected cut l can only be reached through a set of relative raw mass q at $i < l$, then its relative mass at l is at most $s^{-o(1)} q^{2/3}$, up to negligible error. Consequently a power-rare necessary predecessor remains power-rare at the later law. For classical rays the same conclusion holds at the final assigned law $\mathbf{P}^1$.

Proof. At each successive cut insert a regularity peel on plateau-analyzed rows, select an ordinary level which preserves the original terminal certificate under free continuation, and then synthesize. Equation (CE), applied with the fixed finite list, gives Equation (9.2) and saturation of the raw and post-synthesis budgets. The full tests of the selected level give Equation (9.4). Further preserving subdivisions need only stay inside these tests; their individual submasses need not

saturate each test. The choices are made before later insertions, so the certificate with free continuation remains available at every chosen cut.

Restrict the i -rows to the necessary predecessor set and synthesize. The resulting states have total energy at most $s^{-o(1)}qE_i$ and regularity at most $s^{-o(1)}\kappa_i$. The upper cube bound on the segment gives

$$\sum_{\lambda} m_{\lambda}^{3/2} w_{\lambda}^{-1/2} \leq s^{-o(1)} q s^{(1/2-\Gamma)(l-i)} E_i \sqrt{\kappa_i} = s^{-o(1)} q E_l \sqrt{\kappa_l}.$$

The identity uses Equation (9.1). Holder's inequality and Equation (9.4) yield

$$\sum_{\lambda} m_{\lambda} \leq \left(\sum_{\lambda} m_{\lambda}^{3/2} w_{\lambda}^{-1/2} \right)^{2/3} \left(\sum_{\lambda} w_{\lambda} \right)^{1/3} \leq s^{-o(1)} q^{2/3} E_l.$$

Nonconnections from the complementary rows are negligible by successive kernel truncations and complementary projections, using Equation (acc). In the classical case the restriction is positive and raywise, so the same proof applies through the final assignment. All tested families are polynomial on each inner sequence; very small components and tail errors are handled before normalization relative to the original energy. $\square$

We will use deterministic log profiles of finitely many discrete names. Their construction is worth specifying. At a measuring cut a , let $\mathbf{P}_{\text{pre}}^a$ be the normalized raw row law from Lemma 9.6, with all earlier choices fixed but before this additional restriction. To *preflatten* a fixed finite list of names and joint names, partition rows by sufficiently fine logarithmic bins of their probabilities at the sampled values, all computed under $\mathbf{P}_{\text{pre}}^a$. Polynomial cardinalities make the mass with probabilities below a sufficiently small power negligible. There are only a subpower number of retained logarithmic classes. The coherent splitting inequality gives one common class B_a preserving the terminal exponent, and Lemma 9.6 forces $\mathbf{P}_{\text{pre}}^a(B_a) = s^{o(1)}$. Below, $\mathbf{P}^a$ denotes the selected law $\mathbf{P}_{\text{pre}}^a(\cdot \mid B_a)$; at any later preflattening step, its pre-restriction law already includes the earlier fixed choices. If a name has mass $s^{D_*+o(1)}$ before restriction, it has at most $s^{-D_*-o(1)}$ possible values in this class, and each has mass at most $s^{D_*-o(1)}$ under the normalized selected law. Counting the values whose mass is too small shows that its surprise exponent equals $D_* + o(1)$ with probability tending to one. The same argument for joint names gives conditional scores. This conclusion persists after any further restriction of subpower probability, by likelihood comparison for the numerator and denominator. At a fixed strict tolerance the exceptional probabilities can be made power-small by taking the earlier approximations sufficiently accurate.

Lemma 9.7 (Stability of ancestor names). *At a wave cut a , preflatten $C(a, v)$ and $(C(a, v), A_w)$, with $v \leq w \leq u(a)$. At a later sampled cut b with $w \leq u(b)$, the same deterministic exponents hold for their backevaluated names, including after further subpower restrictions of $\mathbf{P}^b$.*

Proof. On a retained correspondence path, the angular discrepancy between the two rows is at most $s^{u(b)-o(1)}$. Backevaluating at T_a gives horizontal error at most $\ell_a s^{u(b)-o(1)}$ and vertical error, tilted by the earlier direction, at most $\ell_a s^{2u(b)-o(1)}$. These statements follow by adding the position errors in Equation (acc) and the linear and quadratic errors from changing the velocity during backflow. The same bounds with the longer time unit apply to earlier anchors. Since $v, w \leq u(b)$, a joint name at either end determines at most $s^{-o(1)}$ possible joint names at the other end. When an angular center changes, recompute its vertical tilt from the horizontal

coordinate: the residual uncertainty is the horizontal grid width times the angular uncertainty, within the stated vertical width.

The earlier name count therefore bounds the later support, apart from negligible probability. A list of too few later names pulls back to a power-rare list at a , because of the individual upper mass bound there. Lemma 9.6 excludes a nonnegligible later probability on such a list. The count bound excludes probabilities which are too small. Apply this to both the root and joint names to get conditional scores. Subpower restrictions preserve these deterministic limits by the preceding likelihood argument. On a boundary use slightly coarser widths and then subpower neighboring lists. $\square$

9.4 Entropy profiles and time spread

For a discrete name U and side data D_* write $i_s(U \mid D_*) = -\log \mathbf{P}(U \mid D_*) / \log(1/s)$, evaluated at the random sample. Side data may be nonatomic; regular conditional probabilities are used. Equation (LP) and its finite-list versions will be applied to these scores.

Proposition 9.8 (Profiles). *Passing to finite-list subsequences and then a diagonal, there are Lipschitz profiles*

$$\begin{aligned} D(a, v) &= \lim \left(i_s(C(a, v)) - \log_{1/s}(e/\kappa) \right), \\ J(a, v, w) &= \lim i_s(A_w \mid C(a, v)), \quad v \leq w \leq u(a), \end{aligned} \tag{prof}$$

in probability on each of the valid measuring laws. For classical rays only $D(a, 0)$ and $J(a, 0, w)$, $0 \leq w \leq 1$, are used. The wave excess satisfies

$$E(a, v) := D(a, v) - 4v - (p + d)a \geq 0, \quad E(a, u(a)) = 0. \tag{excess}$$

Classically $D(a, 0) \geq (p + d)a$ and $D(1, 0) \leq p + d$. Furthermore $0 \leq J(a, v, w) \leq 2(w - v)$, J is nonincreasing in a for fixed v, w , and in the wave case

$$J(a, v, w) \geq J(a, v, x) + J(a, x, w), \quad v < x < w. \tag{9.5}$$

Every bounded-branching score increment in these assertions has the same limit in expectation as in probability.

Proof. A cell $C(a, v)$ restricts one interval to a subpower number of round tests of radius s^v , hence has raw mass at most $s^{-o(1)} \kappa_a s^{4v}$. Division by E_a gives the lower bound for D in Equation (excess). At the wave boundary, Equation (round-u) makes every peeling label determine $C(a, u(a))$ to a subpower list; its weight is $s^{4u(a)+o(1)}$. Equation (9.4) gives the reverse count bound. Classically, transfer too-sparse lists to the final law with Lemma 9.6; the final weight count gives the reverse inequality at $a = 1$.

Increasing v refines the name up to bounded ambiguity. An increment δv costs at most $6\delta v$ branching exponents: two angular, two horizontal, and two vertical, with the tilt recalculated from the horizontal position. Increasing a at fixed v likewise refines the name, by backflow within the angular cell. It costs at most $4\delta a$ exponents, one for time and three for position. The residual tilted vertical velocity is quadratic in $A - A_v$, which is exactly what is needed for this backflow comparison. Adding A_w costs at most two exponents per unit increase of w . Compare names on a common later measuring law using Lemma 9.7; near the domain boundary change widths first

and then the cut. These comparisons give uniform Lipschitz bounds for the shifted profiles and their increments.

Conditional entropy decreases under refinement of the conditioning name, giving monotonicity in a . Since $C(a, x)$ knows $C(a, v)$ and A_x up to bounded lists, the entropy chain rule gives Equation (9.5). All entropy comparisons use bounded-branching increments. Their tails are uniformly integrable: if a discrete variable has at most s^{-B} possible refinements, the probability of surprise exceeding $B + t$ is at most s^t . Thus their expected and typical limiting scores agree, without requiring a uniform bound on a common unconditional offset.

On each finite rational list choose increasingly accurate saturation and preflattening data, then take a subsequence on the bounded shifted scores. The Lipschitz bounds give a compatible diagonal and unique continuous extensions. A finite real query can subsequently be replaced by rational parameters with arbitrarily small error in profile values. This procedure never requires uniformity in the number of actual gates. Extra cuts used to test free continuation from a selected earlier state leave that earlier state and its established profiles in place. $\square$

Lemma 9.9 (Classical angular memory). *For classical rays, at any fixed root name $C(a, 0)$ and any $0 < w < 1$, the typical conditional surprise of A_w is at least $(3 - p)w - o(1)$. Consequently*

$$m := 3 - p = 2 - 2d + 2\Gamma \in (0, 2]. \quad (9.6)$$

Proof. Use the regularized cut $i = 1 - w$. Fix a terminal label and an angular s^w -cell. Their preimage at this cut lies in horizontal widths $\ell_i s^{w-o(1)}$ and tilted vertical width $\ell_i s^{w-o(1)}$, including the terminal position error. It is therefore covered by $s^{-w-o(1)}$ round boxes of radius s^w , of total weight $s^{3w-o(1)}$. Positive transport and cut regularity bound this mass by $\kappa_i s^{3w-o(1)}$. Terminal labels of assigned mass less than $\kappa_1 s^\zeta$ have power-small total mass, for every fixed $\zeta > 0$, by Equation (9.4). Dividing by the label mass and using $\kappa_i/\kappa_1 = s^{-pw+o(1)}$ gives the conditional angular bound given the terminal label. Such a label determines any $C(a, 0)$ to a subpower list by exact backflow; applying the same sparse-list bound gives the assertion given $C(a, 0)$. Equation (9.1) gives the expression for m , and the two-dimensional angular count gives $m \leq 2$. $\square$

Fix a finite tangent query satisfying

$$0 < a < a + Kh \leq b, \quad K = 3, \quad v + Kh < u(b) \quad \text{for waves.} \quad (\text{query})$$

Here $b < 1$ for waves, whereas $b = 1$ and $v = 0$ classically. Set

$$C = C(a, v), \quad C' = C(a + Kh, v), \quad Z = (A - A_v)/R_v, \quad \theta = (T_b - T_a)/\ell_a, \quad \rho = s^h.$$

Let W be the position at T_a , after the central tilt and subtraction of the bin origins of C , divided by its two horizontal and its vertical grid lengths. Then Z, W, θ are bounded, and the position evolves in these coordinates with velocity $V(Z) = (Z, |Z|^2)$. Subscripts below mean ordinary coordinate bins at width ρ^z . Put $Y_\# = (Z, W)_K$.

Lemma 9.10 (Time spread). *For every fixed finite list $0 < z \leq 1$,*

$$i_\rho(\theta_z \mid C, Y_\#) \geq dz - o(1) \quad (9.7)$$

typically. The statement is power-robust at each strict tolerance, including after subpower restrictions of the measuring law.

Proof. Use the actual peel cut $l = a + zh$. Consider a predicted list of at most $\rho^{-dz+\zeta}$ time cells for each $(C, Y_\#)$, where $\zeta > 0$. Use Lemma 3.19 to take a Borel inner version of the tested list event at b , mark the l -rows that have a continuation to it, and retain a Borel inner version of that analytic mark under the full analyzed l -row energy law, inside the current end-state support. For input restriction, take the Borel outer hull of all predecessors under the unrestricted truncated a -to- l geometric correspondence, with the same full analyzed a -row energy. A selected row at a reaches at most $s^{-o(1)}$ possible values of $(C, Y_\#)$ on any such continuation: concatenate the correspondences and use the strict uncertainty slack in Equation (query). Changes of tilt and bin origins are determined by the available coordinates up to the same small lists. A time cell and T_a determine T_l to bounded ambiguity. Therefore this earlier row can feed the marked set at at most $s^{-o(1)}\rho^{-dz+\zeta}$ different T_l . Contraction separately for each such interval bounds total marked raw mass by

$$s^{-o(1)}\rho^{-dz+\zeta}E_a = s^{-o(1)}\rho^\zeta E_l.$$

Complementary nonconnections are negligible. Transfer the marked rarity from l to b with Lemma 9.6. Testing the popular conditional time cells proves Equation (9.7). The same proof with exact positive paths applies classically. Earlier errors can be made smaller than any fixed fraction of $h\zeta$, so the conclusion persists after a subpower restriction. $\square$

Lemma 9.11 (Removal of conditional plane concentration). *At the query in Equation (query), fix $\varepsilon > 0$. In the wave case one may restrict $\mathbf{P}^b$ by subpower mass; in the classical case no additional restriction is necessary. The resulting laws have all the preceding robust score properties and satisfy, typically in (C, C') ,*

$$\sup_{\Pi} \mathbf{P}\{\text{dist}(V(Z), \Pi) \leq \rho^\varepsilon \mid C, C'\} \longrightarrow 0, \quad (9.8)$$

where Π ranges over affine planes in $\mathbb{R}^3$. These choices are made before any conditional product sampling.

Proof. Classically choose one measurable candidate slab for each name pair. The terminal label determines that pair to subpower lists. At $i = 1 - h' > a + Kh$, partition the actual regularized input rays by the pair and restrict to its specified slab. Use the two possible step lengths in Proposition 9.5, one for each discriminant range, including both cuts in the finite list. Choose the large-discriminant h'/h first; then choose λ so small that P^C costs less than a fixed part of $(\Gamma - 4\eta)h'$; finally choose the tiny-cap step. Each case gives a strict gain over exponent Γ , so for some $\sigma > 0$, possibly different for the two steps,

$$\sum m_{\lambda, \text{hit}}^{3/2} \leq \sum_{h'} s^{-o(1)} s^{(1/2-\Gamma+\sigma)h'} E_i \sqrt{\kappa_i}.$$

Weights are $s^{o(1)}$ in these normalized units. Equation (CE) and the terminal weight count force the hit probability to be power-small. The estimate is uniform in the candidate slabs. Taking measurable near maximizers, or a fine finite net of their bounded coefficients with a slightly enlarged width, proves Equation (9.8).

For waves take $v > 0$ in the interior and put $j = 1 - 2v$, so $b < j < 1$. Ignore future planned insertions after b and insert one peel at j on its free preserving continuation. Its raw assignment family saturates the segment bound and has $r_j/R_v = s^{o(1)}$, $f_j = s^{o(1)}$ and the weight count of Lemma 9.6. Keep its assignment multipliers fixed when comparing synthesized contributions.

For each (C, C') , greedily remove disjoint portions of the raw b -rows contained in a slab and having mass greater than ρ^τ relative to the original conditional law. There are at most $\rho^{-\tau}$ portions for each pair. Stop when every slab has residual original mass at most ρ^τ . The plane choices use a countable dense set and an arbitrarily small enlargement of their widths. Take their joint Borel versions under the full analyzed $(C, C', b\text{-row})$ energy law, as in Lemma 3.19, before any later path restriction.

Split removed portions before synthesis by their pair and their plane. A label at j receives contributions from only $s^{-o(1)}$ name pairs: its angular width is $R_v s^{o(1)}$, and Equation (acc) backflows its positions to the two ancestor grids with small lists. Coherent recombination of the removed contributions at that label therefore costs at most $s^{-o(1)} \rho^{-O(\tau)}$. Indeed, after freezing its assignment operator, if $v_{\lambda, \nu}$ are the analyzed contributions and $N_\lambda \leq s^{-o(1)} \rho^{-\tau}$ is their number, then

$$\left\| \sum_{\nu} v_{\lambda, \nu} \right\|_2^3 \leq N_\lambda^2 \sum_{\nu} \|v_{\lambda, \nu}\|_2^3.$$

Nonconnecting portions are negligible by complementary projections.

For one removed component, insert a regularity split at $i = j - h' > b$, using the appropriate step from Proposition 9.5. Its input at b has regularity $s^{-o(1)} \kappa_b$ and energy charged to its raw row mass e_{comp} . A level of raw mass z_i and threshold $\tilde{\kappa}_i$ satisfies

$$z_i \sqrt{\tilde{\kappa}_i} \leq s^{(1/2-\Gamma)(i-b)-o(1)} e_{\text{comp}} \sqrt{\kappa_b}$$

by the prefix cube upper bound. The low-floor remainder obeys the same estimate when the floor is chosen sufficiently small. Its synthesis has regularity at most $s^{-o(1)} \tilde{\kappa}_i$, with summed energy charged to z_i . In Z -coordinates, synthesis at b and at i broadens angular spectral support only by $O(d_{\ell_b} + d_{\ell_i})/R_v$. Since $H/(\ell_i R_v^2) = \Delta s^{o(1)}$, this is within the $s^{-o(1)} \sqrt{\Delta}$ thickening permitted in Proposition 9.5. Changing angular units multiplies input regularity by R_v^4 , canceling the ending weight. Relative to the ordinary free bound on the step $i \rightarrow j$, the tiny-cap branch gains $\Gamma h'$, while the transverse branch gains $(\Gamma - 4\eta)h' - C\lambda h$, for a fixed C absorbing the P^C factors. Both are positive by the choice of h' and then λ . The prefix $b \rightarrow i$ contributes only the baseline factor $s^{(1/2-\Gamma)(i-b)-o(1)}$, so each branch retains its strict gain over the full $b \rightarrow j$ comparison. Sum the components, whose raw budgets sum to at most $s^{-o(1)} E_b$, and the logarithmic levels. Choose τ after the two strict gains so that the recombination factor cannot absorb them. The removed field is then power-insufficient for the fixed j -certificate.

The residual field must still have the full exponent of this certificate. The ordinary cube upper bound applied to its synthesis at b , with inherited regularity, forces its raw mass to be $s^{o(1)} E_b$. Under the normalized residual law, the conditional surviving fraction in a name pair is typically at least $\rho^{\tau/2}$: pairs with smaller surviving fraction contribute at most $\rho^{\tau/2}$ of the original total mass, which is negligible relative to the subpower residual budget. Every conditional residual plane mass is therefore at most $\rho^{\tau/2}$ on the typical pairs. This proves Equation (9.8). Deterministic profiles and Lemma 9.10 survive by their subpower robustness. The removal need only be performed for the one finite tangent query under consideration. $\square$

9.5 Differentiation and stationarity of the angular profile

Lemma 9.12 (Diagonal differentiation of an interval function). *Let $J(x, y)$ be continuous for $x \leq y$ in a compact interval, with $0 \leq J(x, y) \leq M(y - x)$ and $J(x, z) \geq J(x, y) + J(y, z)$. Then*

there is a measurable $m \in [0, M]$ such that

$$\lim_{r \downarrow 0} \frac{J(x, x+r)}{r} = m(x) \quad \text{for almost every } x.$$

Applied to the wave profiles, there is a jointly measurable version $m(a, v) \in [0, 2]$, nonincreasing in a for fixed v , for which

$$\frac{J(a, v, v+zh)}{h} \rightarrow zm(a, v) \quad (9.9)$$

at almost every interior (a, v) , for each fixed $z > 0$.

Proof. Define

$$\mu([x, y]) = \inf_{\mathcal{P}} \sum_{[u, w] \in \mathcal{P}} J(u, w),$$

where $\mathcal{P}$ runs over finite partitions of $[x, y]$. Joining two partitions proves one direction of additivity. For the other, insert the joining point into any partition; by superadditivity this can only decrease its sum. Thus μ is additive and $0 \leq \mu([x, y]) \leq M(y - x)$. Its cumulative function is nondecreasing and M -Lipschitz, so $\mu([x, y]) = \int_x^y m(t) dt$ for some $0 \leq m \leq M$.

Set $K(x, y) = J(x, y) - \mu([x, y])$. It is nonnegative and superadditive, and its partition-sum infimum on every interval is zero. Fix $\epsilon_0 > 0$ and consider $S = \{x : \limsup_{r \downarrow 0} K(x, x+r)/r > \epsilon_0\}$. For any $\delta_0 > 0$, choose a finite partition whose sum of K is less than δ_0 . Away from its finitely many endpoints, the intervals $[x, x+r]$ lying in the same partition cell and satisfying $K(x, x+r) > \epsilon_0 r$ form a Vitali cover of S . A disjoint countable subfamily covers S up to a null set. Superadditivity and nonnegativity, first for finite subfamilies inside each cell, bound the sum of their K values by δ_0 . Hence $|S| \leq \delta_0/\epsilon_0$. Let $\delta_0 \downarrow 0$, then use a countable sequence of $\epsilon_0 \downarrow 0$. The excess $K(x, x+r)/r$ tends to zero almost everywhere. Lebesgue differentiation of μ proves the first assertion with the full $r \downarrow 0$ limit.

For the parameterized profiles define, on their common domain,

$$m(a, v) = \limsup_{\substack{r \downarrow 0 \\ r \in \mathbb{Q}}} \frac{J(a, v, v+r)}{r}.$$

Continuity makes this jointly measurable, and monotonicity of J in a makes this version nonincreasing in a pointwise. The first assertion and Fubini identify it with the diagonal derivative almost everywhere. Substituting $r = zh$ gives Equation (9.9). $\square$

Lemma 9.13 (Dyadic stationarity). *At almost every interior wave point, and almost every classical depth a with $v = 0$, along $h = 2^{-n} \downarrow 0$ one has*

$$J(a, v, v+Kh) - J(a+Kh, v, v+Kh) = o(h). \quad (\text{stat})$$

Proof. Work on a compact rectangle inside the validity domain and let $\delta = Kh$. For fixed v , the function $f(a) = J(a, v, v+\delta)$ is nonincreasing and lies between zero and 2δ . Consequently

$$\int_{a_0}^{a_1} (f(a) - f(a+\delta)) da = \int_{a_0}^{a_0+\delta} f(a) da - \int_{a_1}^{a_1+\delta} f(a) da \leq 2\delta^2.$$

After integration in v , the integral of the nonnegative left side of Equation (stat), divided by h , is $O(h)$. Its sum over dyadic h is finite, so Tonelli's theorem implies convergence to zero almost everywhere. A countable exhaustion by compact rectangles proves the wave statement. For classical $v = 0$ omit the v integral. $\square$

Choose a point where the applicable conclusions of Lemmas 9.12 and 9.13 hold and D is differentiable. Write $o_T = \partial_a D(a, v)$. Classically use instead the constant m of Equation (9.6). For each sufficiently small dyadic h , choose the finite configuration of Equation (query), and perform Lemma 9.11 before sampling additional rows. All profile approximations at this query are taken before letting h tend to zero. Rational perturbations of size $o(h)$ are permitted by the Lipschitz profile bounds.

There is a bounded representative $O \in \mathbb{R}^4$, determined by (C, C') , such that

$$O = (0, W) + \theta e(Z) + O(\rho^K), \quad e(Z) = (1, Z, |Z|^2). \quad (9.10)$$

Take the spacetime position at the finer ancestor and its time, expressed in the moving frame of C . Its discrepancy from the later row position is at most $O(\rho^K)$, since both the time discrepancy and the centered velocity are bounded at that scale. Given C , the unit-depth name O_1 and $C(a + h, v)$ determine one another to bounded lists: move between their times by at most $O(\rho)$ with bounded centered velocity. Thus

$$i_\rho(O_1 \mid C) \longrightarrow o_T, \quad \frac{H(O_1 \mid C)}{\log(1/\rho)} \longrightarrow o_T. \quad (9.11)$$

Here $H(\cdot \mid \cdot)$ denotes Shannon conditional entropy. The angular statements give, for every needed finite $0 < z \leq 1$,

$$i_\rho(Z_z \mid C) \geq mz - o(1). \quad (9.12)$$

For waves the expected score has limit mz . Moreover

$$\frac{I(Z_K; C' \mid C)}{\log(1/\rho)} = o(1). \quad (9.13)$$

Indeed C' determines C to bounded ambiguity, Z_K is equivalent given the roots to A_{v+Kh} , and the two conditional angular entropies differ by the left side of Equation (stat), in s -units. Both profiles remain valid on the residual law by Lemma 9.7. Equation (LP), including the small-information version, therefore preserves Equation (9.12) after conditioning on C' .

Lemma 9.14 (Width derivative). *At almost every interior wave point,*

$$\partial_v D \leq m + o_T - d + 1. \quad (\text{rv})$$

Proof. Given C , the bins $(Z, W)_1$ determine $C(a, v + h)$ except for at most $O(\rho^{-1})$ possibilities. The extra factor is exactly the further vertical precision: the new vertical grid is ρ^2 in the old units, while W_1 has width ρ . The new angular center and the known horizontal bin determine its change of tilt at the required accuracy.

On the other hand, adjoining θ_1 to $(Z, W)_1$ adds conditional entropy at least $(d - o(1)) \log(1/\rho)$. This follows from Lemma 9.10, since $Y_\#$ is a finer name than $(Z, W)_1$, and from entropy monotonicity. Equation (9.10) shows that $(Z, W, \theta)_1$ is determined to bounded lists by (Z_1, O_1) given C . Its entropy is therefore at most $(m + o_T + o(1)) \log(1/\rho)$. Subtract the time contribution, add the one vertical branching exponent, and use the tangent increment of $D(a, v + h) - D(a, v)$. This proves Equation (rv). $\square$

9.6 The outgoing entropy inequality

Proposition 9.15 (Outgoing entropy). *Use the exponent data d, p, Γ of Equation (9.1). For almost every classical depth $a \in (0, 1)$ at which $D(a, 0)$ is differentiable and Lemma 9.13 applies, put $o_T = \partial_a D(a, 0)$ and $m = 3 - p$. For waves, let (a, v) be almost every interior point with $0 < a < 1$ and $0 < v < (1 - a)/2$ at which D is differentiable and Lemmas 9.12 and 9.13 apply, and put $o_T = \partial_a D(a, v)$ and $m = m(a, v)$. In either case,*

$$o_T \geq 4d + \min(1 - d, m/2). \quad (\text{round-out})$$

Proof. Fix the finite incidence queries required by Equations (inc)–(cond) for a prescribed strict final loss. Later we choose the geometric and score tolerances small compared with these queries. Given (C, C') , sample M independent copies of (Z, W, θ) , all sharing the representative O . Let $L_i = (Z_i)_K$ be their fine angular names, represented by bounded grid points, and put $e_i = e(L_i)$. The integer M will be fixed before taking the tangent limit.

The time-spread estimate survives conditioning on all these pins. The own pin is already part of $Y_{\#,i}$, and conditional independence gives

$$\begin{aligned} I(\mathbf{L}_{\neq i}; Y_{\#,i}, \theta_i \mid C) &\leq I(\mathbf{L}_{\neq i}; C' \mid C) \\ &\leq \sum_{j \neq i} I(L_j; C' \mid C). \end{aligned}$$

For clarity, the second inequality follows by writing the mutual information as conditional entropy minus entropy given C, C' : the former is at most the sum of the marginal entropies, and the latter equals their sum by conditional independence. Equation (9.13) makes the whole information cost $o(\log(1/\rho))$ for fixed M . Thus Lemma 9.10 and Equation (LP) give the required time scores even with these pins. The angular score also satisfies, typically,

$$i_\rho((Z_k)_z \mid C, C', \mathbf{L}_{\neq k}) \geq mz - o(1), \quad (9.14)$$

because the conditional law of copy k is unchanged by the other copies when C, C' are given.

Use the law already obtained from Lemma 9.11, with a sufficiently small $\varepsilon > 0$. With probability tending to one, every four distinct e_i have determinant of magnitude at least $\rho^{C\varepsilon}$, and every pair of angular representatives has separation at least $\rho^{C\varepsilon}$. To check this, sample successively in the first-coordinate-one affine slice of $\mathbb{R}^4$. The affine span of the preceding points is contained in an affine plane in its three remaining coordinates. The probability that the next point is within ρ^ε of such a plane tends to zero, uniformly in the previously chosen points. Successive distances and bounded vector lengths give the determinant bound. A plane through one preceding point also bounds the probability of a close pair; V is Lipschitz on the bounded angular range. Pin rounding by $O(\rho^K)$ is negligible. There are finitely many tuples, so all the conditions hold simultaneously. We call this the transverse event.

Four transverse pins already give the lower bound $4d$, for every $m \geq 0$. Choose any four of the sampled pins and apply Gram–Schmidt in their order to produce an orthonormal basis (f_1, f_2, f_3, f_4) , renumbering these pins from 1 to 4 within this calculation. Reveal the bins of $O \cdot f_i$ in reverse order. At step i , adjoin $Y_{\#,i}$ to the conditioning data. It predicts all the previously read components to bounded ambiguity, since their directions are perpendicular to e_i and Equation (9.10) eliminates their time dependence. The new component determines θ_i to at most $\rho^{-C\varepsilon}$ cells: its coefficient $e_i \cdot f_i$ is bounded below by $\rho^{C\varepsilon}$. The time score with all pins consequently contributes at least $d - C\varepsilon - o(1)$ at each step. The chain rule, the list bounds,

and Equation (LP) show that the score of O_1 is at least $4d - C\varepsilon - o(1)$. The change from the orthonormal bins to ordinary O_1 bins has bounded multiplicity given the pins. Drop the pins by Equation (LP) and use Equation (9.11); then let $\varepsilon \downarrow 0$. This proves Equation (round-out) when $m = 0$.

When $m > 0$, we sharpen two of these four time contributions by projecting onto a plane. For $i < j$ let π_{ij} be orthogonal projection onto the two-dimensional quotient by $\text{span}(e_i, e_j)$, with measurable orthonormal coordinates, and define the projective direction

$$g_{ij}(Z) = [\pi_{ij}e(Z)].$$

Only values on the transverse event matter, so the definition at zero may be arbitrary. Color each triple $i < j < k$ by the vector of quantized scores

$$i_\rho((g_{ij}(Z_k))_z \mid C, C', \mathbf{L}_{\neq k})$$

at the finitely many required widths, with color intervals of length ε . Use one overflow color. It is avoided with probability tending to one, since a bounded projective chart has at most $O(\rho^{-z})$ cells and counting controls the upper tail of its surprise. There are a fixed finite number of colors. The same finite Ramsey theorem [24, Theorem B, p. 267] therefore supplies, for sufficiently large fixed M , four indices whose triples all have the same color vector.

Here is the geometric score constraint on such a quadruple. Let $i < j < l < k$. Given the two directions $g_{ij}(Z_k)$ and $g_{il}(Z_k)$ to width ρ^z , and all pins except L_k , the number of possible good $(Z_k)_z$ cells is at most $\rho^{-C\varepsilon}$. Fix one compatible transverse witness Z_0 . The first directional constraint places $e(Z)$ within $O(\rho^{z-C\varepsilon})$ of $\text{span}(e_i, e_j, e(Z_0))$, and the second does the same for $\text{span}(e_i, e_l, e(Z_0))$. The determinant lower bound controls the inverse linear algebra, so their intersection confines $e(Z)$ within $O(\rho^{z-C\varepsilon})$ of $\text{span}(e_i, e(Z_0))$. In the first-coordinate-one slice this is the secant line

$$(1 - t)e(L_i) + te(Z_0).$$

Its quadratic null residual is $t(1 - t)|L_i - Z_0|^2$ (up to the harmless choice of sign). The roots 0, 1 are separated and their derivatives have magnitude at least $\rho^{C\varepsilon}$. Boundedness and the pair separation therefore leave only neighborhoods of radius $O(\rho^{z-C\varepsilon})$ of the two endpoints. Each contains at most $\rho^{-C\varepsilon}$ angular ρ^z -cells. This proves the list bound, also when $z \leq C\varepsilon$, by the trivial ambient angular count.

Apply this list bound and Equation (9.14) with the same conditioning data $C, C', \mathbf{L}_{\neq k}$. The joint score of the two directions is at least $mz - C\varepsilon - o(1)$. Their marginal score sum is at least their joint score up to $o(1)$ outside a vanishing exceptional set, by Equation (LP) and the chain rule. Thus

$$i_\rho((g_{ij}(Z_k))_z \mid C, C', \mathbf{L}_{\neq k}) + i_\rho((g_{il}(Z_k))_z \mid C, C', \mathbf{L}_{\neq k}) \geq mz - C\varepsilon - o(1). \quad (9.15)$$

On a monochromatic quadruple these two scores differ by at most ε . Hence both are at least $(m/2)z - C\varepsilon - o(1)$ simultaneously at every required width. There are finitely many possible quadruples; after passing to a subsequence, select fixed indices $i < j < k$ and an event of fixed positive probability on which all these bounds hold. This restriction has probability bounded below independently of ρ after the parameters are fixed, so the finite score bounds and the small-information consequences survive by likelihood comparison. Retain transversality as well.

Use the side data $D_* = (C, \mathbf{L}_{\neq k})$, the point $Q = \pi_{ij}O$, and the line $\mathcal{L}$ through $\pi_{ij}(0, W_k)$ in direction $\pi_{ij}e(Z_k)$. They are incident up to $O(\rho^K)$. Restrict to a positive-probability bounded

slope chart after a fixed rotation; a finite collection of such charts covers all directions. We verify Equation (cond) with exponents d and $m/2$.

For the point score, adjoin $Y_{\#,k}$. This determines the line at the required conditioning resolution to at most $\rho^{-C\varepsilon}$ choices, by transversality and fine pinning. Conditioning additionally on its line cell therefore costs at most $C\varepsilon + o(1)$ in score. Given these data, the projected point determines θ_k at width $\rho^{z'}$ to at most $\rho^{-C\varepsilon}$ possibilities, because the projected speed is at least $\rho^{C\varepsilon}$. Lemma 9.10, its preservation under the other pins, and Equation (LP) give

$$i_\rho(Q_{z'} \mid \mathcal{L}_z, D_*) \geq dz' - C\varepsilon - o(1), \quad z' \leq z.$$

For the line score, adjoin C' . The point is then known, and the line cell determines its direction at the same resolution, up to the chart's controlled lists. The directional lower bound from Equation (9.15), with the same conditioning $C, C', \mathbf{L}_{\neq k}$, gives

$$i_\rho(\mathcal{L}_{z'} \mid Q_z, D_*) \geq (m/2)z' - C\varepsilon - o(1).$$

These are exactly the finite-grid hypotheses of Equation (cond). Choose ε and the earlier tolerances sufficiently small. Equation (inc) yields

$$i_\rho(Q_1 \mid D_*) \geq d + \min(1, d + m/2) - o(1) \tag{9.16}$$

to any prescribed strict loss.

It remains to recover the other two coordinates of O . Let f_i be the unit vector along e_i , and let f_j be the unit vector along the component of e_j perpendicular to e_i . After Q_1 , reveal $(O \cdot f_j)_1$, then $(O \cdot f_i)_1$. For the former, adjoin $Y_{\#,j}$; it predicts Q_1 to bounded ambiguity because $\pi_{ij}e_j = 0$, and the new coordinate reveals θ_j to $\rho^{-C\varepsilon}$ choices. For the latter, adjoin $Y_{\#,i}$; it predicts both preceding coordinates because they annihilate e_i , and this last coordinate reveals θ_i . Each conditional score is therefore at least $d - C\varepsilon - o(1)$ by the time estimate with the pin data. All these conditioning variables are available: D_* includes the own pins L_i, L_j , and the information argument at the start of the proof permits the other pins.

Combining these two contributions with Equation (9.16) and then dropping D_* down to C by Equation (LP) gives

$$i_\rho(O_1 \mid C) \geq 3d + \min(1, d + m/2) - o(1) = 4d + \min(1 - d, m/2) - o(1)$$

on the retained event of fixed positive probability. Its deterministic limit in Equation (9.11) forces the asserted inequality.

The order of choices is essential. Fix the desired final strict loss and the finite grids in Equation (inc); next choose the geometric and color tolerances; next fix a Ramsey-sufficient M ; then let dyadic $h \downarrow 0$ through the typical point, taking all previous finite-list and inner limits sufficiently accurate for that h . Every restriction made after the stationarity estimate has a positive probability bound at these fixed choices. No arbitrary subpower restriction is used to preserve a mutual-information assertion. $\square$

9.7 The differential contradiction

Lemma 9.16 (A two-direction descent). *Let $\Omega = \{(t, v) : 0 < t < 1, 0 < v < t/2\}$. Suppose that E is locally Lipschitz, nonnegative, and has zero trace on $v = t/2$. Let m be measurable and nondecreasing in t at each fixed v , with this condition allowed outside a null set of v . If $c > 0$ and $q \in \mathbb{R}$, the following pair of almost everywhere inequalities is impossible:*

$$E_t + \frac{1}{2}E_v \leq -c \quad \text{where } m \leq q, \quad E_t \leq -c \quad \text{where } m \geq q.$$

Proof. Fix $0 < t_0 < t_1 < 1$, put $T = t_1 - t_0$, and let N be the null set where differentiation or an applicable inequality fails. Fubini after the affine change $(t, \delta) \mapsto (t, t/2 - \delta)$ shows that almost every line $\gamma_\delta(t) = (t, t/2 - \delta)$ meets N in a null set. Ordinary Fubini gives the same property for almost every vertical section. Include exceptional monotonicity sections in the latter null set. Choose a good $\delta > 0$ sufficiently small that $\delta < t_0/4$ and $E_0 := E(t_0, t_0/2 - \delta) < cT/2$; the trace condition permits this. The function $f(t) = E(\gamma_\delta(t))$ is Lipschitz on $[t_0, t_1]$; let B be a Lipschitz constant.

Set $A = \{t : m(\gamma_\delta(t)) > q\}$. If $|A| = 0$, the first inequality integrates to $f(t_1) \leq E_0 - cT < 0$. Otherwise let τ be the essential first time in A . Then $\tau < t_1$, A has zero measure before τ , and every sufficiently short interval immediately after τ has positive intersection with A . Integration before τ gives $f(\tau) \leq E_0 - c(\tau - t_0)$. Choose $t_* \in A$ arbitrarily close to τ from above, with $v_* = t_*/2 - \delta$ a good vertical section. Such a choice exists because this affine map of t_* preserves null sets. Monotonicity gives $m(t, v_*) > q$ for $t \geq t_*$, so the second inequality integrates along this vertical segment:

$$E(t_1, v_*) \leq f(t_*) - c(t_1 - t_*) \leq E_0 - cT + (B + c)(t_* - \tau).$$

Taking t_* sufficiently close to τ makes the right side negative, again a contradiction. Each segment is a compact subset of Ω , so local Lipschitz regularity suffices. $\square$

Proof of Theorem 9.1. In the classical contradiction, Equation (9.6) gives $m/2 = 1 - d + \Gamma$. Proposition 9.15 therefore implies $\partial_a D(a, 0) \geq 1 + 3d$ almost everywhere. The profile is Lipschitz and $D(0, 0) \geq 0$ by continuity of the lower bound in Proposition 9.8. Integration gives $D(1, 0) \geq 1 + 3d$. But the endpoint bound gives

$$D(1, 0) \leq p + d = 1 + 3d - 2\Gamma,$$

a contradiction. Hence $\Gamma_{\text{cl,round}} \leq 100\eta$.

For waves suppose the exponent is larger than 100η ; the classical conclusion just obtained supplies the strict comparison needed in Proposition 9.3. Put $t = 1 - a$ and use the same letter E for the excess in these coordinates. It is Lipschitz, nonnegative, and zero on $v = t/2$. The version of m from Lemma 9.12 is nondecreasing in t at fixed v . Equations (rv) and (round-out) imply

$$E_t = 1 + 3d - 2\Gamma - o_T, \quad E_v \leq m + o_T - d - 3.$$

If $m \leq 2 - 2d$, then $o_T \geq 4d + m/2 \geq m + 5d - 1$, so

$$E_t + \frac{1}{2}E_v \leq -2\Gamma.$$

If $m \geq 2 - 2d$, then $o_T \geq 1 + 3d$, so $E_t \leq -2\Gamma$. Apply Lemma 9.16 with $c = 2\Gamma > 0$ and $q = 2 - 2d$. This is a contradiction. The free wave exponent is therefore at most 100η . The finite-stack comparison completes all assertions of the theorem. $\square$

10 The physical half-wave and frequency summation

We now deduce Theorem 1.1 from the wave assertion of Theorem 9.1. We use the negative half-wave

$$Wg(x, \tau) = (2\pi)^{-3} \int_{\mathbb{R}^3} e^{i(x \cdot \xi - \tau|\xi|)} \widehat{g}(\xi) d\xi.$$

Complex conjugation, followed by the change of variable $\xi \mapsto -\xi$, interchanges the two signs. The same conjugation commutes with every real radial Sobolev multiplier.

There are two points to verify before the round-cone estimate applies to the physical wave: the packet state must have the required regularity on the original time slice, and its terminal masses must control the physical L^3 norm. We prove both at one frequency scale, then sum over spatial pieces, amplitude levels, and frequency annuli.

10.1 The estimate on one annulus

For $0 < H \leq 1$, let a_H be a smooth multiplier supported where $c \leq H|\xi| \leq C$. All derivatives of $a_H(H^{-1}\cdot)$ are assumed uniformly bounded. Write

$$W_H g(x, \tau) = (2\pi)^{-3} \int e^{i(x \cdot \xi - \tau|\xi|)} a_H(\xi) \hat{g}(\xi) d\xi. \quad (10.1)$$

The constants below may depend on the fixed annulus and on these smoothness bounds. They do not depend on H or on g .

Proposition 10.1 (Annular estimate). *For every $\delta > 0$,*

$$\|W_H g\|_{L^3(\mathbb{R}^3 \times [1,2])} \leq C_\delta H^{-\delta} \|g\|_{L^3(\mathbb{R}^3)}. \quad (10.2)$$

It suffices to prove the proposition for dyadic H : any H is comparable to a dyadic parameter, and replacing it by that parameter changes the fixed support and derivative bounds by fixed constants. We prove the proposition in several steps. A finite smooth angular partition and rotations reduce its proof to multipliers for which

$$\rho = \frac{|\xi| + \xi_3}{2}, \quad A = \frac{\xi_\perp}{2\rho}, \quad c_0 \leq H\rho \leq C_0, \quad |A| \leq A_0. \quad (10.3)$$

Such a partition exists because each direction has a neighborhood on which, after a rotation, $|\xi| + \xi_3$ is comparable to $|\xi|$.

We can also restrict attention to bounded input and output regions. Indeed, if $K_{H,\tau}$ is the convolution kernel of Equation (10.1), then, for every N , there is a fixed R such that

$$|K_{H,\tau}(x)| \leq C_N H^N (1 + |x|)^{-N} \quad (|x| \geq R, \ 1 \leq \tau \leq 2). \quad (10.4)$$

To see this, substitute $\xi = H^{-1}\omega$. The gradient of the phase in ω is $H^{-1}(x - \tau\omega/|\omega|)$, whose magnitude is comparable to $H^{-1}(1 + |x|)$ in the indicated region. Repeated integration by parts absorbs the initial factor H^{-3} and proves the estimate. Partition output space into unit cubes Q_j , and let Q_j^* be fixed enlargements large enough that $|x - z| \geq R$ whenever $x \in Q_j$ and $z \notin Q_j^*$. On each Q_j replace g by $g\mathbf{1}_{Q_j^*}$. The error is pointwise bounded by the convolution of $|g|$ with the right side of Equation (10.4). Its L^3 norm is $O_N(H^N)\|g\|_3$ by Young's inequality. The cubes Q_j^* have bounded overlap. Consequently a uniform local estimate can be cubed and summed over j . Translation reduces these local estimates to input supported in one fixed compact set K and output in one fixed bounded set.

10.2 The null-coordinate initial state

Set

$$t = \tau + x_3, \quad b = (x_1, x_2), \quad y = \tau - x_3. \quad (10.5)$$

The absolute spacetime Jacobian is $dt db dy = 2 d\tau dx$. In the frequency variables $(\eta, \rho) = (\xi_\perp, (|\xi| + \xi_3)/2)$, one has

$$\xi_3 = \rho - \frac{|\eta|^2}{4\rho}, \quad |\xi| = \rho + \frac{|\eta|^2}{4\rho}, \quad J(\eta, \rho) := \frac{\partial \xi_3}{\partial \rho} = 1 + \frac{|\eta|^2}{4\rho^2} = 1 + |A|^2. \quad (10.6)$$

Thus the same function, denoted by v , is exactly

$$v(t, b, y) = (2\pi)^{-3} \int e^{i(b \cdot \eta - y\rho - t|\eta|^2/(4\rho))} J(\eta, \rho) a_H(\xi(\eta, \rho)) \hat{g}(\xi(\eta, \rho)) d\eta d\rho. \quad (10.7)$$

It is therefore a free round-cone wave. Its energy is independent of t and satisfies

$$e := \|v(t)\|_2^2 = (2\pi)^{-3} \int J(\eta(\xi), \rho(\xi)) |a_H(\xi) \hat{g}(\xi)|^2 d\xi \leq C \|g\|_2^2. \quad (10.8)$$

In particular the map from g to any plateau-analyzed state $V_1 v(T)$ has uniformly bounded L^2 norm. This assertion remains true after replacing g by its restriction to any measurable subset of K .

The bounded physical output region is covered by finitely many unit intervals $[T, T+1]$ in t , with bounded T . We work on one such interval. Write $B = \|g\|_\infty$, and suppose first that $g \in L^\infty$ is supported in K . If $B = 0$ or $e = 0$, the asserted estimate is immediate. The frequency support in $(\eta, -\rho)$ has volume $O(H^{-3})$, so Cauchy–Schwarz in Fourier space gives

$$\|v(t)\|_\infty \leq CH^{-3/2} \sqrt{e}. \quad (10.9)$$

If $e < H^{10} B^2$, then

$$\int_T^{T+1} \int |v|^3 \leq CH^{-3/2} e^{3/2} \leq CB \|g\|_2^2. \quad (10.10)$$

We may therefore assume

$$H^{10} B^2 \leq e \leq C \|g\|_2^2 \leq C_K B^2. \quad (10.11)$$

10.3 Input regularity on the physical initial slice

The physical initial slice $\tau = 0$ is the spacelike plane $y = -t$. We verify the full regularity family on this plane, including the nonround tests. After a kernel truncation uniform over all rows, let $E \subset K$ comprise the physical initial points joined to a test by retained kernel pairs. Equation (10.8) then bounds the test's row mass by $CB^2|E|$, up to an error negligible relative to $\|g\|_2^2$. We therefore need a uniform kernel-localization estimate and a bound for $|E|$ by the test weight $r^4 f^{1-\eta}$, allowing an arbitrarily small power loss.

Lemma 10.2 (Uniform packet localization from the initial slice). *Fix $\gamma_0 > 0$ and put $\Lambda = H^{-\gamma_0}$. The operator $g \mapsto V_1 v(T)$ can be truncated to the following relation between an input point $z = (z_\perp, z_3) \in K$ and a row (A, b, y) :*

$$\begin{aligned} |b - z_\perp - (T - z_3)A| &\leq C\Lambda\sqrt{H}, \\ |y + z_3 - (T - z_3)|A|^2 - 2A \cdot (b - z_\perp - (T - z_3)A)| &\leq C\Lambda H. \end{aligned} \quad (10.12)$$

For every N , the discarded operator has norm at most $C_N H^N$ from $L^2(K)$ to the row space. The constant is independent of any regularity test subsequently imposed on the output rows. The kernel vanishes unless $|A| \leq A_0 + C_\psi \sqrt{H}$, where C_ψ is a radius containing the support of the fixed angular window ψ .

Proof. Insert the Fourier representation of $\hat{g}$ into Equation (10.7) and the plateau analysis. The kernel has phase

$$(b - z_\perp) \cdot \eta - (y + z_3)\rho - (T - z_3) \frac{|\eta|^2}{4\rho}.$$

Its amplitude is supported in $H\rho \asymp 1$ and $|\eta/(2\rho) - A| \leq C_\psi \sqrt{H}$. Together with $|\eta/(2\rho)| \leq A_0$ from Equation (10.3), this proves the stated angular support bound. On making the substitutions $\eta = 2\rho A + H^{-1/2}\omega$ and $\sigma = H\rho$, the frequency Jacobian is H^{-2} , and the amplitude has uniformly bounded derivatives on a fixed compact set. The quadratic remainder in the phase also has bounded derivatives, because $T - z_3$ is bounded. Fourier integration by parts therefore gives

$$|\mathcal{K}_T(A, b, y; z)| \leq C_L H^{-2} \left(1 + \frac{|\Delta b|}{\sqrt{H}} + \frac{|\Delta y|}{H} \right)^{-L},$$

where Δb and Δy are the two differences in Equation (10.12). Only bounded A occur. The row measure at length 1 is $H^{-1} dA db dy$. For fixed A, z , integration in (b, y) costs the packet volume H^2 . Squaring the kernel and then integrating over all rows and $z \in K$ thus gives a Hilbert–Schmidt bound of size at most a fixed power of H^{-1} . Outside the displayed relation, the decay factor adds an arbitrarily large inverse power of Λ . Since γ_0 is fixed, taking L sufficiently large proves any prescribed H^N bound. This estimate was made over all output rows, which proves the asserted uniformity. $\square$

Lemma 10.3 (Volume of physical predecessors). *Fix $0 < \eta < 1$. Consider any round-cone regularity test at time T and length 1, of angular width r and relative thickness f , where $0 < f \leq 1$ and $rf \geq \sqrt{H}$; for a round test set $f = 1$. Let E be the set of $z \in K$ related by Equation (10.12) to some row of the test whose angular coordinate satisfies $|A| \leq A_0 + C_\psi \sqrt{H}$. There is a fixed C_1 such that*

$$|E| \leq C \Lambda^{C_1} r^4 f^{1-\eta}. \quad (10.13)$$

The estimate is uniform in the center, r , and f of the test.

Proof. Use its normalized coordinates

$$u = t - T, \quad x = \frac{b - b_* - uA_*}{r}, \quad w = \frac{y - y_* - 2A_* \cdot (b - b_*) + u|A_*|^2}{r^2}, \quad V = \frac{A - A_*}{r}.$$

At $u = 0$, the test has $|x|, |w|, |V| \leq 1$. The prediction at T from a compatible input point differs from its tested row by

$$|\delta x| \leq C \Lambda \frac{\sqrt{H}}{r} \leq C \Lambda f, \quad |\delta w| \leq C \Lambda \left(\frac{H}{r^2} + |V| \frac{\sqrt{H}}{r} \right) \leq C \Lambda f.$$

Backflow to $t = z_3$ has bounded duration. Consequently the compatible normalized coordinates are bounded by a fixed power of Λ . For a nonround test, write its polynomial as

$$F = c + du + \alpha \cdot x + \beta w + k(uw - |x|^2), \quad G = F_u + V \cdot F_x + |V|^2 F_w.$$

The coefficients are bounded, their discriminant is 1, G is constant on rays, and F is affine on rays. Expanding at the tested row and then backflowing shows that, at a compatible input point,

$$|F| + |G| \leq C\Lambda^{C_2} f. \quad (10.14)$$

Here and below the exponents of Λ are fixed geometric constants.

On the physical input plane the normalized variables satisfy

$$(1 + |A_*|^2)u = -T - y_* - 2rA_* \cdot x - r^2w. \quad (10.15)$$

The absolute Jacobian of $(x, w) \mapsto (z_\perp, z_3)$ is

$$\frac{r^4}{1 + |A_*|^2}. \quad (10.16)$$

Indeed $z_3 = T + u$ and $z_\perp = b_* + uA_* + rx$; subtracting A_* times the last row from the first two rows of their derivative matrix leaves determinant $r^2\partial_w u$. The crude volume of the bounded normalized box proves the lemma for round tests. It also proves it whenever f is larger than a sufficiently small fixed inverse power of Λ .

For the remaining f , the normalized coordinates are bounded by constants, and Equation (10.14) is as small as needed. The discriminant identity reads

$$|F_x + 2VF_w|^2 = 1 + 4(\beta + ku)G - 4kF. \quad (10.17)$$

It implies $|F_x + 2VF_w| \geq 1/2$. Let $h(x, w)$ be the restriction of F to Equation (10.15). Put

$$a = \frac{2rA_*}{1 + |A_*|^2}, \quad c_* = \frac{r^2}{1 + |A_*|^2}.$$

Then $h_x = F_x - aF_u$, $h_w = F_w - c_*F_u$, and

$$G = \frac{1 + |A_* + rV|^2}{1 + |A_*|^2} F_u + V \cdot h_x + |V|^2 h_w. \quad (10.18)$$

If r is bounded by a fixed constant, a nonempty compatible set has bounded A_* , because its actual row velocities $A = A_* + rV$ are bounded. The coefficient of F_u in Equation (10.18) is then bounded below. Together with Equation (10.17), this gives $|\nabla h| \geq c > 0$ once $|G|$ is sufficiently small.

For large r , replace the angular center by 0. This is a boost of size A_*/r in normalized coordinates; that size is bounded because $|A_*| \leq r + A_0 + C_\psi$. The boost preserves the discriminant and changes all bounded coordinates and coefficients by bounded amounts. Now $|V| \leq C/r$, while $|F_w| = |\beta + ku| \leq C$. Equation (10.17) gives $|F_x| \geq 1/4$ for all sufficiently large r . In these coordinates u in Equation (10.15) is independent of x , so $h_x = F_x$. Choosing the cutoff between the two ranges once and for all proves a uniform lower bound for the restricted gradient.

The polynomial h has degree at most two. In a fixed bounded box, the set $\{|h| \leq q, |\nabla h| \geq c\}$ has volume at most Cq , uniformly in its coefficients: partition according to a coordinate derivative of magnitude at least $c/\sqrt{3}$, hold the other coordinates fixed, and split the one-dimensional fiber at the zeros of that derivative. There are at most two monotonic pieces, on each of which the mean value theorem gives length $O(q)$. Applying this with $q = C\Lambda^{C_2} f$, and then using Equation (10.16) (or r^4 after recentering), gives $|E| \leq C\Lambda^{C_3} r^4 f$ in the small- f case. Since $f \leq f^{1-\eta}$, this suffices. In the complementary case the missing factor $f^{1-\eta}$ costs only another fixed power of Λ , as already noted. $\square$

Proposition 10.4 (Regularity of the initial analyzed state). *Fix $0 < \eta < 1$ as in Definition 8.2. For every $\gamma > 0$, the state in Equation (10.7) has input regularity constant*

$$\kappa = C_{\eta,\gamma} H^{-\gamma} B^2 \quad (10.19)$$

at each of the bounded starting times T . Under Equation (10.11), it also satisfies the polynomial-size and negligible-tail hypotheses of the round-wave experiment with terminal length $s = H$.

Proof. Choose $\gamma_0 > 0$ so small that $C_1 \gamma_0 < \gamma/2$ in Lemma 10.3. For a test $\mathcal{T}$, restrict g to its predecessor set E . The truncated operator of Lemma 10.2 sends $g \mathbf{1}_{K \setminus E}$ to zero on $\mathcal{T}$. The unrestricted operator on $g \mathbf{1}_E$ has uniformly bounded L^2 norm by Equation (10.8). Thus, for every N ,

$$\mu_T(\mathcal{T}) \leq C \|g \mathbf{1}_E\|_2^2 + C_N H^N \|g\|_2^2 \leq C B^2 H^{-C_1 \gamma_0} r^4 f^{1-\eta} + C_N H^N B^2.$$

The uncertainty condition gives the uniform weight floor

$$r^4 f^{1-\eta} = (rf)^4 f^{-3-\eta} \geq H^2. \quad (10.20)$$

Taking N large proves regularity with Equation (10.19), including every admissible width.

The ratios e/κ and κ/e are bounded above by fixed powers of H^{-1} by Equation (10.11). The kernel proof gives negligible phase tails outside a fixed polynomial box: the input is compact, all velocities and durations are bounded, and arbitrary decay is available. Its pointwise kernel bound also gives a phase-density bound by a fixed power of H^{-1} times B^2 , and hence times e . All these estimates are uniform along any sequence of the present inputs. $\square$

10.4 Recovering the physical norm from leaf masses

Lemma 10.5 (Synthesis at the terminal scale). *Let v be the wave in Equation (10.7), and let Ω be a fixed bounded subset of (b, y) space. At each left endpoint $T_j = T + jH$ in $[T, T + 1]$, use a plateau analysis of length H , and group its row masses into spatial H -grid cubes. Restrict to a fixed enlargement of Ω , and denote the resulting masses by $m_{j,Q}$. They are admissible round-test assignments with weights bounded above and below by positive constants. For every $0 < \sigma < 1$ and N ,*

$$\int_T^{T+1} \int_{\Omega} |v|^3 \leq C_{\sigma} H^{-C_4 \sigma} H^{-1/2} \sum_{j,Q} m_{j,Q}^{3/2} + C_{N,\sigma} H^N e^{3/2}. \quad (10.21)$$

Here H is dyadic, so the time partition is exact.

Proof. At length H , the angular aperture is $d_H = 1$ and both spatial packet widths are H . All row angles lie in a fixed bounded set. A fixed round test with $r \geq 1$, centered at a given cube, contains that cube and all these angles. Its weight r^4 is a fixed constant, and different cube assignments are disjoint in row space.

Synthesize the plateau analysis and propagate for $0 \leq h \leq H$. The same frequency calculation as in Lemma 10.2, now with frequency boxes of volume $O(H^{-3})$, gives a kernel bounded by

$$C_L H^{-3} \left(1 + \frac{|(b, y) - (b', y')|}{H} \right)^{-L}.$$

The bounded ray displacement during h is $O(H)$ and has been absorbed in this bound. Its squared integral over source rows is $O(H^{-3})$. For an output point in a cube Q , retain only rows

in cubes within distance $CH^{1-\sigma}$ of Q . Cauchy–Schwarz then bounds the retained contribution by

$$CH^{-3/2} \left(\sum_{Q' \sim Q} m_{j,Q'} \right)^{1/2}.$$

The omitted contribution is at most $C_{N,\sigma} H^N \sqrt{e}$ pointwise, by arbitrary kernel decay. For H sufficiently small, all retained cubes lie in the chosen fixed enlargement of Ω . There are $O(H^{-3\sigma})$ neighbors of any Q , with the same bound on overlap. The spacetime cell over Q has volume H^4 . Cubing the pointwise estimate, integrating over that cell, and using

$$\left(\sum_{Q' \sim Q} m_{j,Q'} \right)^{3/2} \leq \#\{Q' \sim Q\}^{1/2} \sum_{Q' \sim Q} m_{j,Q'}^{3/2}$$

proves Equation (10.21); for example, any fixed $C_4 \geq 5$ suffices after harmless enlargement. There are only $O(H^{-4})$ spacetime cells, and arbitrary decay absorbs their number in the error term. $\square$

We explain explicitly how to use the exponent formulation of Theorem 9.1. Its wave bound, with regularity parameter η , is $\Gamma_{\text{wav,round}} \leq 100\eta$. Hence for the uniform class of experiments constructed above and every $\zeta > 0$,

$$H^{-1/2} \sum_{j,Q} m_{j,Q}^{3/2} \leq C_{\eta,\gamma,\zeta} H^{-100\eta-\zeta} e\sqrt{\kappa}. \quad (10.22)$$

The bounded weights have been absorbed into the constant. For completeness, at each fixed H the normalized functional is at most $CH^{-3/2} \sqrt{e/\kappa} \leq CH^{-3/2}$: there are H^{-1} cuts and each has total assigned mass at most e . Thus, if no constant in Equation (10.22) existed, there would be a sequence $H \downarrow 0$ along which the normalized functional exceeds $H^{-100\eta-\zeta}$. The regularity and tail estimates of Proposition 10.4 give one admissible polynomial sequence with all fixed setup constants uniform. Its upper growth exponent would be at least $100\eta + \zeta$, contrary to Theorem 9.1.

Combining Equations (10.21), (10.22), and (10.19) gives

$$\int_T^{T+1} \int_{\Omega} |v|^3 \leq CH^{-100\eta-\zeta-\gamma/2-C_4\sigma} B \|g\|_2^2. \quad (10.23)$$

The error term is absorbed here using $e^{3/2} \leq CB \|g\|_2^2$. The small-energy case was already handled in Equation (10.10). Given any $\beta > 0$, first take $100\eta < \beta/4$, then $\zeta < \beta/4$, $\gamma/2 < \beta/4$, and $C_4\sigma < \beta/4$; finally choose γ_0 as in Proposition 10.4. Thus the loss in Equation (10.23) is smaller than any prescribed β . The finitely many null-time intervals and the constant Jacobian in Equation (10.5) do not change this conclusion.

Proof of Proposition 10.1. It remains to pass from bounded amplitude pieces to arbitrary localized input. Normalize $\|g\|_3 = 1$ with $\text{supp } g \subset K$. Fix an integer $M > 4$. The parts where $|g| < H^M$ and where $|g| > H^M$ have L^1 norms at most $|K|H^M$ and H^{2M} , respectively. The elementary bound $\|K_{H,\tau}\|_{\infty} \leq CH^{-3}$ shows that their contributions on the bounded output region are uniformly bounded, indeed tend to zero. No frequency restriction has been applied to g before making these spatial and amplitude restrictions.

Split the remaining part into disjoint dyadic amplitude pieces g_ν . Their number is $L = O_M(1 + \log H^{-1})$, and, with $B_\nu = \|g_\nu\|_\infty$,

$$B_\nu \|g_\nu\|_2^2 \leq 2 \|g_\nu\|_3^3, \quad \sum_\nu \|g_\nu\|_3^3 \leq 1.$$

Equation (10.23), with parameters making its exponent less than δ , and the inequality

$$\left\| \sum_{\nu=1}^L h_\nu \right\|_3^3 \leq L^2 \sum_\nu \|h_\nu\|_3^3$$

give the local cubed bound $CL^2 H^{-\delta}$. Since $L^2 \leq C_\delta H^{-\delta}$, taking a cube root yields a loss at most $H^{-2\delta/3} \leq H^{-\delta}$. Homogeneity removes the normalization. Sum the cubed estimates over the output cubes, using the bounded overlap of their enlarged input cubes and Equation (10.4). Finally sum the fixed angular partition. This proves Proposition 10.1. $\square$

10.5 Sobolev summation and the remaining exponents

Proof of Theorem 1.1. Let $h = J^\varepsilon f$. Use a smooth radial partition into a low-frequency cutoff and annuli $|\xi| \asymp 2^n$, $n \geq 0$. On the n th annulus put the multiplier $J^{-\varepsilon}$ into a_H , where $H = 2^{-n}$. After factoring out H^ε , the rescaled symbols have uniformly bounded derivatives. Proposition 10.1, applied with $\delta = \varepsilon/2$, gives

$$\|W(P_n J^{-\varepsilon} h)\|_{L_{x,\tau}^3(\mathbb{R}^3 \times [1,2])} \leq C_\varepsilon 2^{-n\varepsilon/2} \|h\|_3.$$

The series is summable by the triangle inequality.

We also record a direct justification at frequency zero. For a smooth cutoff χ supported in $|\xi| \leq 2$, decompose

$$\chi(\xi) e^{-i\tau|\xi|} = \chi(\xi) + \chi(\xi)(e^{-i\tau|\xi|} - 1).$$

The first term has an integrable convolution kernel. Split the second into annuli of radius $\lambda = 2^{-j} \leq 1$. In variables $\xi = \lambda\omega$, its symbol and its derivatives through order 4 are $O(\lambda)$ on a fixed annulus, uniformly for $1 \leq \tau \leq 2$. Integration by parts using $(1 - \Delta_\omega)^2$ shows that its inverse Fourier transform has L^1 norm $O(\lambda)$. These norms sum. The smooth low-frequency factor $J^{-\varepsilon}$ preserves the same bounds. Young's inequality therefore handles the low-frequency term. The resulting inequality is exactly the negative-sign version of Theorem 1.1; complex conjugation proves the stated sign. $\square$

Corollary 10.6 (The positive-regularity local-smoothing range). *For every $2 < p < \infty$ and $\alpha > \max\{0, 1 - 3/p\}$,*

$$\|Uf\|_{L^p(\mathbb{R}^3 \times [1,2])} \leq C_{p,\alpha} \|J^\alpha f\|_{L^p(\mathbb{R}^3)} \quad (f \in \mathcal{S}(\mathbb{R}^3)). \quad (10.24)$$

Proof. Take a smooth radial annular cutoff $P_\lambda = \phi(|D|/\lambda)$, $\lambda \geq 1$. Plancherel gives

$$\|UP_\lambda\|_{L_x^2 \rightarrow L_{x,\tau}^2} \leq C,$$

and Proposition 10.1 gives, for every $\delta > 0$, $\|UP_\lambda\|_{L_x^3 \rightarrow L_{x,\tau}^3} \leq C_\delta \lambda^\delta$. We first prove the remaining endpoint bound

$$\|UP_\lambda\|_{L_x^\infty \rightarrow L_{x,\tau}^\infty} \leq C\lambda. \quad (10.25)$$

The radial kernel, at $r = |x| > 0$, is a fixed constant times

$$\frac{1}{r} \int_0^\infty e^{i\tau\rho} \rho \phi(\rho/\lambda) \sin(r\rho) d\rho.$$

Put $I(s) = \int_0^\infty e^{is\rho} \rho \phi(\rho/\lambda) d\rho$. For every N ,

$$|I(s)| \leq C_N \lambda^2 (1 + \lambda|s|)^{-N}, \quad |I'(s)| \leq C_N \lambda^3 (1 + \lambda|s|)^{-N}.$$

The kernel is a constant times $r^{-1}(I(\tau+r) - I(\tau-r))$. If $0 < r \leq 1/2$, the mean value theorem and $1 \leq \tau \leq 2$ bound it by $C_N \lambda^{3-N}$, which also removes the apparent singularity at $r = 0$. If $r \geq 1/2$, its absolute value is at most

$$C_N \frac{\lambda^2}{r} ((1 + \lambda|\tau-r|)^{-N} + (1 + \lambda(\tau+r))^{-N}).$$

Multiplying by r^2 and integrating in r gives $C\lambda$: the first summand uses $\int_{1/2}^\infty r(1 + \lambda|\tau-r|)^{-N} dr \leq C/\lambda$, and the second is smaller. Hence the kernel has L^1 norm at most $C\lambda$, proving Equation (10.25).

Interpolation between the L^2 and L^3 estimates gives an arbitrarily small positive power of λ for every $2 < p \leq 3$. For $p > 3$, interpolate the L^3 bound with Equation (10.25); the L^3 interpolation weight is $3/p$, giving

$$\|UP_\lambda\|_{L_x^p \rightarrow L_{x,\tau}^p} \leq C_{p,\delta} \lambda^{1-3/p+3\delta/p}.$$

Choose δ small enough that this exponent, or its counterpart for $p \leq 3$, is strictly below α . The same estimates hold for uniformly smooth radial annular symbol families, as both the kernel proof and Proposition 10.1 are uniform for those families. We may thus insert $J^{-\alpha}$ into each annular symbol, factor out $\lambda^{-\alpha}$, and sum the resulting geometric series. The low-frequency kernels have uniformly bounded L^1 norms by the preceding proof, which applies to every L^p . This proves Equation (10.24). $\square$

The same estimate describes the wave equation with independent initial displacement u_0 and velocity u_1 . For Schwartz data its solution is

$$u(t) = \cos(t|D|)u_0 + \frac{\sin(t|D|)}{|D|}u_1,$$

and for the exponents in Corollary 10.6,

$$\|u\|_{L^p(\mathbb{R}^3 \times [1,2])} \leq C_{p,\alpha} (\|J^\alpha u_0\|_p + \|J^{\alpha-1} u_1\|_p).$$

At high frequency this follows from the two signs of the half-wave estimate and the order -1 velocity multiplier. At low frequency, retain the combined multiplier $\sin(t|\xi|)/|\xi|$: it is smooth at zero and, after a smooth frequency cutoff, has a uniformly integrable kernel for $1 \leq t \leq 2$. Density extends the solution estimate to the indicated Sobolev data.

10.6 Maximal half-waves and convergence to the initial data

The spacetime estimate also controls the largest value of the wave over a finite time interval, at the cost of additional regularity. This gives a pointwise interpretation of the evolution for Sobolev initial data. For $s \in \mathbb{R}$ and $1 < p < \infty$, write $W^{s,p}(\mathbb{R}^3) = \{f \in \mathcal{S}'(\mathbb{R}^3) : J^s f \in L^p(\mathbb{R}^3)\}$, with norm $\|J^s f\|_p$.

Corollary 10.7 (Maximal half-wave and pointwise convergence). *Let $3 \leq p < \infty$ and $s > 1 - 2/p$. Then*

$$\left\| \sup_{0 < t < 1} |Uf(\cdot, t)| \right\|_{L^p(\mathbb{R}^3)} \leq C_{p,s} \|J^s f\|_{L^p(\mathbb{R}^3)} \quad (f \in \mathcal{S}(\mathbb{R}^3)). \quad (10.26)$$

The family has a bounded maximal extension to $W^{s,p}(\mathbb{R}^3)$, and in that extension every $f \in W^{s,p}(\mathbb{R}^3)$ satisfies

$$\lim_{t \downarrow 0} Uf(x, t) = f(x) \quad \text{for almost every } x \in \mathbb{R}^3. \quad (10.27)$$

Proof. For $p \geq 3$, Corollary 10.6 gives the $1/p$ -local smoothing gain relative to the fixed-time loss $1 - 2/p$. The established maximal implication [2, Section 3.5, Equation (3.24)] therefore gives (10.26). At each frequency, Sobolev embedding in time costs arbitrarily more than $1/p$ derivatives; together with local smoothing, this requires regularity arbitrarily close to $1 - 2/p$ from above. The strict inequality on s absorbs the remaining positive losses. The implication uses scaling to cover the time intervals approaching zero.

For Schwartz data, the Fourier integral converges uniformly to the initial data as $t \downarrow 0$. Schwartz functions are dense in $W^{s,p}$, and the maximal bound gives the extension and carries this convergence to every such datum. More explicitly, for a Schwartz approximation g to f , the measure of the set where $\limsup_{t \downarrow 0} |Uf - f| > \eta$ is at most $C_{p,s} \eta^{-p} \|J^s(f - g)\|_p^p$. Letting $g \rightarrow f$ proves (10.27). $\square$

11 Bochner–Riesz consequences

We apply the established local-smoothing implications of Beltran–Hickman–Sogge to Corollary 10.6. The maximal result concerns the family in (1.3); the subsequent nonmaximal result concerns the individual linear means. These means recover a function by gradually admitting higher frequencies, with order δ smoothing the boundary of the Fourier ball. Uniform L^p bounds make each approximation stable. A maximal bound is stronger: it controls the error for all radii at once, which allows the elementary convergence for Schwartz data to pass to every L^p datum by density.

For $p \geq 3$, the local-smoothing input has an arbitrarily small positive loss beyond $1 - 3/p$. The strict inequality $\delta > 1 - 3/p$ absorbs that loss in the established implication. Thus the conclusion below requires only L^p integrability of the data, in contrast to the Sobolev hypothesis in the maximal half-wave corollary.

Theorem 11.1 (Maximal Bochner–Riesz bounds and almost-everywhere summation). *Let $3 \leq p < \infty$ and $\delta > 1 - 3/p$. Then*

$$\|B_*^\delta f\|_{L^p(\mathbb{R}^3)} \leq C_{p,\delta} \|f\|_{L^p(\mathbb{R}^3)} \quad (f \in \mathcal{S}(\mathbb{R}^3)). \quad (11.1)$$

The family has a bounded maximal extension to $L^p(\mathbb{R}^3)$, and in that extension every $f \in L^p(\mathbb{R}^3)$ satisfies

$$\lim_{R \rightarrow \infty} B_R^\delta f(x) = f(x) \quad \text{for almost every } x \in \mathbb{R}^3. \quad (11.2)$$

In particular, every $\delta > 0$ is allowed when $p = 3$.

Proof. For $p \geq 3$, the sharp fixed-time Sobolev loss is $1 - 2/p$. Corollary 10.6 permits every exponent $\alpha = 1 - 3/p + \varepsilon$, $\varepsilon > 0$, and hence gives the $1/p$ -local smoothing hypothesis for the Euclidean half-wave in [2, Proposition 3.3]. That proposition applies for $2n/(n-1) \leq p < \infty$ and order above

$$\max \left\{ n \left| \frac{1}{p} - \frac{1}{2} \right| - \frac{1}{2}, 0 \right\}.$$

For $n = 3$ and $p \geq 3$ this threshold is $1 - 3/p$. The arbitrary positive loss in the local-smoothing input is absorbed by the strict inequality on δ .

The cited proposition uses the multiplier $(1 - t|\xi|)_+^\delta$. Write $\mathcal{L}_R^\delta$ for the same means with $t = R^{-1}$ and $\mathcal{L}_*^\delta f = \sup_{R>0} |\mathcal{L}_R^\delta f|$. It gives the asserted strong bound for $\mathcal{L}_*^\delta$. Here is the transfer to the squared-radius convention in (1.3). Choose $\chi \in C_c^\infty((0, \infty))$ equal to one on a neighborhood of $[1/2, 1]$, and put

$$a(\xi) = \chi(|\xi|)(1 + |\xi|)^\delta, \quad b(\xi) = (1 - \chi(|\xi|))(1 - |\xi|^2)_+^\delta.$$

Both a and b are smooth and compactly supported on $\mathbb{R}^3$, and

$$\left(1 - \frac{|\xi|^2}{R^2} \right)_+^\delta = a(\xi/R) \left(1 - \frac{|\xi|}{R} \right)_+^\delta + b(\xi/R).$$

The inverse Fourier transforms of a and b are Schwartz functions. Their dilates are bounded in absolute value by $C_N R^3 (1 + R|x|)^{-N}$ for any fixed $N > 3$, so the Hardy–Littlewood maximal operator $\mathcal{M}$ gives the pointwise estimate

$$B_*^\delta f \leq C_\delta (\mathcal{M}(\mathcal{L}_*^\delta f) + \mathcal{M}f).$$

Its L^p boundedness for $p > 1$ proves (11.1).

For $g \in \mathcal{S}(\mathbb{R}^3)$, dominated convergence in the Fourier integral gives $B_R^\delta g \rightarrow g$ uniformly as $R \rightarrow \infty$. For $f \in L^p$, choose $g_j \in \mathcal{S}$ converging to f in L^p with $\sum_j \|g_{j+1} - g_j\|_p < \infty$. The maximal inequality places $\sum_j B_*^\delta(g_{j+1} - g_j)$ in L^p . Thus $B_R^\delta g_j$ converges uniformly in R for almost every x , defining the usual maximal extension; for each fixed R it agrees with the bounded L^p extension of B_R^δ . For $g \in \mathcal{S}$,

$$\limsup_{R \rightarrow \infty} |B_R^\delta f - f| \leq B_*^\delta(f - g) + |f - g| \quad \text{almost everywhere.}$$

Chebyshev's inequality and (11.1) bound the measure where this limsup exceeds $\eta > 0$ by $C_{p,\delta} \eta^{-p} \|f - g\|_p^p$. Density then proves (11.2). $\square$

Corollary 11.2 (The full strict nonmaximal range). *For $1 \leq p \leq \infty$, with $1/\infty = 0$, set*

$$\delta_{\text{BR}}(p) = \max \left\{ 3 \left| \frac{1}{p} - \frac{1}{2} \right| - \frac{1}{2}, 0 \right\}.$$

If $\delta > \delta_{\text{BR}}(p)$, the operators B_R^δ extend boundedly to $L^p(\mathbb{R}^3)$ with a bound uniform in $R > 0$:

$$\|B_R^\delta f\|_{L^p(\mathbb{R}^3)} \leq C_{p,\delta} \|f\|_{L^p(\mathbb{R}^3)}. \quad (11.3)$$

Proof. Proposition 3.2 of [2], with the same local-smoothing input and the same smooth-factor transfer, gives (11.3) for $3 \leq p < \infty$ and $\delta > 1 - 3/p$; dilation makes the constants uniform in R . For $2 \leq p \leq 3$, fix any $\delta > 0$ and interpolate the $p = 3$ bound at that order with the L^2 bound from Plancherel. This gives the required threshold $\delta_{\text{BR}}(p) = 0$ in this interval. The multiplier is real and even, so duality gives the range $1 < p < 2$, since $\delta_{\text{BR}}(p) = \delta_{\text{BR}}(p')$. Finally, at $p = 1$ and $p = \infty$ the stated condition is $\delta > 1$. The inverse Fourier transform K^δ of $(1 - |\xi|^2)_+^\delta$ obeys $|K^\delta(x)| \leq C_\delta(1 + |x|)^{-2-\delta}$ and is therefore integrable at these orders. Its dilates have the same L^1 norm, so Young's inequality supplies both endpoint bounds. $\square$

For $1 \leq p < \infty$ in this strict range, the uniform bounds also give $B_R^\delta f \rightarrow f$ in L^p as $R \rightarrow \infty$. One first checks this for Schwartz functions with compact Fourier support and then uses their density in L^p together with the uniform operator bound.

The interpolation and duality in this corollary apply to the fixed-radius linear operators; the maximal conclusion remains the range of Theorem 11.1. For comparison, the separate article [20, Theorem 1.1 and Corollary 12.4] proves positive-order L^3 boundedness and the same strict nonmaximal range by a different argument. That article is not an input to the deduction above.

11.1 Spherical restriction and Keakeya maximal estimates

The strict multiplier range has further established consequences for Fourier extension and tube geometry. Let $d\sigma$ be surface measure on S^2 , and define the spherical extension operator by

$$Eg(x) = \int_{S^2} e^{ix \cdot \omega} g(\omega) d\sigma(\omega).$$

Tao's Bochner–Riesz-to-restriction implication [29], also described in [2, Section 3.1], applies to Corollary 11.2. By duality it gives

$$\|Eg\|_{L^r(\mathbb{R}^3)} \leq C_r \|g\|_{L^r(S^2, d\sigma)}, \quad 3 < r < \infty. \quad (11.4)$$

Thus the multiplier bounds also control the physical-space concentration of waves with Fourier support on the sphere. The strict family of positive-order Bochner–Riesz bounds supplies the premise; no order-zero multiplier assertion is needed.

For $0 < h < 1$ and $\omega \in S^2$, let $\mathcal{T}_h(\omega)$ be the family of all unit-length tubes of radius h parallel to ω , at arbitrary spatial positions, and put

$$K_h F(\omega) = \sup_{T \in \mathcal{T}_h(\omega)} \frac{1}{|T|} \int_T |F(x)| dx.$$

The restriction-to-Keakeya implication of [31, Section 4, Proposition 4.1] applies to the diagonal extension range (11.4). For $r > 3$ sufficiently close to 3, it first gives restricted weak type (a, a) , where $a = r/(r - 2) < 3$, with norm $O(h^{-2(3/a-1)})$ for the tubes above. Interpolation with the trivial L^∞ bound gives strong type $(3, 3)$ with loss $h^{-2(1-a/3)}$. Since $a \uparrow 3$ as $r \downarrow 3$, this yields the Keakeya maximal estimate, for every $F \in L^3(\mathbb{R}^3)$,

$$\|K_h F\|_{L^3(S^2, d\sigma)} \leq C_\varepsilon h^{-\varepsilon} \|F\|_{L^3(\mathbb{R}^3)}, \quad \varepsilon > 0, \quad 0 < h < 1. \quad (11.5)$$

This quantitatively controls averages over thin tubes in every direction. In particular, a compact set containing a unit line segment in each direction has Hausdorff dimension three. Wang–Zahl already proved the Hausdorff and Minkowski dimension assertions [30, Theorem 1.1]; the maximal estimate is a stronger quantitative conclusion. The deductions here use the established Bochner–Riesz-to-restriction-to-Keakeya chain and also apply to the independent Bochner–Riesz result cited above.

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